Implements

Overview of all implements for the Bruut OpenAgbot: the 90 that are worked out in FreeCAD, with every variant on its own page, followed by the 0 ideas that are not worked out yet.

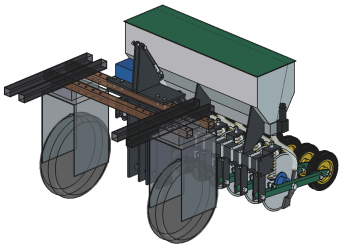
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    Weed & Crop Protection
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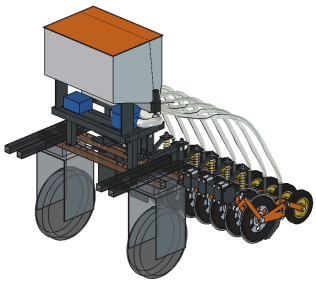
IMPLEMENT · SOWING & PLANTING

Overseeder

Grassland slit seeder. 1 m modules with 8 rows at 125 mm and coupling flanges. A thin disc cuts a 15 mm slot, the seed is placed about 12 mm deep and a sprung press wheel closes the slot. Every row follows the ground on its own arm; two gas springs push the floating lift frame down with robot weight.



Simple

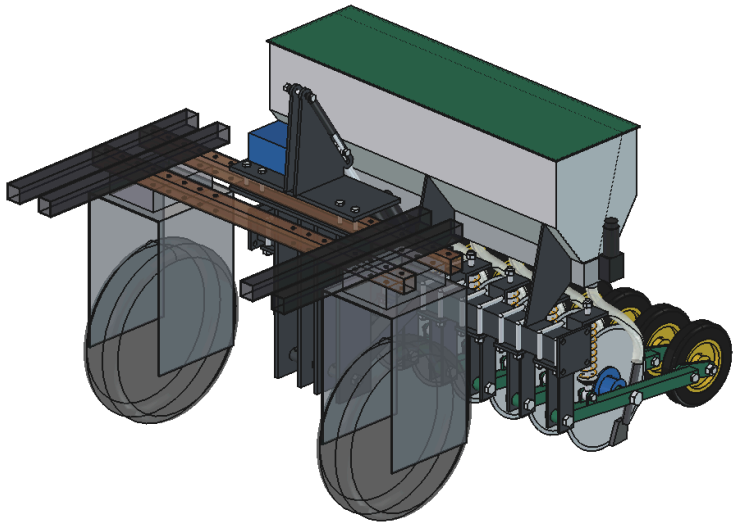


Advanced

2 variants: Simple · Advanced
Folder: `overseeder/`

OVERSEEDER · VARIANT 1 OF 2

Simple



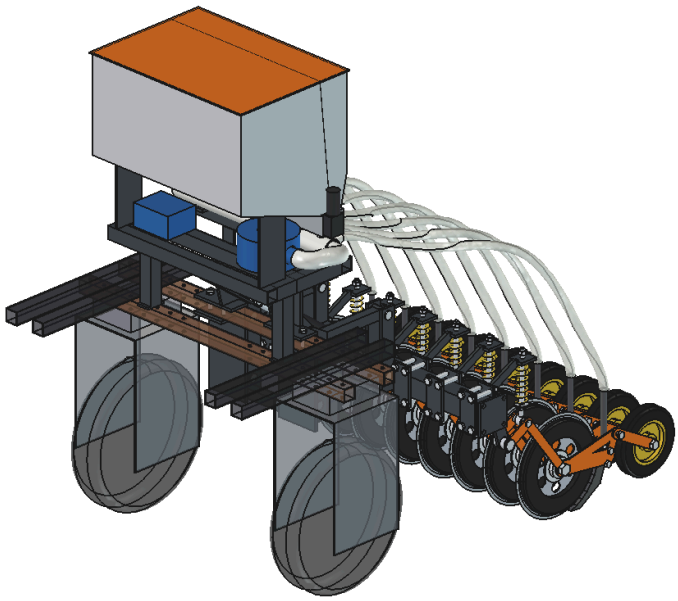
Angled single disc (7°, Ø300 × 3) with a seed boot in its lee and a depth ring. Gravity-fed seed hopper with a compartment for grass (41 l) and one for fine seed (18 l), each with its own cam roller and a motor that follows the driving speed. Built on the headstock and lift frame of the simple applicator v2.

- 8 rows at 125 mm = 1 m per module
- Draft force approx. 335 N (normal)
- Depth within ±3 mm: 94 %
- About €1,675 in parts

Folder: `overseeder/simple/`

OVERSEEDER · VARIANT 2 OF 2

Advanced



Parallel linkage so the toolbar stays parallel, straight Ø300 × 3 disc with depth rings on both sides, a curved seed coulter directly behind the disc and a press wheel in a fork. Air seeder (62 + 17 l hopper, 12 V fan) on a frame above the rear axle blows the seed through 8 hoses to the coulters.

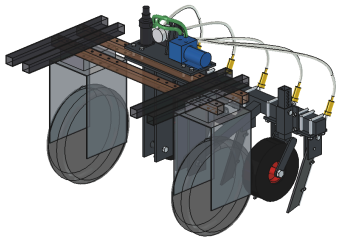
- 8 rows at 125 mm = 1 m per module
- 2 gas springs in the slotted hole
- Depth within ±3 mm: 98 %
- About €2,280 in parts

Folder: `overseeder/advanced/`

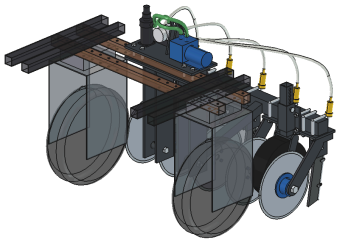
IMPLEMENT · SPRAYING, FERTILIZING & IRRIGATION

Liquid fertilizer applicator

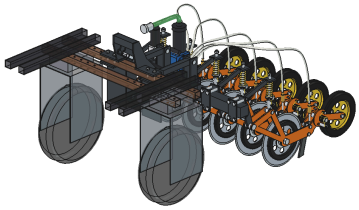
Injects liquid fertilizer (such as RENURE) just under the sward, without mist or drift. Concept versions (simple, simple v2, advanced) and the precision applicator, the first production version with a roller pump.



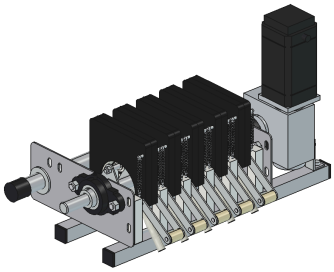
Simple



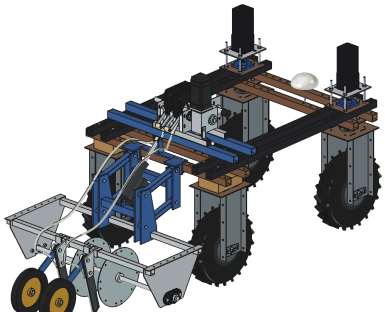
Simple v2



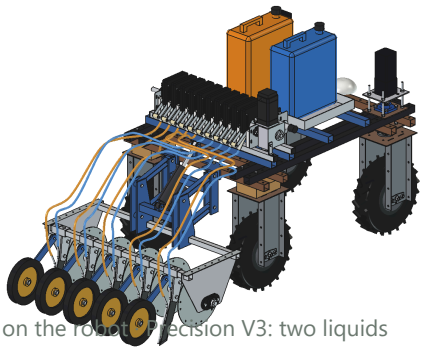
Advanced



Precision: roller pump



Precision V2 on the robot



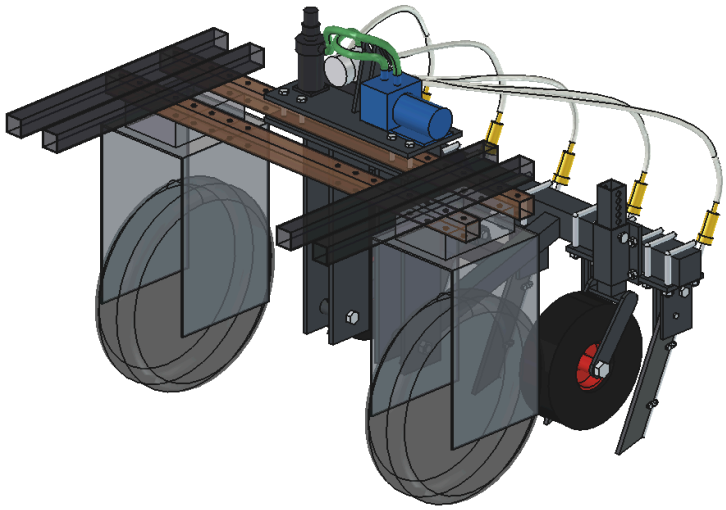
Precision V3: two liquids

6 variants: Simple · Simple v2 · Advanced · Precision: roller pump · Precision V2 on the robot · Precision V3: two liquids

Folder: `liquid fertilizer applicator/`

LIQUID FERTILIZER APPLICATOR · VARIANT 1 OF 6

Simple



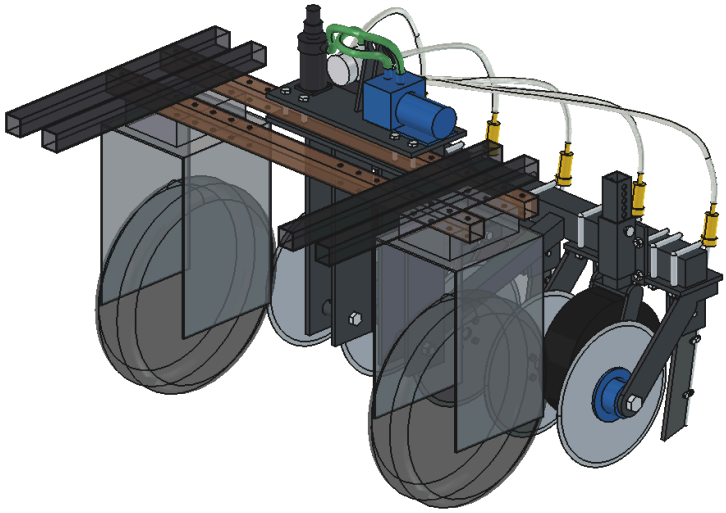
Cheap and robust: 5 fixed injection knives with a stainless steel tube, two wheelbarrow wheels as depth wheels, a lift frame with a single low pivot and one cheap linear actuator, a 12 V diaphragm pump with fixed regulator and orifice plates. Only flat bar, tube and standard parts.

- 5 knives at 200 mm
- 12 V diaphragm pump, dose depends on speed
- About €600 in materials

Folder: `liquid fertilizer applicator/concept versions/simple/`

LIQUID FERTILIZER APPLICATOR · VARIANT 2 OF 6

Simple v2

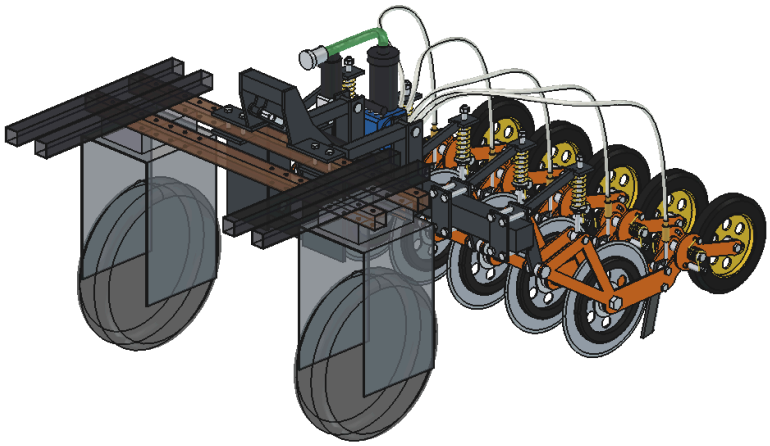


Simple version plus a flat coulter disc Ø300 in front of each knife, cutting 45 mm deep; the knife runs 14 mm behind it in the same plane. Costs 26 kg extra on the lift frame (70 kg), a 1500 N actuator and up to 50 kg of ballast in hard sward.

- Disc + knife per row
- Lift frame 70 kg
- About €915 in materials

Folder: `liquid fertilizer applicator/concept versions/simple_v2/`

Advanced

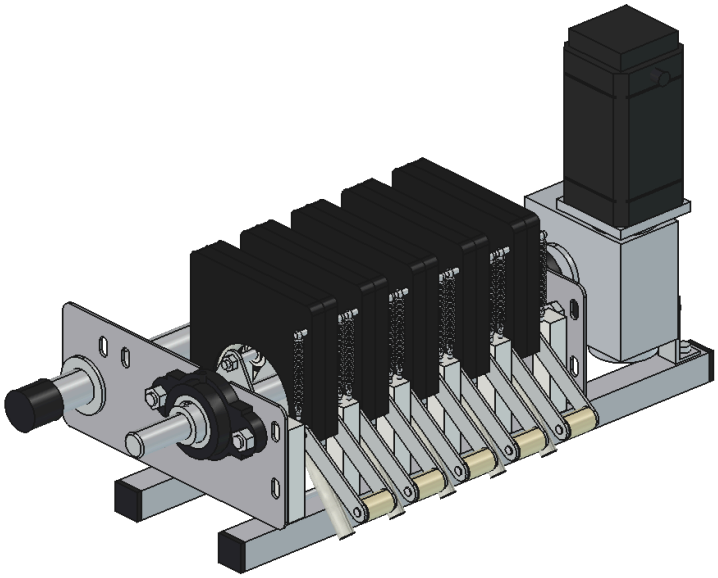


Disc, knife, tube and press wheel per element on spring arms; a 5-channel peristaltic pump driven by a worm-gear motor whose speed follows the robot's driving speed, and an electrically lifted parallel linkage that floats while working (slotted hole). No ground wheel.

- 5 elements at 200 mm
- 89–1,316 l/ha, 18 settings
- Includes ground-following animation

Folder: `liquid fertilizer applicator/concept versions/advanced/`

Precision: roller pump

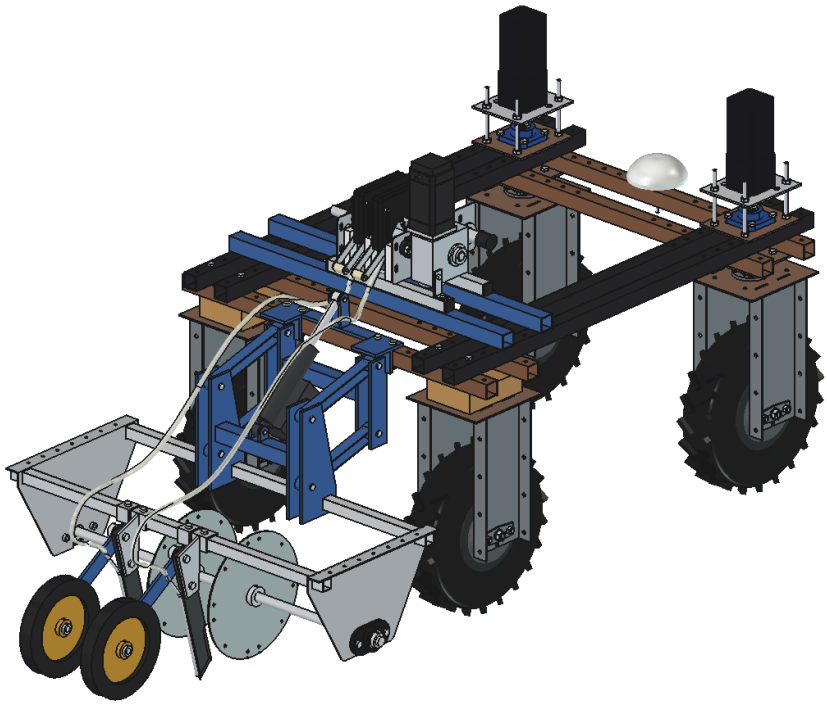


The first production version, built with Herre Bartlema from measured parts. A modular peristaltic roller pump on the robot, stepper-driven so the dose follows the driving speed; 5 or 10 sections.

- 10 ml per revolution per tube
- 5 or 10 sections
- Includes a running animation

Folder: `liquid fertilizer applicator/precision_applicator/roller_pump/`

Precision V2 on the robot

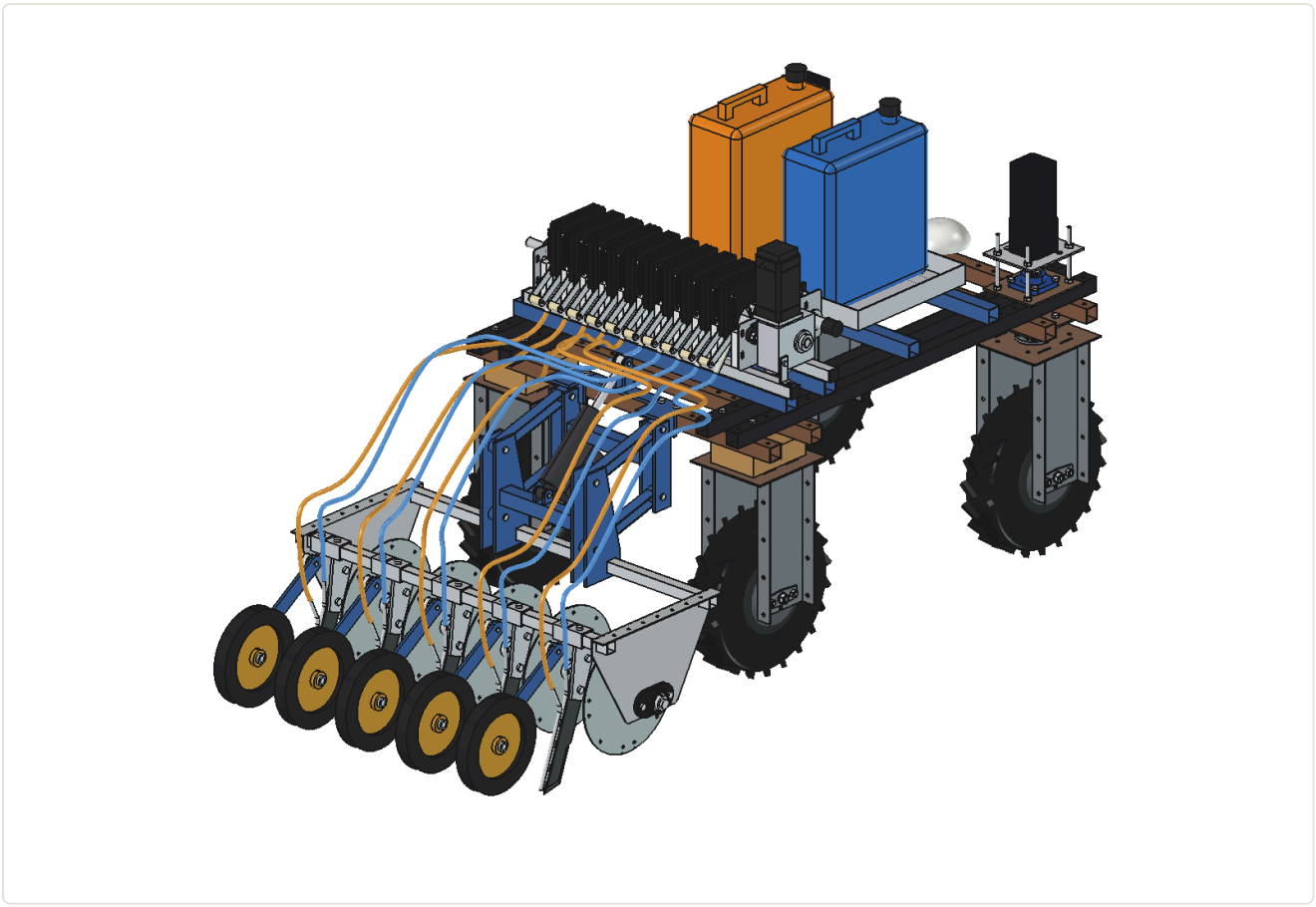


Applicator V2 with the roller pump on the robot: Ø 325 discs with injector knives behind them on a parallel linkage with an electric actuator, and a press wheel behind every knife that closes the slot. 5 rows, or 2 rows in the middle (applicator V1, combined).

- Disc, knife and press wheel per row
- Electric actuator lift
- Includes hoses and an animation

Folder: `liquid fertilizer applicator/precision_applicator/combined_v2/`

Precision V3: two liquids



Applicator V3 with 5 rows of trailing double knives (both outlets 50 mm deep) and press wheels, fed by the 10-section roller pump: 5 sections per liquid, so two products go in at once.

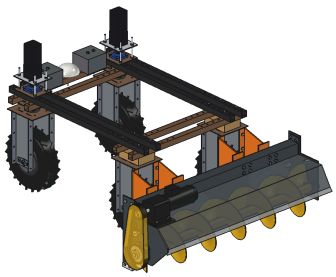
- 5 rows, double knives
- 10-section roller pump, 2 liquids
- Includes an animation

Folder: `liquid fertilizer applicator/precision_applicator/combined_v3/`

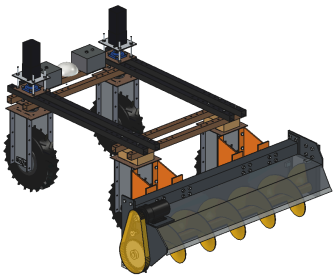
IMPLEMENT · TRANSPORT & GENERAL

Feed pusher auger

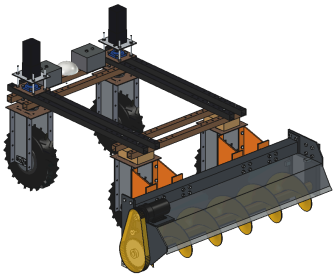
Pushes feed back to the feed fence. Sits at the back of the robot; the robot drives with the auger first. The auger picks up the pushed-away feed and carries it sideways to the fence.



V1



V2

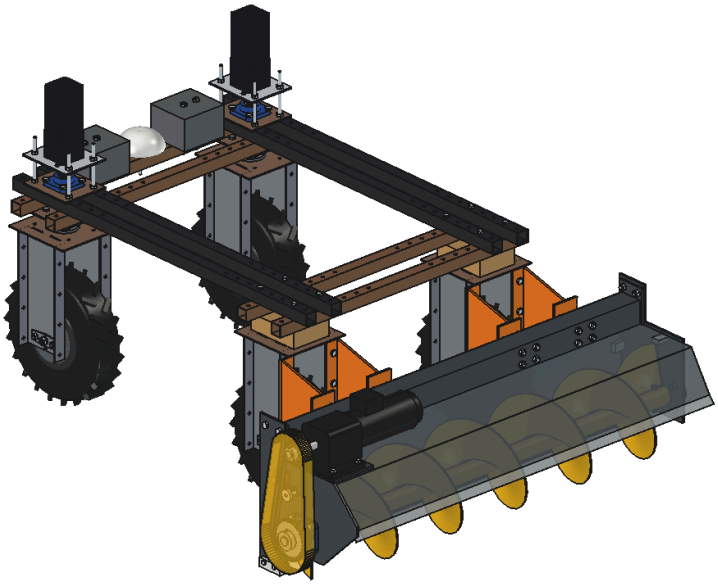


V3

3 variants: V1 · V2 · V3
Folder: `feed pusher auger/`

FEED PUSHER AUGER · VARIANT 1 OF 3

V1



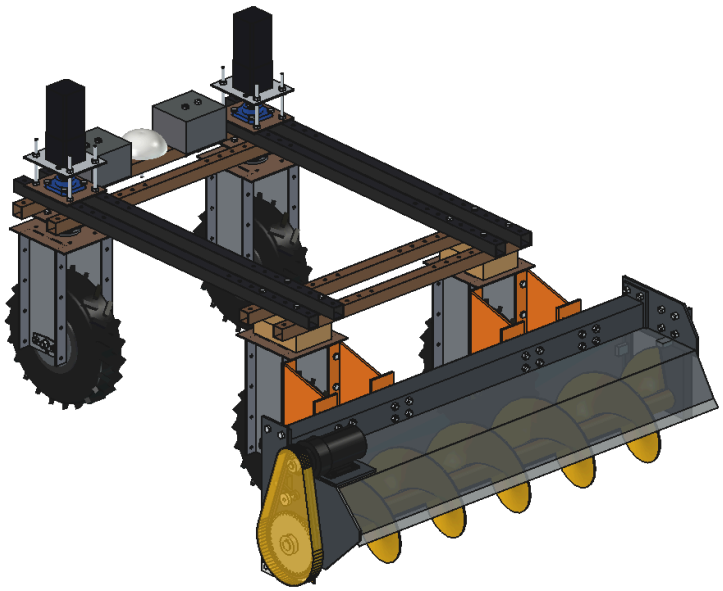
Ø320 auger under a 2 mm folded hood with 4 bends. Foot-mounted gearmotor on top of the hood, 08B-1 chain with tensioner along the left side plate. Mounted with 4 arms on the rear flange holes of the rear wheel modules. Open discharge end. Includes a feed-transport model and a force animation.

- 24 V gearmotor with chain drive
- Open discharge end
- Feed-transport model and force animation

Folder: `feed pusher auger/V1/`

FEED PUSHER AUGER · VARIANT 2 OF 3

V2



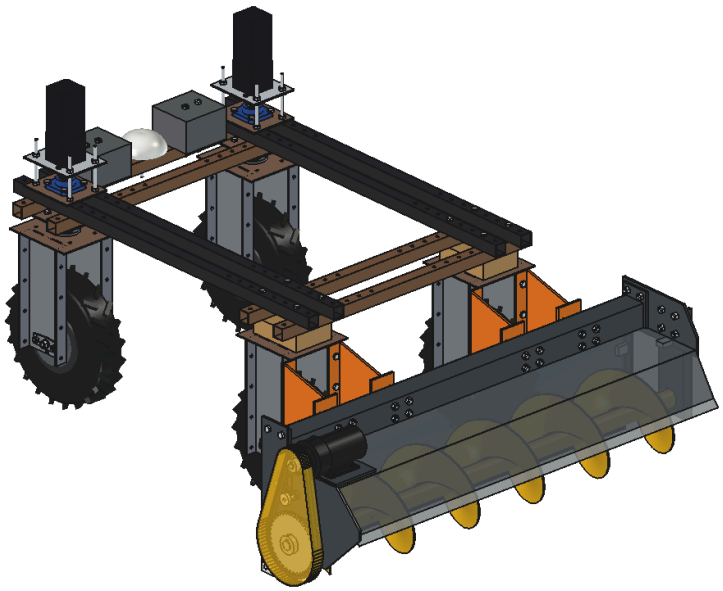
V1 without a through shaft: the free end hangs on a strap with a Ø25 pin and bearings inside the auger, the drive end has a stepped drive bush with an M6 shear bolt. The auger slides out through a cutout in the end plate.

- Auger removable without opening the frame
- Frame, hood and drive as V1
- M6 shear bolt breaks at 252 Nm

Folder: `feed pusher auger/V2/`

FEED PUSHER AUGER · VARIANT 3 OF 3

V3



V2 with the existing, measured auger: Ø300 flight on a Ø60 x 3 core tube, M12 shear bolt between welded rings, and a wear-resistant plastic ring on a fixed Ø20 pin at the free end instead of bearings.

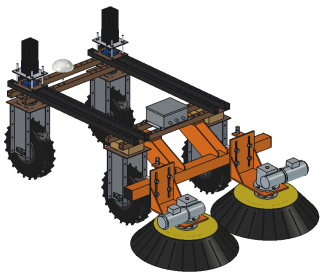
- Measured existing auger
- No bearings at the free end
- Frame, hood and drive as V2

Folder: `feed pusher auger/V3/`

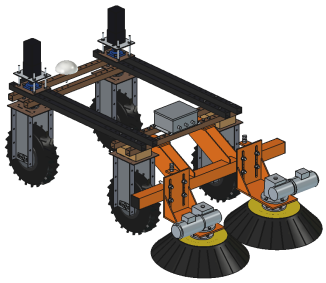
IMPLEMENT · TRANSPORT & GENERAL

Disc brushes

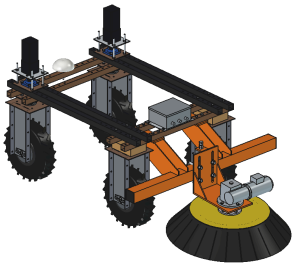
Sweeper disc brushes for loose material on concrete. Each brush on its own 48 V worm gearmotor (NMRV063, 550 W); reversing the motors makes a windrow in the middle or spreads material wide. Mounted on an 80 × 80 toolbar under the rear beams with U-bolts. Includes an animation with a particle model.



Double 750



Double 650

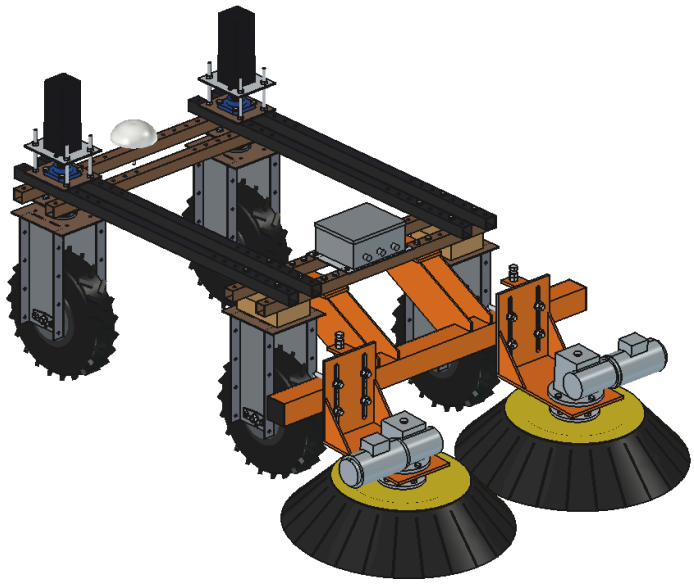


Single 850

3 variants: Double 750 · Double 650 · Single 850
Folder: `brushes/`

DISC BRUSHES · VARIANT 1 OF 3

Double 750



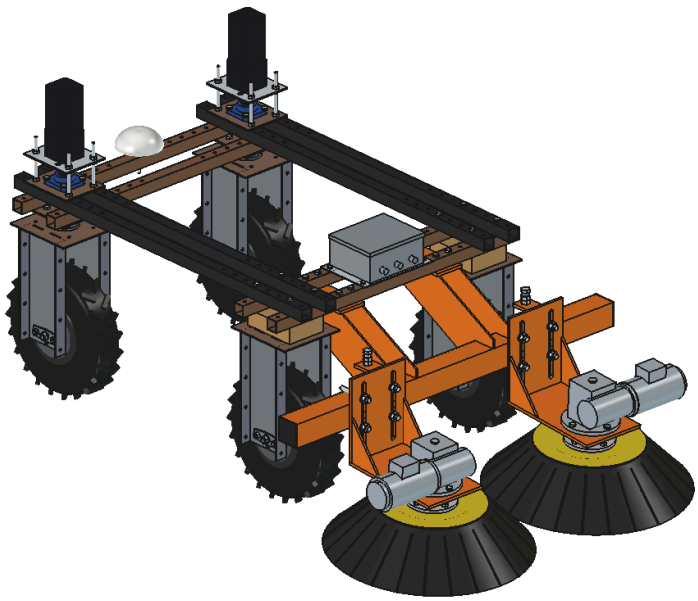
Two Ø750 brushes (standard). Working width 1.50 m with brushes touching to 1.70 m with 200 mm between them. Spread wide or make a windrow.

2 × Ø750	1.50–1.70 m

Folder: `brushes/simple/`

DISC BRUSHES · VARIANT 2 OF 3

Double 650



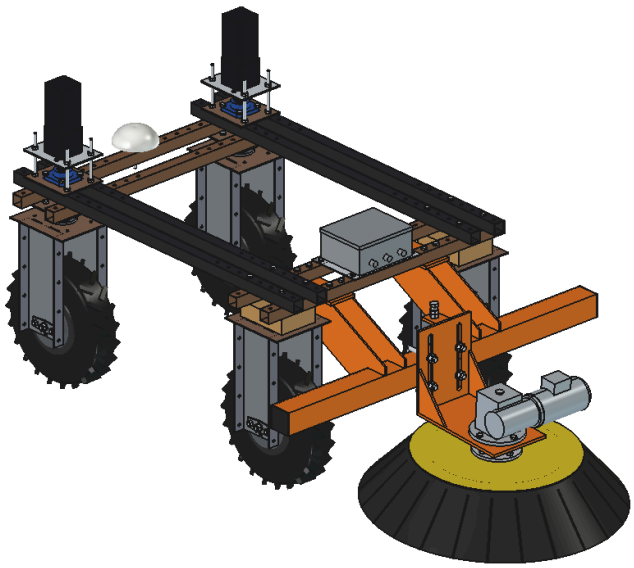
Two Ø650 brushes: the same job, narrower and lighter.

2 × Ø650	1.30–1.50 m

Folder: `brushes/simple/`

DISC BRUSHES · VARIANT 3 OF 3

Single 850



One Ø850 brush covering the robot's own width, sweeping the strip in front of the robot to one side.

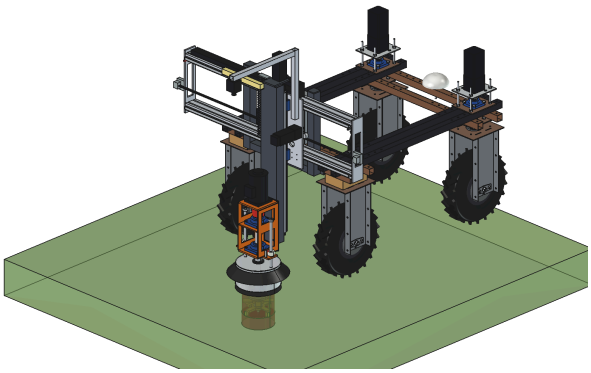
1 × Ø850	0.85 m

Folder: `brushes/simple/`

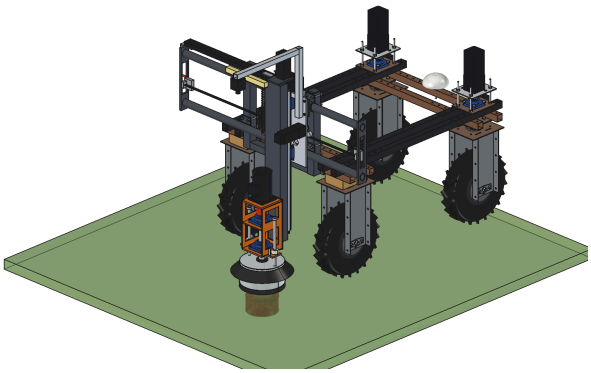
IMPLEMENT · SOIL CULTIVATION & WEED CONTROL

Dockweed drill

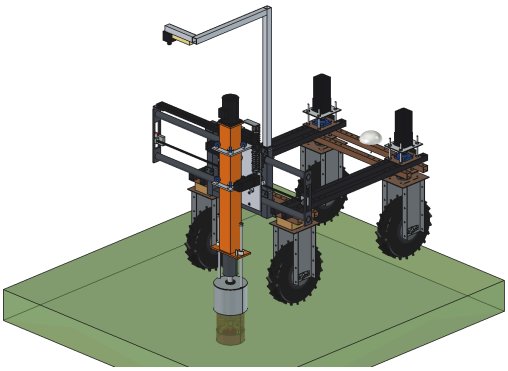
Finds a dock plant (broad-leaved dock) and mills out its root on a CNC gantry behind the robot, chemical-free. Eight numbered versions that each change one part of the machine.



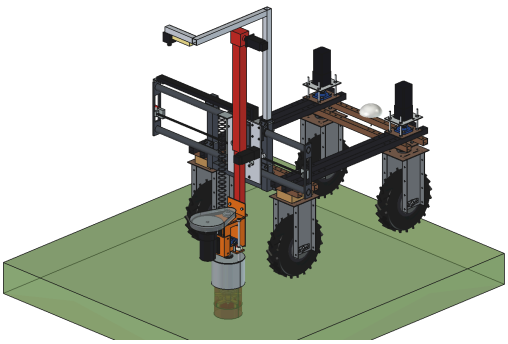
v1



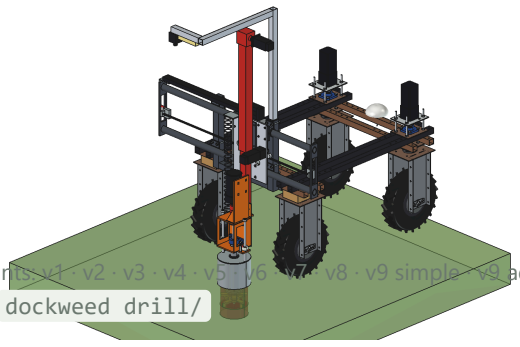
v2



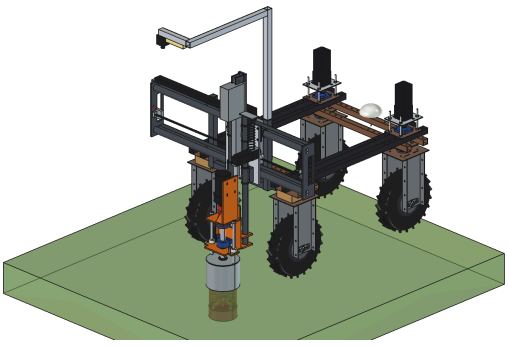
v3



v4



v5

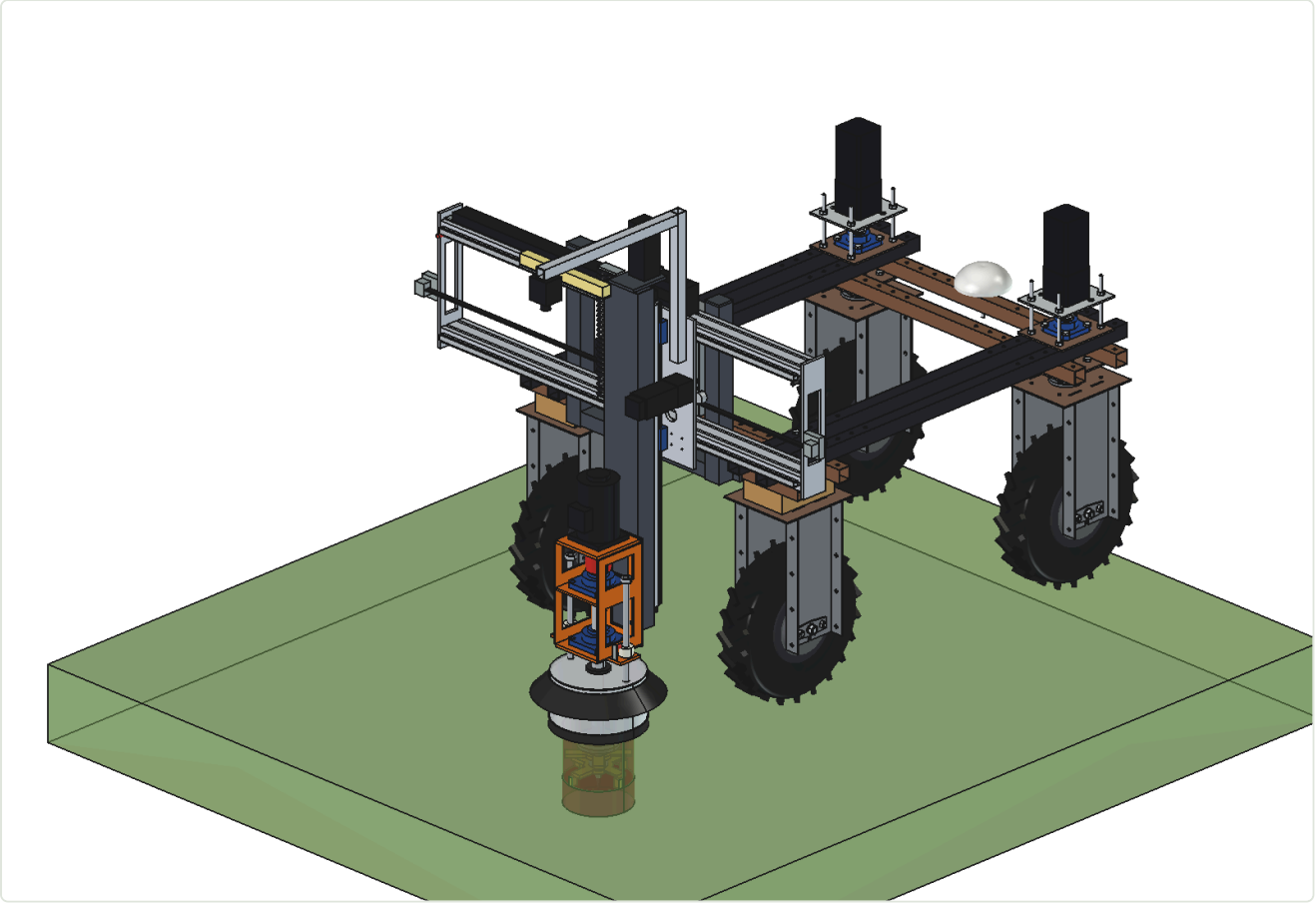


v6

10 variants: v1 · v2 · v3 · v4 · v5 · v6 · v7 · v8 · v9 simple · v9 advanced

Folder: `dockweed drill/`

v1

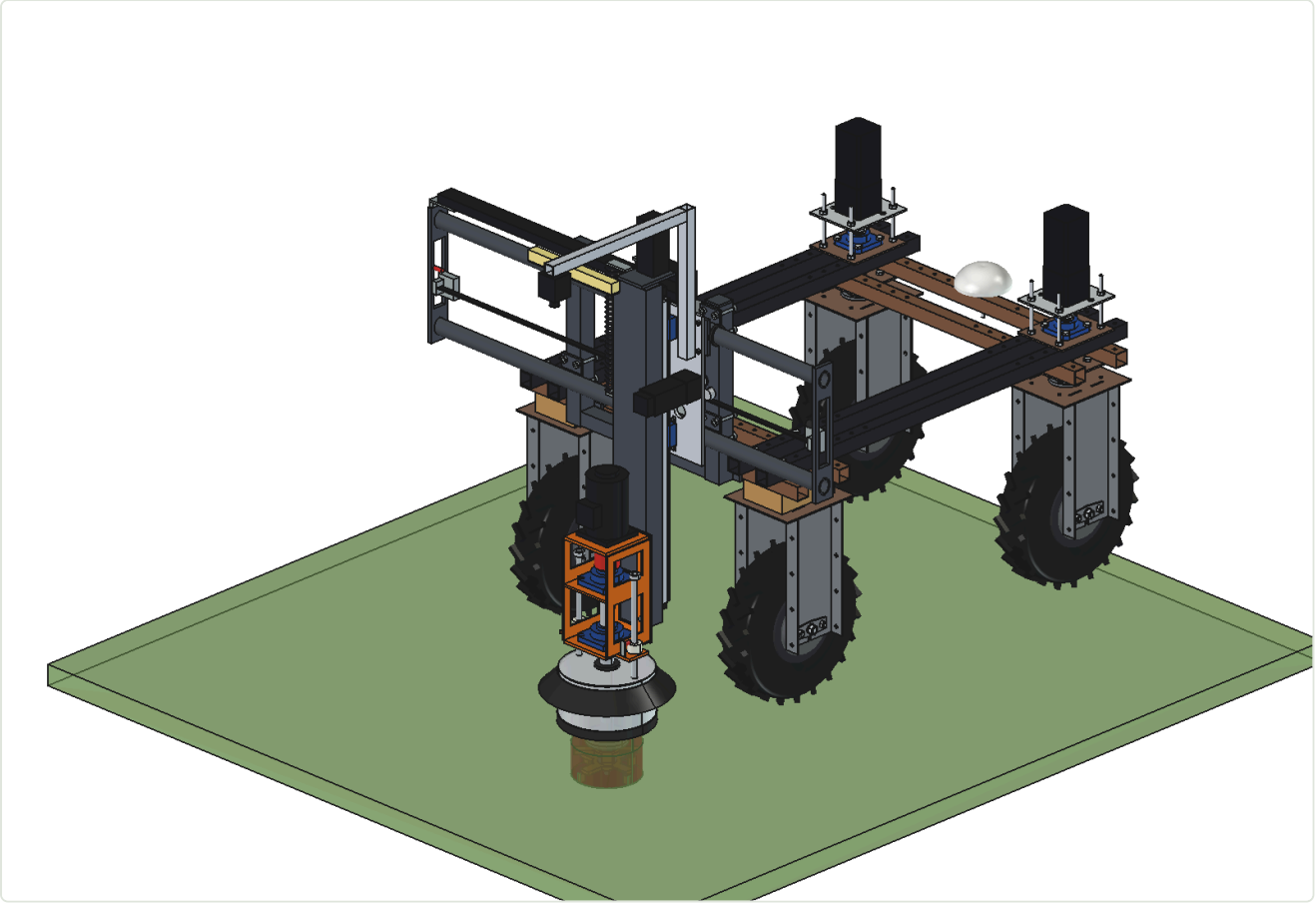


X slide on 2 aluminium profiles 40 x 80 with HGR15 rails and 4 carriages; cutter Ø180 at 1500 rpm in a pot Ø250 and a mowing disc as second tool.

- X stroke 1000 mm
- HGR15 rails
- Includes an animation

Folder: `dockweed drill/v1/`

v2

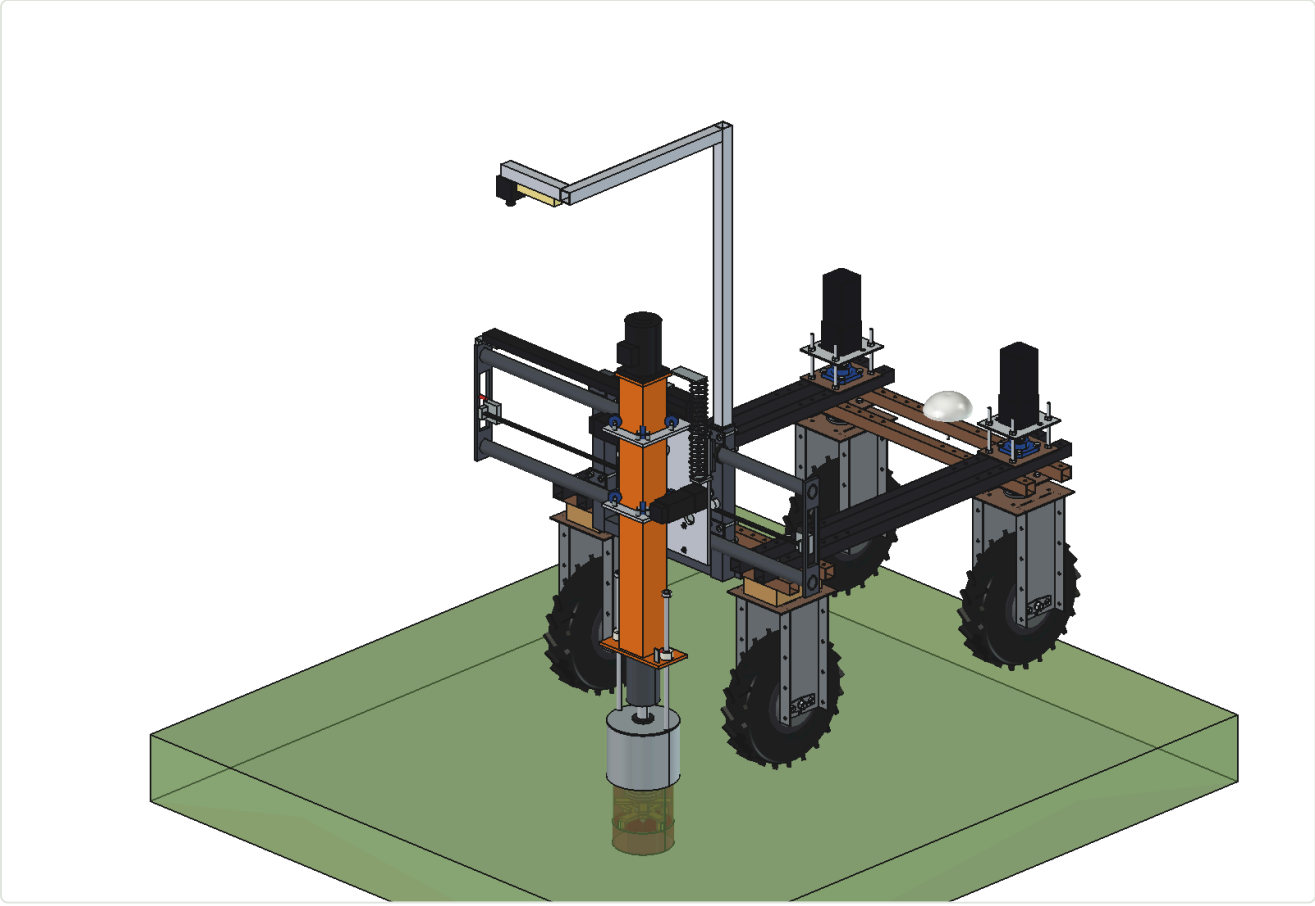


X slide on PTO-shaft tubes. The robot is the Y axis and stops with the plant under the spindle; the cutter goes 150 mm deep and grinds up the root crown and upper taproot.

- X slide on PTO tubes
- Cutter 150 mm deep
- Animations: three plants, nettle mowing

Folder: `dockweed drill/v2/`

v3

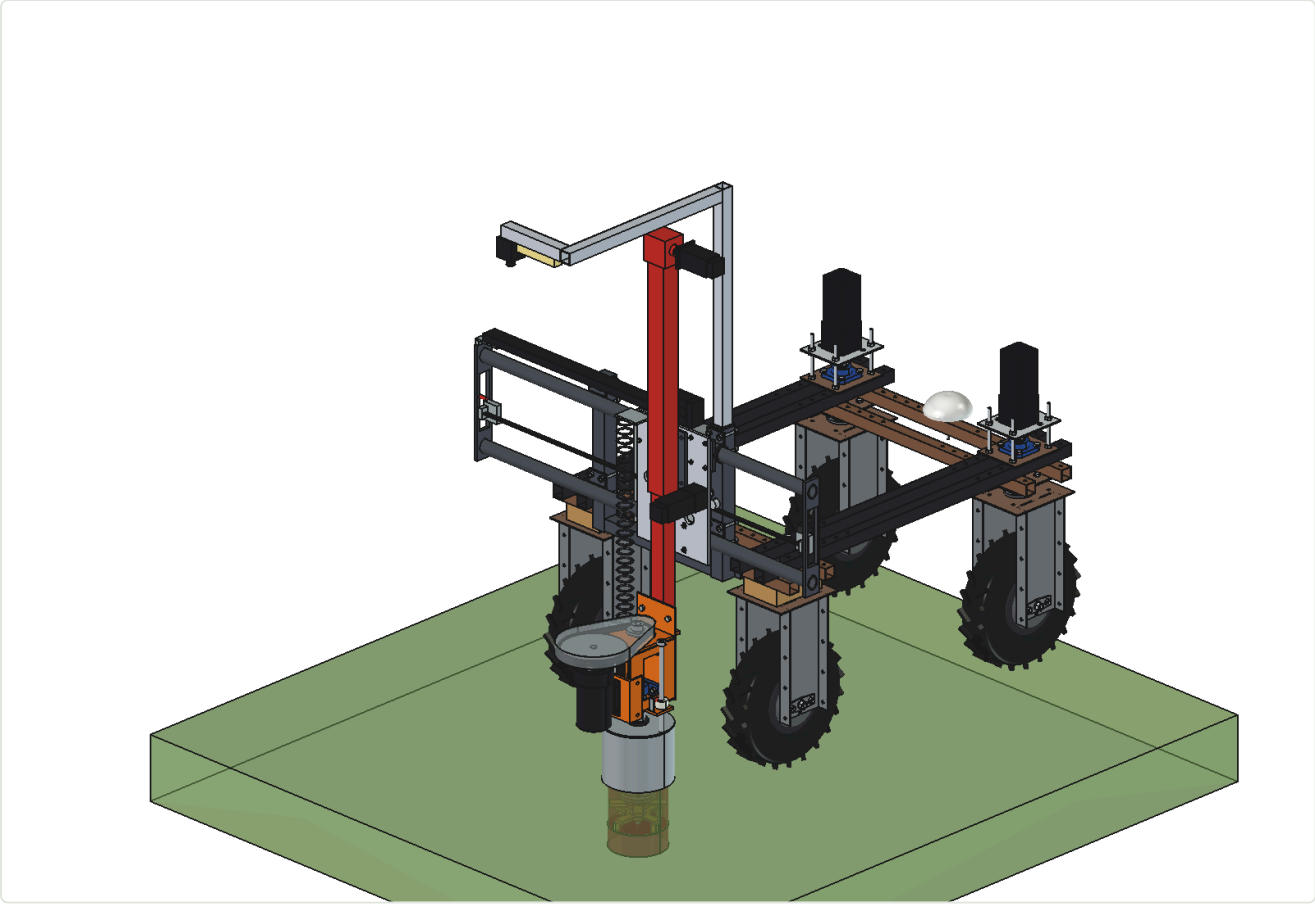


v2 with a cheaper Z axis: a 100 x 100 mast with the spindle inside, rolling on 8 V-groove corner rollers, and a steel can like Robot Ruud's.

- Z mast on corner rollers
- Ruud-style can
- Includes an animation

Folder: `dockweed drill/v3/`

v4

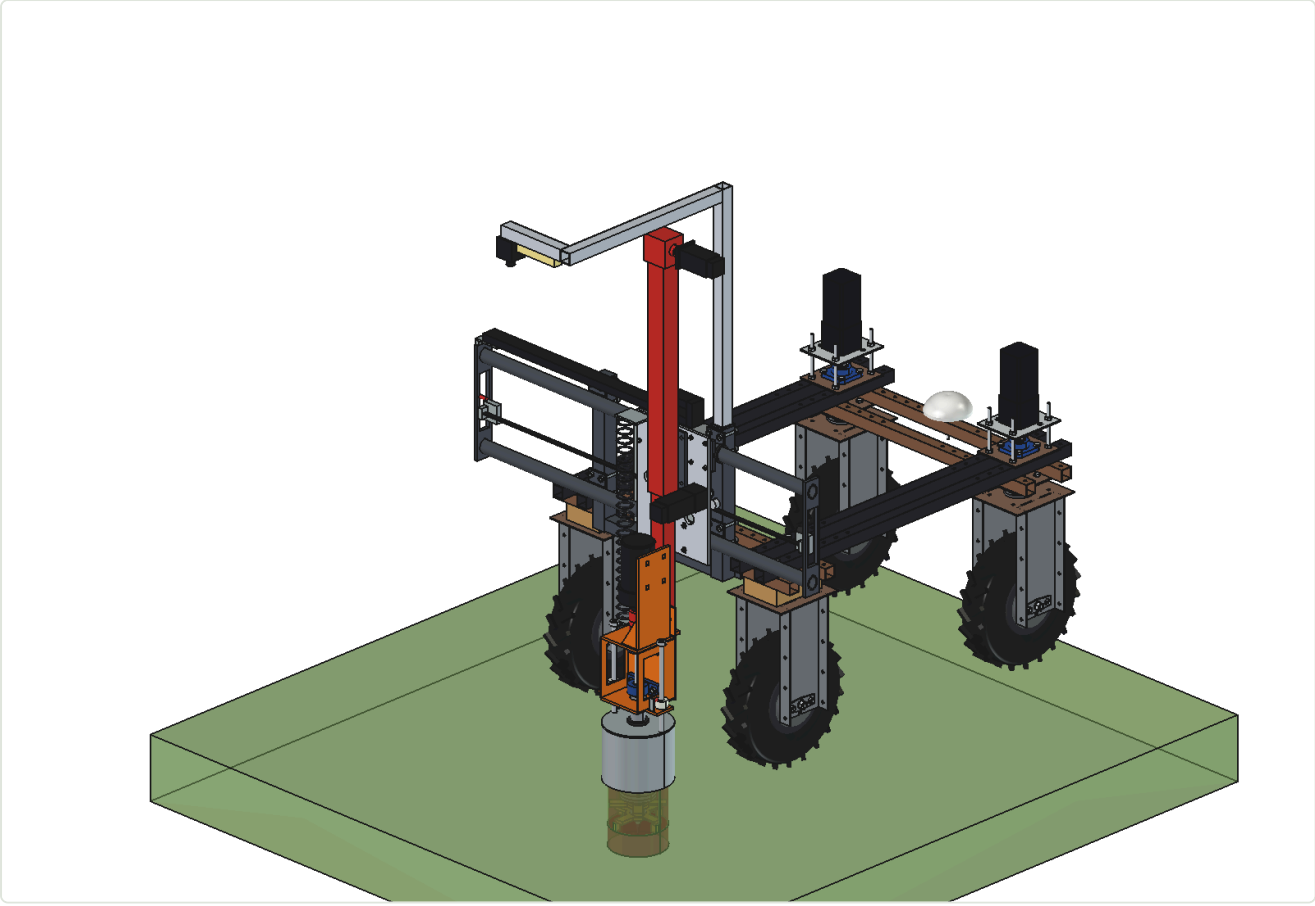


v3 with a drop-leg trailer jack as the Z axis (stroke 380 mm, NEMA 23) and the 800 W gearmotor with chain on 2 x UCP 204.

- Trailer jack as Z
- 800 W gearmotor + chain
- Includes an animation

Folder: `dockweed drill/v4/`

v5

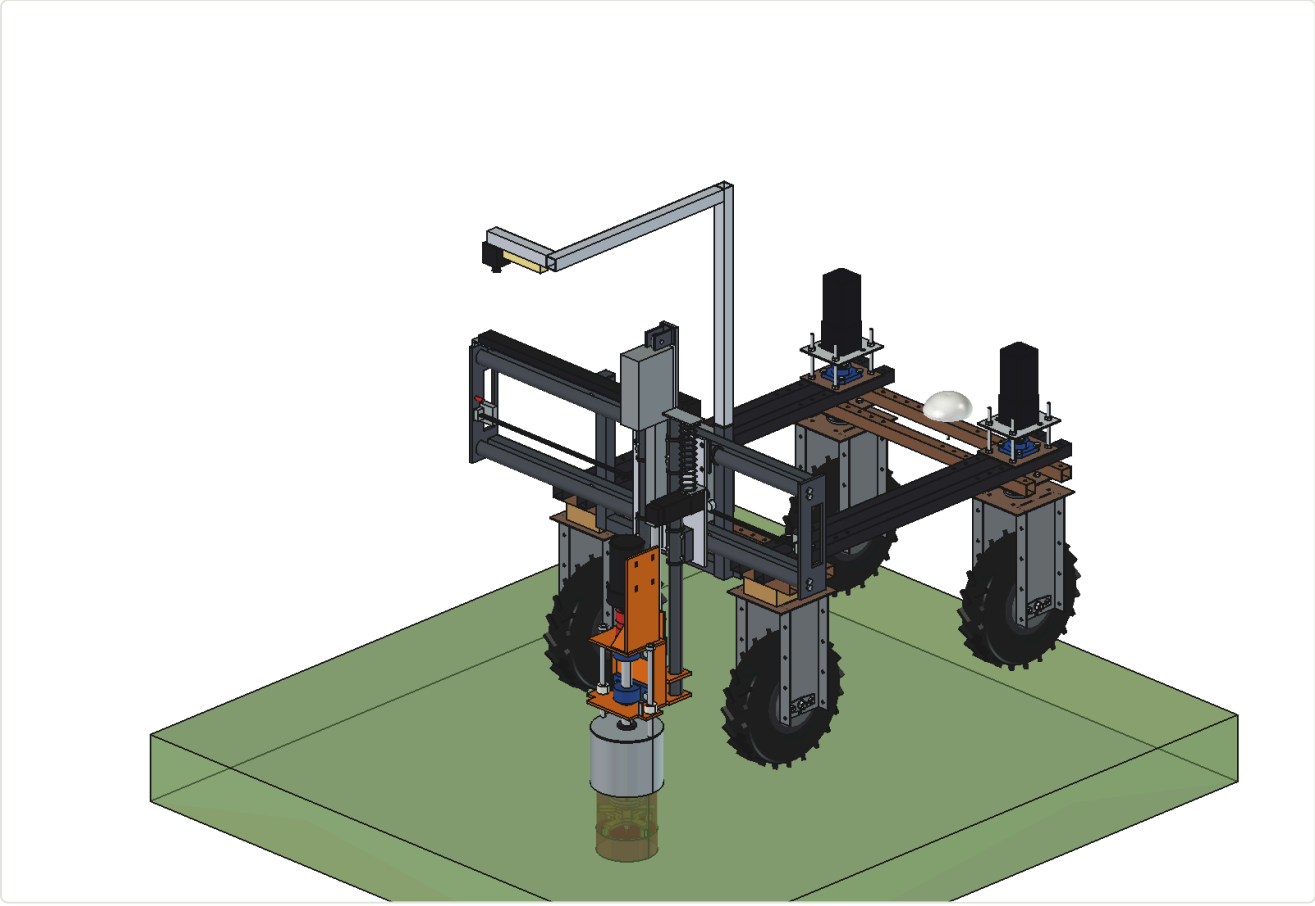


v4 with the 800 W gearmotor directly above the spindle through a flexible jaw coupling: no chain, no sprockets.

- Direct drive, jaw coupling
- No chain guard
- No animation

Folder: `dockweed drill/v5/`

v6

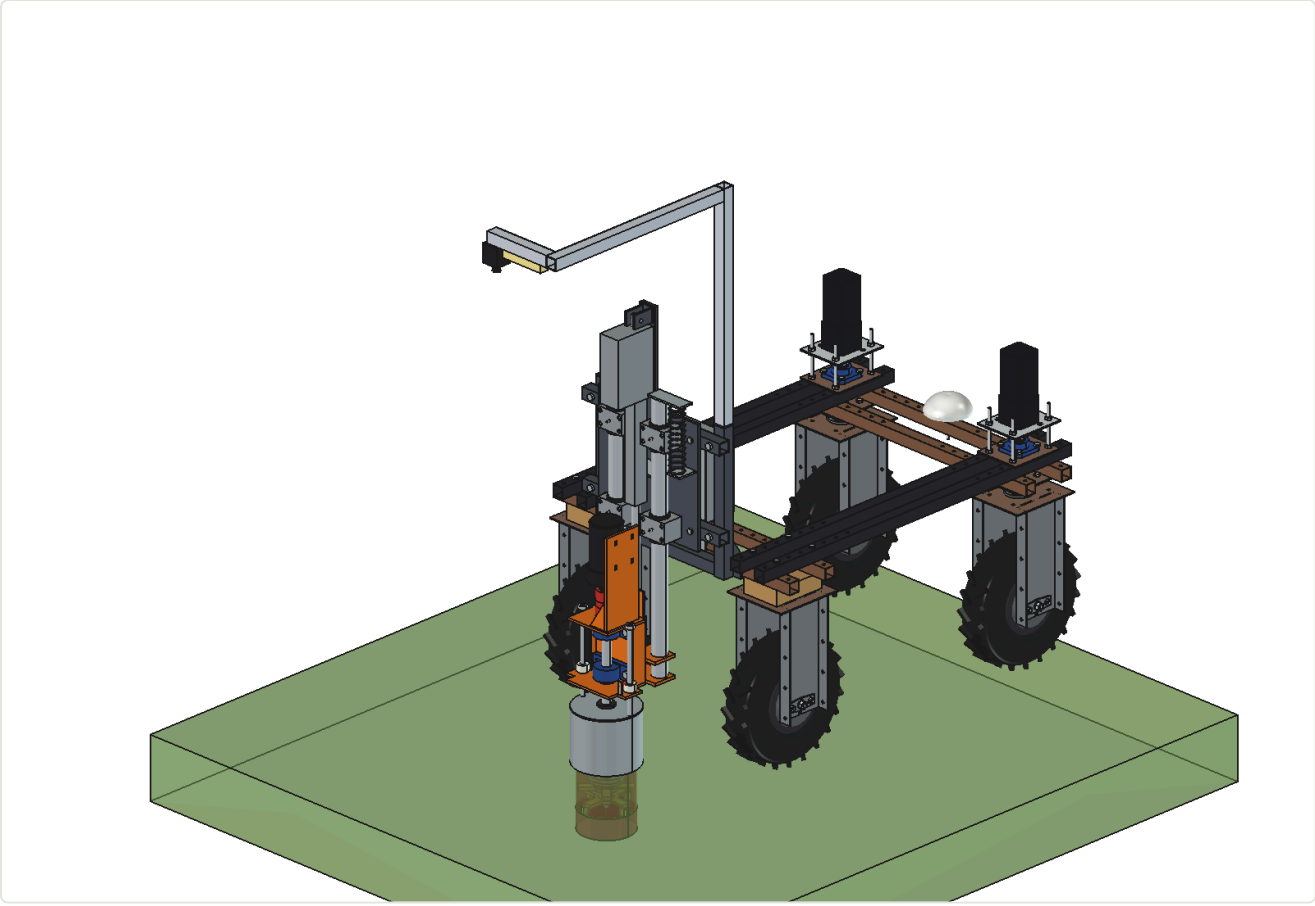


v5 with a 40 x 40 frame, PTO tubes bolted between 12 mm side plates, a Z tube on HGR15 rails driven by a 24 V actuator, and a Ø30 spindle in 2 x UCP 206.

- Bolted PTO tubes
- 24 V actuator Z axis
- Ø30 spindle in UCP 206

Folder: `dockweed drill/v6/`

v7

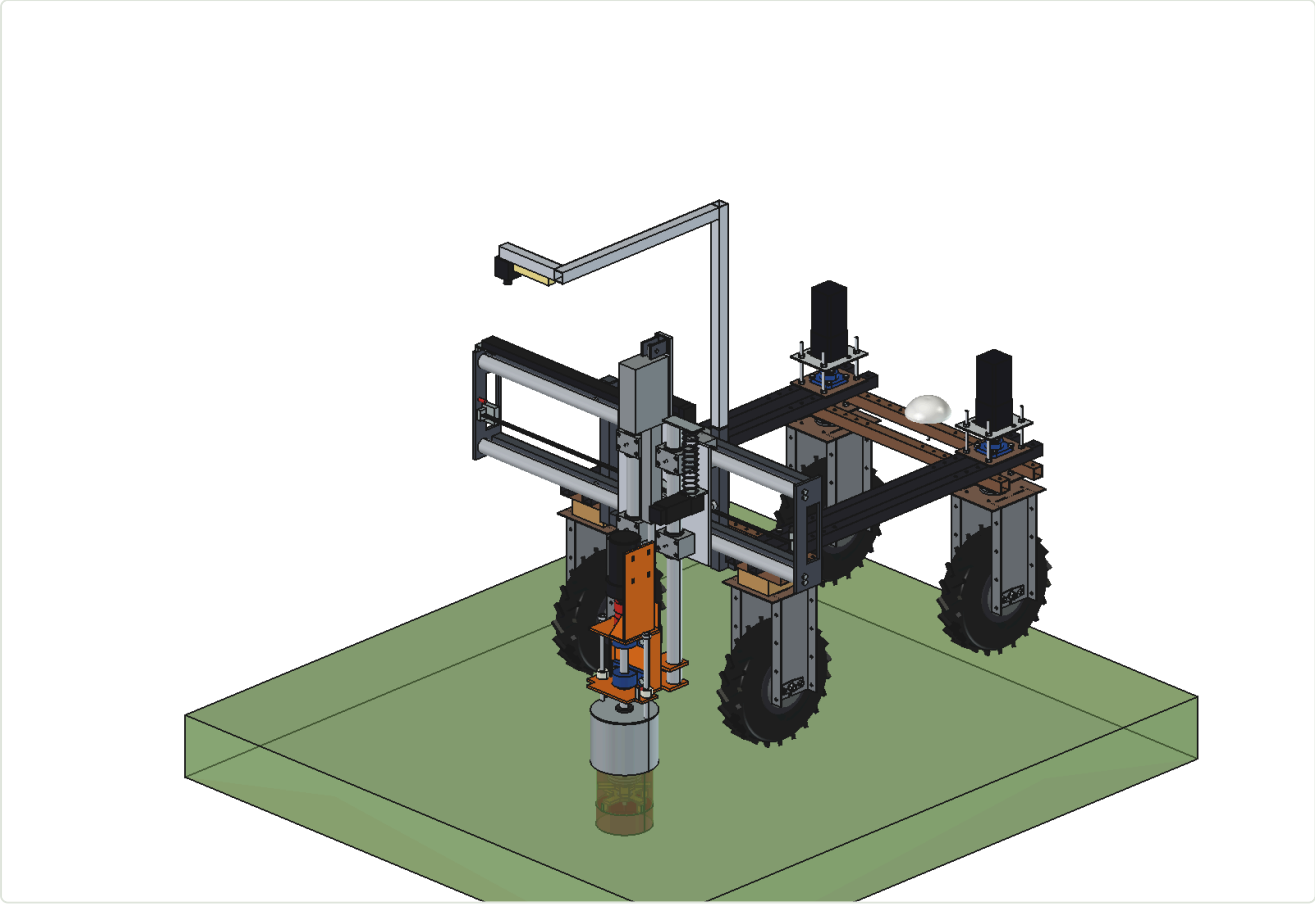


v6 without the X axis: the spindle is fixed at x 0 and the robot steers to the plant. Z on 2 round precision tubes Ø50 x 4 in plain-bush housings with the Linak LA36 in the middle; 23 kg lighter.

- No X axis, robot steers
- Round tubes in plain bushes
- 99.9 kg
- Steering and exploded-view animations

Folder: `dockweed drill/v7/`

v8

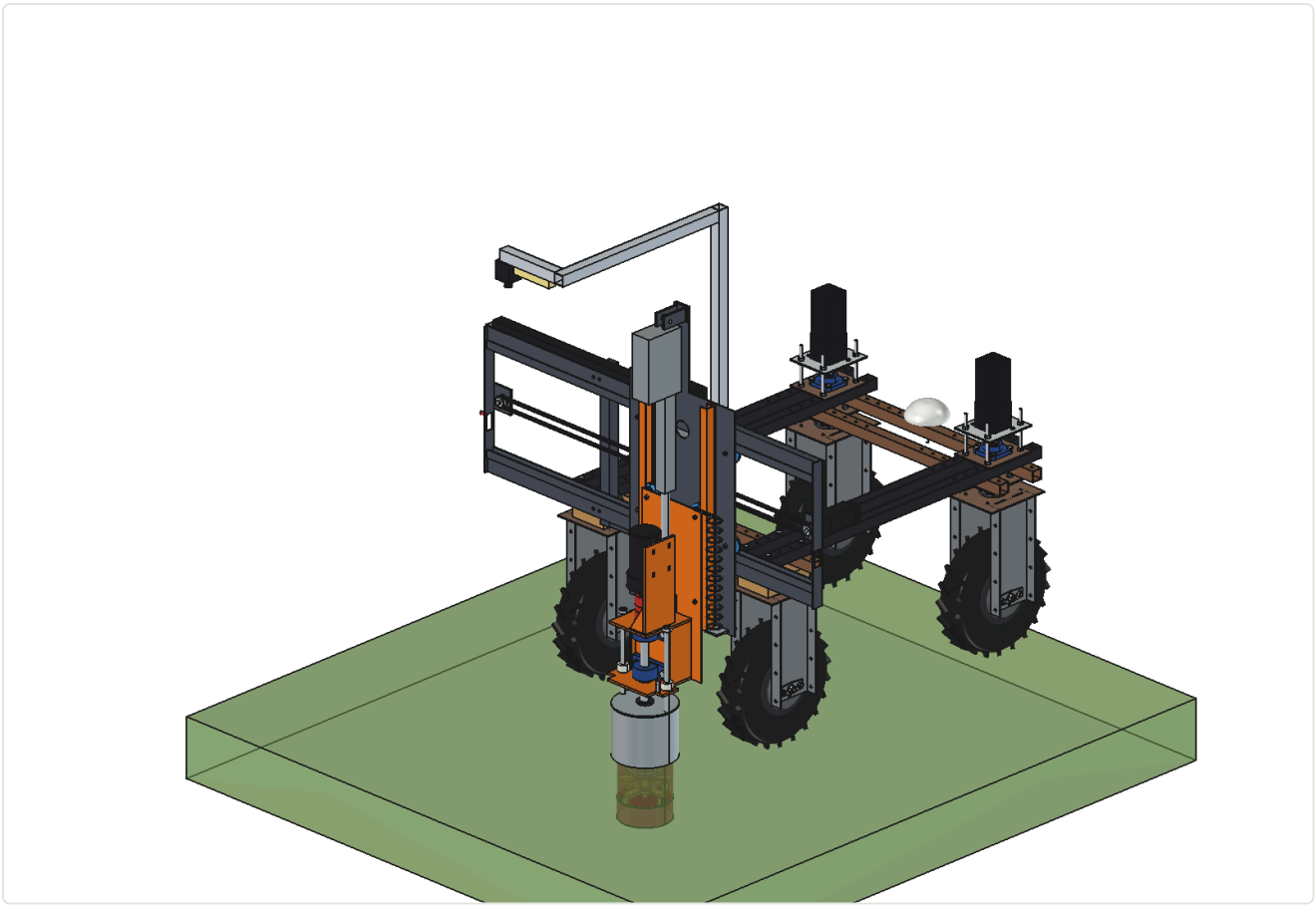


The Z part of v7 on the X axis of v6, with round precision tubes Ø50 x 4 instead of PTO tubes: the same tube and plain-bush housing on both axes, and the v7 mount plate becomes the X slide.

- X stroke 1000 mm on round tubes
- One tube and housing type for X and Z
- 127 kg
- Includes an animation

Folder: `dockweed drill/v8/`

v9 simple

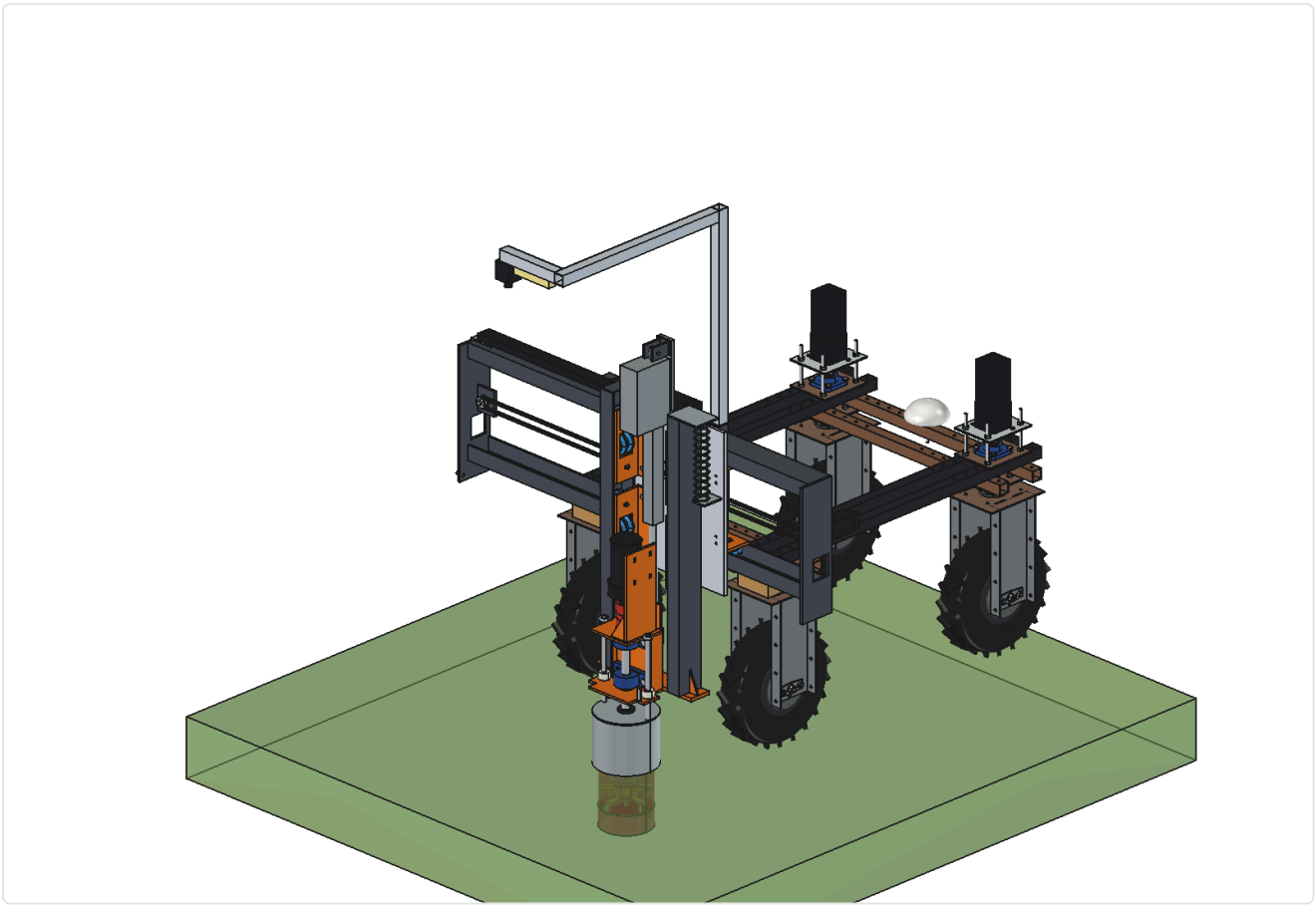


v8 with both axes on 80 mm 90A skate wheels in bent 4 mm U tracks (25 mm inside): 2 wheels per track, the Z tracks fixed on the X slide with the LA36, the drill plate rolls down in them; X tracks on 50 x 30 x 4 beams.

- 8 skate wheels in 4 mm U25 tracks
- Z tracks fixed on the X slide
- 117.6 kg
- Includes an animation

Folder: `dockweed drill/v9/simple/`

v9 advanced



v8 with both axes on 80 mm 90A skate wheels running in UPE 100 channels that face each other: 8 identical bolt-on wheel blocks (2 flange wheels + 2 web wheels each), the X frame bolted straight onto the uprights and the X motor fixed at the end.

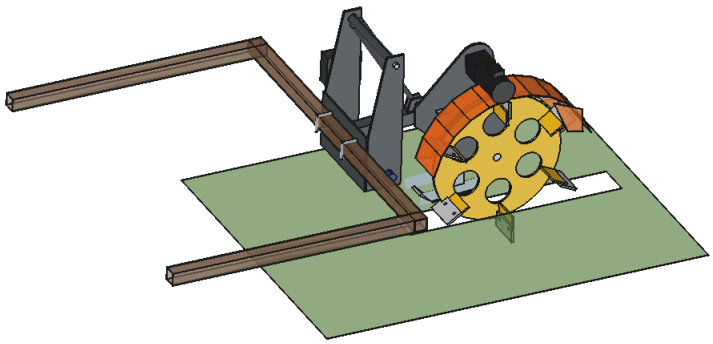
- 32 skate wheels 80 x 24 in UPE 100
- One wheel block for X and Z
- 170 kg, counterweight needed
- Includes an animation

Folder: `dockweed drill/v9/advanced/`

IMPLEMENT · SOIL CULTIVATION & WEED CONTROL

Trencher

Wheel trencher for cutting narrow trenches.

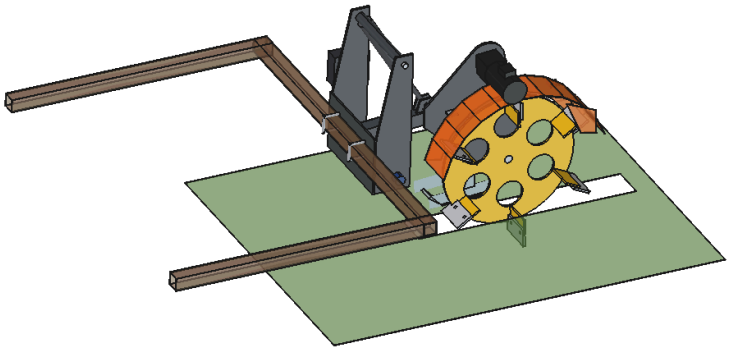


One design

1 variant: One design
Folder: `trencher/`

TRENCHER · VARIANT 1 OF 1

One design



Wheel trencher mounted on the Bruut. Renders and STEP export. An older single-script design (trencher_design.py).

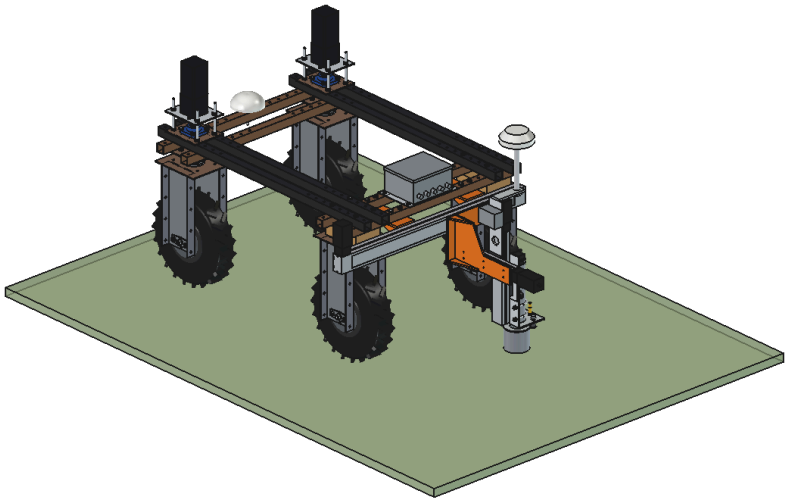
> Renders and STEP export

Folder: `trencher/`

IMPLEMENT · MEASURING & SENSORS

Field marker

Stakes out points from a coordinate list (piles, posts, planting positions, plot corners). The robot stops roughly behind the point; an XYZ stage puts a spray-paint head exactly on it.

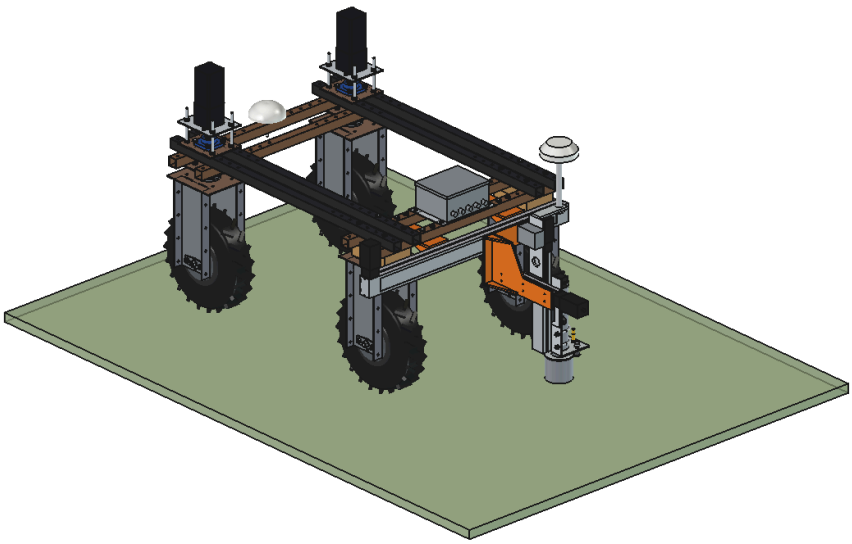


Simple

1 variant: Simple
Folder: `marker/`

FIELD MARKER · VARIANT 1 OF 1

Simple



Three bought linear modules: X 800 mm belt module, Y 200 mm and Z 300 mm ball screws. Paint head: upside-down marking-spray can in a PVC drain pipe pushed by a servo; sprung PVC 110 shroud with an inductive sensor finds the ground. Second GNSS antenna on a mast straight above the nozzle.

- Robot may stop ±100 mm off
- NEMA 23 closed-loop motors
- About €1,450 in parts
- Concept lists flag, peg, punch and soil-probe heads

Folder: `marker/simple/`

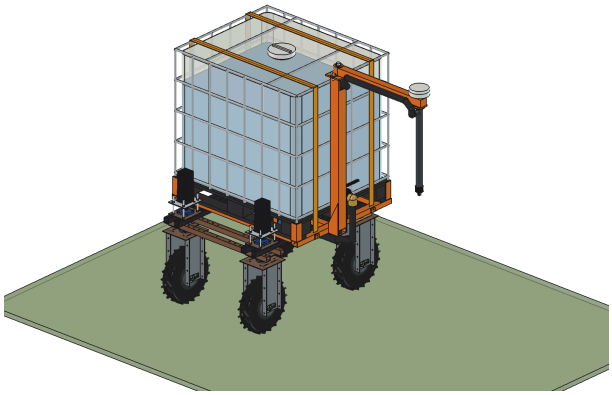
IMPLEMENT · SPRAYING, FERTILIZING & IRRIGATION

Mobile refiller

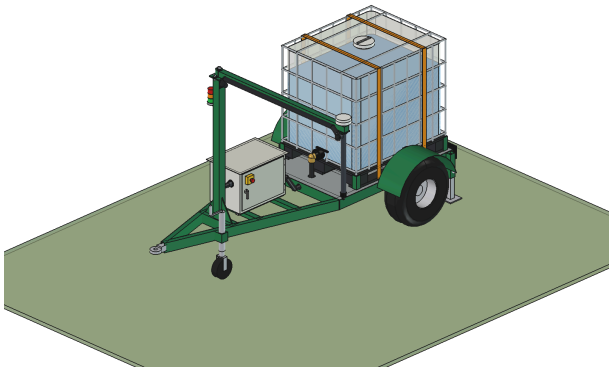
Support robot that brings liquid (and in concept seed, granular fertilizer, water, batteries) to the working robots, so they never leave the field.

Mobile refiller Liquid

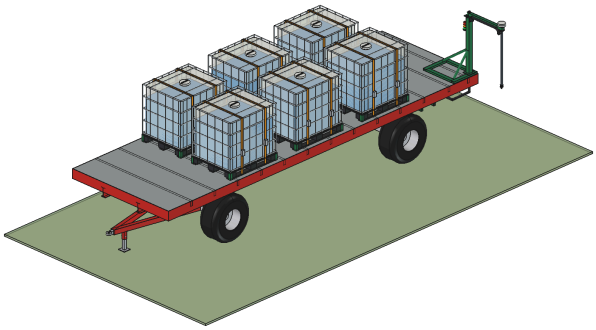
Liquid



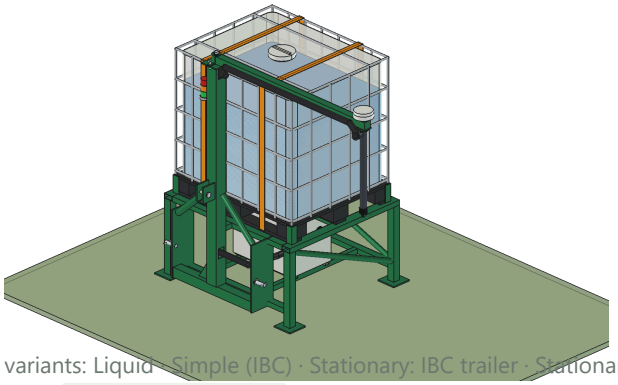
Simple (IBC)



Stationary: IBC trailer



Stationary: flat wagon



5 variants: Liquid · Simple (IBC) · Stationary: IBC trailer · Stationary: flat wagon · Stationary: three-point frame

Folder: `mobile refiller/`

Stationary: three-point frame

MOBILE REFILLER · VARIANT 1 OF 5

Liquid



268 l T-shaped tank with a pump below the liquid level, a slewing boom with a trolley and a spout, an RTK antenna on the spout and a camera for confirmation, and a powered trailer of 246 l (6 driven wheels). Tops up the working robot in the straight reverse of its headland turn while both drive.

> About 25 l per turn, ≈ 3 s standstill

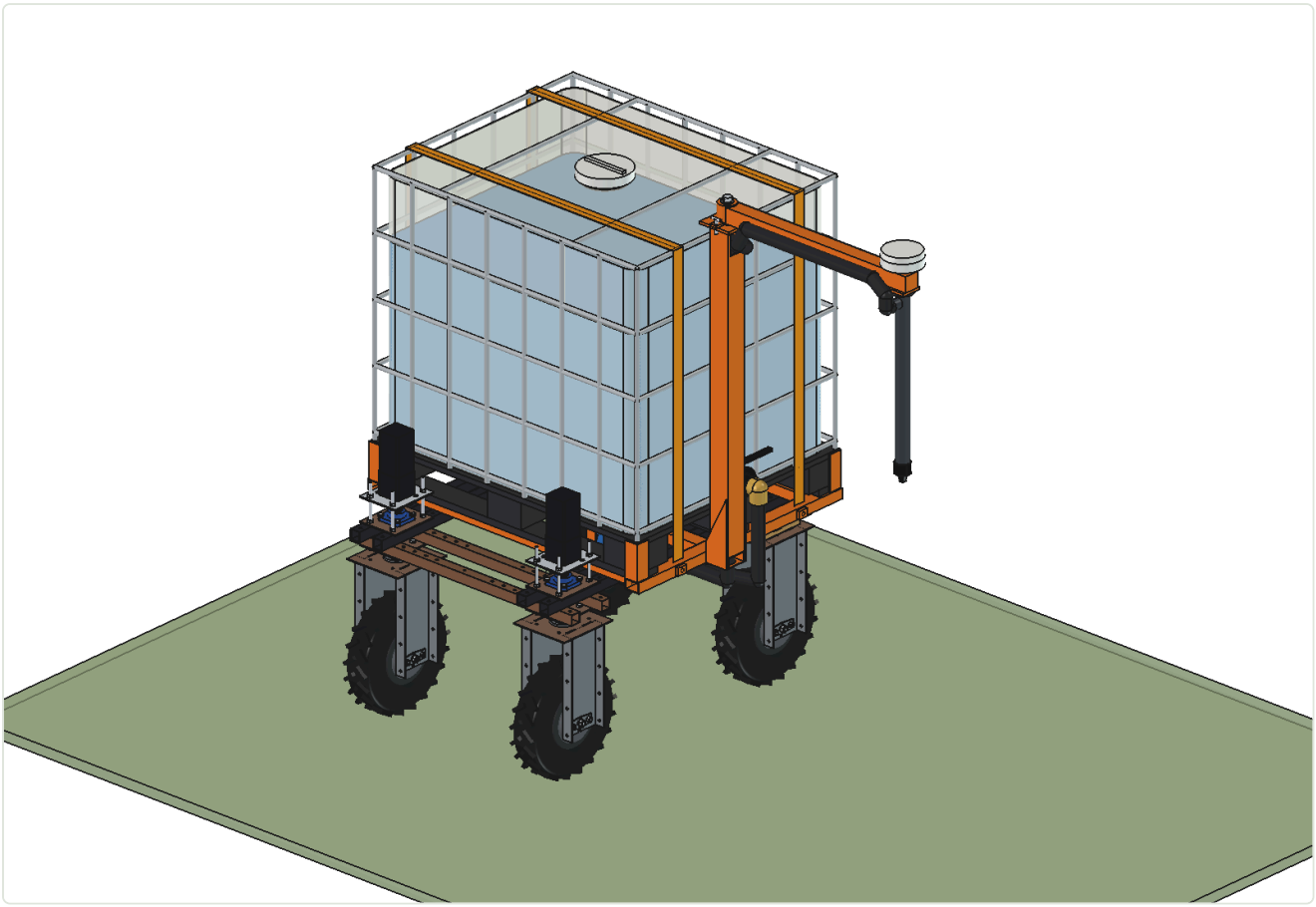
> Pump 150 l/min

> About €6,000 in parts

Folder: `mobile refiller/liquid/`

MOBILE REFILLER · VARIANT 2 OF 5

Simple (IBC)



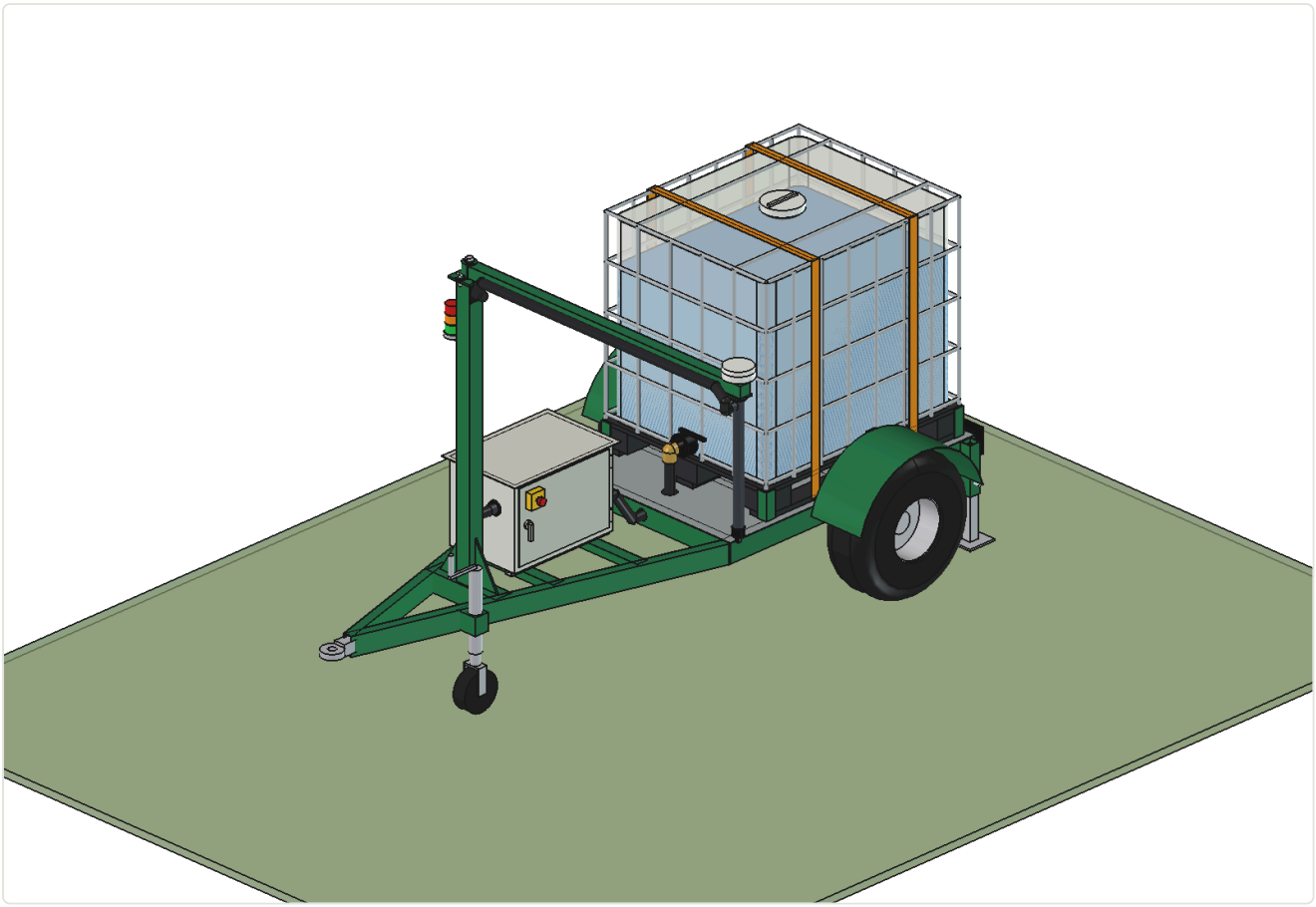
A standard 1000 l IBC set on with pallet forks, pump under the IBC, a fixed mast with a hanging hose and the RTK antenna above the nozzle. The working robot skips passes so every turn is a U-turn; the refiller stands inside the turn and the robot stops ≈ 13 s under the nozzle; 1 m further per refill.

- Standard 1000 l IBC
- About €1,650 in parts
- No boom, trailer or camera

Folder: `mobile refiller/simple/`

MOBILE REFILLER · VARIANT 3 OF 5

Stationary: IBC trailer



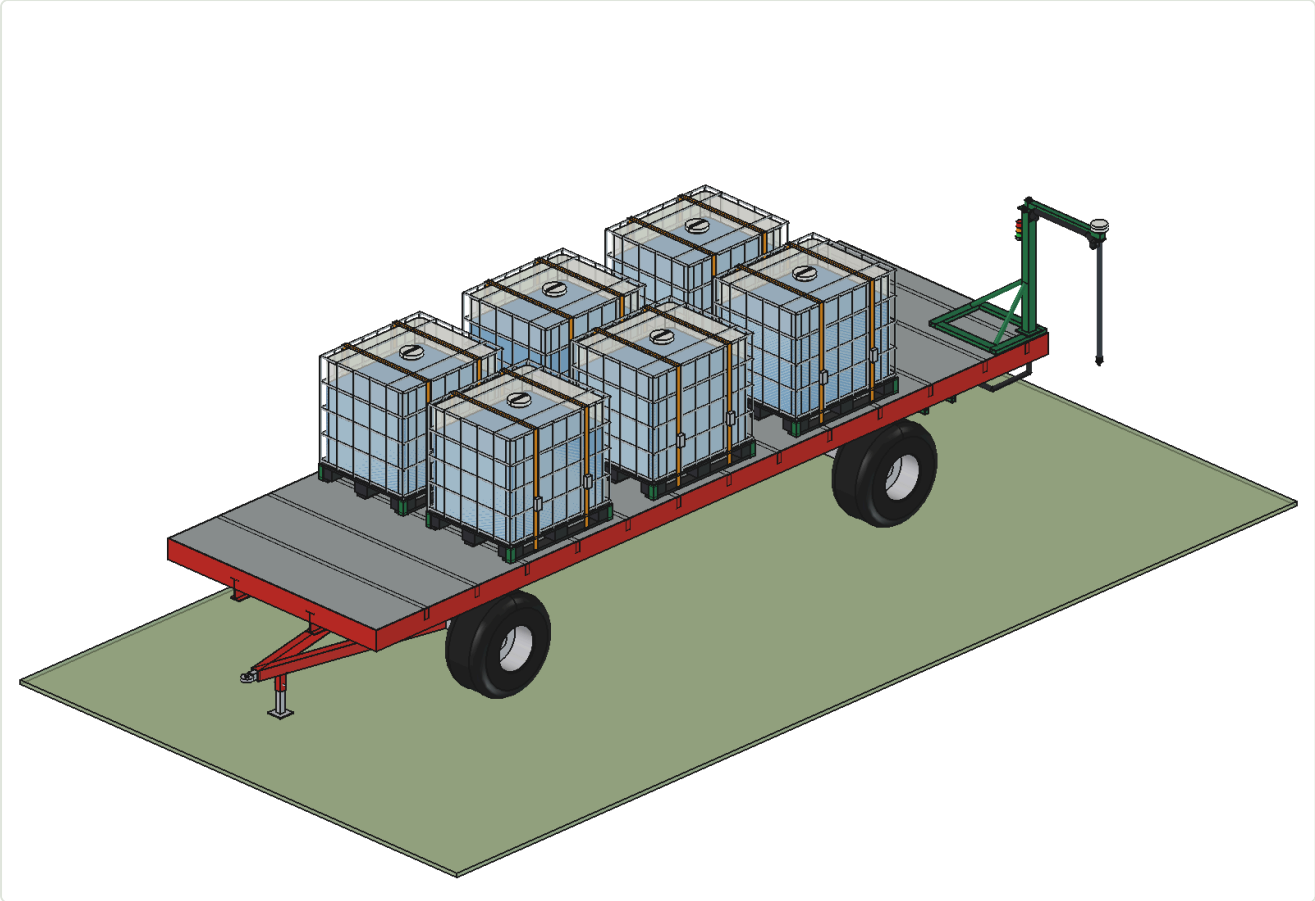
Single-axle trailer for one or two 1000 l IBCs, parked on the headland by a tractor. Control box (pump, battery, controller, RTK receiver, radio) on the A-frame drawbar and a mast with swinging arm, drop pipe, drip-stop nozzle and RTK antenna straight above the nozzle.

- 1 or 2 IBCs
- 120 l per visit
- The working robot drives to it

Folder: `mobile refiller/stationary/ibc_trailer/`

MOBILE REFILLER · VARIANT 4 OF 5

Stationary: flat wagon



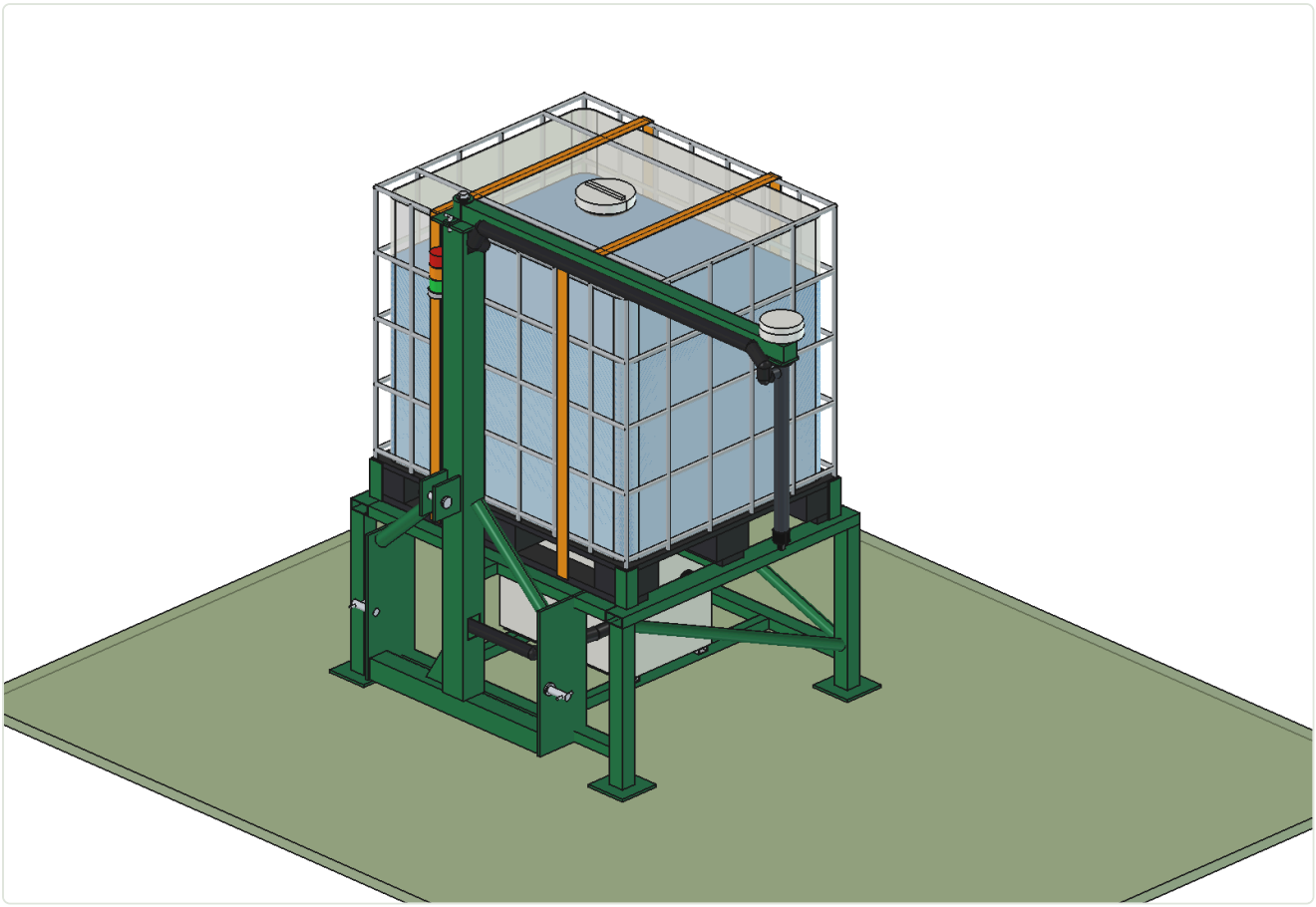
Standard two-axle flat bale wagon (8 × 2.48 m) with 6 IBCs, two across. Only two holes in the deck and two box hangers; suction header along the deck, control box under the deck, mast on a skid at the rear left corner.

- 6 IBCs
- Pump 0.8 m below the IBC bottoms
- Wagon itself unchanged

Folder: `mobile refiller/stationary/flat_wagon/`

MOBILE REFILLER · VARIANT 5 OF 5

Stationary: three-point frame



Welded frame on four legs for one 1000 l IBC, picked up on a category 2 three-point linkage and set down on the headland. The mast doubles as the hitch tower; control box under the platform.

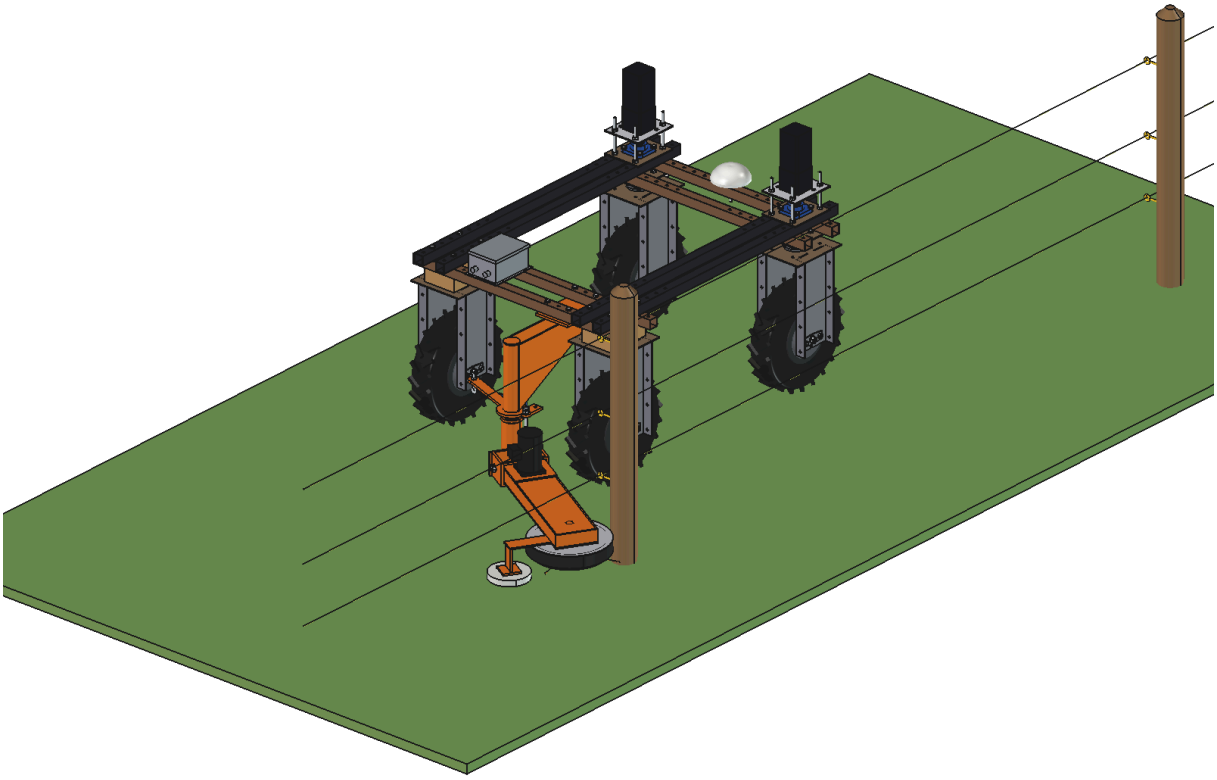
- 1 IBC
- Category 2 three-point
- Pump 535 mm below the IBC bottom

Folder: `mobile refiller/stationary/three_point/`

IMPLEMENT · MOWING

Around-pole fence mower

Mows the strip under an electric fence and goes around the posts by itself.

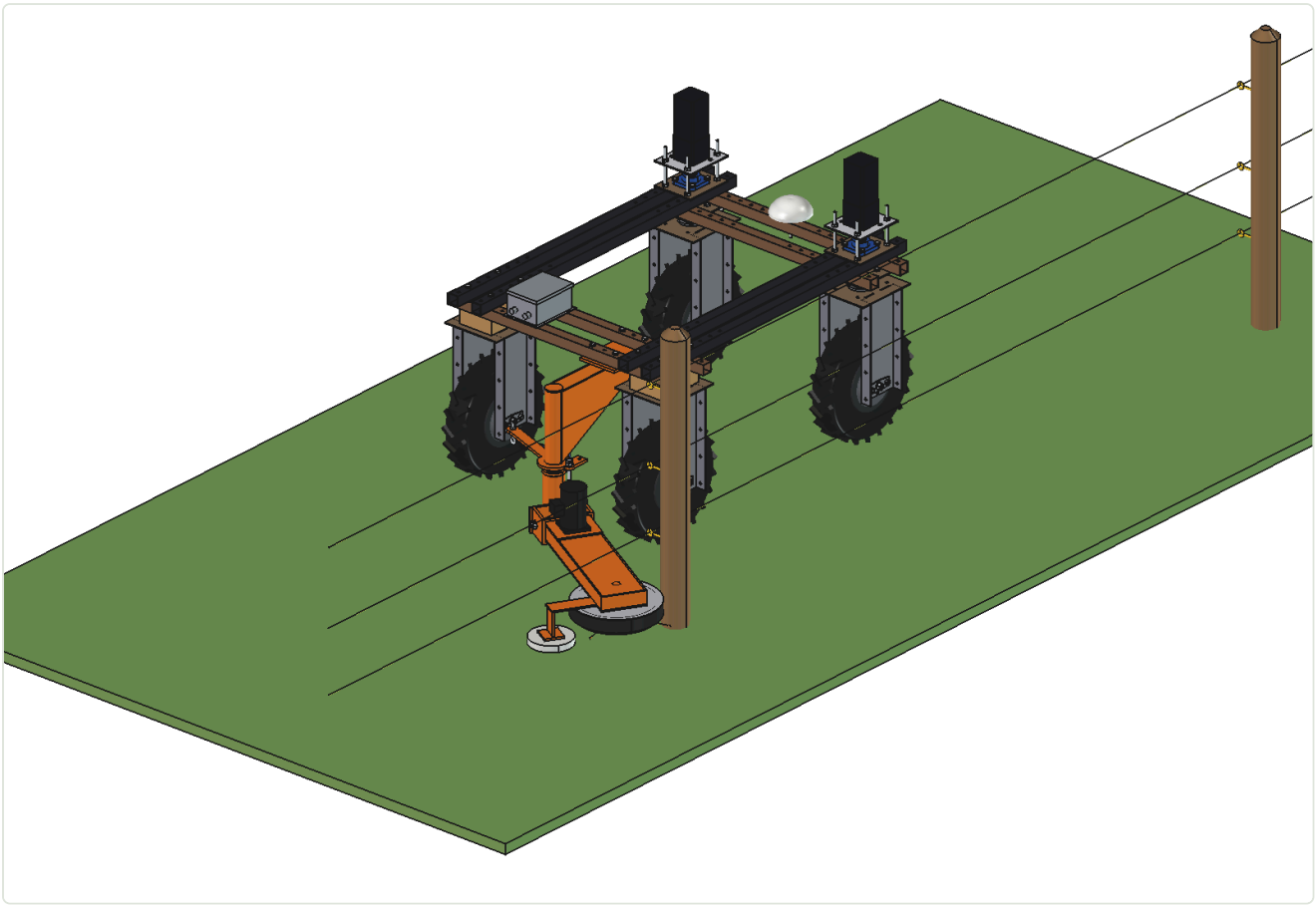


Simple

1 variant: Simple
Folder: `around_pole_mower/`

AROUND-POLE FENCE MOWER · VARIANT 1 OF 1

Simple



Trailing swing arm on a post behind the right rear wheel, held on a rest stop by a spring. A Ø400 nylon-string head under a PE-pipe guard ring slides around each post. 48 V BLDC motor at the root of the arm, V-belt inside the arm to a bought lawn-tractor deck spindle; head 182 mm high on an HDPE skid. No sensor, no actuator, no software.

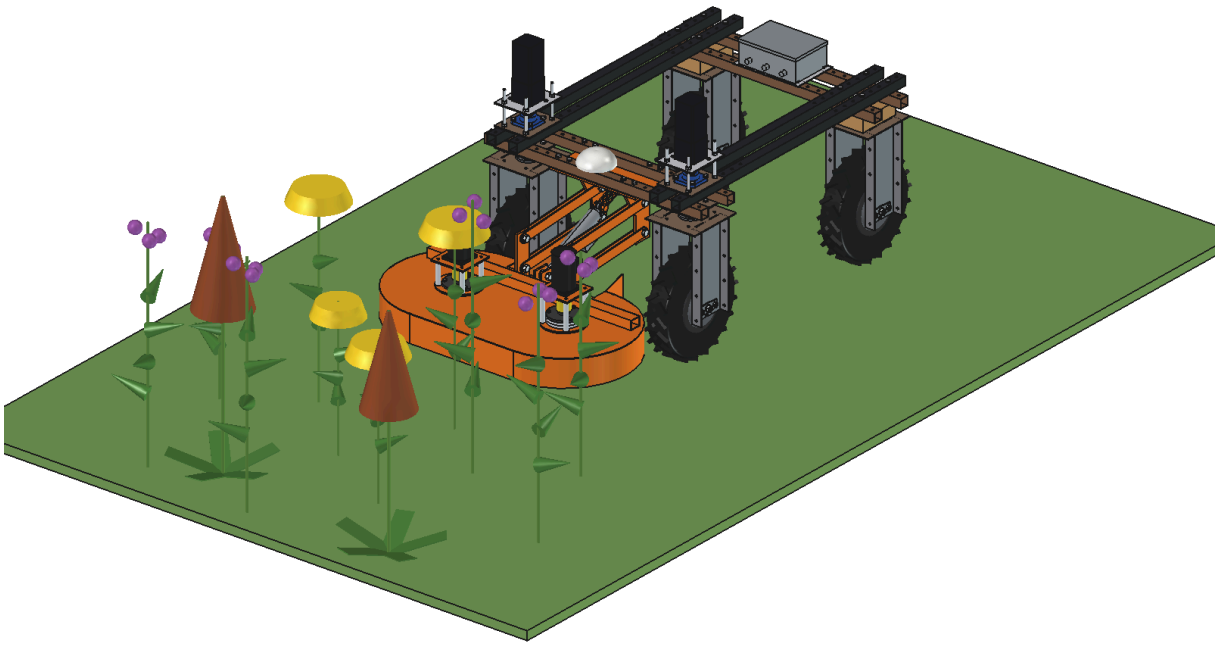
- ≈ 615 W at 0.5 m/s
- 1.8 km of fence per hour
- About €575 in parts
- Arm dynamics and animation with grass

Folder: `around_pole_mower/simple/`

IMPLEMENT · MOWING

Weed picker

Tops thistles, ragwort and dock above the grass and gathers them in one windrow, no bin. Ragwort in particular has to leave the field.

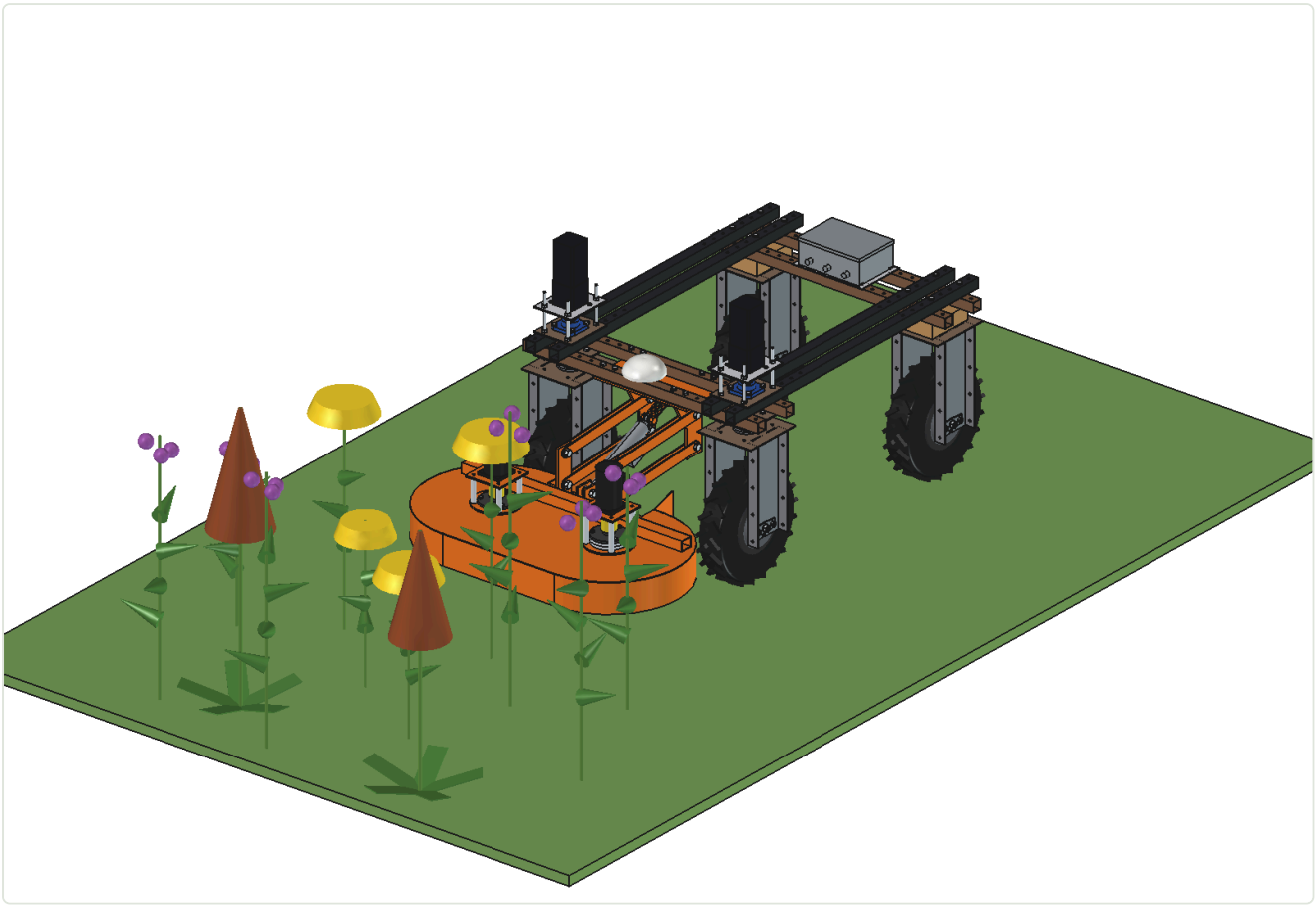


Mowing deck

1 variant: Mowing deck
Folder: `weed picker/`

WEED PICKER · VARIANT 1 OF 1

Mowing deck



Twin-rotor mowing deck in front of the robot: two rotors with free-swinging blades (Ø510, 990 mm cut) turn inwards and throw the chopped weed out of a centre opening as a windrow ≈ 320 mm wide. Each rotor: 48 V BLDC motor on a lawn-tractor deck spindle; the left rotor runs 24 mm higher. Cut height 100–400 mm from a 4000 N actuator on a parallel linkage, with a float link for humps.

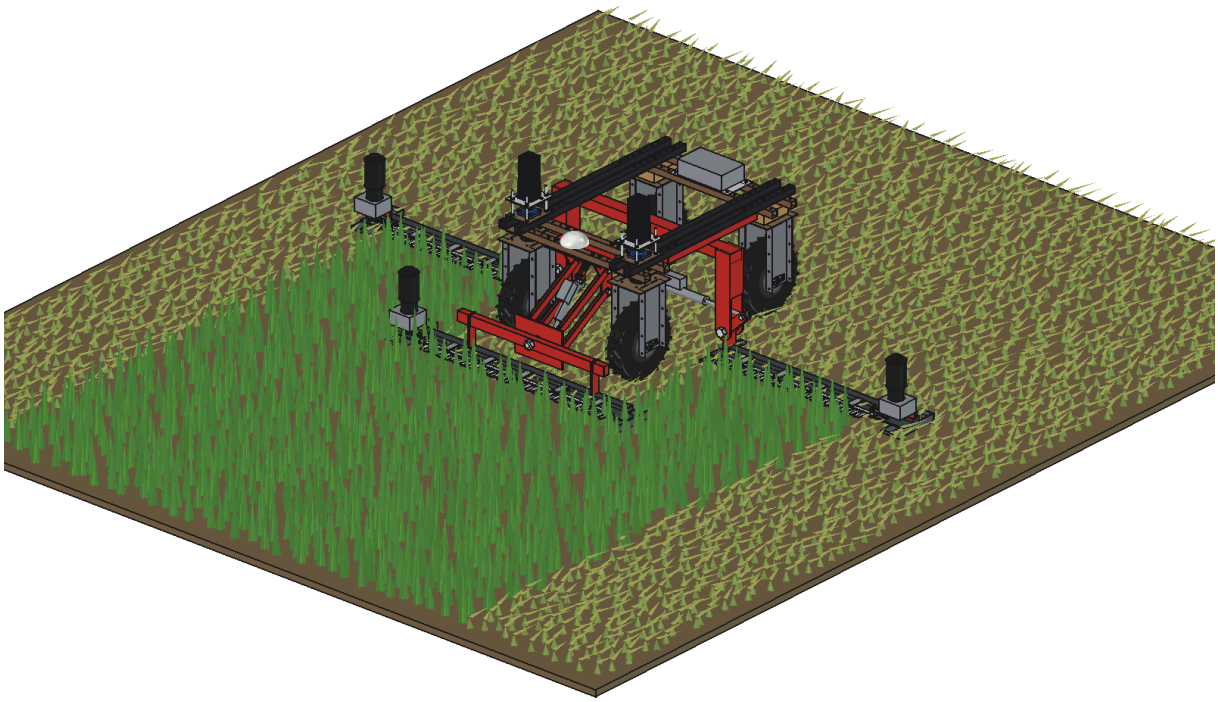
- ≈ 1 kW at 0.5 m/s
- 82 kg (rear counterweight advised)
- About €1,040 in parts
- Animation in a weed patch

Folder: `weed picker/mowing_deck/`

IMPLEMENT · MOWING

Sickle bar mower

Mows grass with double-knife cutter bars instead of rotating blades: a clean cut at low power, the grass left lying in a swath to dry. The bars follow the ground on their shoes.

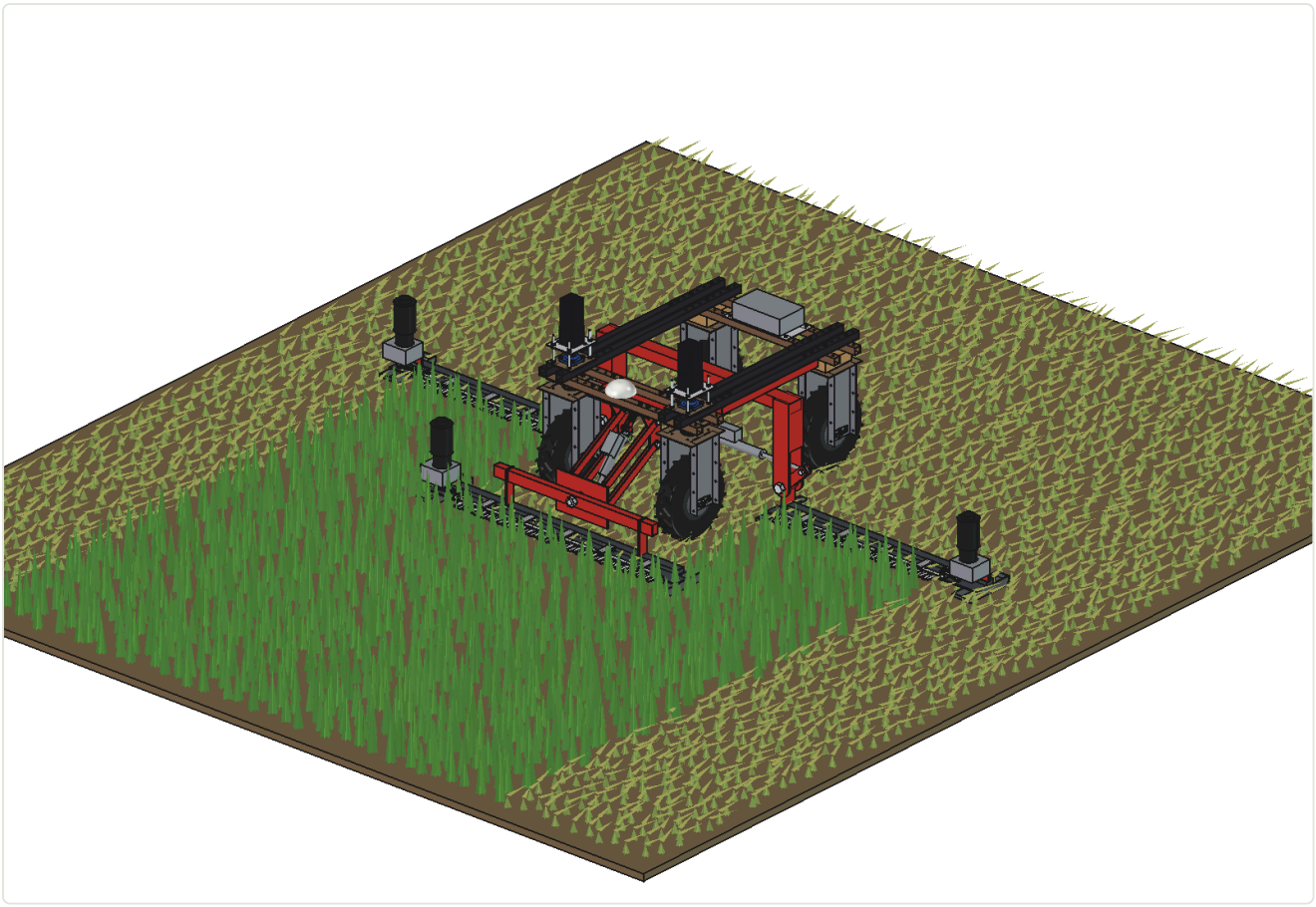


Triple

1 variant: Triple
Folder: `sickle bar mower/`

SICKLE BAR MOWER · VARIANT 1 OF 1

Triple



Three double-knife bars: 1.25 m in front of the robot on a parallel linkage with a float link and a roll pin, and two 1.0 m wings in the middle of the frame on a mid frame under the top beams. Each bar: upper and lower knife (50 mm pitch, stroke 50, 600 rpm), a crank gearbox with two cranks 180° apart so the knife forces cancel, and a 48 V 1000 W BLDC motor. The wings fold up with a 4000 N actuator each; the anchor slides in a slot so they float with the ground.

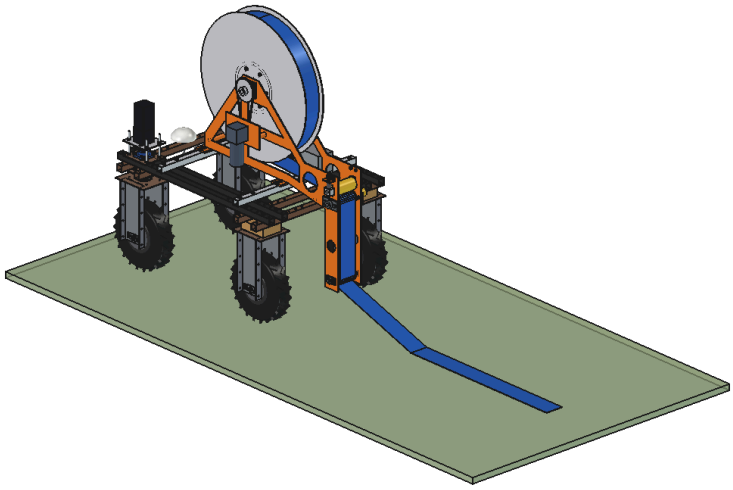
- 3.25 m working width
- ≈ 2.2 kW mowing at 1 m/s, 1.0 ha/h
- 210 kg, 163 kg on the shoes while mowing
- About €3,200 in parts

Folder: `sickle bar mower/triple/`

IMPLEMENT · SPRAYING, FERTILIZING & IRRIGATION

Irrigation

Irrigates grassland by flooding at low pressure from a pump at the ditch: little energy, no wind drift, no evaporation in the air.

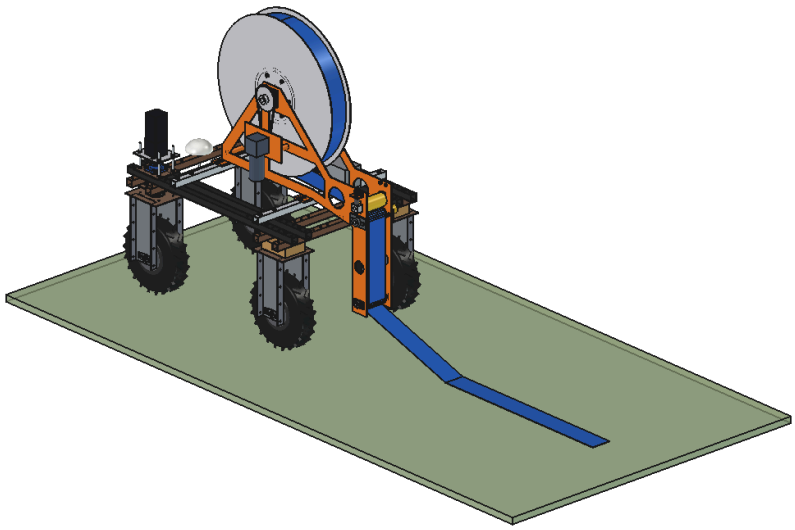


Stationary pump + hose reel

1 variant: Stationary pump + hose reel
Folder: `irrigation/`

IRRIGATION · VARIANT 1 OF 1

Stationary pump + hose reel



The robot carries a reel with 150 m of 3" lay-flat hose, lays it out empty, the petrol pump at the ditch floods a 4 m strip (outlets every 2 m), the hose runs empty and the robot winds it up again. Pinch rollers keep the reel dry and squeeze the last water out.

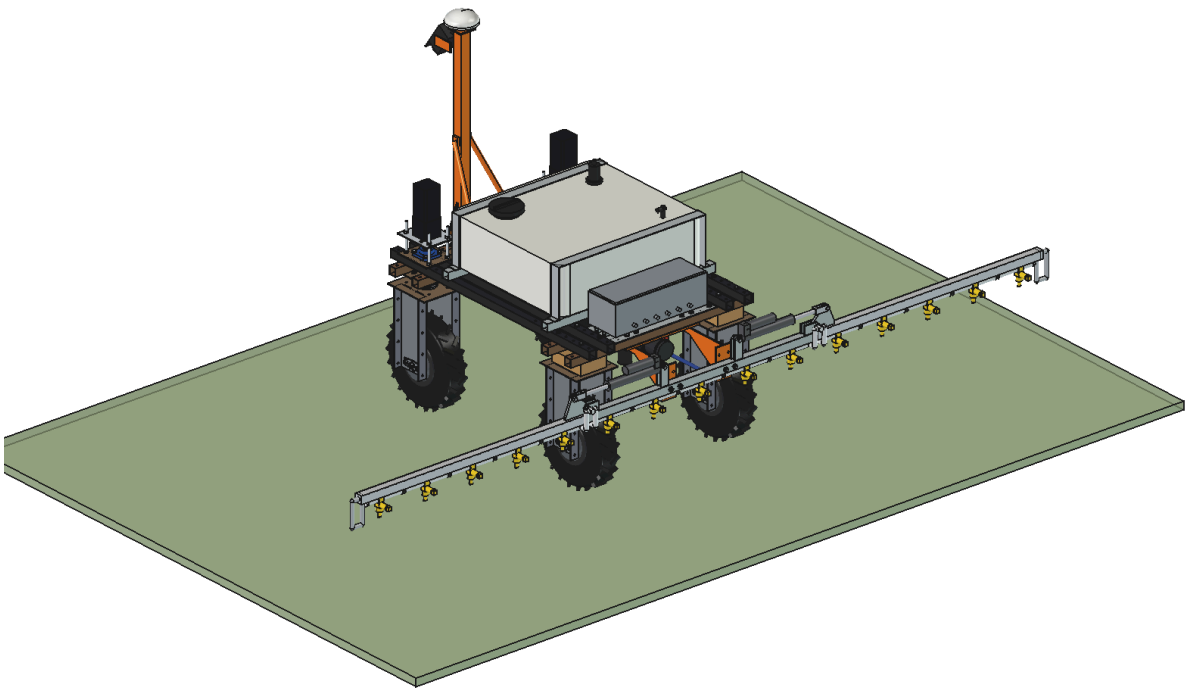
- 7.4 m^3/h at 0.17 bar, CU ≈ 99 %
- 122 min per 500 m^2 strip at 25 mm
- 146 kg, about €1,900 in parts

Folder: `irrigation/stationary + roller/`

IMPLEMENT · SOIL CULTIVATION & WEED CONTROL

Spot sprayer

Sprays weeds plant by plant instead of the whole field. A camera in front finds the weeds; the boom behind opens only the nozzles above a plant.

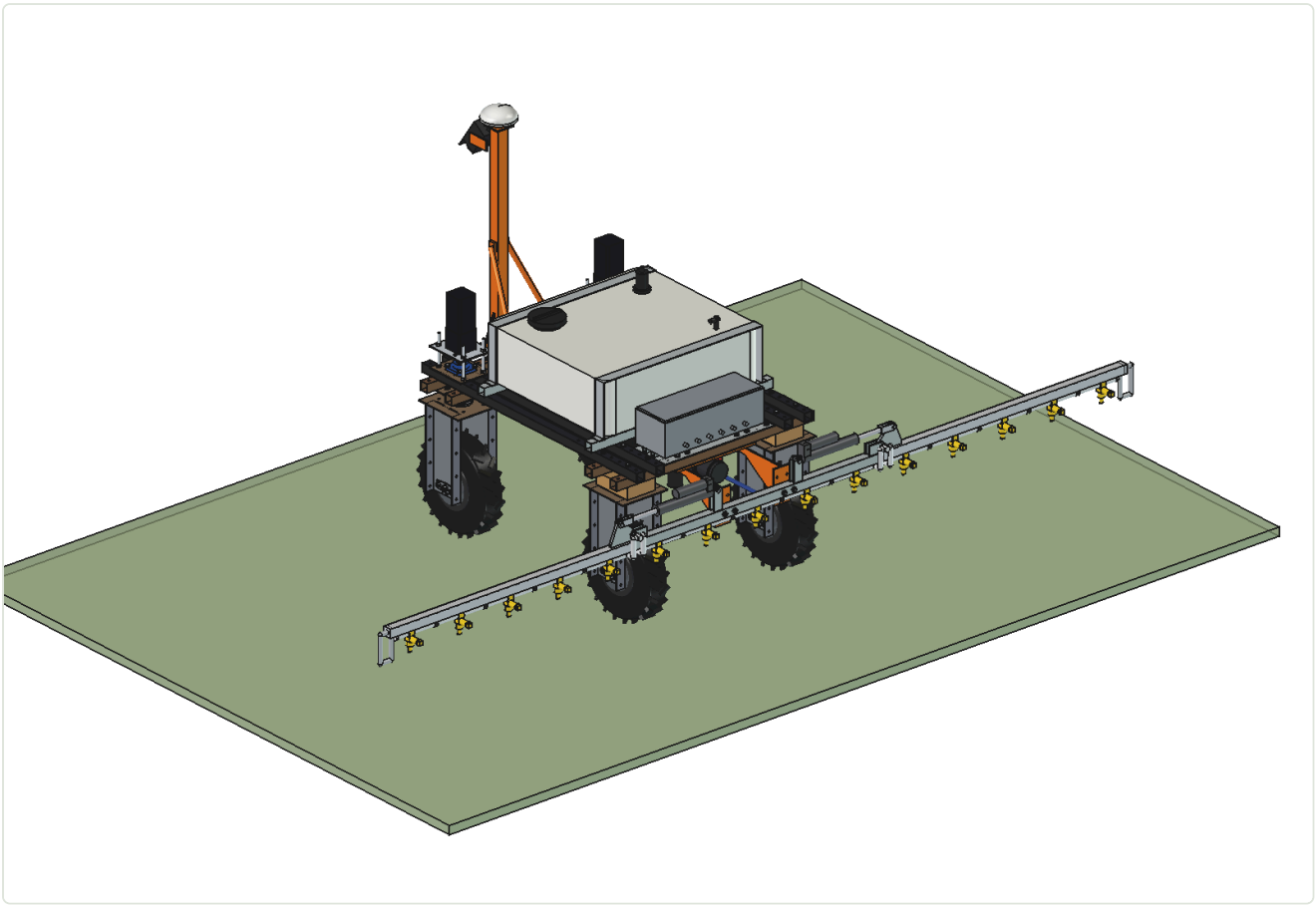


Folding 3.75 m boom

1 variant: Folding 3.75 m boom
Folder: `spotspray unit/`

SPOT SPRAYER · VARIANT 1 OF 1

Folding 3.75 m boom



Wide-angle global-shutter camera on a mast (1.48 m, 55° down) sees the whole strip from 1.8 m ahead; 2.65 s later 15 PWM nozzles at 250 mm on a stainless pipe open above the plant. Centre section 1.25 m and two 1.25 m wings that fold up in 3.2 s with a 24 V actuator each, also in the field. 120 l tank and diaphragm pump on the robot.

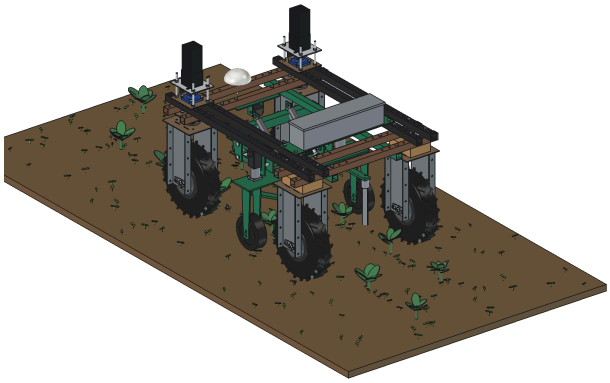
≈ 20 l/ha at 5 % weed cover, 1.1 ha/h	91 kg
About €3,580 in parts	Animation with docks and a fence post

Folder: `spotspray unit/boom/`

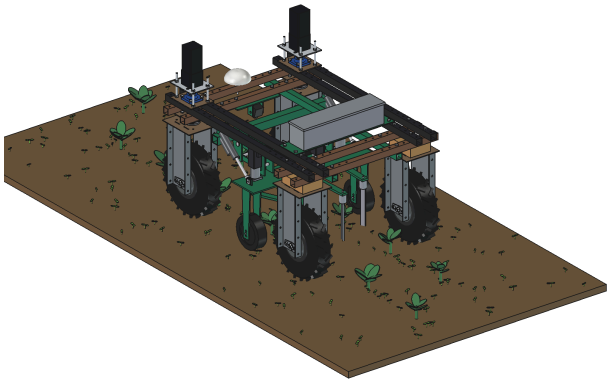
IMPLEMENT · SOIL CULTIVATION & WEED CONTROL

In-row weeder

Hoes inside two rows of broccoli (375 mm apart, wheels in the inter-rows). Camera under the front beams finds the plants 0.84 s ahead; knives swing out around each plant.



Finger, 2 rows

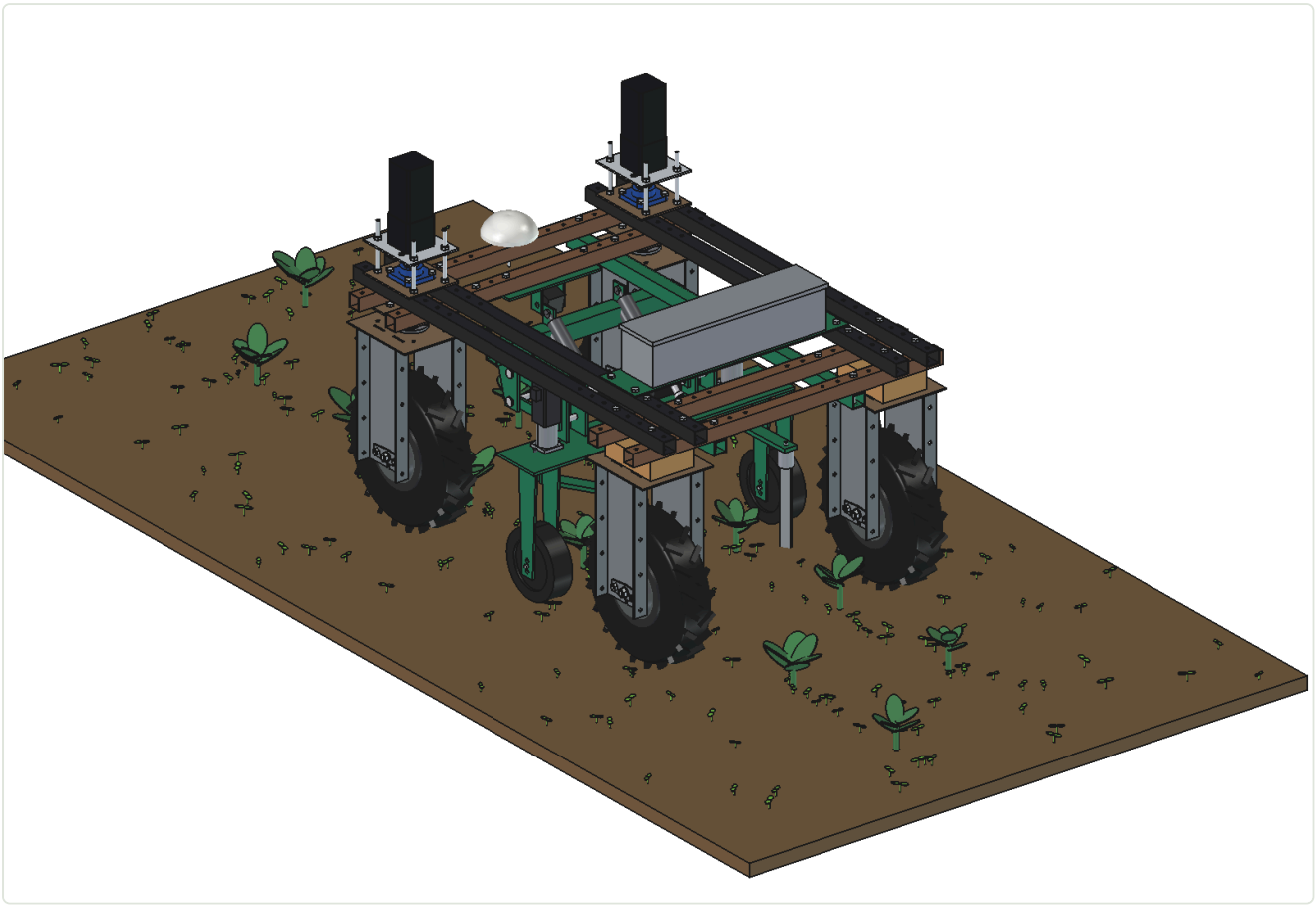


Double arms

2 variants: Finger, 2 rows · Double arms
Folder: `in row weeder/`

IN-ROW WEEDER · VARIANT 1 OF 2

Finger, 2 rows



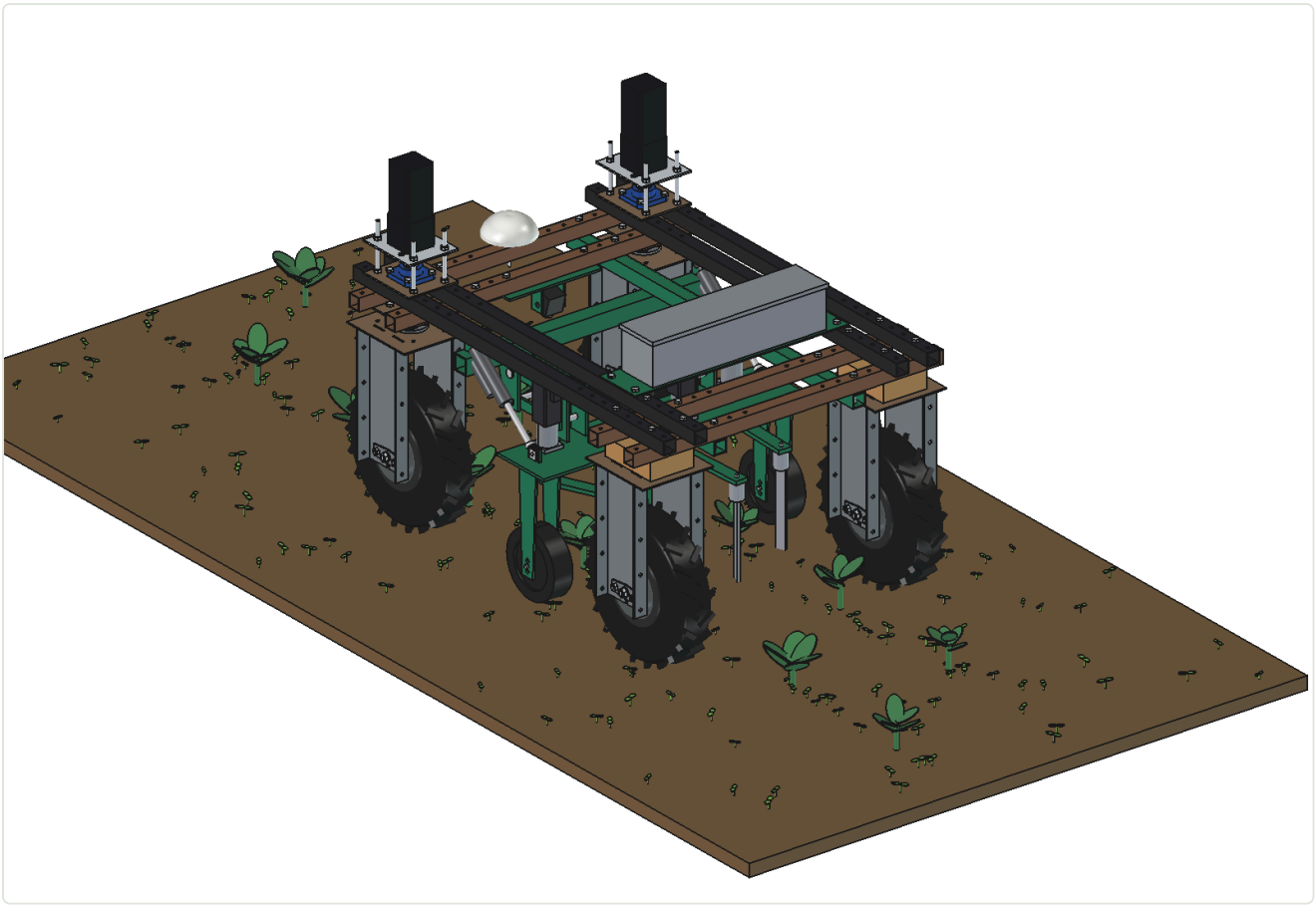
Two row units under the belly on parallelograms with gauge wheels in the robot's wheel track. A NEMA 23 closed-loop stepper with a 10:1 gearbox swings a 220 mm arm with a spring-steel finger and a flat knife 20 mm deep. Out in 98 ms, back in 148 ms; never closer than 50 mm to a stem.

- 0.5 m/s, 6,400 plants/h
- 55 kg
- About €2,240 in parts

Folder: `in row weeder/finger/`

IN-ROW WEEDER · VARIANT 2 OF 2

Double arms



Two knives per row (4 steppers) that open to both sides of each plant and close behind it, timed by the real distance to the stem. Each knife moves only 78 mm: out in 89 ms, back in 118 ms.

> 90 % of the row band outside the safe zones hoed

> 68 kg

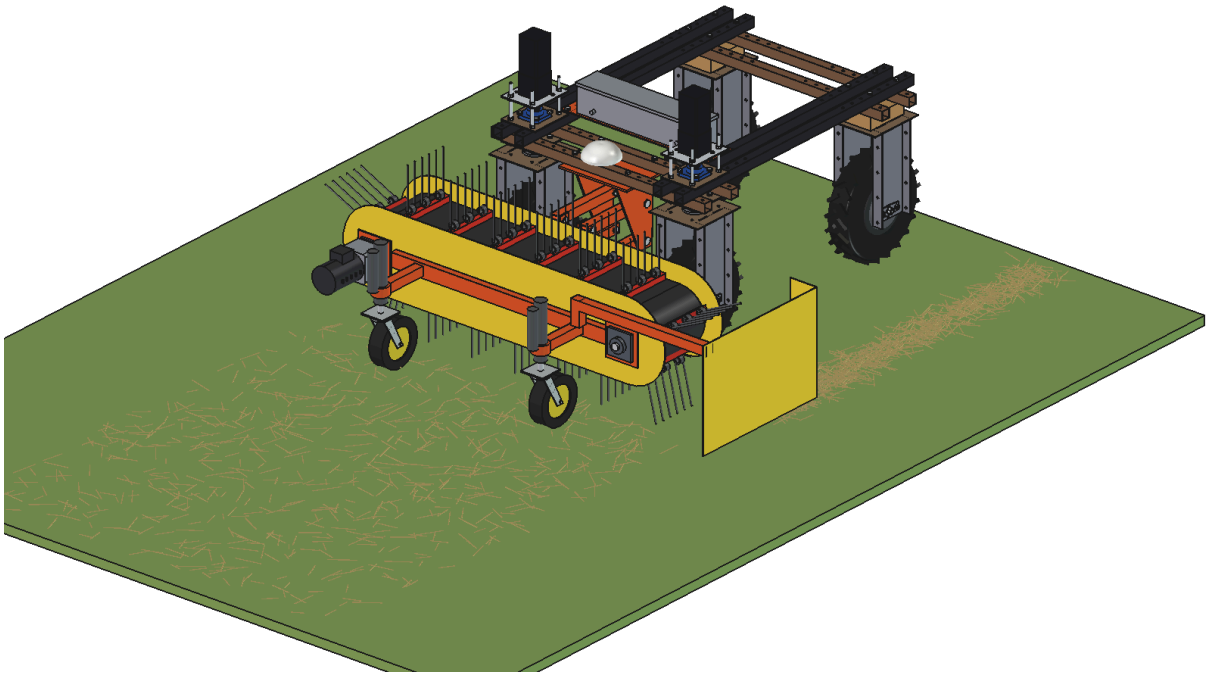
> About €2,820 in parts

Folder: `in row weeder/double_arm/`

IMPLEMENT · SOIL CULTIVATION & WEED CONTROL

Belt rake

Band rake in front of the robot: a rubber belt with 15 tine bars around two drums (1.52 m working width). Combs weed seedlings out of nursery beds between young trees or rakes moss and thatch out of grassland into one windrow.

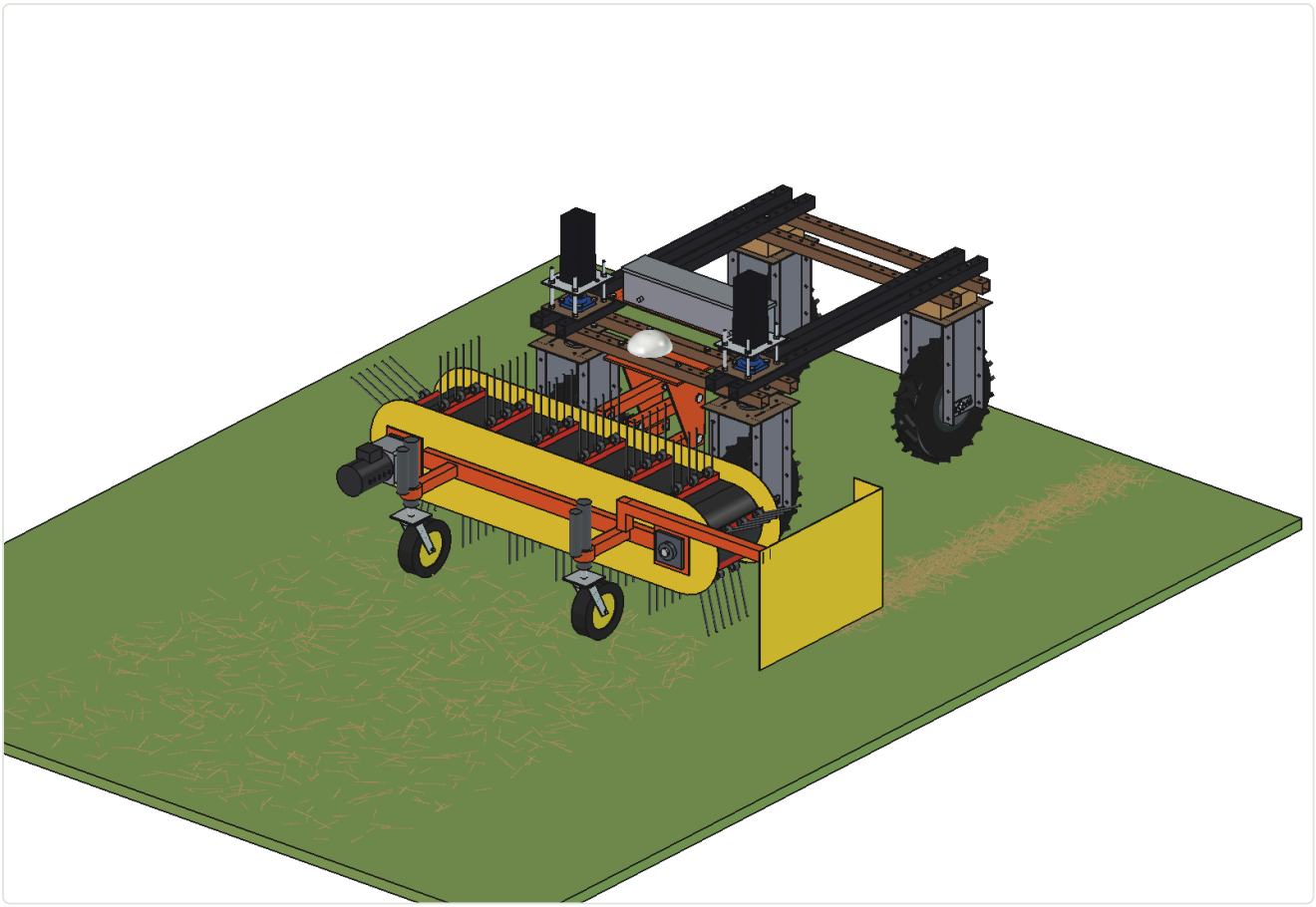


Electric

1 variant: Electric
Folder: `bandrake/`

BELT RAKE · VARIANT 1 OF 1

Electric



48 V 750 W gearmotor with a 20:1 planetary gearbox on the right drum. Two swivel castors on 24 V actuators carry the 110 kg rake, set the depth and lift it 100 mm for the headland; a free parallelogram pushes it.

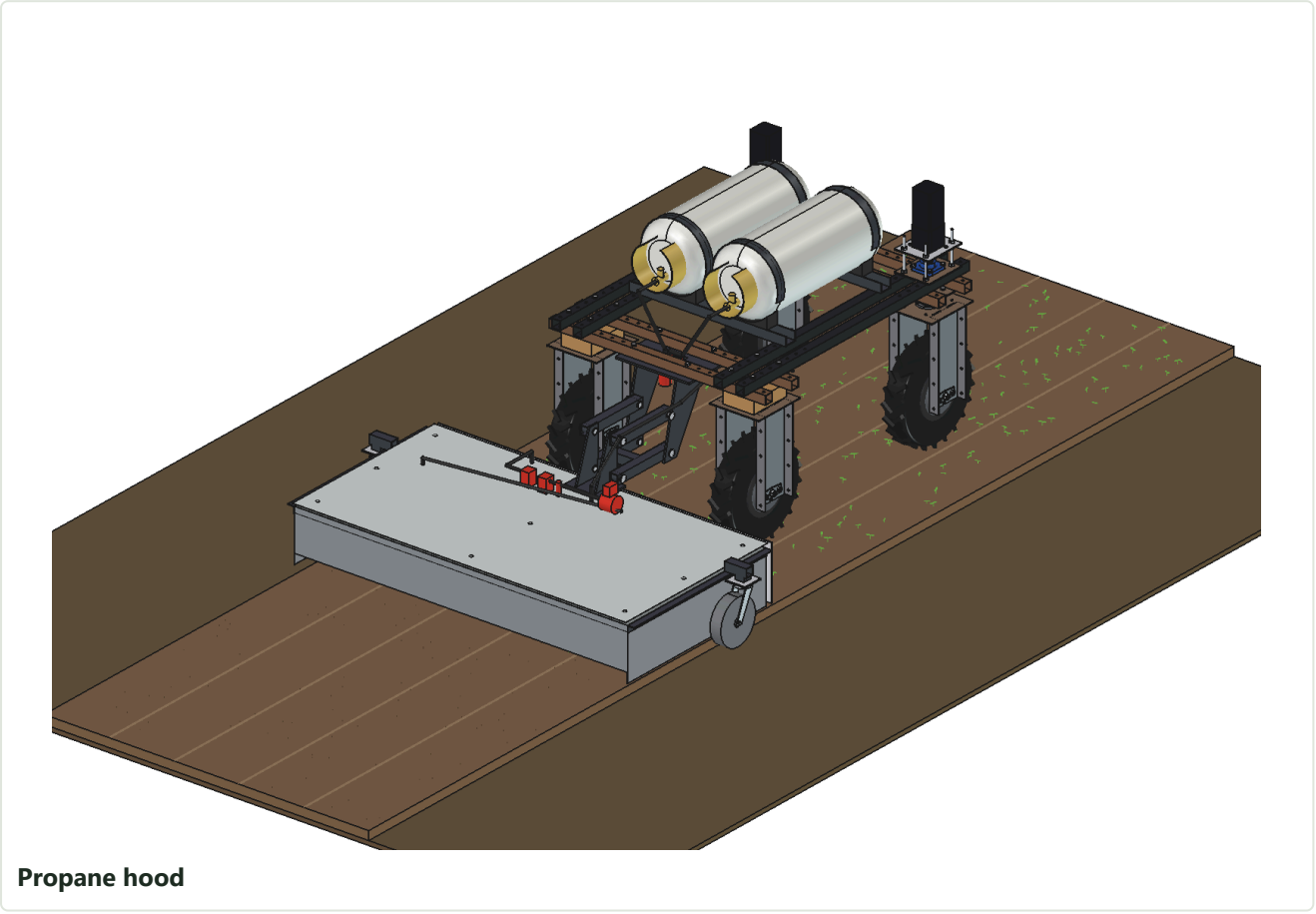
- Comb mode 0.6 m/s, rake mode 1.2 m/s, 0.49 ha/h
- About €2,380 in parts
- Tine forces and animation

Folder: `bandrake/electric/`

IMPLEMENT · SOIL CULTIVATION & WEED CONTROL

Weeds burner (flame weeder)

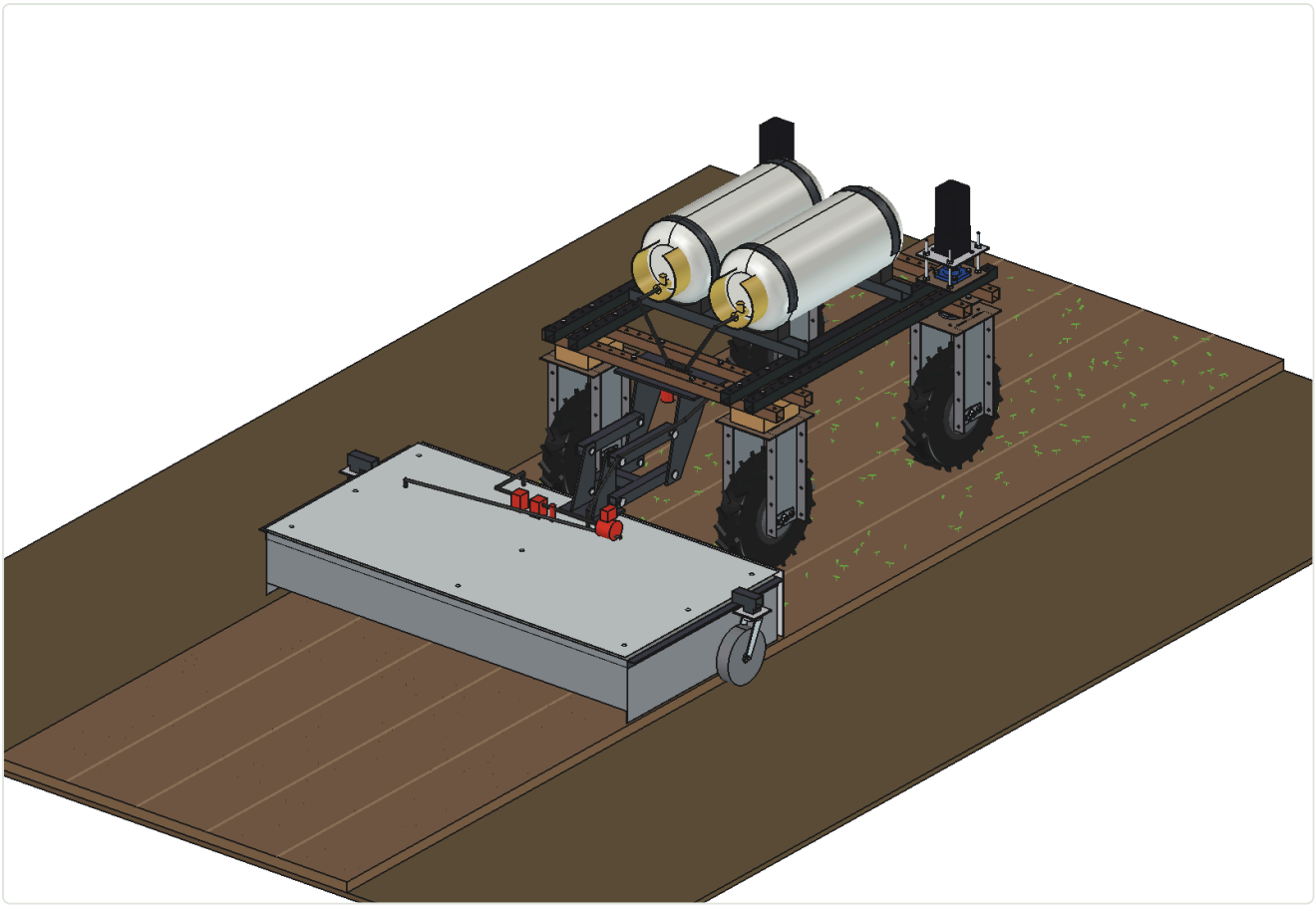
Pre-emergence flame weeder for freshly sown beds (carrot, onion, beetroot, parsnip): kills the emerged seedlings without moving the soil.



Propane hood

1 variant: Propane hood
Folder: `weeds burner/`

Propane hood



Insulated 1.5 × 0.8 m hood behind the robot with 6 flat-flame burners (13 kg/h, 167 kW). Two 18 kg forklift cylinders with liquid withdrawal on the robot, a vaporiser coil in the hood, two main solenoid valves in series, a motorised regulator, pilots with igniters and flame sensors; the valves close at once when the robot stops.

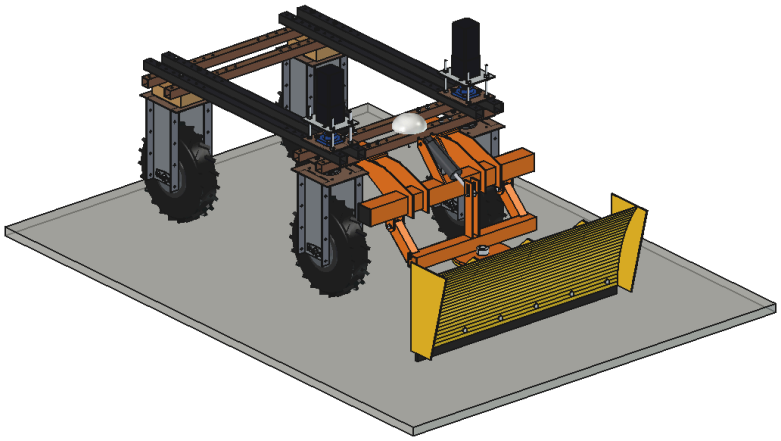
- 34 kg/ha at 0.7 m/s, 0.84 ha per filling
- Gas ≈ €55/ha
- 131 kg empty, about €3,400 in parts

Folder: `weeds burner/propane_hood/`

IMPLEMENT · SOIL CULTIVATION & WEED CONTROL

Push blade

Front-mounted dozer / grader blade: pushes feed back to the feed fence, scrapes slurry and snow, grades soil and paths. Covers the manure scraper idea as well.

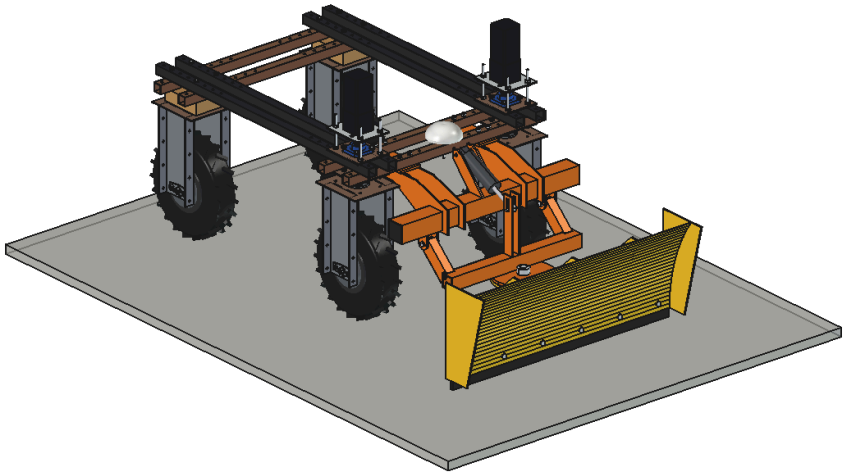


Simple

1 variant: Simple
Folder: `push blade/`

PUSH BLADE · VARIANT 1 OF 1

Simple



1100 x 450 moldboard on R700 with a bolted rubber edge (steel edge for soil), angle +-25 degrees on an M30 king pin with a lock pin, two lift arms on a front toolbar under the front lower beams, 24 V 1500 N actuator with a float slot.

> Net push 801 N on dry concrete, 348 N with manure

> Lift 159 mm, clears the steered front tyres by 148 mm

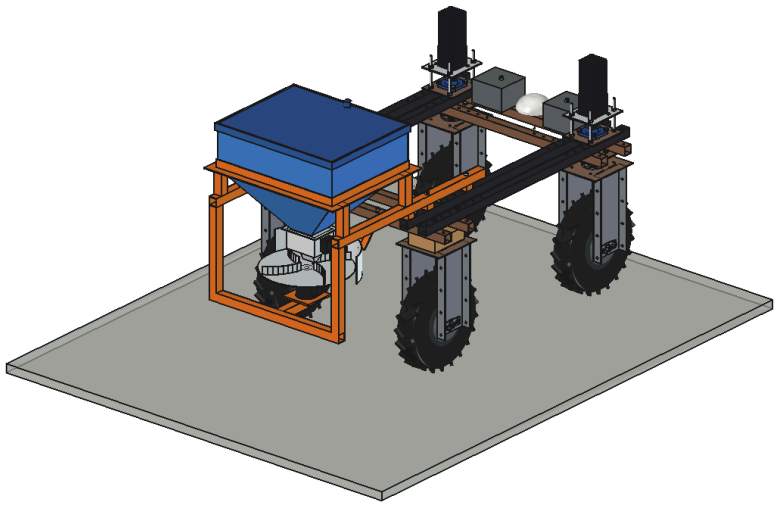
> 81 kg

Folder: `push blade/simple/`

IMPLEMENT · SPRAYING, FERTILIZING & IRRIGATION

Fertilizer spreader

Spinning-disc spreader for granular fertilizer (and granular lime with a coarser roller). Dose follows the driving speed and a task map.

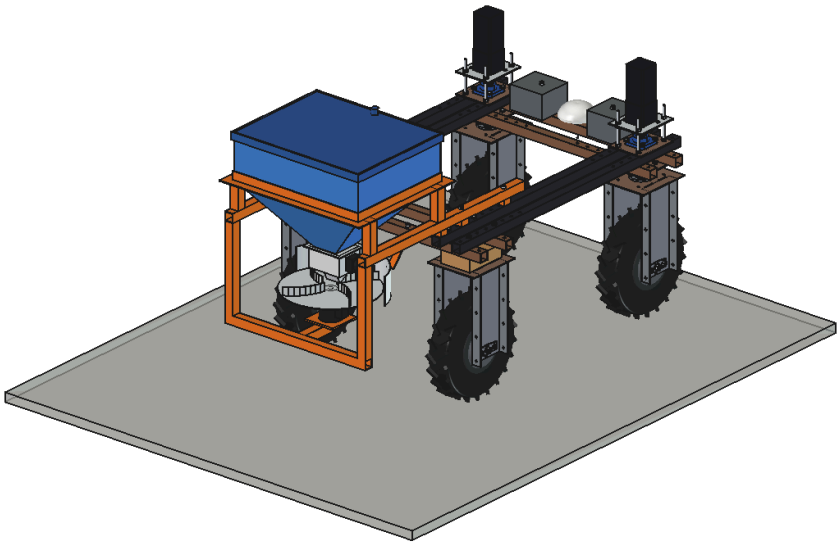


Simple

1 variant: Simple
Folder: `fertilizer spreader/`

FERTILIZER SPREADER · VARIANT 1 OF 1

Simple



88 l PE hopper over and behind the rear axle on rails on the upper beams, fluted metering roller O80 on a 12 V gearmotor, vaned disc O400 on a 48 V BLDC motor behind the robot with a guard towards the robot. Optional 2 x 20 kg front counterweight.

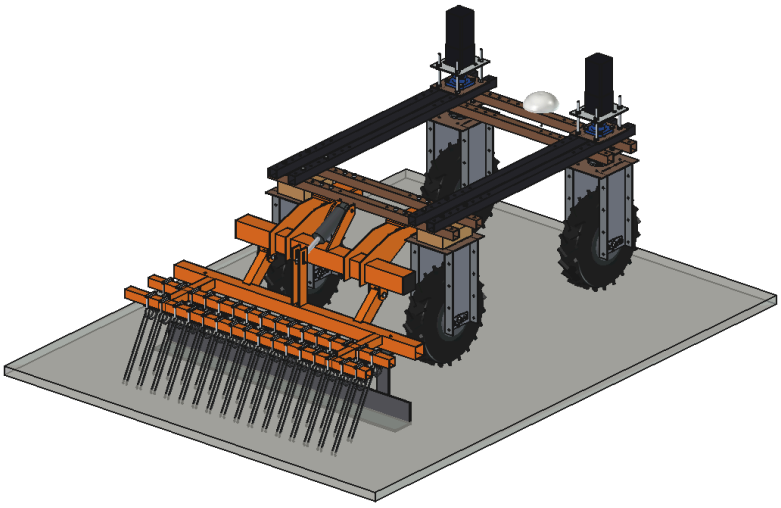
- Driving width 4.1..6.7 m (500..900 rpm)
- 100..541 kg/ha at 1 m/s
- ~1.6 ha/h, 130 kg full

Folder: `fertilizer spreader/simple/`

IMPLEMENT · CROP & SOIL

Grassland harrow

Pasture harrow behind the robot: a levelling board spreads molehills and dung pats, two rows of spring tines rake out moss and work in seed. Covers the molehill spreader idea.

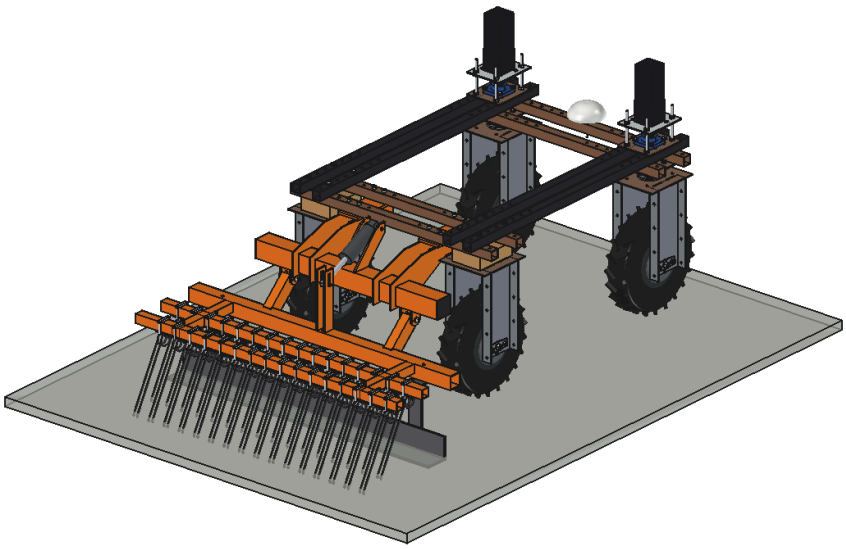


Simple

1 variant: Simple
Folder: `grassland harrow/`

GRASSLAND HARROW · VARIANT 1 OF 1

Simple



1300 mm harrow on two lift arms from a rear toolbar (the push blade geometry, mirrored): angle-iron levelling board, 32 double spring tines O8 (a track every 37.5 mm), floats on the sward, lifted by a 24 V actuator.

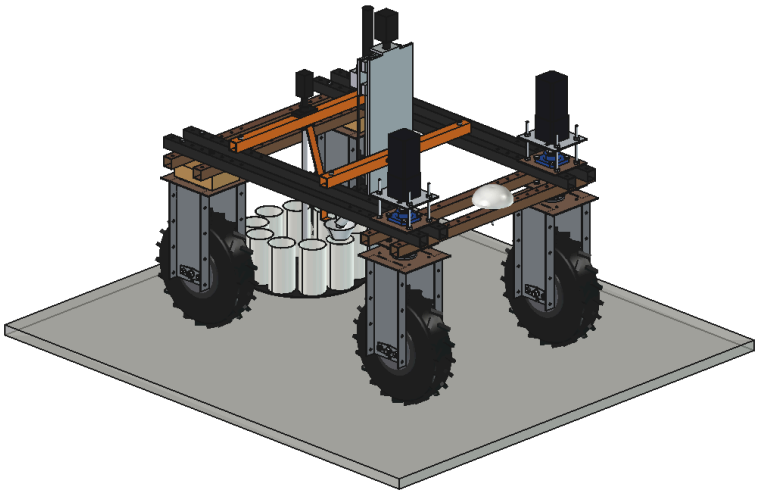
- Pull 629 N on grass, grip margin 1.44
- 0.56 ha/h at 1.5 m/s
- 85 kg

Folder: `grassland harrow/simple/`

IMPLEMENT · CROP & SOIL

Soil sampler

Takes soil cores at GPS points with a gouge on a ball-screw Z axis in the middle of the robot and collects 40 cores per plot in a cup of a 10-cup carousel.

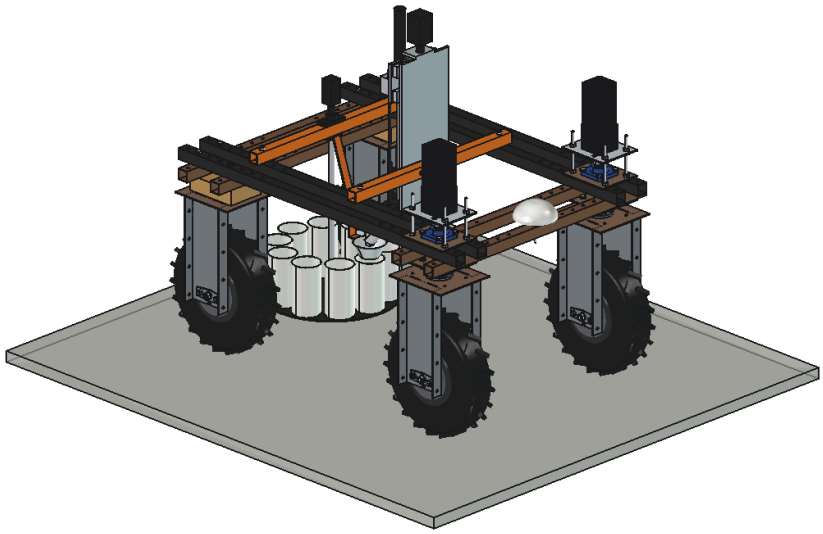


Simple

1 variant: Simple
Folder: `soil sampler/`

SOIL SAMPLER · VARIANT 1 OF 1

Simple



SBR20 rails and an SFU1605 ball screw with a NEMA23 on a mast plate between two bridges on the upper beams, O25 gouge with ejector, sprung foot, swing chute and funnel, carousel of 10 cups of 1.3 l.

> 0-10 cm grass needs 0.5..0.8 kN, 1.3 kN available

> 38 s per core, 25 min per plot

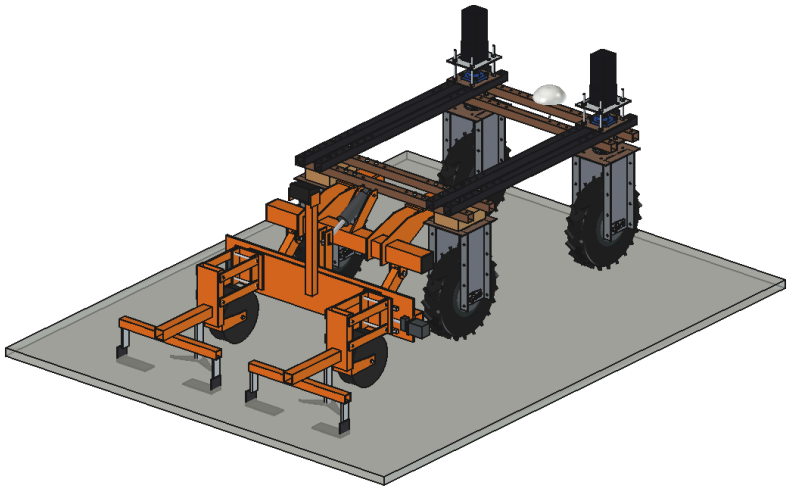
> 33 kg

Folder: `soil sampler/simple/`

IMPLEMENT · SOIL CULTIVATION & WEED CONTROL

Inter-row hoe

Single-axis guided hoe behind the robot for row crops such as maize: two hoe units in the inter-rows behind the wheels, kept on the rows by a camera-steered ball-screw side shift.

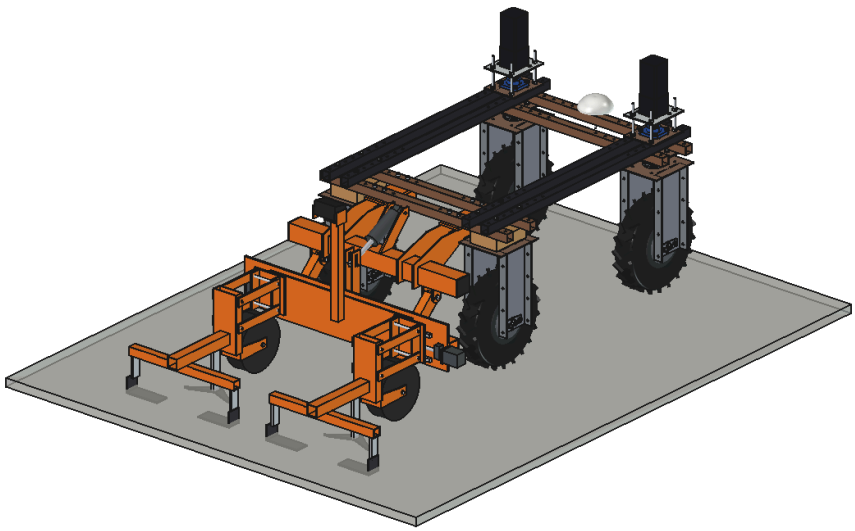


Simple

1 variant: Simple
Folder: `inter row hoe/`

INTER-ROW HOE · VARIANT 1 OF 1

Simple



Rear toolbar and lift arms of the grassland harrow (3000 N actuator), carrier with 2 HGR15 rails, 8 mm shift plate moved +-75 mm by an SFU1610 ball screw and a closed-loop NEMA23. Two parallelogram units with a gauge wheel O250, a goosefoot 250 and two L-blades (550 mm per inter-row, 100 mm band each side of the row). Camera on a mast above the middle row.

> 1.5 m per pass (2 x 750), ~0.43 ha/h at 1 m/s

> Pull 764 N at 20 mm, grip margin 1.23

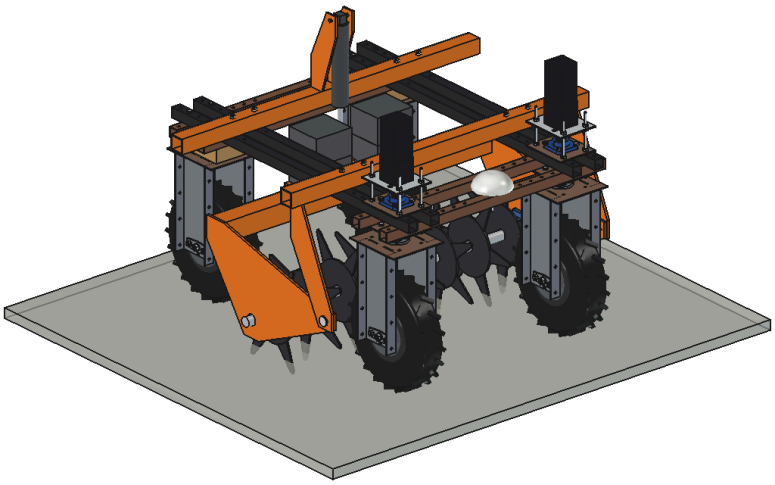
> Side shift 80 mm/s, ~119 kg, needs the 2 x 20 kg front counterweight

Folder: `inter row hoe/simple/`

IMPLEMENT · SOIL CULTIVATION & WEED CONTROL

Grassland aerator

Slitter aerator in the middle of the robot, between the axles: a passive roller with star discs cuts slits into grassland, pressed in by frame weight and ballast, both axles sharing the reaction.

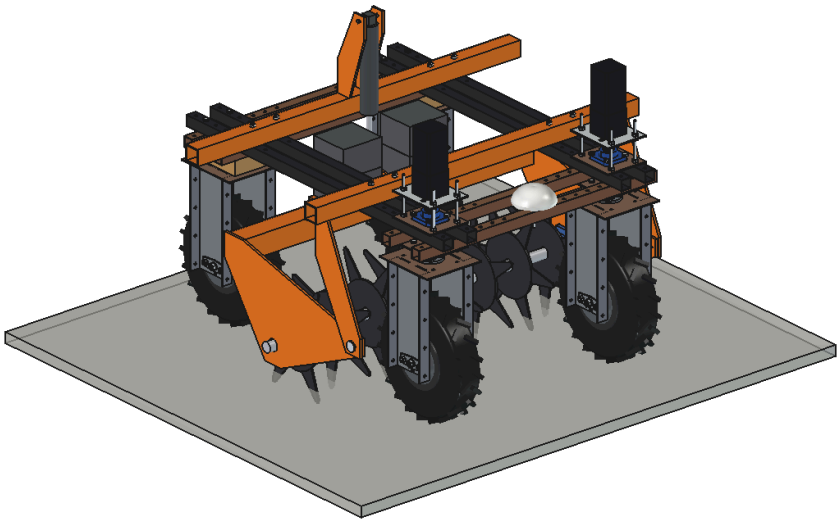


Simple

1 variant: Simple
Folder: `grassland aerator/`

GRASSLAND AERATOR · VARIANT 1 OF 1

Simple



2 bridges on the upper beams, side arms on pivots outside the steering sweep, roller with 8 stars of 6 slitting knives (O450) in 2 UCFL208 bearings, cross beam with 4 x 20 kg ballast, 24 V 2000 N actuator from a mast on the rear bridge (lifts 14 degrees, can press).

> 1.2 m, slits 150 x 236 mm, 70 mm deep

> ~0.41 ha/h at 1.2 m/s, pull 325 N

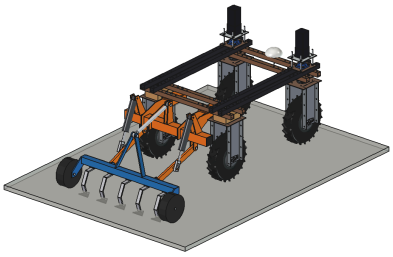
> 201 kg incl. 80 kg ballast, 47 % on the front axle

Folder: `grassland aerator/simple/`

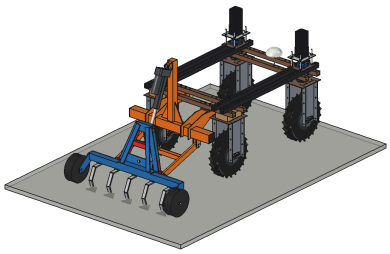
IMPLEMENT · TRANSPORT & GENERAL

Three-point hitch

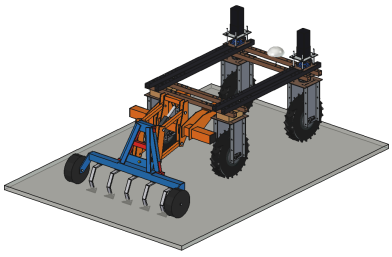
Ways to hang implements and machines at the rear: a category 0 three-point linkage with two electric lift rods, and three variants with an A-frame (triangle) quick coupler sized for the robot and a Linak LA36: the LA36 as top link, a parallel linkage as on the liquid fertilizer applicator, and a lift mast for picking up and setting down machines. Drawn with a small 5-tine cultivator (covers the small cultivator idea) and a demo machine on parking legs.



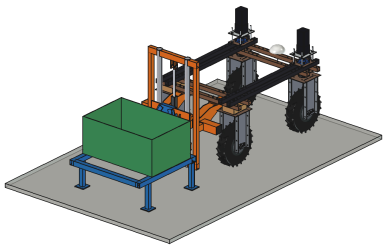
Simple



A-frame



Parallel

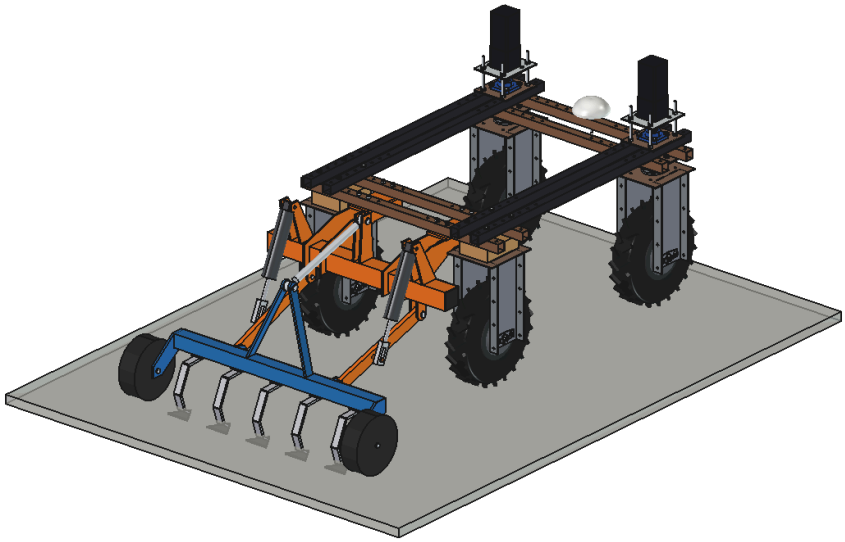


Lift mast

4 variants: Simple · A-frame · Parallel · Lift mast
Folder: `three point hitch/`

THREE-POINT HITCH · VARIANT 1 OF 4

Simple



Rear toolbar of the grassland harrow with lugs for 2 lower links (50 x 20, O16 pins, 500 spacing), a top link bracket and 2 x 24 V 3000 N actuators as lift rods with a slotted eye (the implement floats). Demo implement: 1.0 m cultivator, 5 S-tines with duckfoot shares, 2 gauge wheels.

> Lower pins 220 .. 425 mm, implement level within 2 degrees

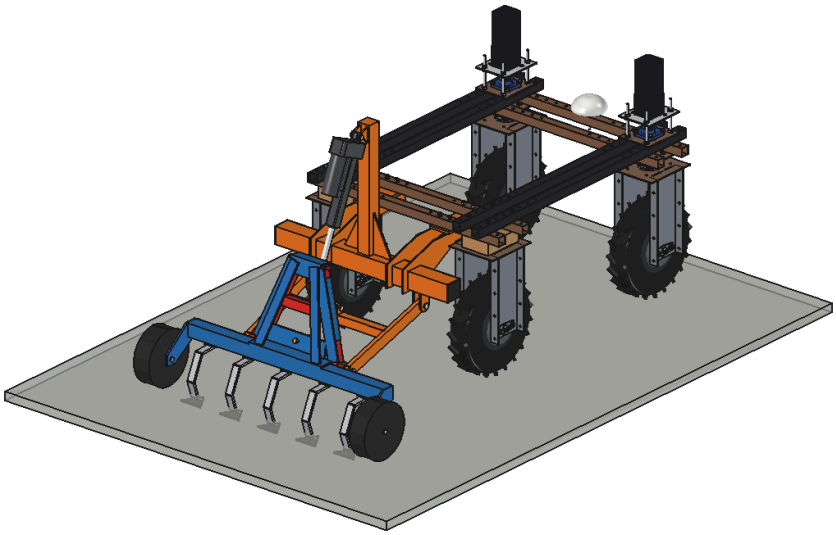
> Carries ~56 kg lifted with the 2 x 20 kg counterweight (front axle >= 20 %)

> Cultivator: 808 N draft at 40 mm, grip margin 1.29, 0.29 ha/h

Folder: `three point hitch/simple/`

THREE-POINT HITCH · VARIANT 2 OF 4

A-frame



Male A-frame triangle (40 x 40 x 3, 400 / 120 mm, 390 high) on 2 lower arms with a cross tube and a pitch brace: one lift frame turning about the lower pivots of the simple hitch. A Linak LA36 (20F, 1700 N, stroke 200) in the place of the top link, from a tower on the toolbar to the apex, its rod eye in a 30 mm slot so the frame floats. The cultivator carries a female triangle.

> Lower pins 230 .. 431 mm in 2.5 s, the implement tilts 25.6 degrees

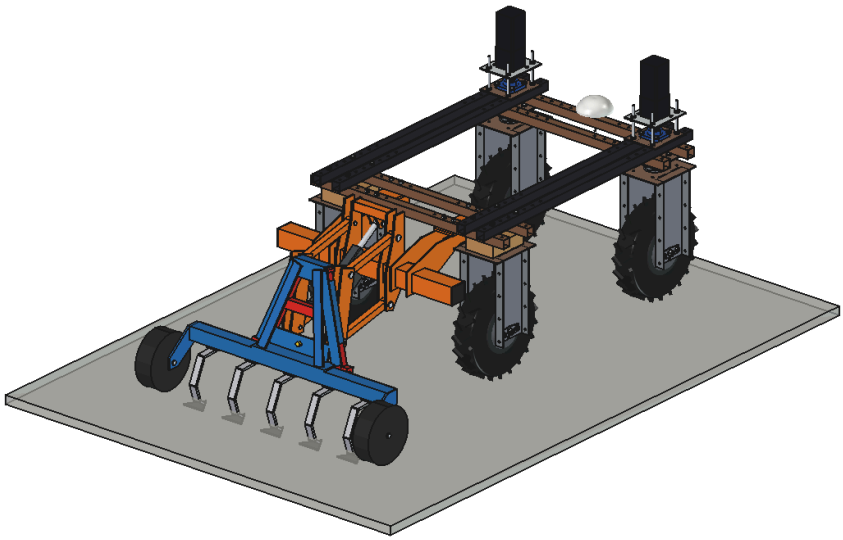
> LA36 678 .. 901 N (53 %), float +-19 mm

> ~48 kg lifted with the 2 x 20 kg counterweight, 84 kg in all

Folder: `three point hitch/a_frame/`

THREE-POINT HITCH · VARIANT 3 OF 4

Parallel



The suspension of the liquid fertilizer applicator as a general hitch: a headstock welded on the toolbar, 4 links 30 x 30 x 3 (300 mm), a carrier with the male A-frame triangle, a Linak LA36 (16F, 2000 N, stroke 150) on the diagonal with a 31 mm slot in the headstock.

> Straight up 203 mm, level, in 2.4 s

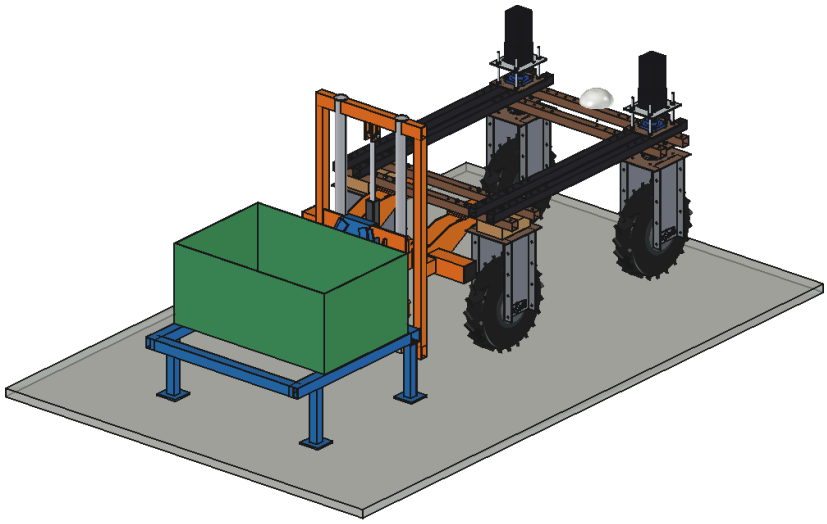
> Floats -40 / +41 mm, LA36 up to 1306 N (65 %)

> ~49 kg lifted with the 2 x 20 kg counterweight, 85 kg in all

Folder: `three point hitch/parallel/`

THREE-POINT HITCH · VARIANT 4 OF 4

Lift mast



Mast with 2 fixed columns O50 x 4, a carriage on 4 iglidur bush housings with the male A-frame triangle and a vertical Linak LA36 (20F, stroke 300, motor turned along x). Picks up a machine on parking legs by reversing under it, lifts, carries and sets it down again.

> 280 mm travel: machine 220 mm off its legs in 4.6 s

> LA36 957 N with the 50 kg demo machine (incl. 253 N bush friction)

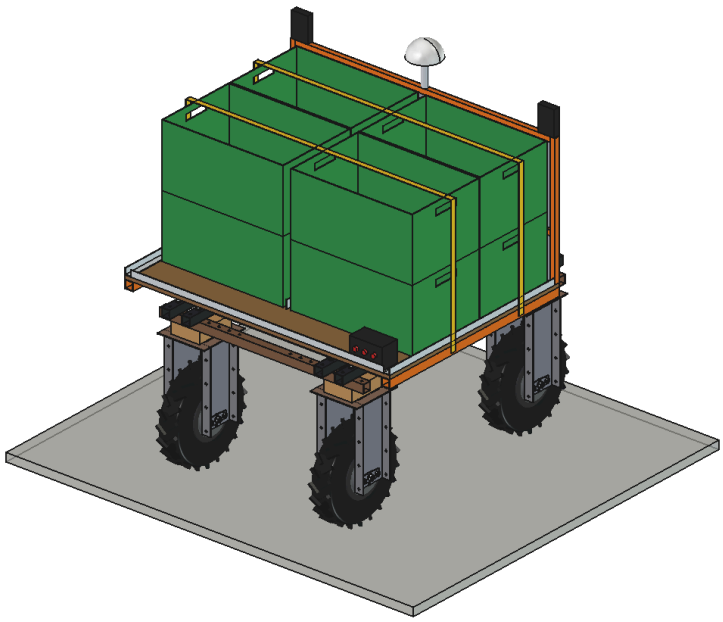
> Demo machine needs ~58 kg front counterweight

Folder: `three point hitch/lift/`

IMPLEMENT · TRANSPORT & GENERAL

Crate carrier

Flat deck on the upper beams for 8 harvest crates, carried from the rows to the field edge following a picker (UWB) or on an RTK route. The same deck carries tools and material (covers the tool cart idea).

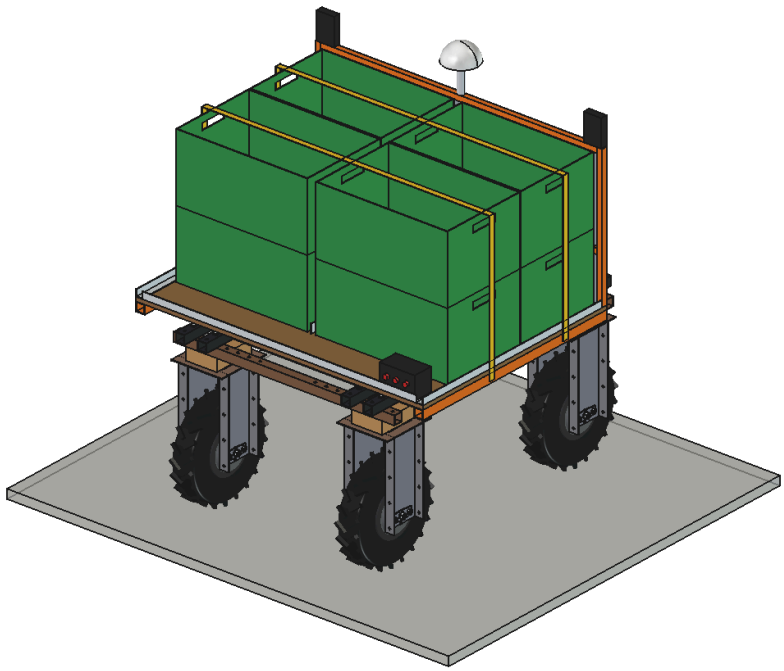


Simple

1 variant: Simple
Folder: `crate carrier/`

CRATE CARRIER · VARIANT 1 OF 1

Simple



3 bearers across the upper beams, side rails and 18 mm plywood (1300 x 1000, loading height 757), angle guides, divider, mesh headboard in front of the crates, 2 ratchet straps; 2 UWB anchors, a stop/go/follow box and the GNSS antenna moved to a mast above the stack.

> 8 crates 600 x 400 x 300 in 2 layers

> Apples: 141 kg, 48 % front axle, side tip angle 28 degrees

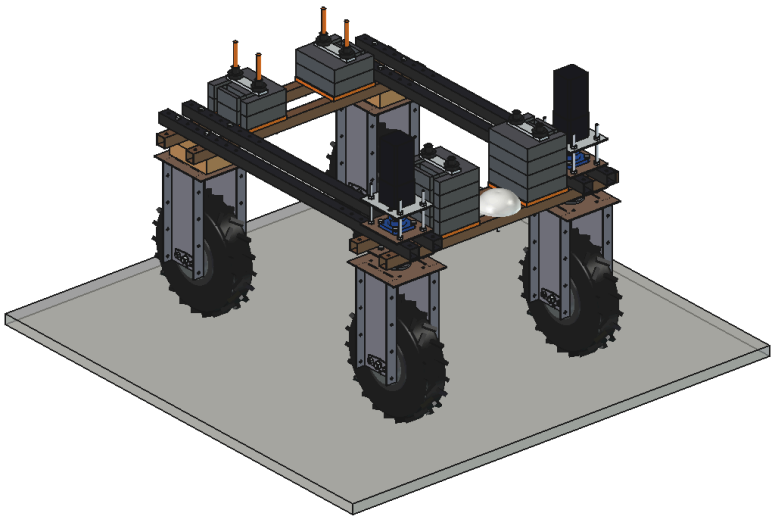
> ~107 crates per hour on a 150 m shuttle, 51 kg

Folder: `crate carrier/simple/`

IMPLEMENT · TRANSPORT & GENERAL

Counterweights

Stackable 11 kg ballast plates on the lower cross beams, front and rear, to keep enough load on the front axle with rear implements (and on the rear axle with the push blade).

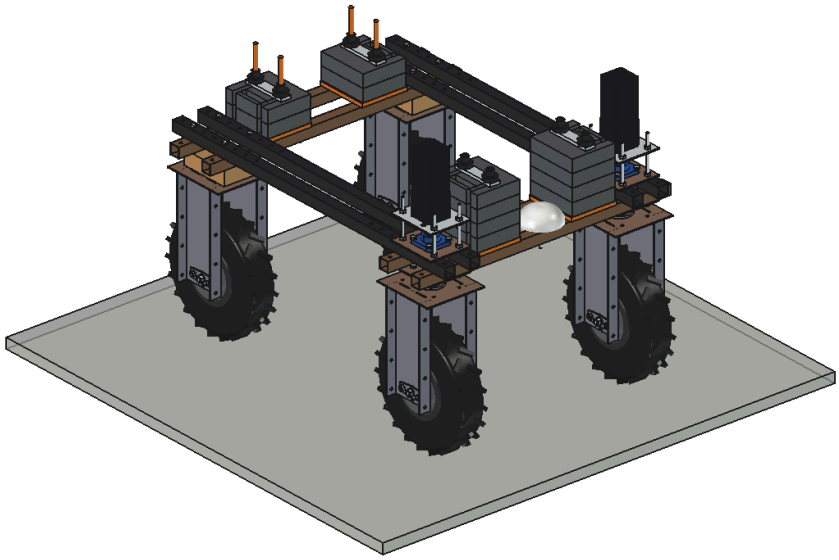


Simple

1 variant: Simple
Folder: `counterweights/`

COUNTERWEIGHTS · VARIANT 1 OF 1

Simple



Per stack (4 stacks): a 15 mm base plate on the two lower beams with 2 M16 studs, up to 4 plates of 170 x 200 x 50 with open slots that slide on from the side, a clamp bar and 2 knob nuts. Fits together with the push blade and the rear toolbar.

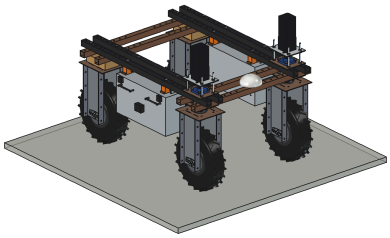
- 11 kg per plate, 0..99 kg per axle end
- Harrow, hitch, spreader: 1 plate per stack front
- Inter-row hoe: 2 plates per stack front, 5 % -> 21 % front axle

Folder: `counterweights/simple/`

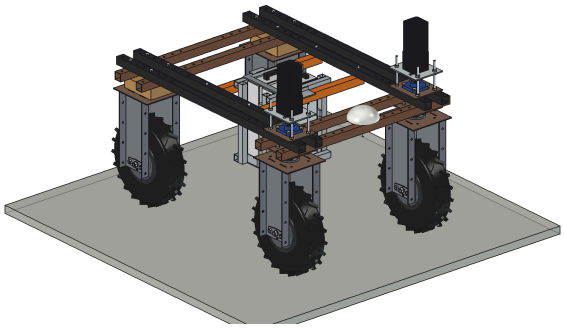
IMPLEMENT · POWER & DRIVE

Battery pack

Two exchangeable 48 V LiFePO4 pods (2.56 kWh each) hanging under the upper beams between the wheels: low, centred, and free of all toolbars and decks.



Simple

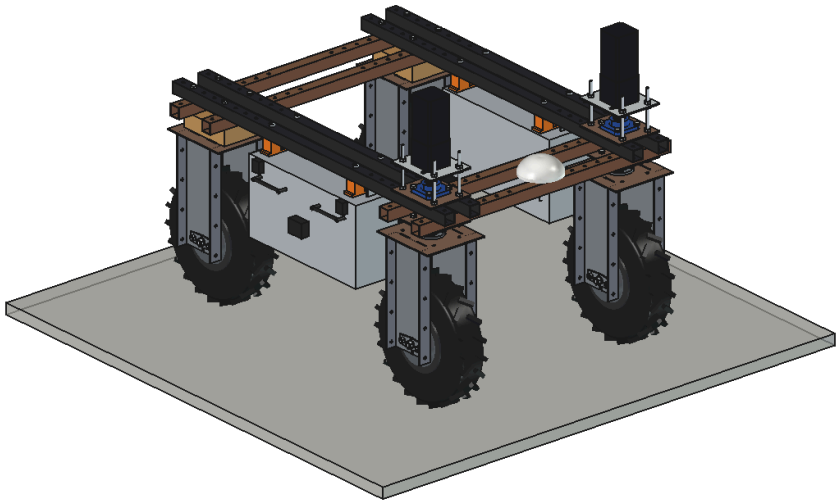


MBH3+ box

2 variants: Simple · MBH3+ box
Folder: `battery pack/`

BATTERY PACK · VARIANT 1 OF 2

Simple



Per side: 2 hangers (flat bar 40 x 8, M10 through the upper beams) and an aluminium pod with 16 LiFePO4 cells 50 Ah, a 16S BMS, hook strips, latches, handles and an Anderson SB50 connector.

> 51.2 V, 2 x 2.56 kWh, 26 kg per pod

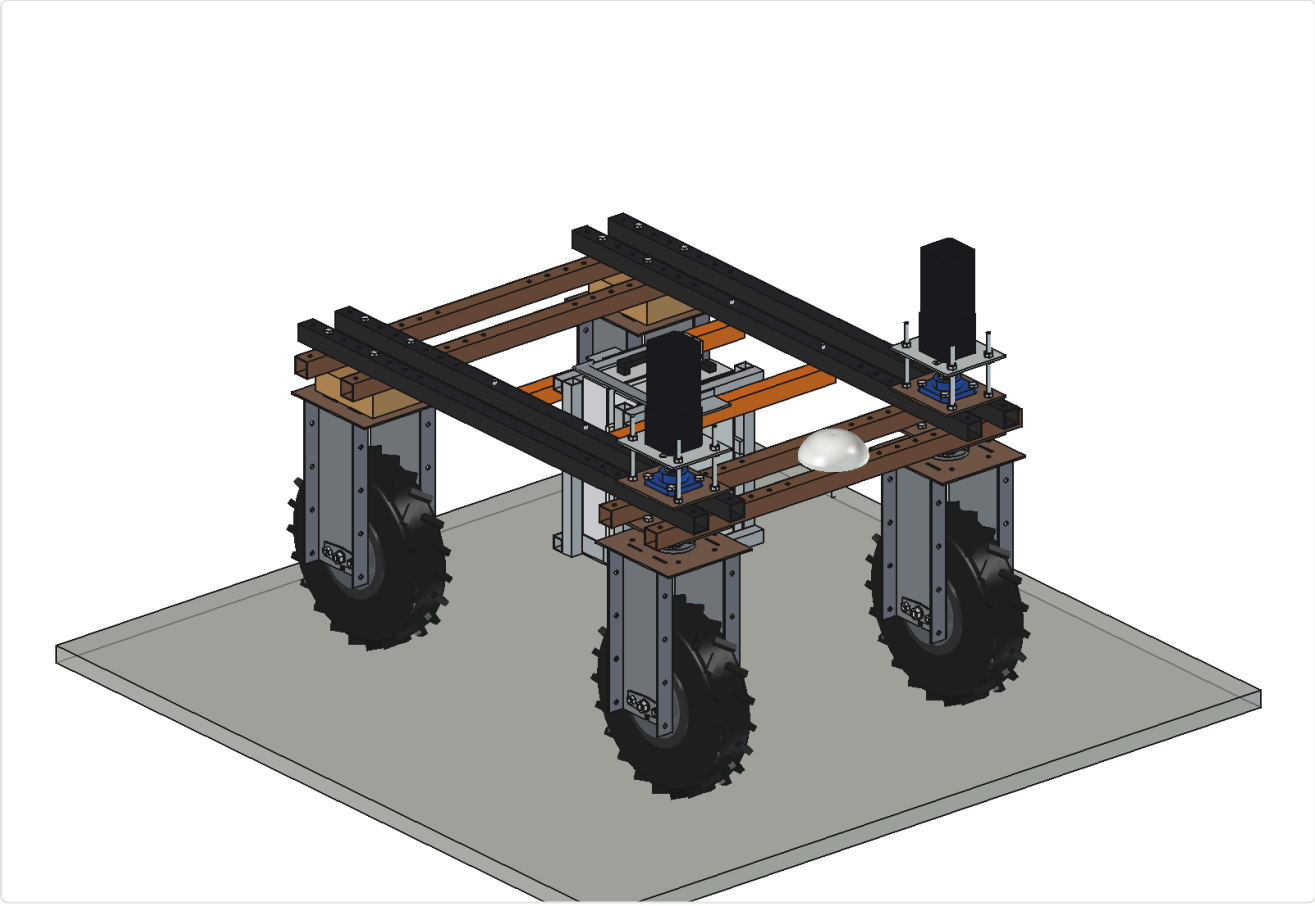
> ~230 N extra grip, 52 mm from the steered tyres

> Inter-row hoe 1.8 -> 6.0 h, harrow 1.4 -> 4.9 h

Folder: `battery pack/simple/`

BATTERY PACK · VARIANT 2 OF 2

MBH3+ box



Open rack for BatteryLabs MBH3+ drop-in batteries (14S 48 V 50 Ah, 2.56 kWh each) in tray segments of 100 mm, so 2 batteries grow to 3 or 4; on top, sunk in the centre well (recommended) or in side cradles. Also a closed alu cabinet and a wooden box version, with placeholders for MBH3+ and MU3+.

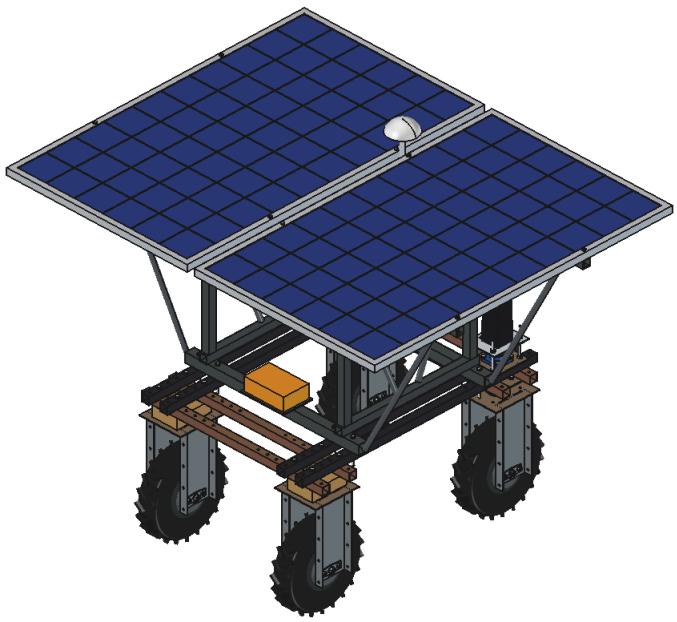
- 2 / 4 batteries: 4.6 / 9.2 kWh usable
- Centre well: cg 502 mm
- Inter-row hoe 4.3 / 8.5 h

Folder: `battery pack/mbh3_box/`

IMPLEMENT · POWER & DRIVE

Solar canopy

Roof of solar panels on a tube frame above the robot that charges the battery through an MPPT controller and carries the GNSS antenna.

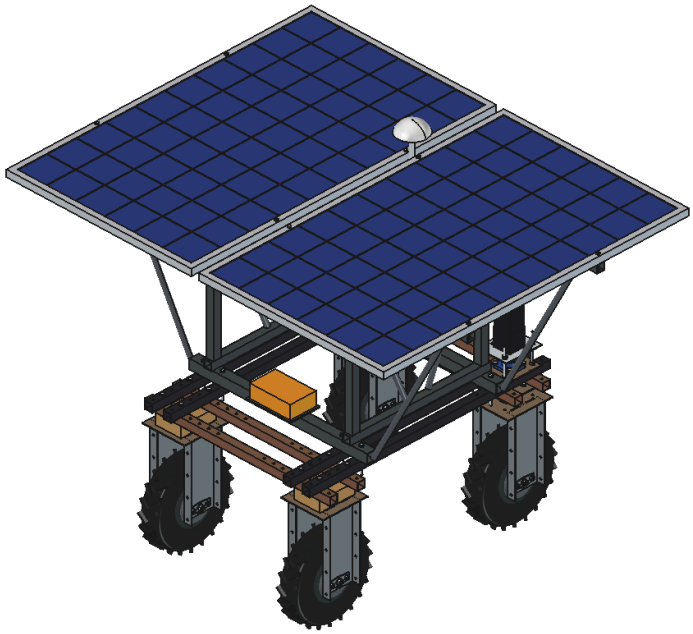


Canopy

1 variant: Canopy
Folder: `solar/`

SOLAR CANOPY · VARIANT 1 OF 1

Canopy



2 panels 1000 x 1650 mm (320 Wp each, 3 possible) on a steel tube frame: base tubes on the upper beams, 4 posts, top rails and cross members, diagonal struts to the overhangs; MPPT box, GNSS antenna on a mast between the panels.

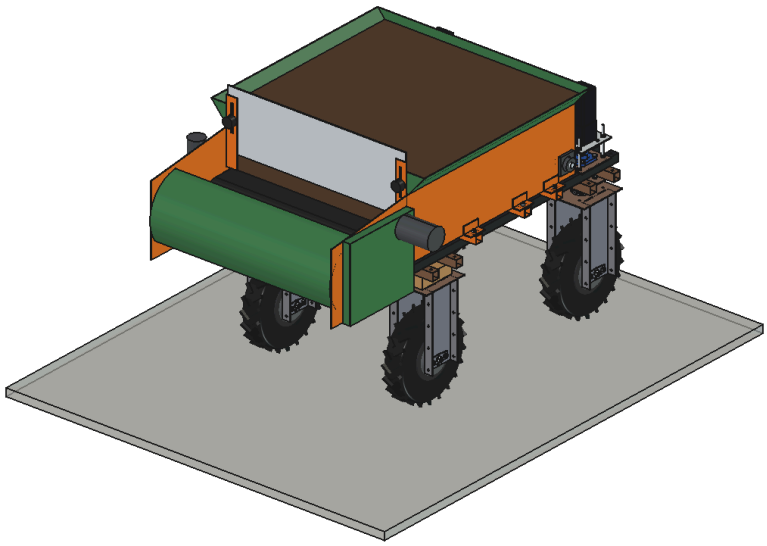
- 640 Wp, ~480 kWh per year (NL)
- 2.3 kWh per day in June, 0.2 in December
- 91 kg, tips over at ~29 degrees static

Folder: `solar/`

IMPLEMENT · SPREADING & BEDDING

Compost spreader

Bin on the upper beams with a moving floor belt and a beater behind the robot: top-dresses beds, orchard strips and grassland with compost in an even 1.1 m layer, dose from the belt speed.

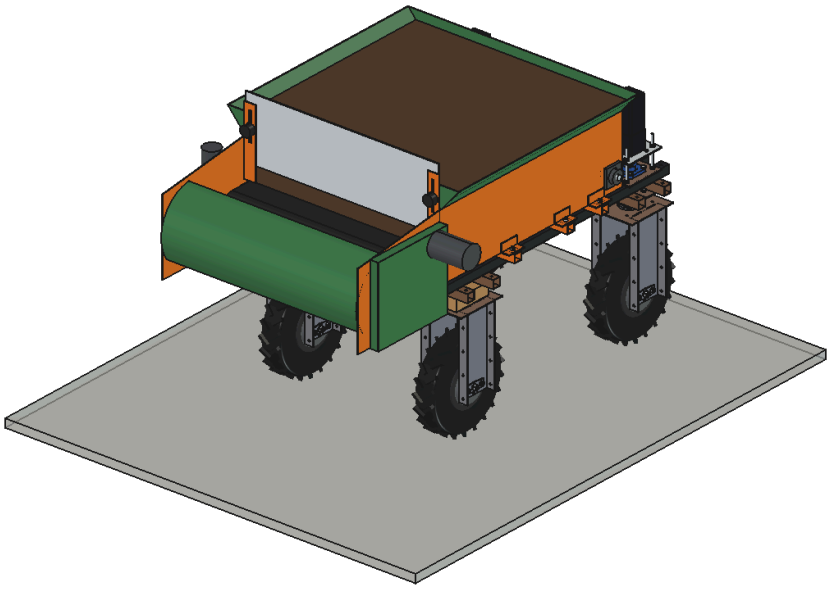


Simple

1 variant: Simple
Folder: `compost spreader/`

COMPOST SPREADER · VARIANT 1 OF 1

Simple



299 l bin (150 kg payload) on 3 bearers, floor belt 920 wide on a UHMW slide bed round 2 drums O100 driven by a 24 V worm gearmotor, slide gate 30..150 mm, paddle beater O280 at 300 rpm on a 48 V 1 kW BLDC with a poly-V belt and a deflector hood.

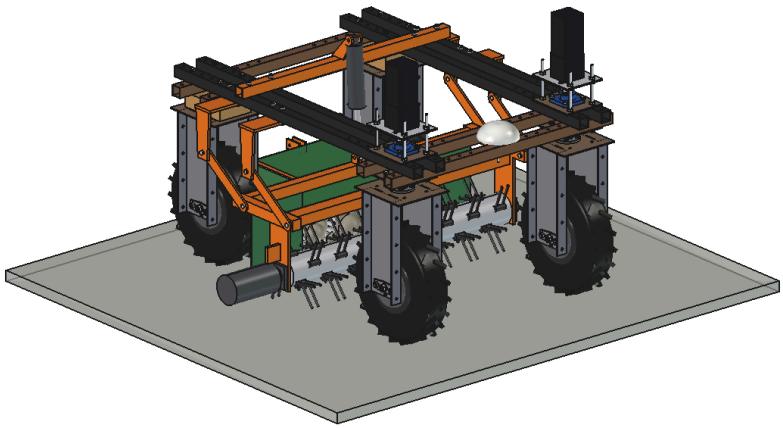
- 1..3 kg/m2, 50..150 m2 per bin, ~1.1 t/h with refilling
- Needs 1 counterweight plate per stack (front axle 33 %)
- 190 kg empty, climbs 8 degrees full

Folder: `compost spreader/simple/`

IMPLEMENT · CROP & SOIL

Stone picker

Mid-mounted under the robot between the axles: a 1 m tine rotor flicks stones over a comb into a sieve basket, on a parallelogram lift; the floor drops open at the heap.

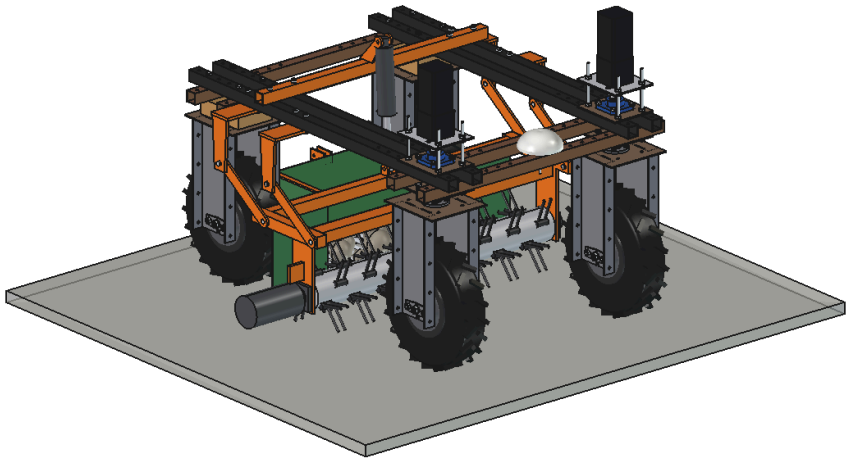


Simple

1 variant: Simple
Folder: `stone picker/`

STONE PICKER · VARIANT 1 OF 1

Simple



Rotor O114 with 48 double spring tines (tip radius 150, 20 mm deep) on a 48 V 750 W gearmotor at 250 rpm, comb of 12 mm bars, 40 l basket with a sieve floor (25 mm gaps) opened by a 24 V actuator, frame on 4 swing links outside the wheel line lifted 100 mm by one 1500 N actuator.

> 1 m wide, 1200..1700 m2/h, ~1 ha per day

> 44 kg of stones per basket, front axle 41..45 %, no ballast

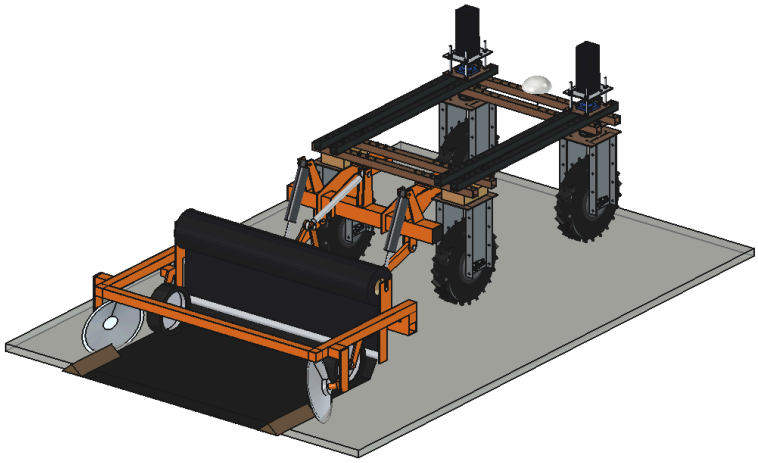
> 92 kg, 525 W rotor

Folder: `stone picker/simple/`

IMPLEMENT · CROP & SOIL

Mulch layer

Category 0 implement for the three-point hitch that lays 1.2 m mulch foil over the bed behind the robot: press wheels push the edges down, angled discs bury them.

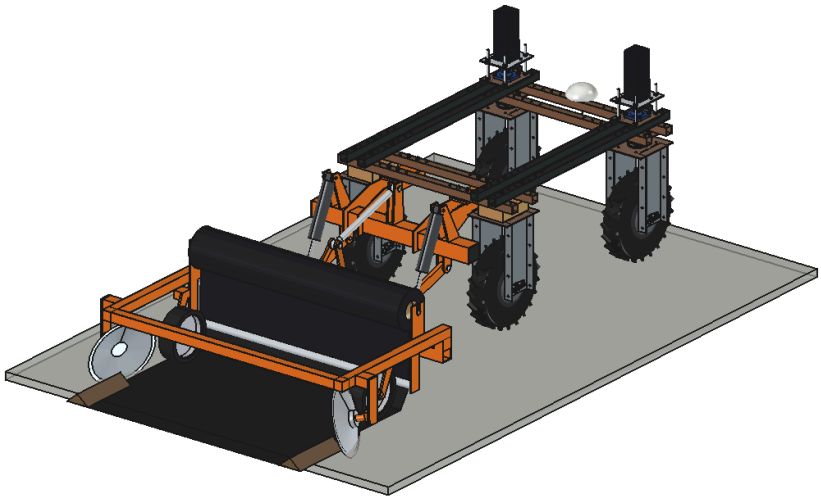


Simple

1 variant: Simple
Folder: `mulch layer/`

MULCH LAYER · VARIANT 1 OF 1

Simple



Frame with a 60 x 60 toolbar, A-frame mast and 50 x 50 side beams on the hitch of three point hitch/simple (imported), foil roll O200 x 1200 in open V-holders with a band brake, guide bar, 2 semi-pneumatic press wheels O250 on the edges and 2 concave covering discs O350 at 22 degrees, 50 mm deep.

> 12 beds of 100 m per hour, a 1075 m roll lasts ~50 min

> Pull 730 N, grip margin 1.51

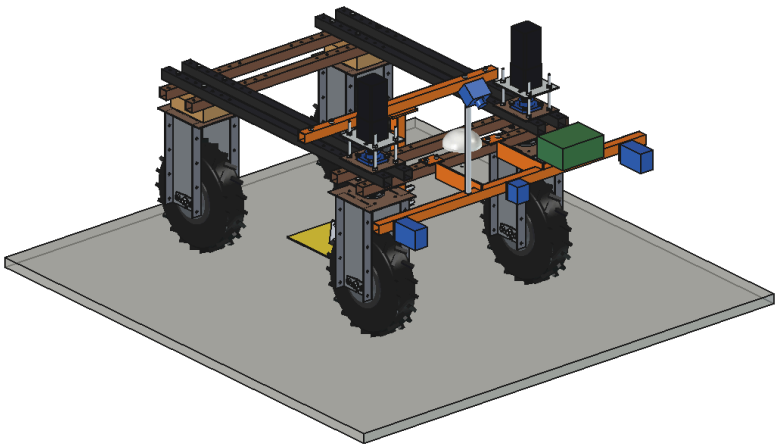
> 94 kg with a full roll: needs 3 counterweight plates per front stack

Folder: `mulch layer/simple/`

IMPLEMENT · MEASURING & SENSORS

Grass meter

Maps grass cover while driving: front boom with 2 active NDVI sensors, a lidar and a camera, plus a sledge plate meter between the axles for compressed height and dry matter. Covers the NIR sensor idea.

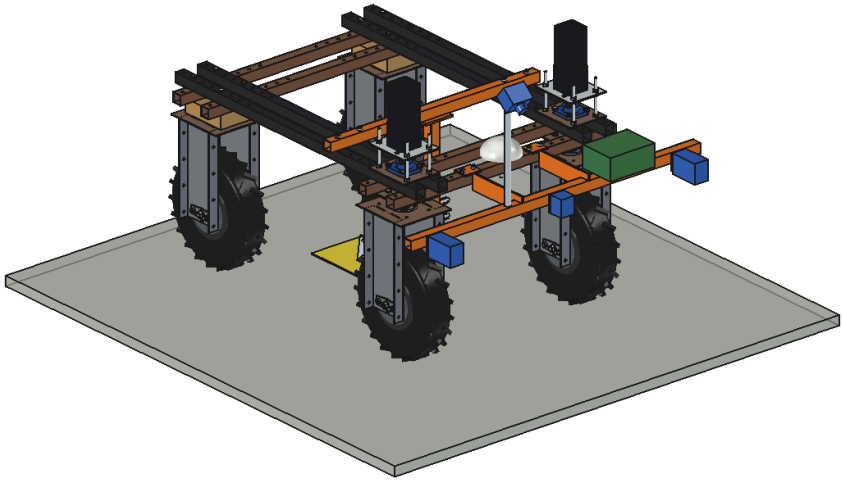


Simple

1 variant: Simple
Folder: `grass meter/`

GRASS METER · VARIANT 1 OF 1

Simple



Boom 1300 on two flat brackets on the front outer lower beam (above the tyres), 2 active NDVI sensors at x +-550, a lidar and an RGB camera; plate 0.1 m2 at 5 kg/m2 on a 350 mm parallelogram from a post under the upper beams, angle sensor on the pivot, 24 V actuator to lift it.

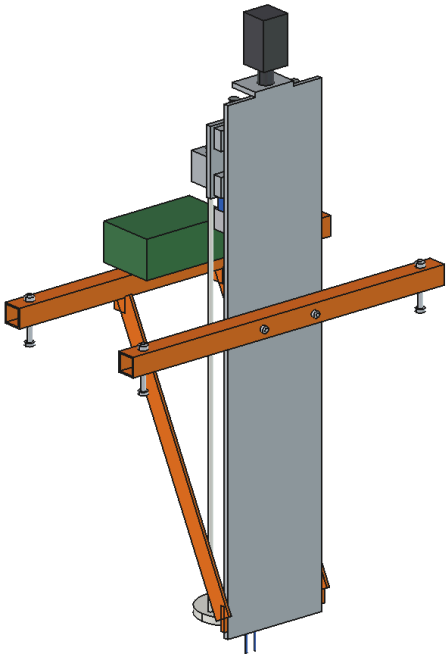
- 4.4 ha/h on 10 m transects, ~35 ha per battery
- Plate 0..200 mm, 0.3..0.5 mm resolution
- 22 kg, stays on with most implements

Folder: `grass meter/simple/`

IMPLEMENT · MEASURING & SENSORS

Soil compaction meter

Automatic cone penetrometer in the middle of the robot: a ball-screw Z axis pushes an ASABE cone at 30 mm/s to 600 mm, a load cell reads the force; compaction maps per depth layer.

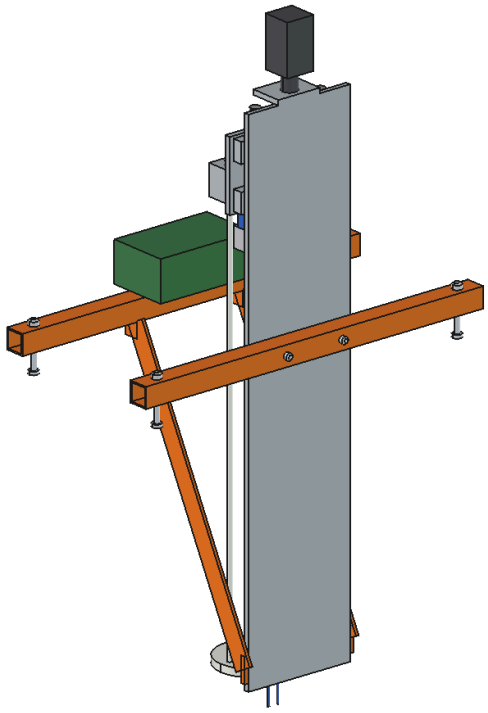


Simple

1 variant: Simple
Folder: `soil compaction meter/`

SOIL COMPACTION METER · VARIANT 1 OF 1

Simple



Bridges, mast, SBR20 rails and SFU1605 screw as the soil sampler (mast z 120..1150 with stays), S-type 2 kN load cell, rod O9.5 x 700 with a 30 degree cone O12.83, sprung foot with a PTFE rod bush and a capacitive moisture probe.

> Up to 10.2 MPa with 70 % of the robot weight

> 35 s per 600 mm profile, ~47 points per hour

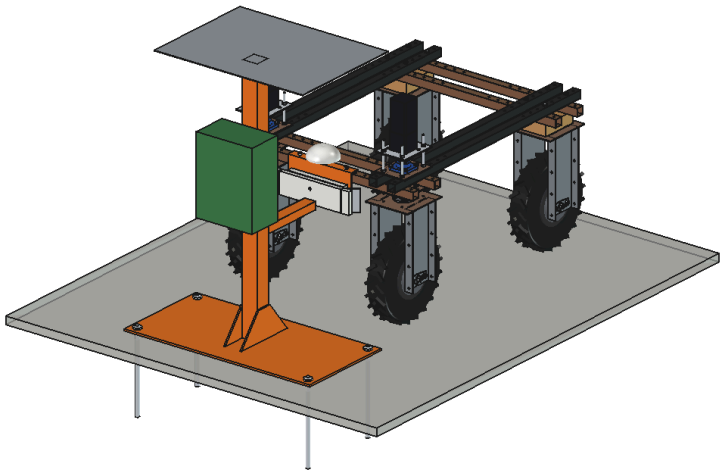
> 32 kg

Folder: `soil compaction meter/simple/`

IMPLEMENT · POWER & DRIVE

Charging dock

Fixed docking station at the field edge or in the shed: the robot drives in and charges through sprung copper contacts (the contact version of the wireless charging dock idea).

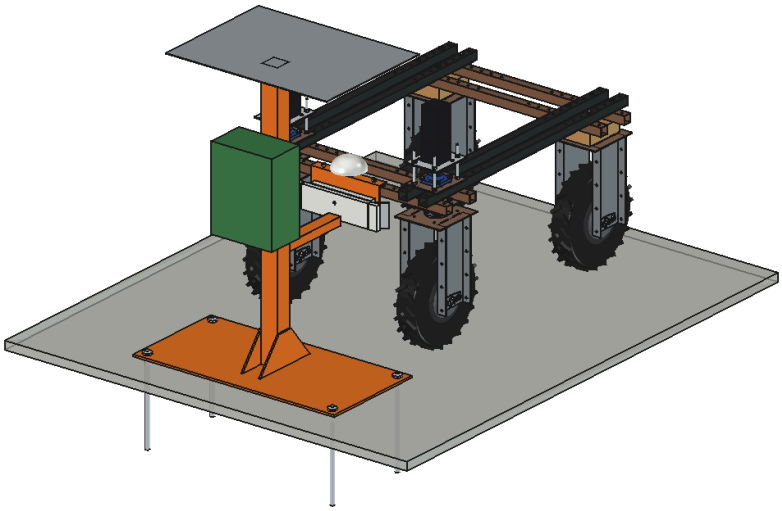


Simple

1 variant: Simple
Folder: `charging dock/`

CHARGING DOCK · VARIANT 1 OF 1

Simple



Robot side: a 3.9 kg contact bracket on the front outer lower beam with 2 sprung copper pads and a pilot pin. Station: base plate with ground anchors, post with a 48 V 1 kW IP65 charger and a roof, an arm back between the front wheels to a PE contact head with 2 wide copper bars and a V-funnel.

> 17 A at 54 V, 0.3 W contact loss

> Robot 20 -> 100 % in 2 h

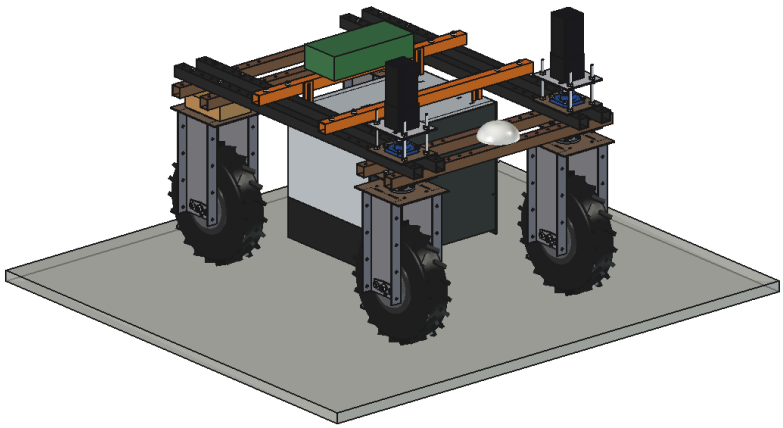
> Catches 75 mm sideways; 80 kg station

Folder: `charging dock/simple/`

IMPLEMENT · WEED & CROP PROTECTION

UV-C mildew light

Reflecting hood with 6 UV-C lamps under the robot between the axles: drives over the bed at night and gives a low UV-C dose against powdery mildew, without chemicals.

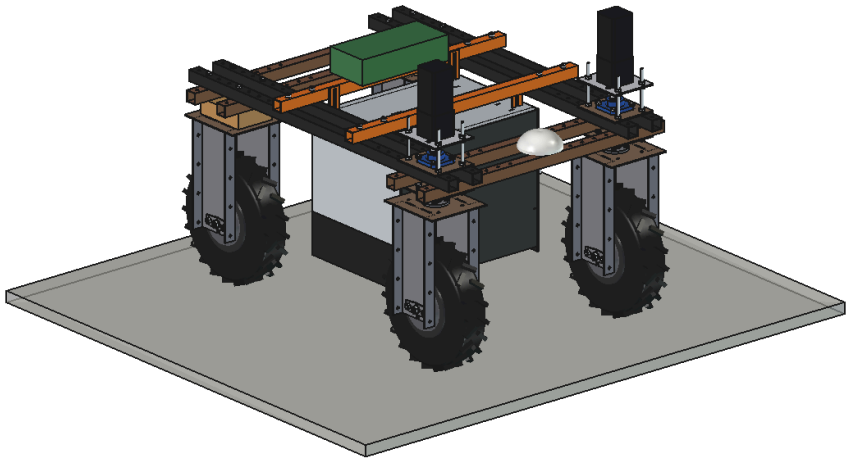


Simple

1 variant: Simple
Folder: `uv mildew light/`

UV-C MILDEW LIGHT · VARIANT 1 OF 1

Simple



Aluminium hood 580 x 440 on 4 slotted hangers from 2 bearers on the upper beams, rubber side skirts and brush curtains, 6 compact UV-C tubes (PL-L 55 W class) behind guard wires, ballasts and interlock in a box on the rear bearer.

> 200 J/m2 at 0.58 m/s, ~1020 m2/h

> 489 W, ~4000 m2 per battery

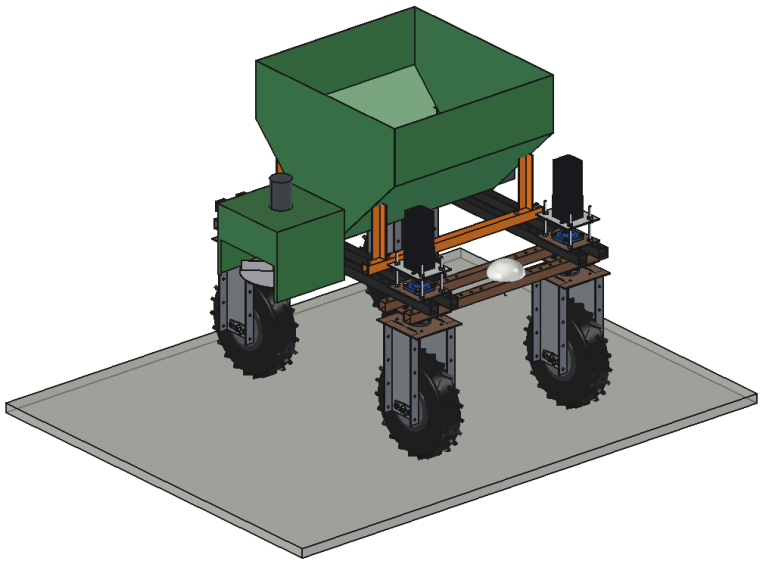
> 21 kg, 25 mm from the steered tyres

Folder: `uv mildew light/simple/`

IMPLEMENT · SPREADING & BEDDING

Sawdust spreader

Bedding spreader for cubicles: a 308 l bin with an auger across the robot feeds a spinner on the right side that throws sawdust, chopped straw or a lime mix into the cubicles from the alley.

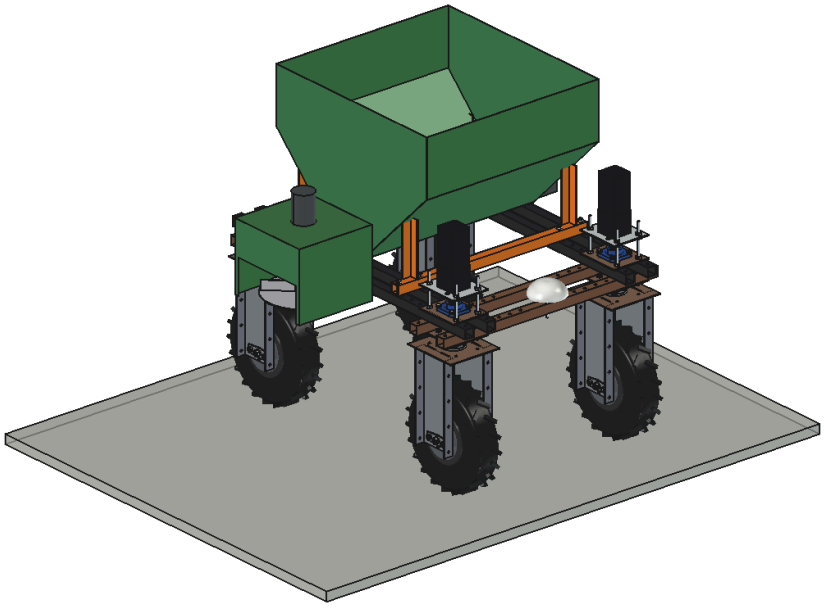


Simple

1 variant: Simple
Folder: `sawdust spreader/`

SAWDUST SPREADER · VARIANT 1 OF 1

Simple



Bin 2 mm with 45 degree walls to a trough on 2 bearers on the upper beams, auger O150 across the robot on a 24 V worm gearmotor, spinner housing outside the right outer upper beam with a O320 disc (4 vanes, 350 rpm) open to the right.

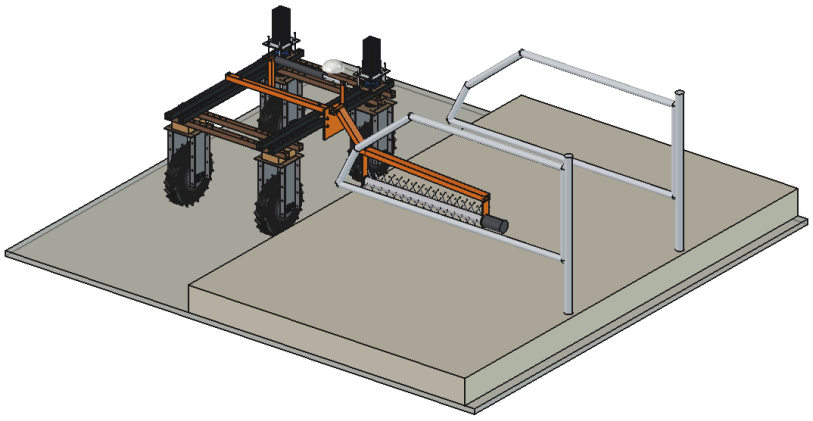
- 61 cubicles per fill of sawdust, ~570 per hour
- Throws ~1.3 m beyond the wheel line
- 75 kg empty

Folder: `sawdust spreader/simple/`

IMPLEMENT · LIVESTOCK & FARM

Bedding rake

Cubicle groomer on a side arm: reaches from the alley over the curb, under the divider loops, with a tine rotor along the bed that loosens and levels the bedding in the rear metre of each cubicle.



Simple

1 variant: Simple
Folder: `bedding rake/`

BEDDING RAKE · VARIANT 1 OF 1

Simple

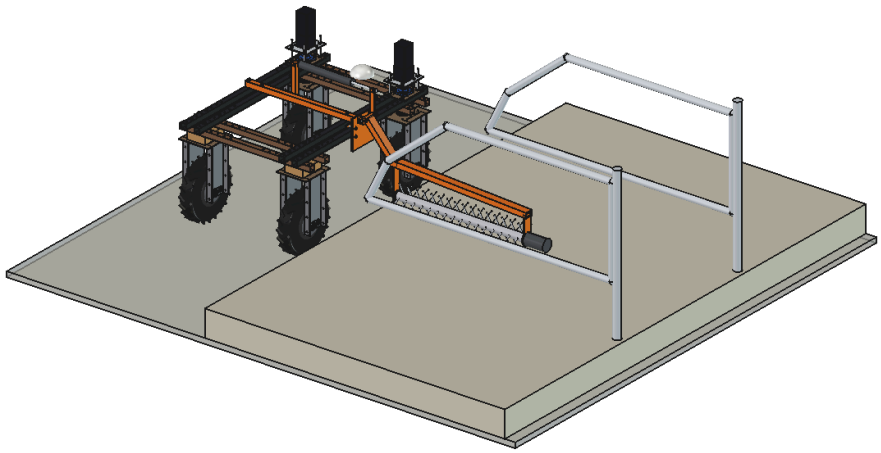


Plate bracket with pivot lugs on the right outer upper beam, arm 60 x 40 from a pivot above the beams down past the curb and out over the bed, tine rotor O240 x 970 along the bed on a 48 V 400 W gearmotor, 24 V actuator across the robot top to a crank lifts the arm 40 degrees. A row of cubicles is modelled for the checks.

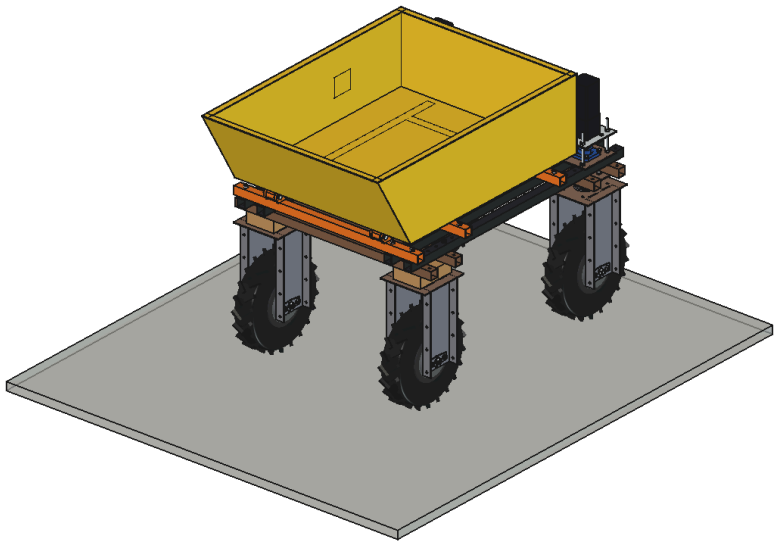
- Grooms 100..1070 mm behind the curb, 90 mm under the loops
- ~655 cubicles per hour, ~300 W
- 37 kg, side tip angle 27 degrees

Folder: `bedding rake/simple/`

IMPLEMENT · TRANSPORT & GENERAL

Dumper barrow

Tipping skip on the upper beams: a 353 l steel bin hinged at the rear end of the robot, tipped backwards to 48 degrees by two actuators so the load slides out behind the rear wheels.

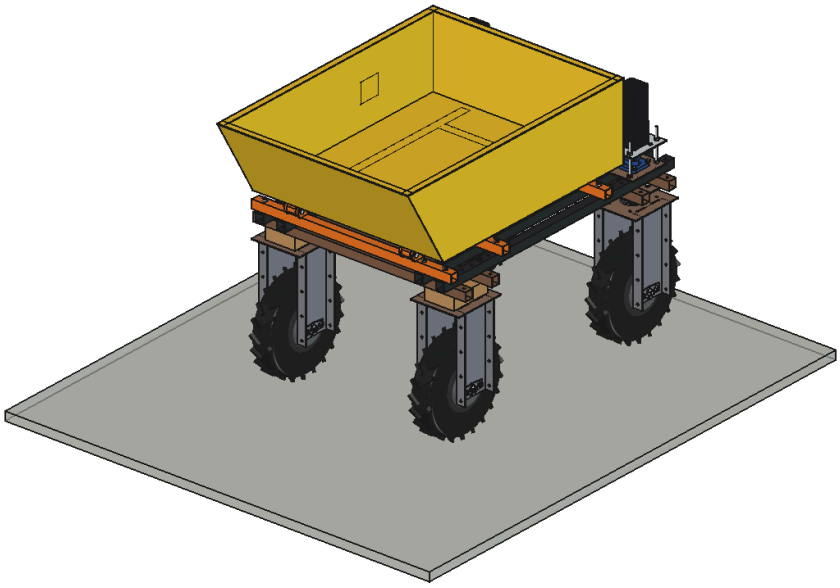


Simple

1 variant: Simple
Folder: `dumper barrow/`

DUMPER BARROW · VARIANT 1 OF 1

Simple



3 bearers on the upper beams (rest pads, actuator lugs, hinge lugs), subframe of 2 rails 40 x 40 with hinge bosses and a cross tube, 2 mm bin 900 x 1020 x 350 with a 30 degree rear lip, 2 x 24 V 3000 N actuators (stroke 300).

> 150 kg of soil (100 l) or 353 l of chips / green waste

> Tips 48 degrees in ~36 s, cg stays in front of the rear axle

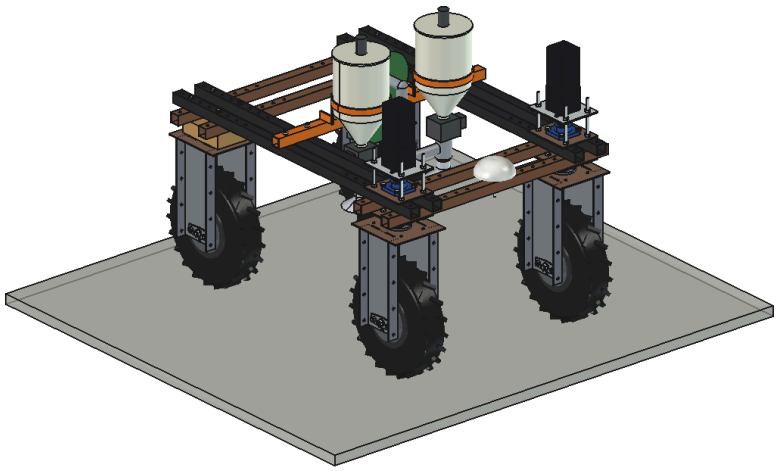
> 72 kg, ~950 kg of soil per hour on 100 m

Folder: `dumper barrow/simple/`

IMPLEMENT · WEED & CROP PROTECTION

Biocontrol spreader

Releases predatory mites and parasitic wasps (in bran, vermiculite or sawdust) over two rows: 2 hoppers with metering wheels and a blower that carries the product through hoses to deflector nozzles.

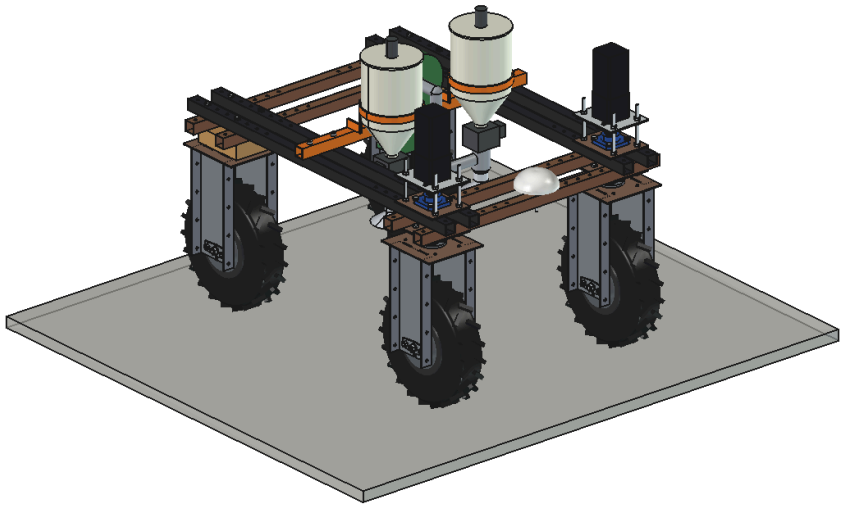


Simple

1 variant: Simple
Folder: `biocontrol spreader/`

BIOCONTROL SPREADER · VARIANT 1 OF 1

Simple



Bearer across the upper beams with 2 PE hoppers of 7.9 l (agitator, NEMA17 metering wheel) over the rows at x +-187.5, a 24 V 60 W centrifugal fan, venturis, hoses O40 and deflector nozzles 450 mm above the ground.

> Predatory mites 2 ml/m2: 0.8 ha per fill

> ~1350 m2/h, ~12 h per battery

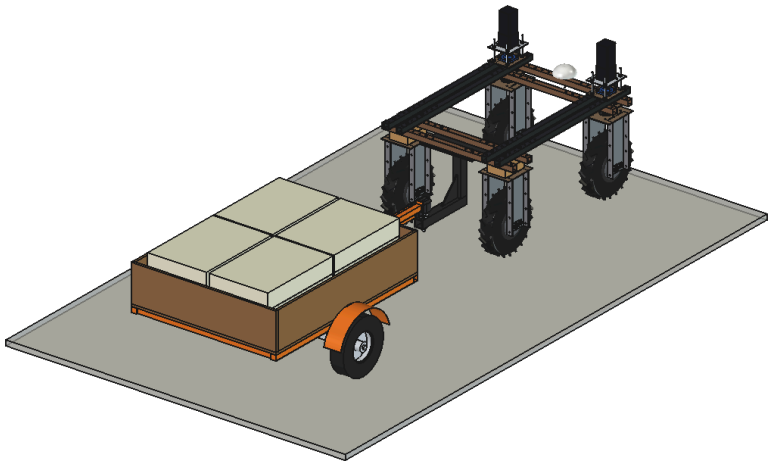
> 18 kg

Folder: `biocontrol spreader/simple/`

IMPLEMENT · TRANSPORT & GENERAL

Bag trailer

Two-wheeled trailer on the 50 mm tow ball of the cambridge roll (hitch, coupling and drawbar imported): a plywood deck with boards and a drop tailgate for 12 bags of 25 kg.

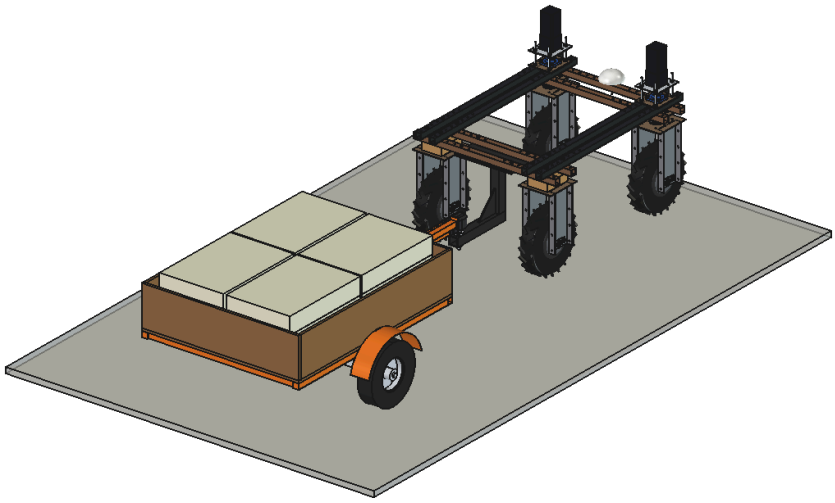


Simple

1 variant: Simple
Folder: `bag trailer/`

BAG TRAILER · VARIANT 1 OF 1

Simple



Frame of 40 x 40 rails and cross members behind the cbr cross beam and drawbar, plywood deck 1110 x 1410 at 358 mm with side and front boards and a drop tailgate, axle tube on hangers, 2 x 4.00-8 wheels with mudguards.

> 12 bags of 25 kg, 24 kg on the ball

> Steepest slope loaded: 3.5 deg wet grass, 6.5 dry

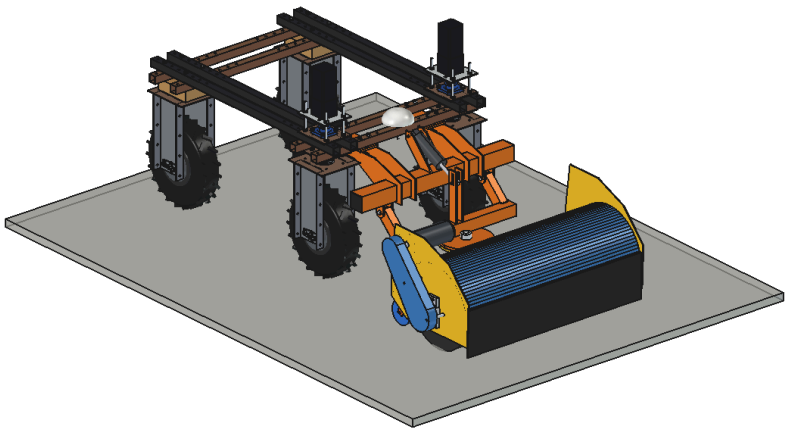
> 110 kg empty

Folder: `bag trailer/simple/`

IMPLEMENT · TRANSPORT & GENERAL

Broom / sweeper

Front angle broom on the push blade's toolbar, lift arms and king pin (imported): a roller brush O400 x 1000 under a hood, 48 V belt drive, on 2 castors, angled +-25 degrees to throw the material aside.

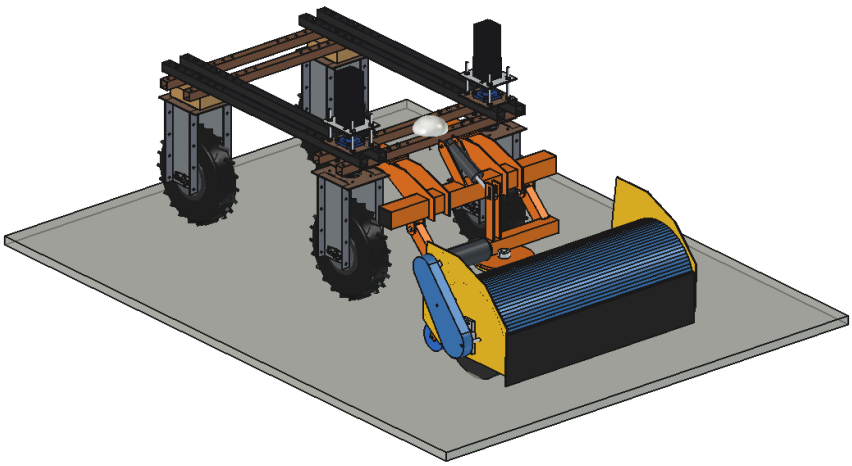


Simple

1 variant: Simple
Folder: `broom sweeper/`

BROOM / SWEEPER · VARIANT 1 OF 1

Simple



Head frame with the swivel plate, cross beam 80 x 60 and 5 mm end plates, PP brush O400 x 1000 in UCFL 208 bearings at 400 rpm, 48 V 1000 W gearmotor with an HTD belt, aluminium hood and rubber flap, 2 swivel castors O125; 3000 N version of the blade actuator.

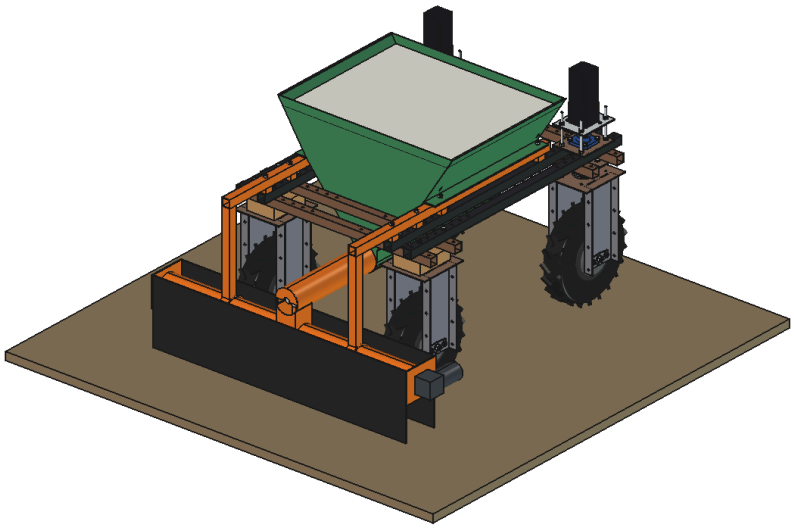
- ~2900 m2/h angled, ~4900 m2 per charge
- ~910 W brush, rear axle 54 % working
- Lifting needs 4..9 counterweight plates; 124 kg

Folder: `broom sweeper/simple/`

IMPLEMENT · SPRAYING, FERTILIZING & IRRIGATION

Lime spreader

Drop spreader for granulated or ground lime: a 178 l V-hopper centred over the robot, a trough auger to a 1.3 m cross auger behind the rear wheels that lets the lime fall through a row of holes inside a rubber dust skirt.

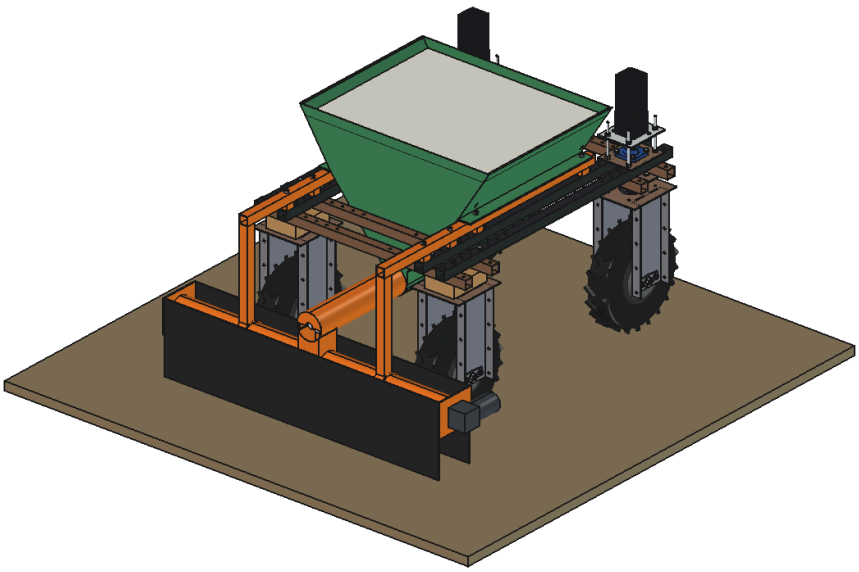


Simple

1 variant: Simple
Folder: `lime spreader/`

LIME SPREADER · VARIANT 1 OF 1

Simple



Rails on the inner upper beams, steel V-hopper down between the beams to a U-trough, trough auger O100 in a tube under the rear lower beams, cross auger O100 x 1300 with 25 outlet holes O20 and flights from the middle outward, 2 x 24 V worm gearmotors, rubber skirt 20 mm above the ground.

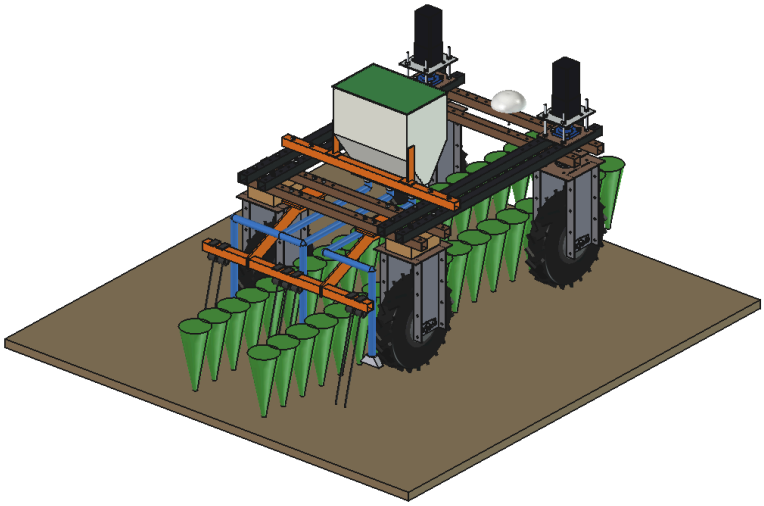
- 205 kg granulated lime per fill: 0.21 ha at 1000 kg/ha
- Up to 3.5 t/ha, 0.30 ha/h, no dust cloud
- 97 kg empty, rear axle 60 % full

Folder: `lime spreader/simple/`

IMPLEMENT · CROP & SOIL

Cover crop undersower

Sows a cover crop into a standing 2-row crop: a 28 l hopper with 3 fluted metering cells feeds hoses to fishtail deflectors in the free middle inter-row and behind both rear tyres; spring tines on a bar high over the crop cover the seed.

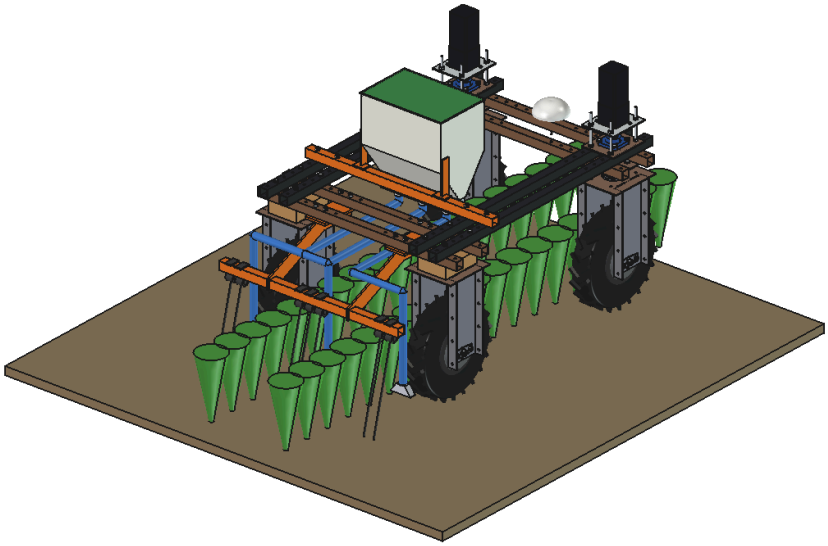


Simple

1 variant: Simple
Folder: `cover crop undersower/`

COVER CROP UNDERSOWER · VARIANT 1 OF 1

Simple



Bearer across the upper beams with a PE V-hopper, aluminium metering housing with 3 fluted rollers on one 12 V speed-following shaft, hoses O32 routed over the crop rows, fishtail deflectors 40 mm above the ground, tine bar at 430 mm on 2 hangers under the rear lower beams with 7 spring tines O8.

> Ryegrass 0.43 ha, clover 3.6 ha per fill

> 0.22 ha/h, ~1.1 ha per charge

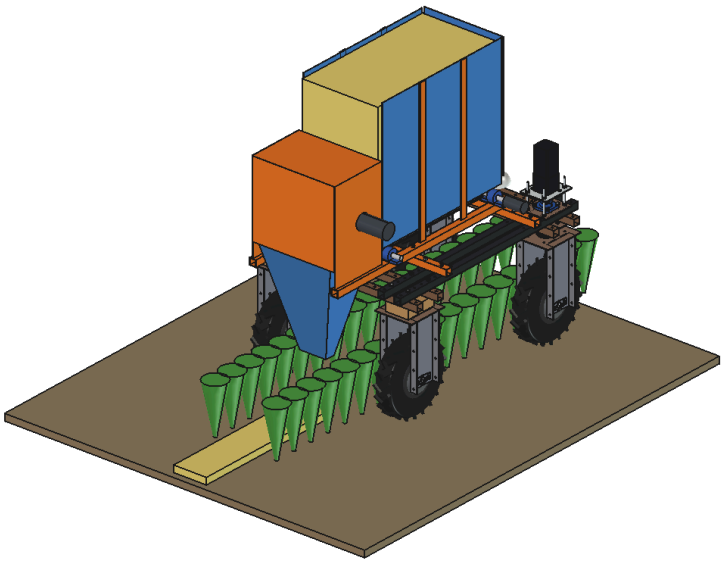
> 26 kg; crop rows untouched (checked)

Folder: `cover crop undersower/simple/`

IMPLEMENT · SPREADING & BEDDING

Straw spreader

Lays a straw mulch strip in the free middle inter-row: 2 small bales on a slatted floor chain are fed against a spiked shredder drum, the loose straw falls through a chute to a 160 mm outlet between the crop rows.

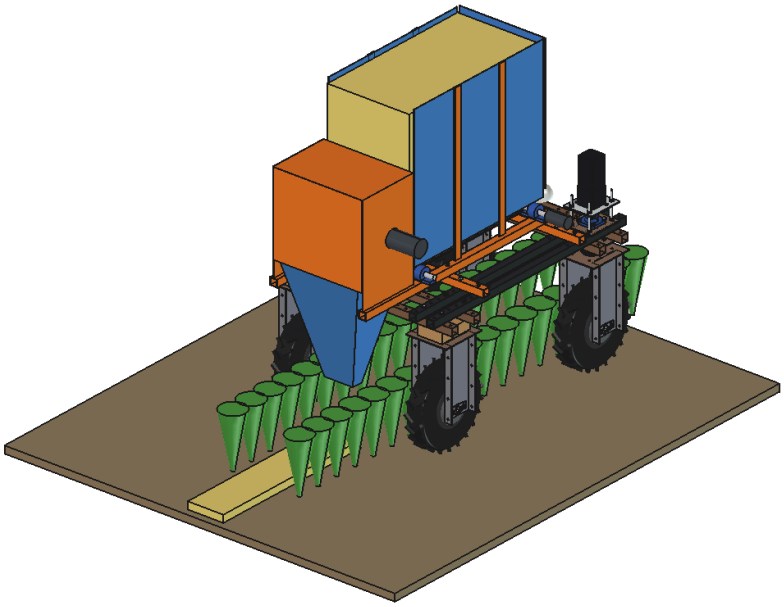


Simple

1 variant: Simple
Folder: `straw spreader/`

STRAW SPREADER · VARIANT 1 OF 1

Simple



2 bearers and 2 rails on the upper beams, slatted floor of 2 chains 08B in UCP 204 shafts with a 24 V gearmotor, aluminium bale chamber, spiked drum O320 at 250 rpm on a 48 V 750 W motor in a 3 mm housing, aluminium chute to z 300.

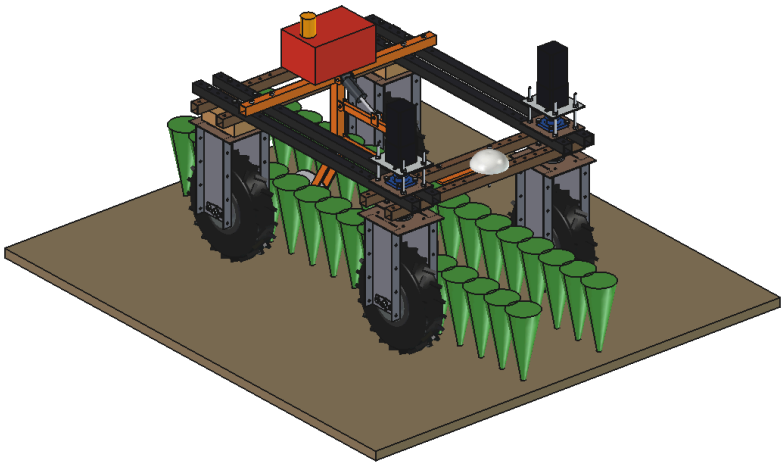
- 260 m of strip per fill at 0.5 kg/m2
- ~440 W drum, ~3.5 km of strip per charge
- 102 kg empty; crop rows untouched (checked)

Folder: `straw spreader/simple/`

IMPLEMENT · WEED & CROP PROTECTION

Electric weeder

Spot electric weed control in the free middle inter-row: a camera finds weeds, a 5 kV generator switches on a comb of spring-steel fingers only while a weed is under it, the current returns through a coulter disc.

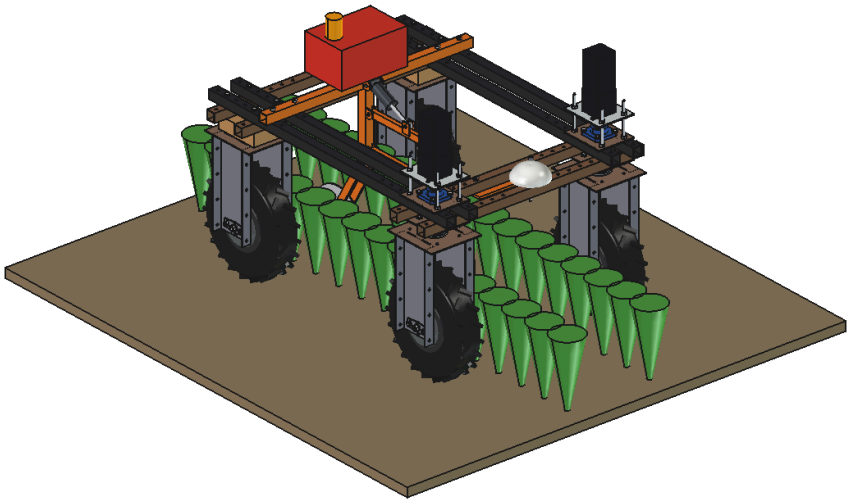


Simple

1 variant: Simple
Folder: `electric weeder/`

ELECTRIC WEEDER · VARIANT 1 OF 1

Simple



Bearer on the upper beams with the HV box and beacon, post with a parallel-arm lift (24 V actuator), PE holder with 9 spring-steel fingers and rubber curtains, coulter disc O250 as earth return, camera under the inner front lower beam.

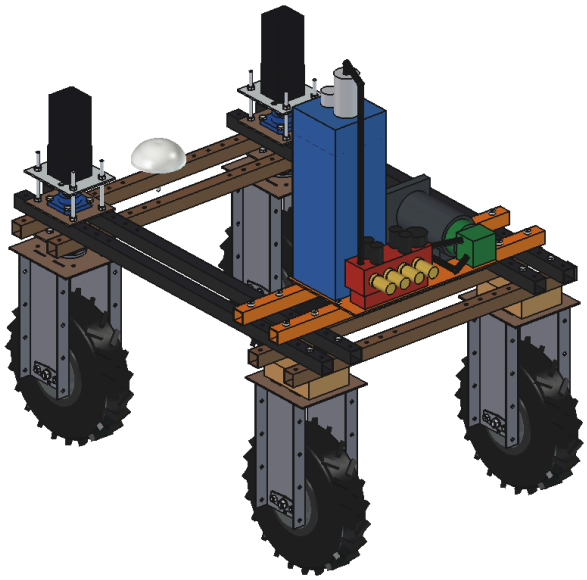
- 4-leaf weeds at 0.2 m/s: 1.8 kJ per weed
- ~3000-3800 weeds per charge
- 27 kg; crop rows untouched (checked)

Folder: `electric weeder/simple/`

IMPLEMENT · TRANSPORT & GENERAL

Hydraulic power pack

48 V hydraulic power unit on the rear half: BLDC motor with a 4 cc gear pump (11 l/min, 150 bar), 16 l tank, 2-section solenoid valve block and 2 pairs of ISO-A quick couplers for hydraulic implements.

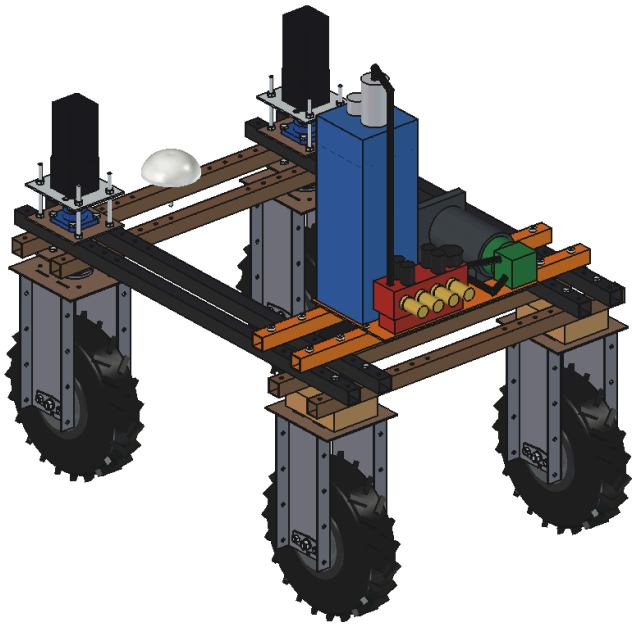


Simple

1 variant: Simple
Folder: `hydraulic pump/`

HYDRAULIC POWER PACK · VARIANT 1 OF 1

Simple



2 bearers and a 5 mm base plate on the upper beams, 16 l steel tank with return filter, BLDC 48 V 3.5 kW with bell housing and group-1 gear pump, motor controller, valve block with relief valve, gauge and 2 double-acting 24 V sections, couplers screwed into its rear face.

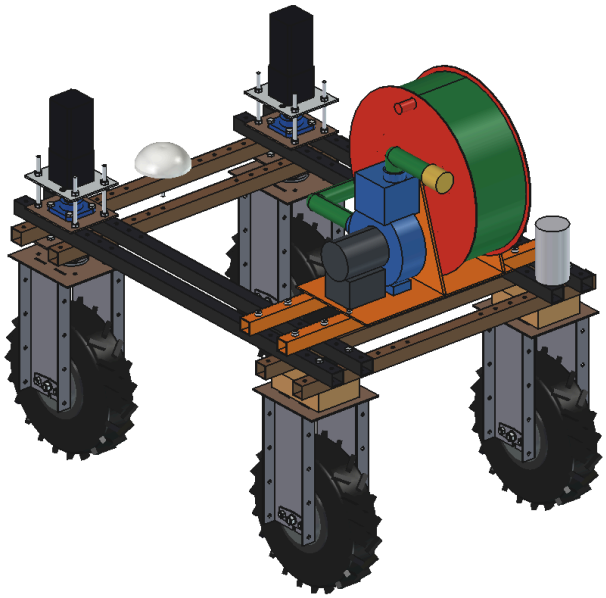
- 11 l/min, 150 bar: 19-30 kN cylinders
- 44 min of full flow at 100 bar per charge
- 49 kg + 12 kg oil, rear axle 62 %

Folder: `hydraulic pump/simple/`

IMPLEMENT · SPRAYING, FERTILIZING & IRRIGATION

Irrigation pump

Direct pump from ditches and ponds: a self-priming 1.5" pump on a 48 V 1.5 kW motor with an 8 m suction hose on a reel, delivering to a Camlock coupling at the rear; mount of the hydraulic power pack.

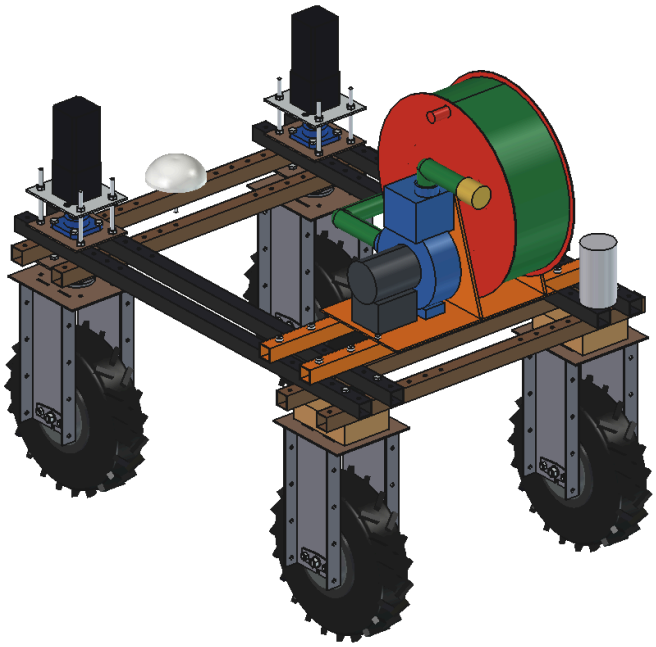


Simple

1 variant: Simple
Folder: `irrigation pump/`

IRRIGATION PUMP · VARIANT 1 OF 1

Simple



Bearers and base plate from the hydraulic pack, self-priming centrifugal pump with priming chamber and BLDC 1.5 kW, reel O520 with 8 m of spiral suction hose, rotary union and strainer with foot valve, discharge to a 1.5" Camlock at the rear edge.

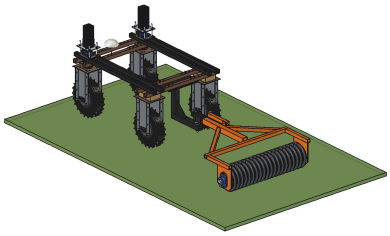
- 8.5 m3/h at 3 m lift into a drip line
- 10-17 m3 per charge
- 62 kg

Folder: `irrigation pump/simple/`

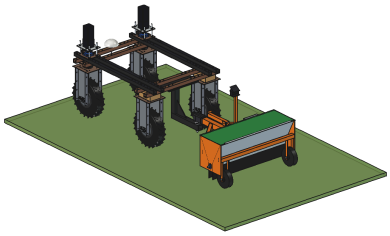
IMPLEMENT · SOWING & PLANTING

Cambridge roll and overseeder

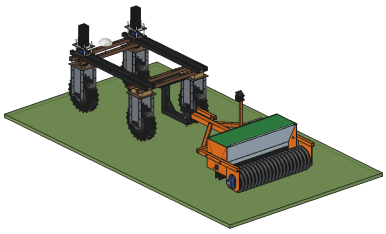
Simple trailed lawn-repair implements on a 50 mm tow ball under the rear beams: a Cambridge roll, a drop seeder that a camera switches on over bare soil, and the two combined.



Cambridge roll



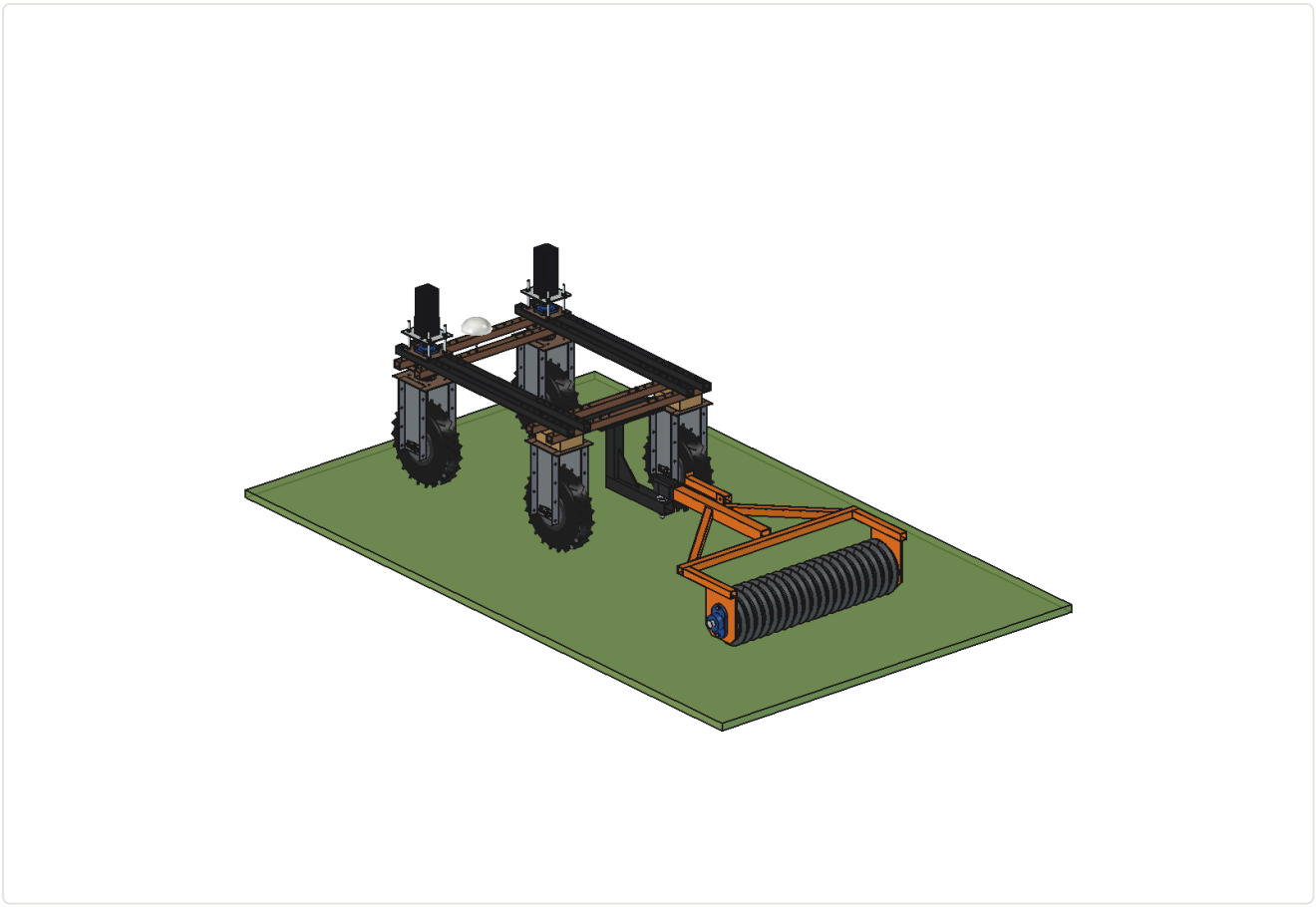
Overseeder (drop seeder)



Overseeder with Cambridge roll

3 variants: Cambridge roll · Overseeder (drop seeder) · Overseeder with Cambridge roll
Folder: `overseeding/`

Cambridge roll

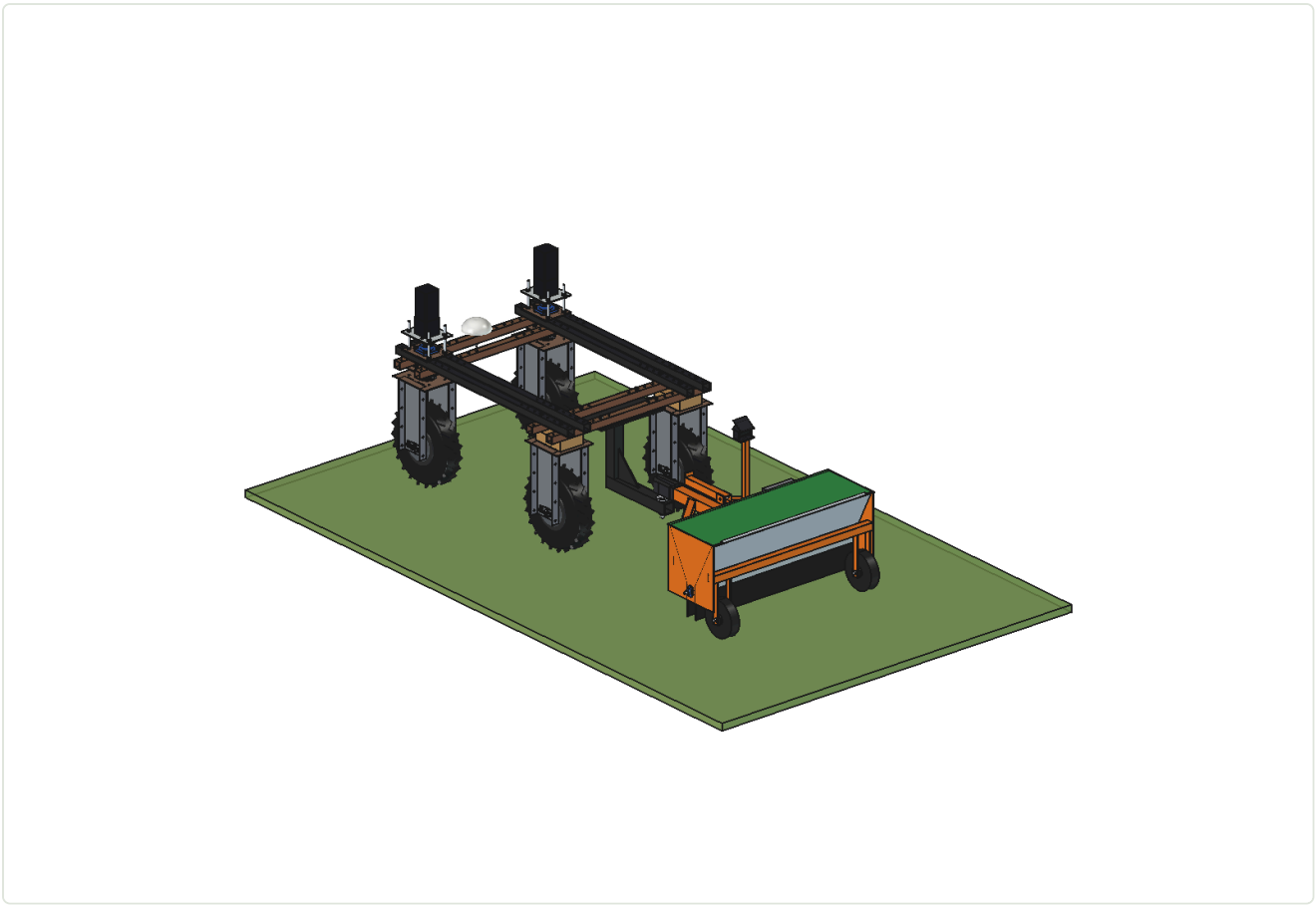


1 m ring roller with 20 cast-iron rings Ø280 on a Ø40 shaft, on the tow hitch with coupling head and drawbar.

- 178 kg
- 100 ... 250 N draft
- About €850 with the hitch

Folder: `overseeding/cambridge roll/`

Overseeder (drop seeder)

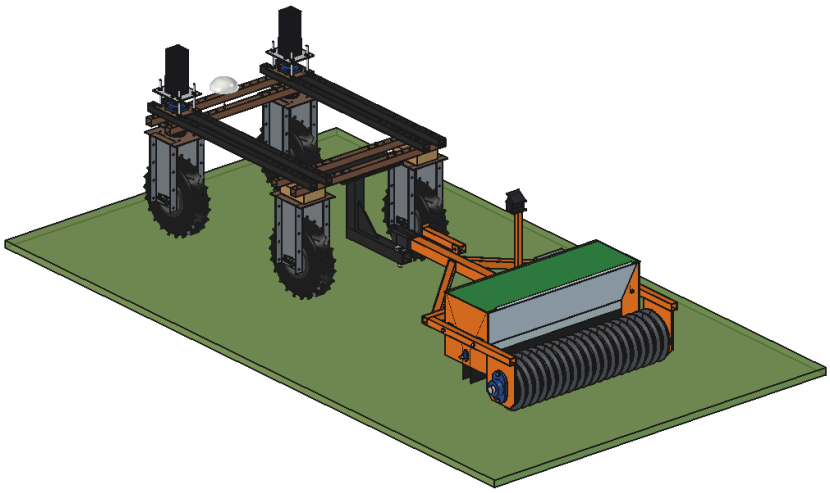


1 m drop seeder with a 49 l hopper and a fluted roller on a 24 V worm gearmotor that a camera switches on only over bare soil, on two small wheels.

- 61 kg
- About 60 N draft
- About €500

Folder: `overseeding/overrseeder/`

Overseeder with Cambridge roll



The seeder unit hangs between the arms of the roll: seed falls 300 mm in front of the rings and is rolled in. Includes the turn kinematics of the trailer and an animation with bare patches.

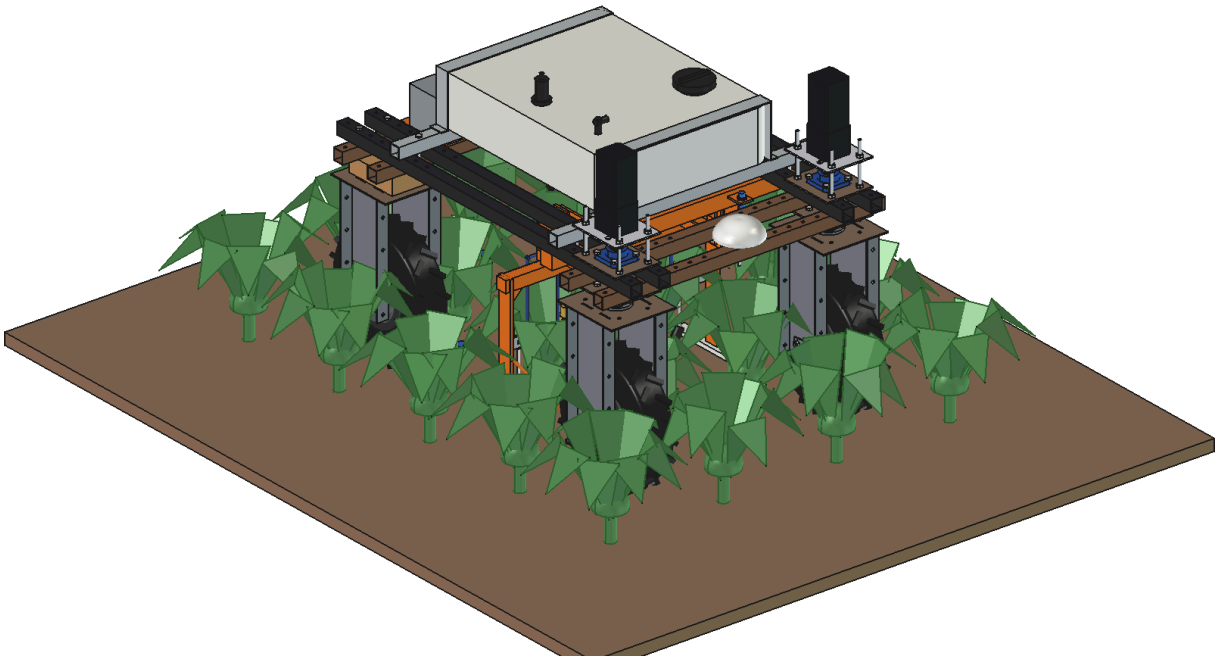
- 216 kg (234 kg full)
- 185 ... 310 N draft
- About €1,200

Folder: `overseeding/overseeder with cambridge roll/`

IMPLEMENT · WEED & CROP PROTECTION

Cauliflower root-crown applicator

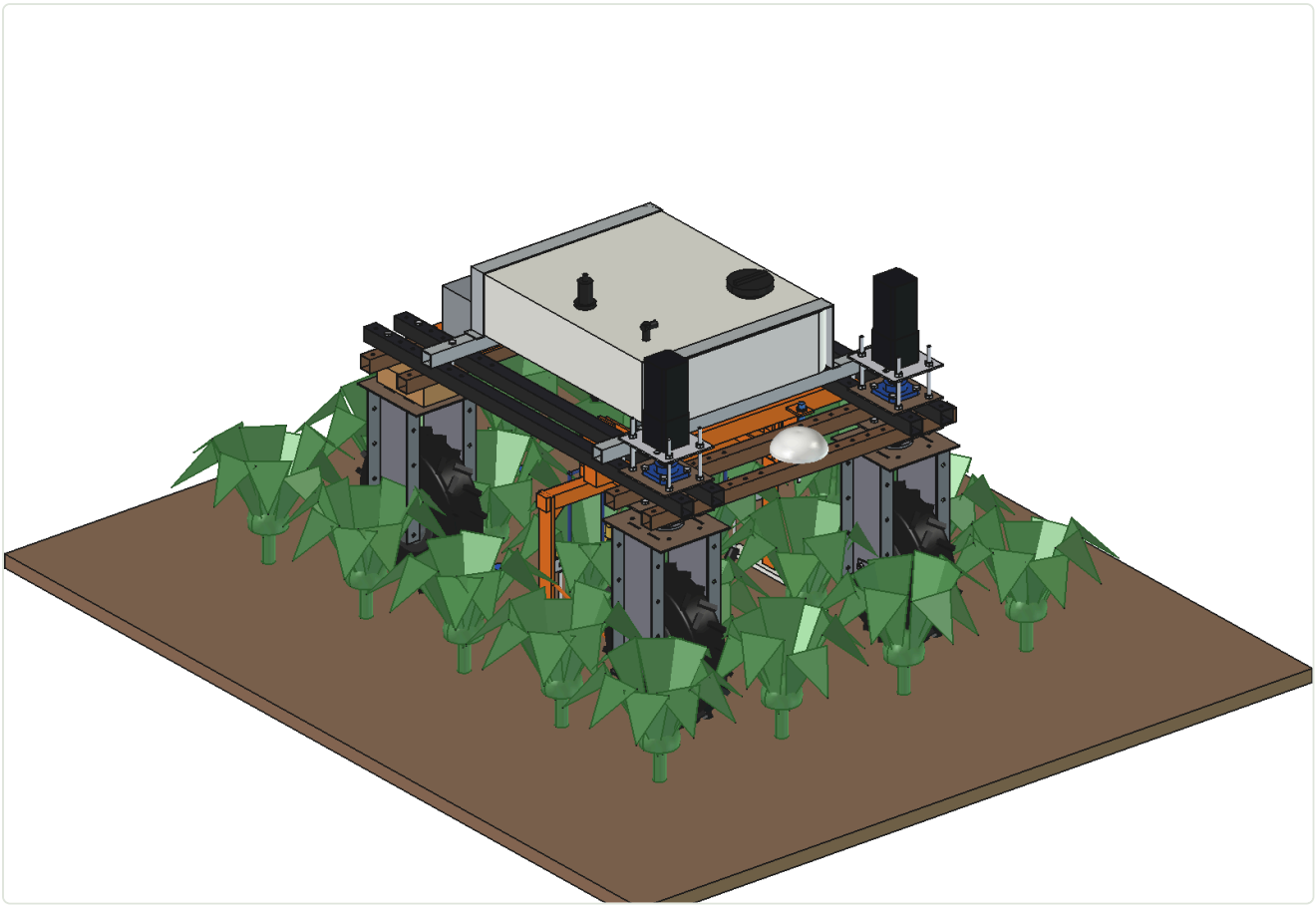
Doses insecticide in pulses onto the soil 40 mm beside the stem of every cauliflower plant (mid-season, 40 x 40 cm), not over the crop.



Simple

1 variant: Simple
Folder: `cauliflower applicator/`

Simple



Two rows pass under the robot; per row a parallelogram unit on UHMW skids follows the soil, leaf lifter rods clear the stem base and two 24 V solenoid valves fire 30° drop nozzles inside splash shrouds. A distance sensor finds the crown, wheel odometry times the pulse. 60 l tank, diaphragm pump, regulator and accumulator on the rear beams.

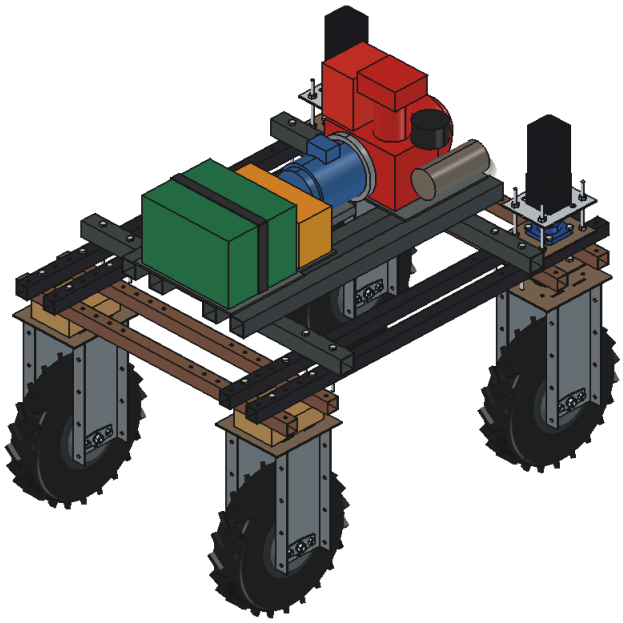
- 4 ml/plant = 250 l/ha
- 0.11 ha/h, 0.24 ha per tank
- 56.5 kg

Folder: `cauliflower applicator/simple/`

IMPLEMENT · POWER & DRIVE

Hybrid range extender

Petrol generator set on a steel tube base on top of the robot that charges the battery while driving; the battery delivers the peak power for heavy pulling (the worked-out generator / engine idea).

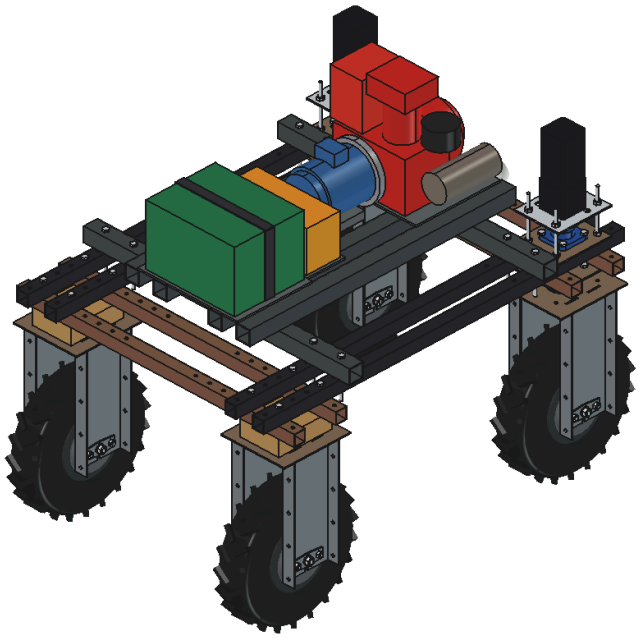


Range extender

1 variant: Range extender
Folder: `Hybrid/`

HYBRID RANGE EXTENDER · VARIANT 1 OF 1

Range extender



In-line train on 50 x 50 tubes bolted to the upper beams: 212 cc petrol engine (4.8 kW), jaw coupling, 3 kW permanent-magnet generator, controller box and a 16S 59 V 40 Ah battery (2.4 kWh). Clear of the steering stacks.

> About 76 kg, cg z 868 mm

> About 1.5 l/h at full output

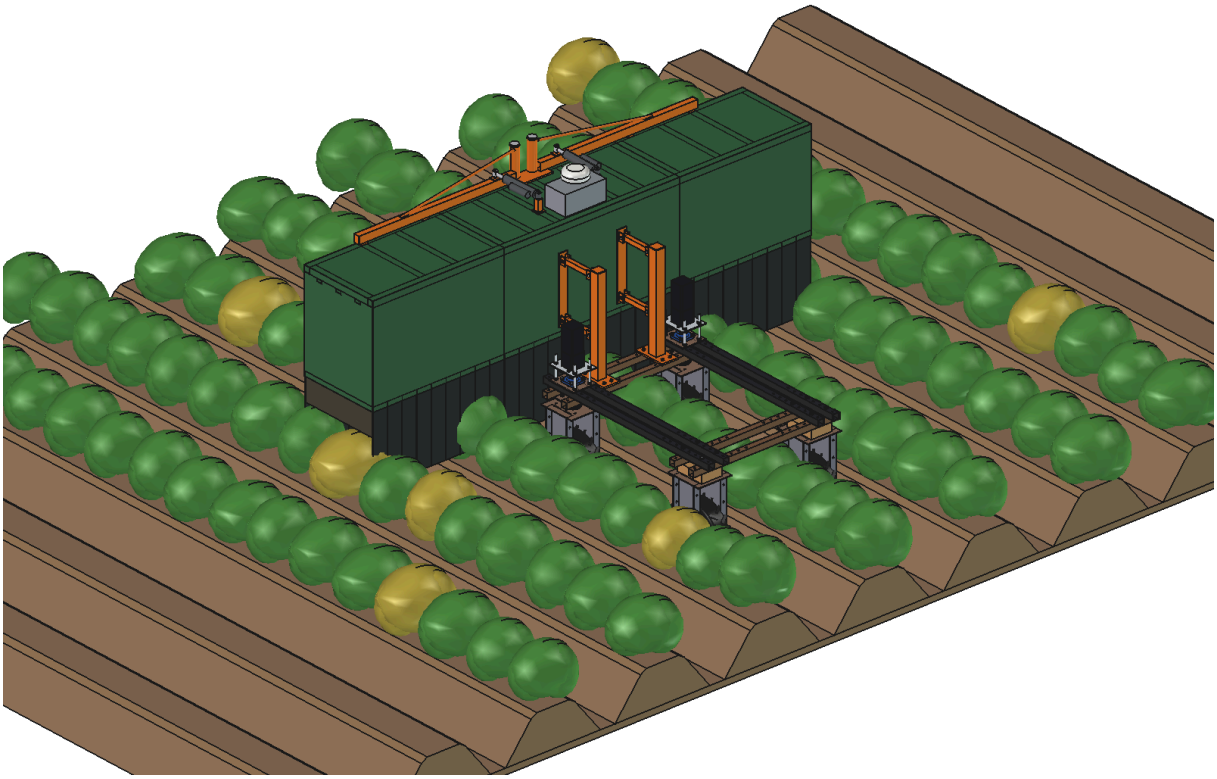
> Light task: battery +2.1 kW

Folder: `Hybrid/`

IMPLEMENT · MEASURING & SENSORS

Potato disease scout

A light-tight camera hood in front of the robot, 3.0 m over 4 potato rows: 10 downward cameras and 20 LED strobe bars photograph every plant, day and night, for virus (PVY, leaf roll), blackleg and late blight; suspect plants are stored with their GNSS position (cf. Doorgrond Croptiscan).



Potato

1 variant: Potato
Folder: `disease scout/`

POTATO DISEASE SCOUT · VARIANT 1 OF 1

Potato



Centre section of 1.2 m and two 0.9 m wings that swing forward round a kingpin on the roof (24 V actuator each): 1.20 m wide in transport. The hood rides on 4 castors in the furrows and is pushed by a parallelogram from two posts on the front beams, so the robot carries almost nothing. Strip curtains let the haulm through; 12 mm lenses give 0.16-0.24 mm/px, a 200 µs flash freezes the motion.

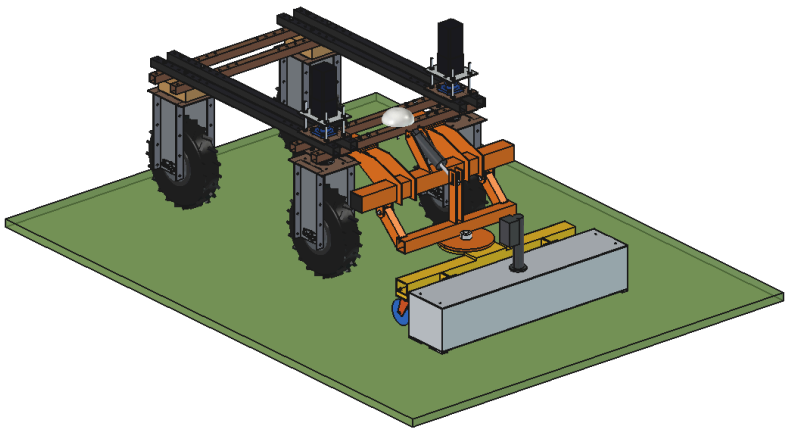
- 3.0 m, 0.86 ha/h at 1 m/s, 3.9 ha per charge
- Folded 1.20 m wide
- 135 kg (hood 120 kg on castors)

Folder: `disease scout/potato/`

IMPLEMENT · TRANSPORT & GENERAL

Magnet sweeper

Flat permanent magnet 20 mm over grass, festival fields and yards that picks up nails, screws, tent pegs, wire and bottle caps. Emptied over a flat tray at the field edge by lifting the magnet core inside its box; on the push blade's arms and the broom's castors (imported).

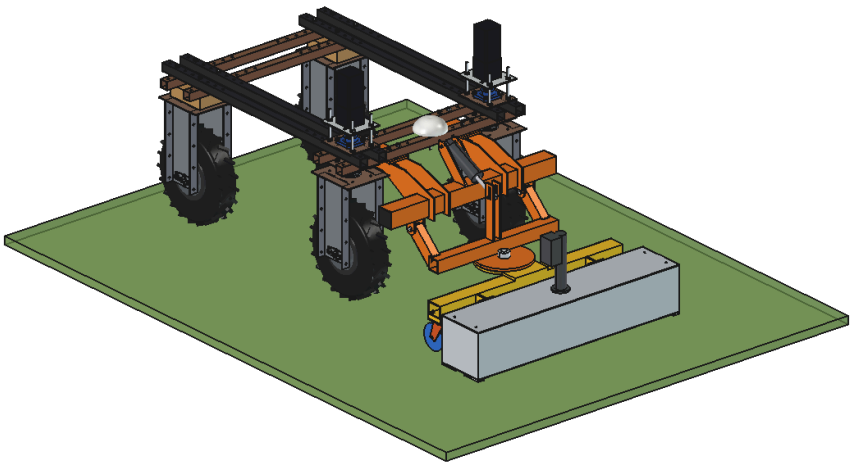


Simple

1 variant: Simple
Folder: `magnet sweeper/`

MAGNET SWEEPER · VARIANT 1 OF 1

Simple



Aluminium box 1100 x 240 with a 3 mm stainless skin; inside a steel back plate with 2 rows of 10 NdFeB N35 blocks 100 x 100 x 10 (opposite poles for a strong field gradient) on 4 guide rods. A 3000 N actuator on the lid lifts the core 100 mm: the field at the skin drops from 140 to 20 mT and the iron falls into a 10 mm high tray, without lifting the arms.

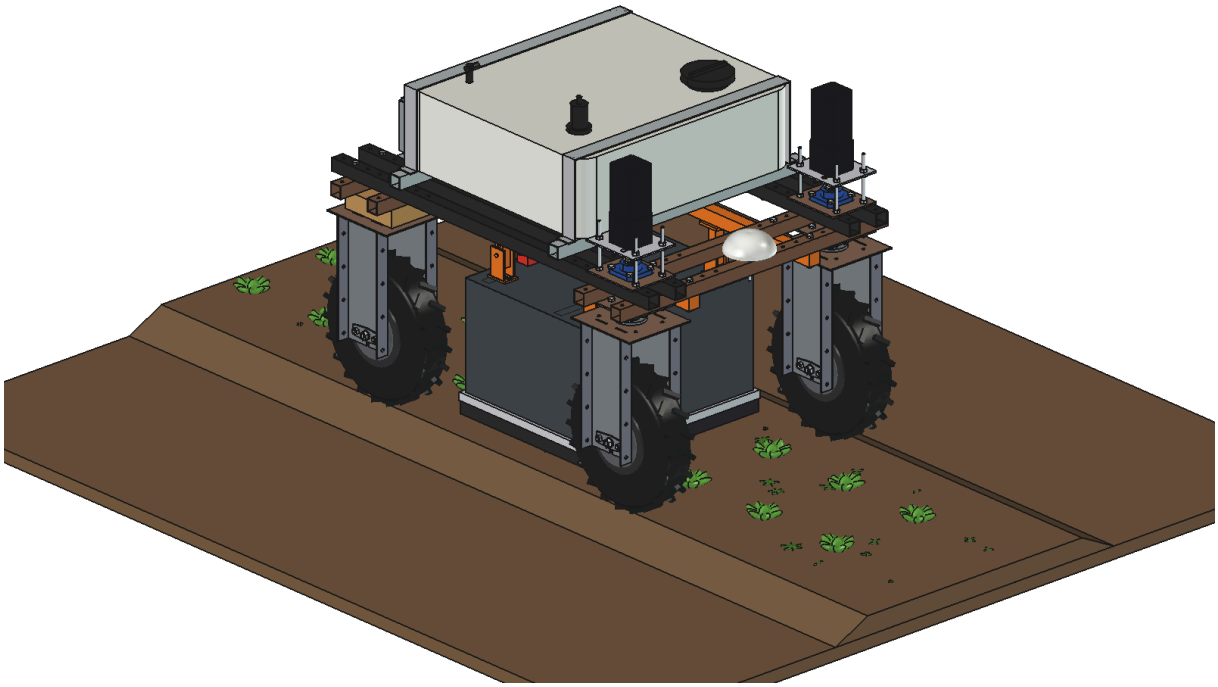
- Nails picked up 35-56 mm from the magnet
- ~0.41 ha/h, ~2.2 ha per charge
- No ballast needed to empty; 125 kg

Folder: `magnet sweeper/simple/`

IMPLEMENT · SOIL CULTIVATION & WEED CONTROL

Drop-on-demand boom

A light-tight hood under the belly of the robot, over one bed of 0.6 m: a camera finds the weeds and a rail of 24 micro-injectors fires single 5 ul drops onto them, inside the hood (no drift). ~3 l/ha against 200 l/ha broadcast; tank, pump and control box of the spot sprayer (imported).

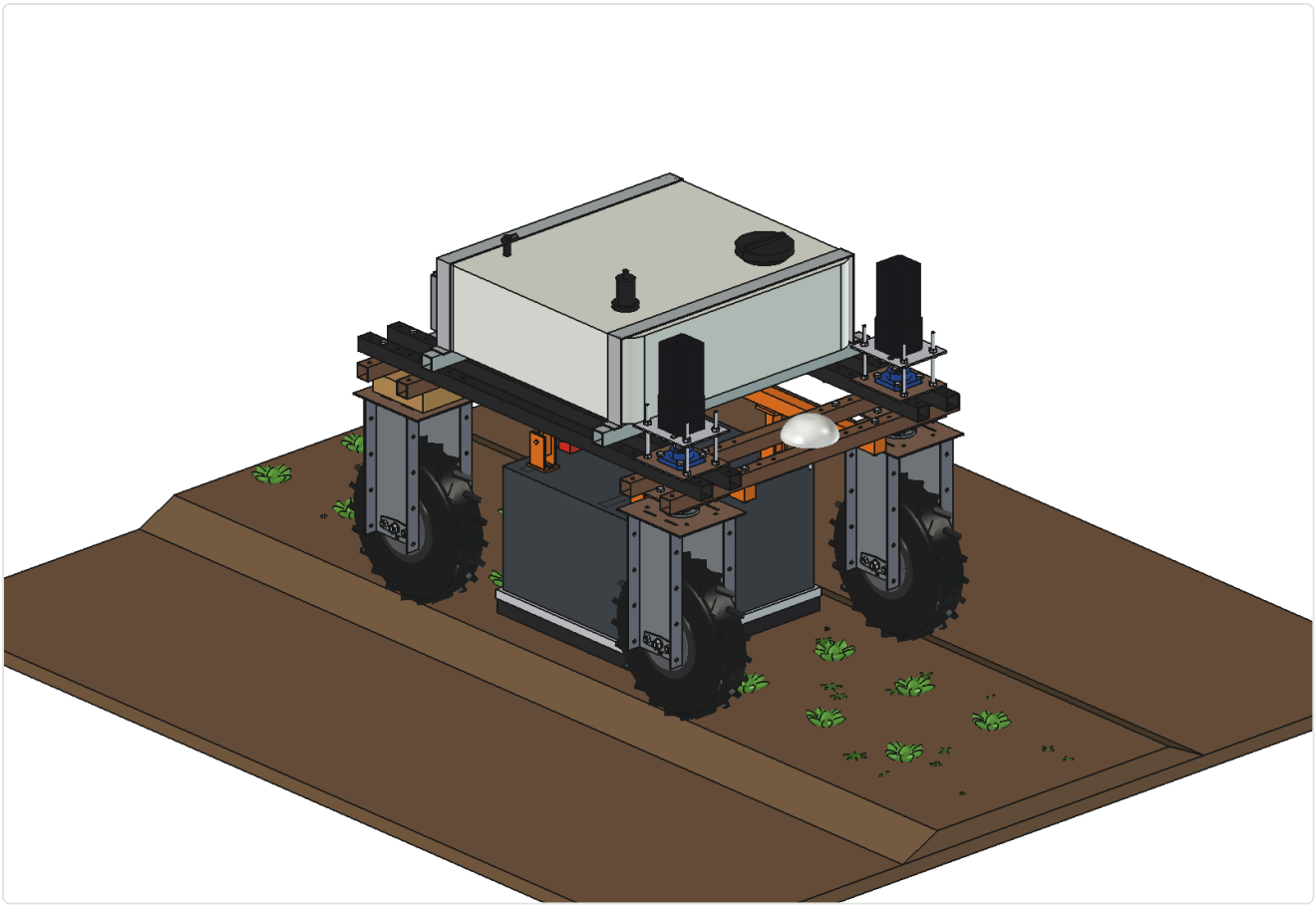


Simple

1 variant: Simple
Folder: `drop on demand boom/`

DROP-ON-DEMAND BOOM · VARIANT 1 OF 1

Simple



Aluminium hood 600 x 524 x 360 with a brush strip, on 2 rails under the lower beams and 4 posts (3 height steps of 25 mm). 5 MP camera straight down with 2 LED bars, 0.29 mm/px; 465 mm behind the image edge a manifold with 24 solenoid micro-injectors at 25 mm pitch, tips 80 mm above the soil, 0.5 bar from a second regulator and accumulator. Drops fall in 62 ms and hit at 1.6 m/s (no splash); none within 40 mm of a crop plant.

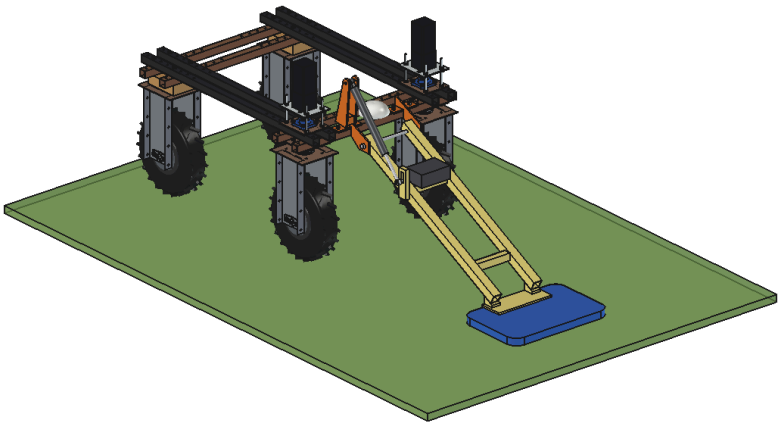
- ~3 l/ha (98.5 % less), 30 l = ~10 ha
- 0.08 ha/h at 0.5 m/s, ~4.4 h per charge
- 74 kg empty

Folder: `drop on demand boom/simple/`

IMPLEMENT · MEASURING & SENSORS

Metal detector

Pulse-induction metal detector that maps buried metal (cans, wire, horseshoes, bolts, plough points) and logs a GNSS point per hit. A 600 x 400 search coil slides on PE skids 1.2 m in front of the robot on a floating glass-fibre boom, away from the robot's steel.

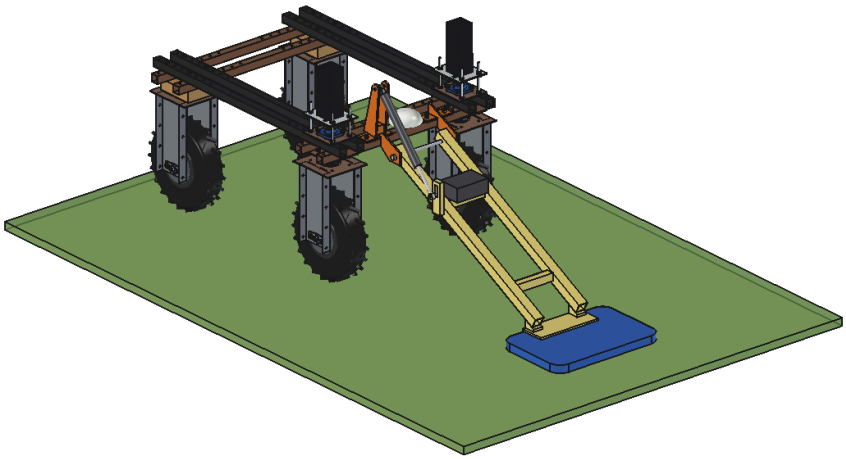


Simple

1 variant: Simple
Folder: `metal detector/`

METAL DETECTOR · VARIANT 1 OF 1

Simple



2 steel root brackets on the front outer lower beam with a stainless pivot pin, a boom of 2 GFRP tubes 50 x 50 with GFRP cross tubes and carrier, a mono loop 600 x 400 in an ABS housing on 2 PE skids, a PI board in an IP67 box and a 500 N actuator with a float slot that lifts the boom 14 degrees at the headland. No metal in front of the pivot.

> Can ~0.49 m, horseshoe ~0.42 m, bolt ~0.15 m deep

> 0.18 ha/h, ~1.8 ha per charge

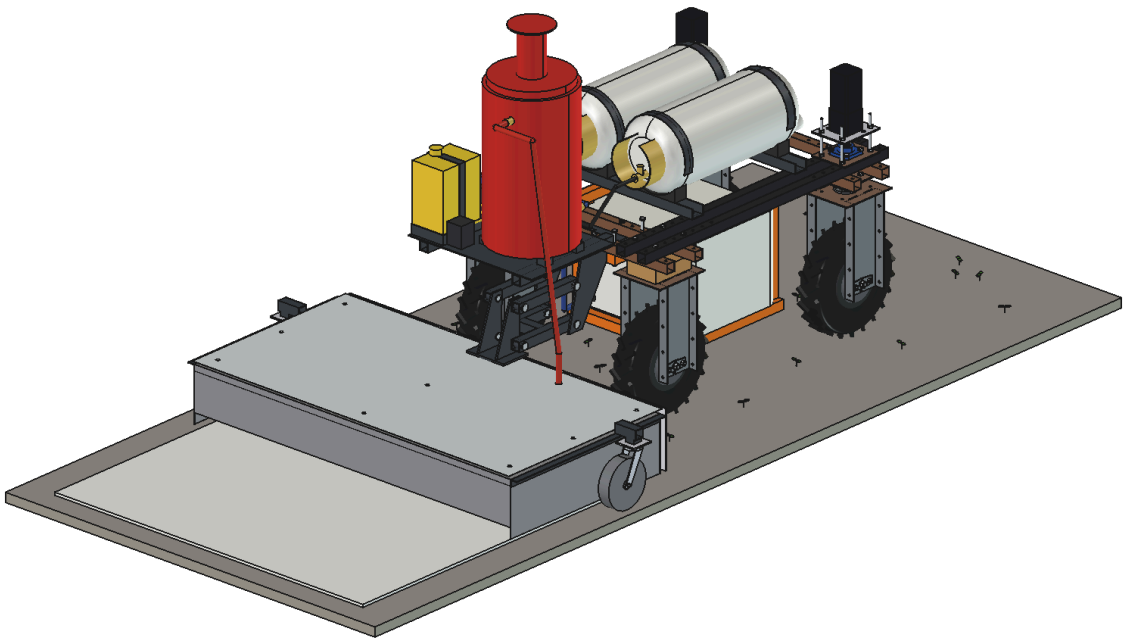
> Hit position ~0.1 m; 18 kg

Folder: `metal detector/simple/`

IMPLEMENT · WEED & CROP PROTECTION

Hot foam weeder

Kills weeds on paths, yards, fence lines and tree rows with hot water (96 C) and 0.5 % foam under the flame weeder's insulated hood: no flame on the ground, so also where flaming is a fire risk. Hood, castors, parallelogram and propane of the flame weeder (imported).

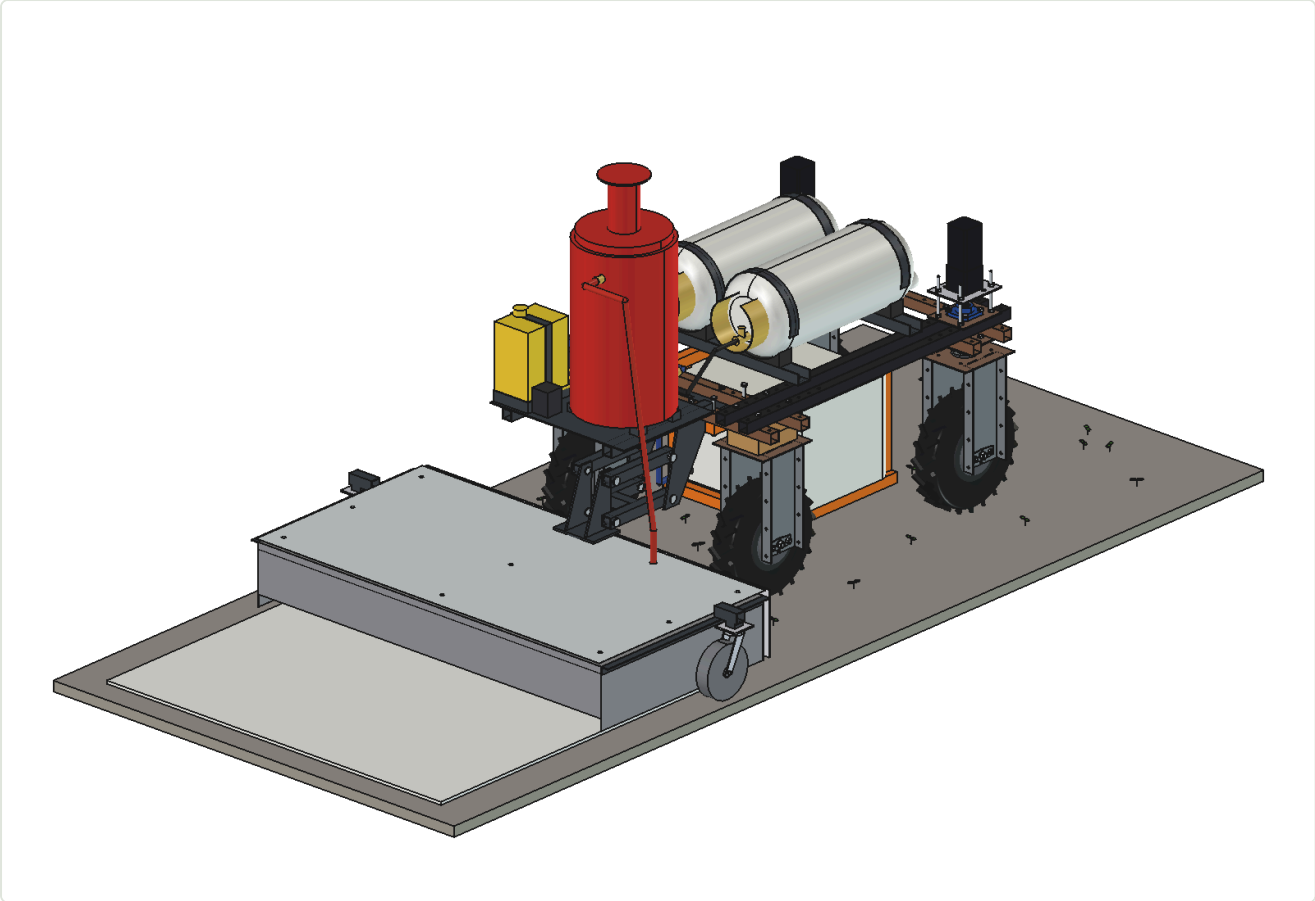


Simple

1 variant: Simple
Folder: `hot foam weeder/`

HOT FOAM WEEDER · VARIANT 1 OF 1

Simple



A 40 kW propane coil boiler, the water pump and a foam can with a dosing pump on a rear rack; a ~95 l water tank under the belly on the drop-on-demand rails; a foam bar with 12 nozzles in the 1.5 m hood instead of the burners. 6.8 l/min at 96 C gives 0.4 l/m2 at 0.19 m/s; the foam keeps the leaves hot for ~20 s.

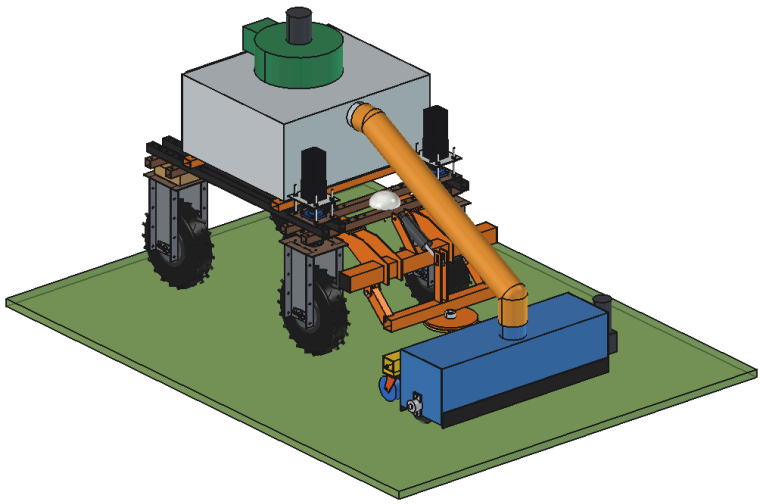
- ~1000 m2/h, 240 m2 per tank of water
- Propane ~36 kg/ha (as flaming)
- 224 kg empty, hood on its castors

Folder: `hot foam weeder/simple/`

IMPLEMENT · TRANSPORT & GENERAL

Debris vacuum

Vacuums litter after a festival (cups, cans, bottles, caps, butts) and solar-panel glass shards from grassland. A 1 m nozzle with a brush roller on the magnet sweeper's head and castors, a O125 hose to a 270 l bin on the upper beams, and a clean-air fan on the lid so glass never passes the impeller.

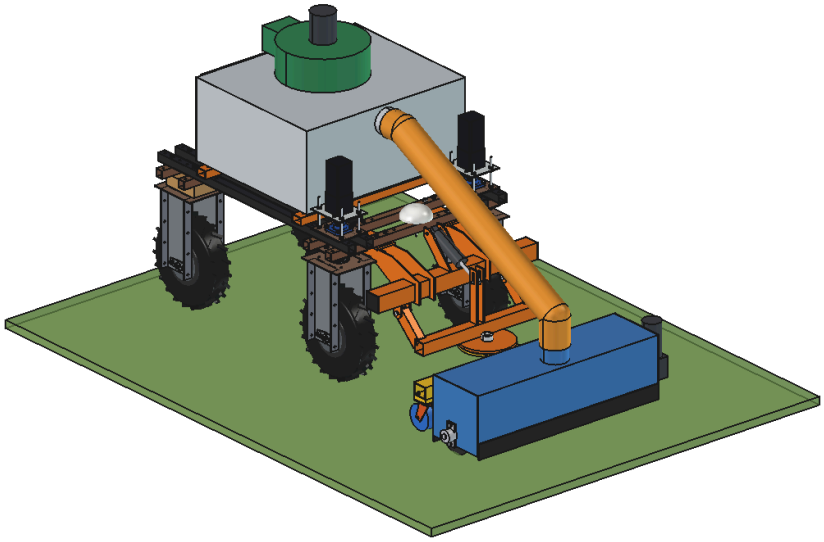


Simple

1 variant: Simple
Folder: `debris vacuum/`

DEBRIS VACUUM · VARIANT 1 OF 1

Simple



Steel nozzle 1030 x 300 with rubber flaps and a PP brush roller O200 (48 V 250 W worm gearmotor), PU hose O125, aluminium bin 900 x 750 x 400 with a liner bag and a rear door on 2 cross bars bolted to the upper beams, centrifugal fan O420 with a 48 V 1.2 kW motor behind a screen in the lid.

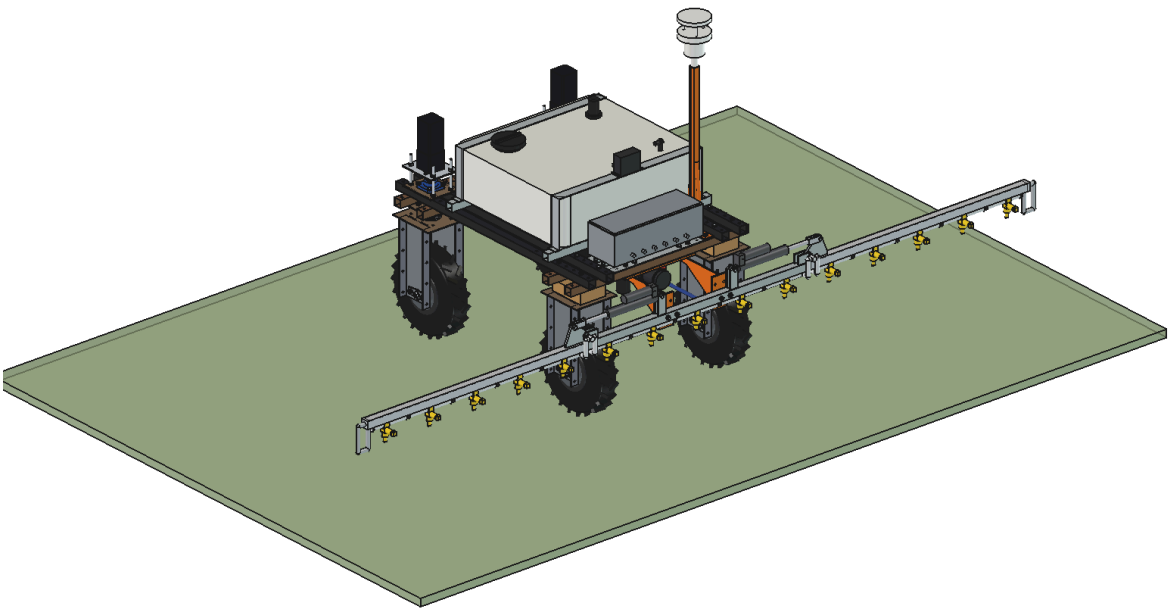
- 0.27 m3/s at 22 m/s: carries PV glass shards
- 0.26 ha/h, ~1.4 kW: 0.34 ha per charge
- 156 kg; range extender for big sites

Folder: `debris vacuum/simple/`

IMPLEMENT · SPRAYING, FERTILIZING & IRRIGATION

Biostimulant sprayer

Sprays compost tea, seaweed extract or nitrogen-fixing bacteria over the whole field at the right moment, and logs every pass with weather and GNSS track. The spot sprayer (tank, pump, 3.75 m folding boom) imported as it is, with coarse tips, an aerator and a weather mast.

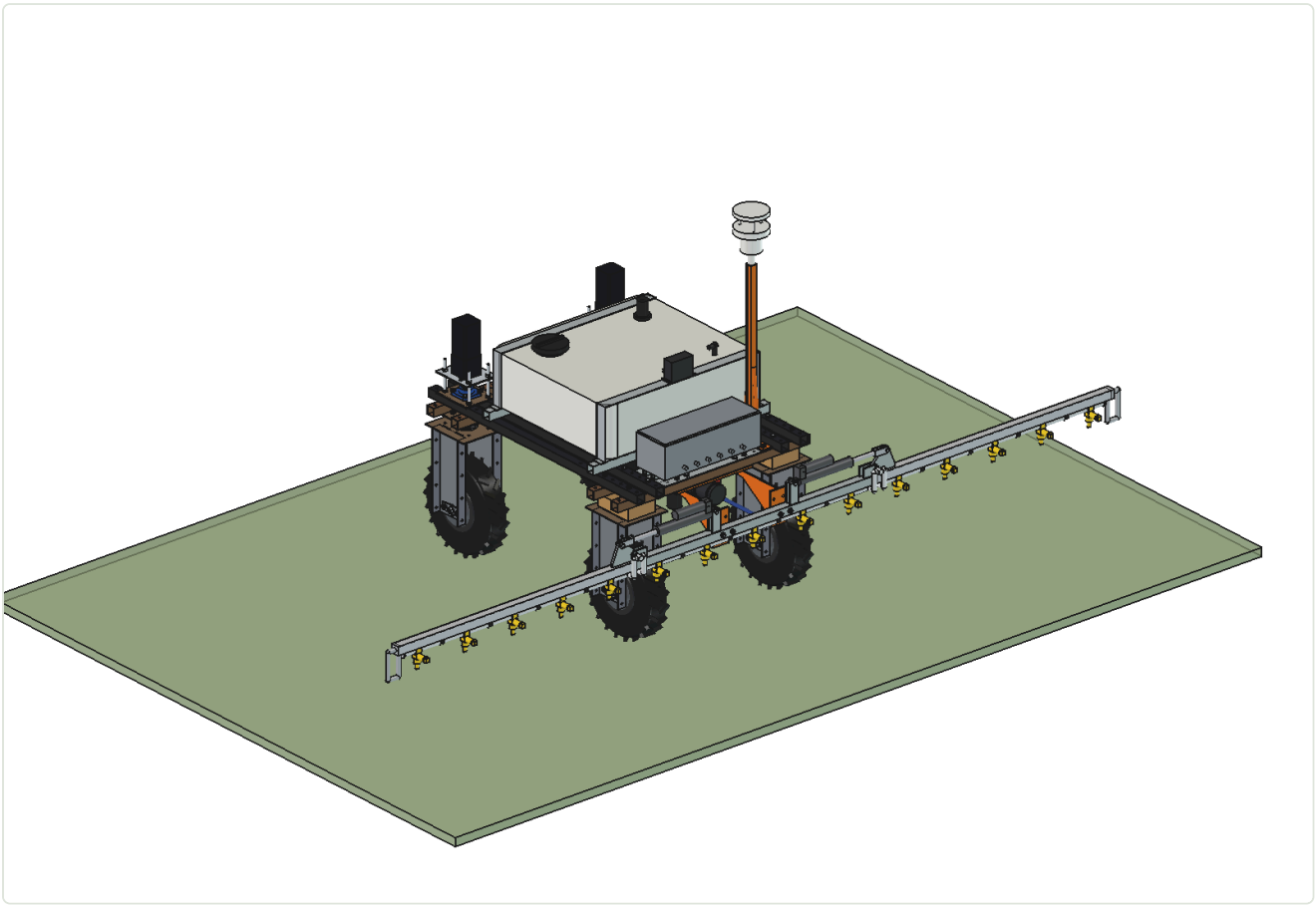


Simple

1 variant: Simple
Folder: `biostimulant sprayer/`

BIOSTIMULANT SPRAYER · VARIANT 1 OF 1

Simple



All 15 nozzles open with size 02 twin flat-fan turbulence tips at 1.5 bar (coarse drops, gentle for living organisms, no blocking behind the 50-mesh filter); a 30 l/min air pump and a diffuser ring keep compost tea aerobic; an all-in-one weather sensor on a mast on the rear beams lets the controller spray only inside the weather window.

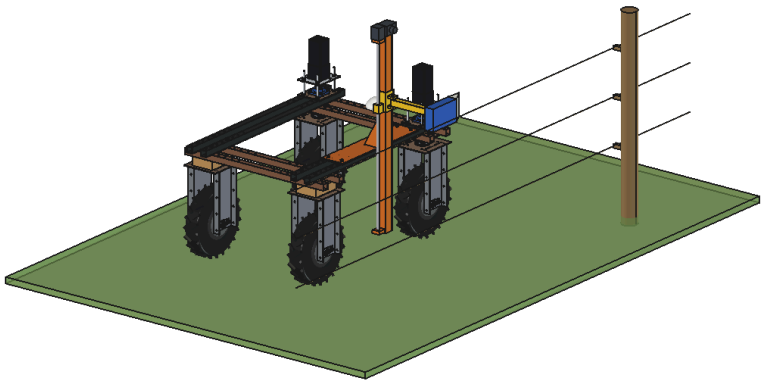
- 186 l/ha at 2 m/s, 0.64 ha per tank
- ~1.5 ha/h with refills, ~3.4 ha per charge
- 83 kg empty

Folder: `biostimulant sprayer/simple/`

IMPLEMENT · LIVESTOCK & FARM

Fence inspector

Drives along an electric fence and reads the voltage and current of every pulse without touching the wire (E-field plate and pickup coil 80 mm away), one reading per metre with GNSS: breaks, leaks and shorts on a map, and a photo of every post.

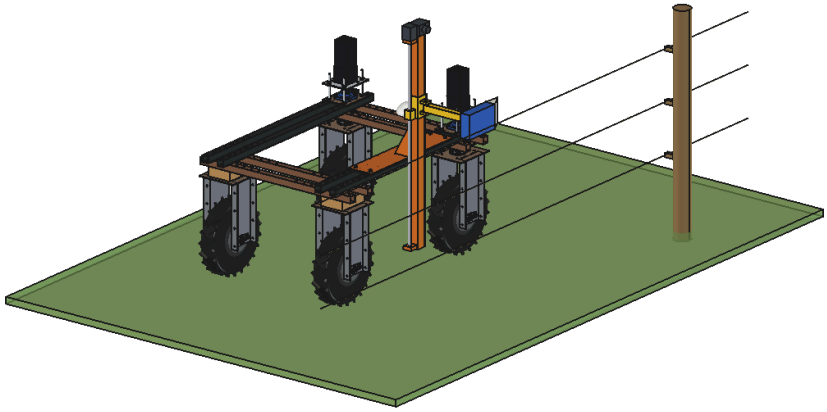


Simple

1 variant: Simple
Folder: `fence inspector/`

FENCE INSPECTOR · VARIANT 1 OF 1

Simple



Base plate on the right upper beams with a mast 60 x 60, a carriage on a Tr16 lead screw with a stepper (wires 300-1100 mm), a side arm on a spring hinge that swings back on obstacles, and a sensor head with an E-field plate, a pickup coil and a ToF distance sensor; a camera on the mast top for the posts.

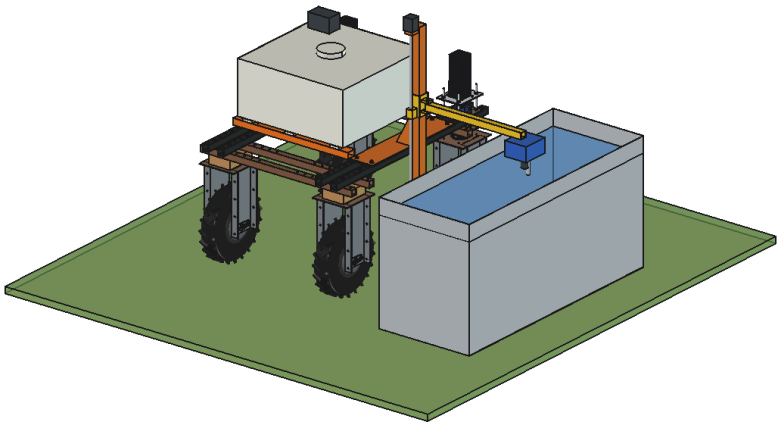
- Voltage +-9 % at 80 mm, contactless
- 1 reading per metre, ~42 km per charge
- 23 kg

Folder: `fence inspector/simple/`

IMPLEMENT · LIVESTOCK & FARM

Drinking trough check

Visits drinking troughs at stored GNSS positions, measures the water level with an ultrasonic sensor, photographs the water and tops the trough up from its own 160 l tank when the float valve or supply has failed. Uses the fence inspector's mast and carriage with a longer arm.

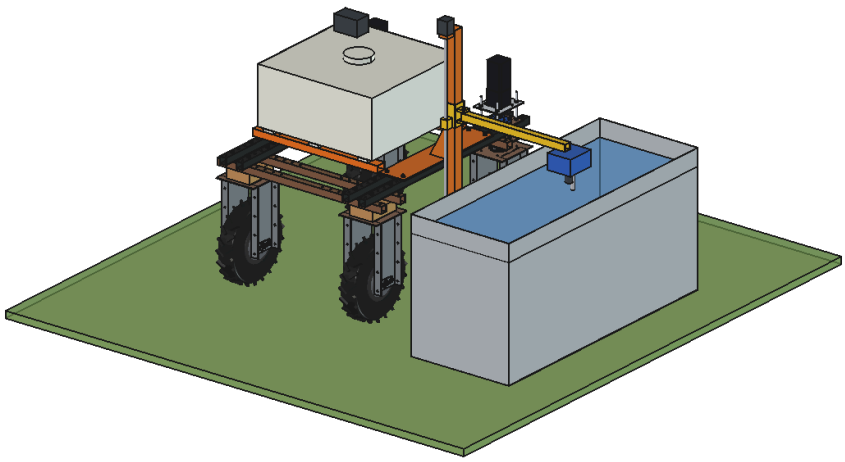


Simple

1 variant: Simple
Folder: `drinker check/`

DRINKING TROUGH CHECK · VARIANT 1 OF 1

Simple



Fence-inspector mast with lead screw and spring hinge (imported), a 635 mm arm with a head holding an ultrasonic level sensor, a camera and a 3/4 inch fill nozzle, and a 160 l PE tank with a 24 V 20 l/min pump on cross bars on the left upper beams.

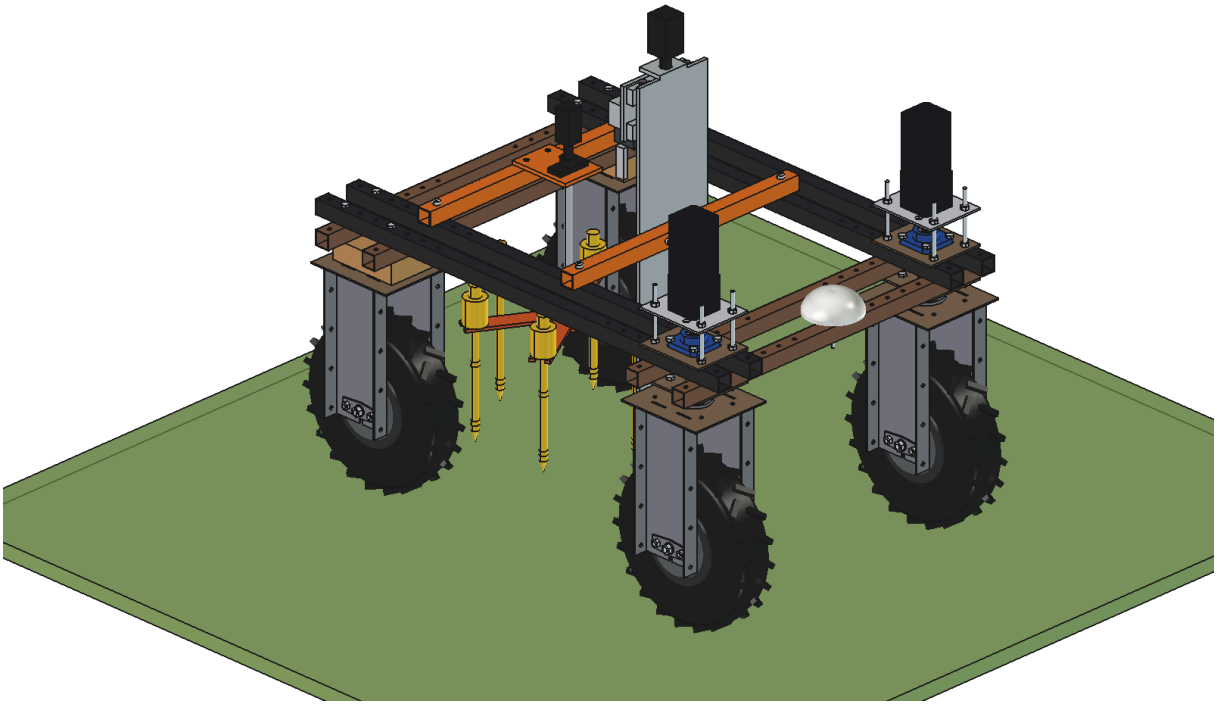
- Level +-5 mm (+-4 l)
- 85 l top-up in ~4 min
- ~11 rounds of 6 troughs per charge

Folder: `drinker check/simple/`

IMPLEMENT · MEASURING & SENSORS

Soil moisture sensor placer

Places wireless soil moisture sensors at set GNSS points, reads them by radio and picks them up again at the end of the season. The soil sampler's Z axis (imported) with a gripper, and a magazine star of 6 sensors.

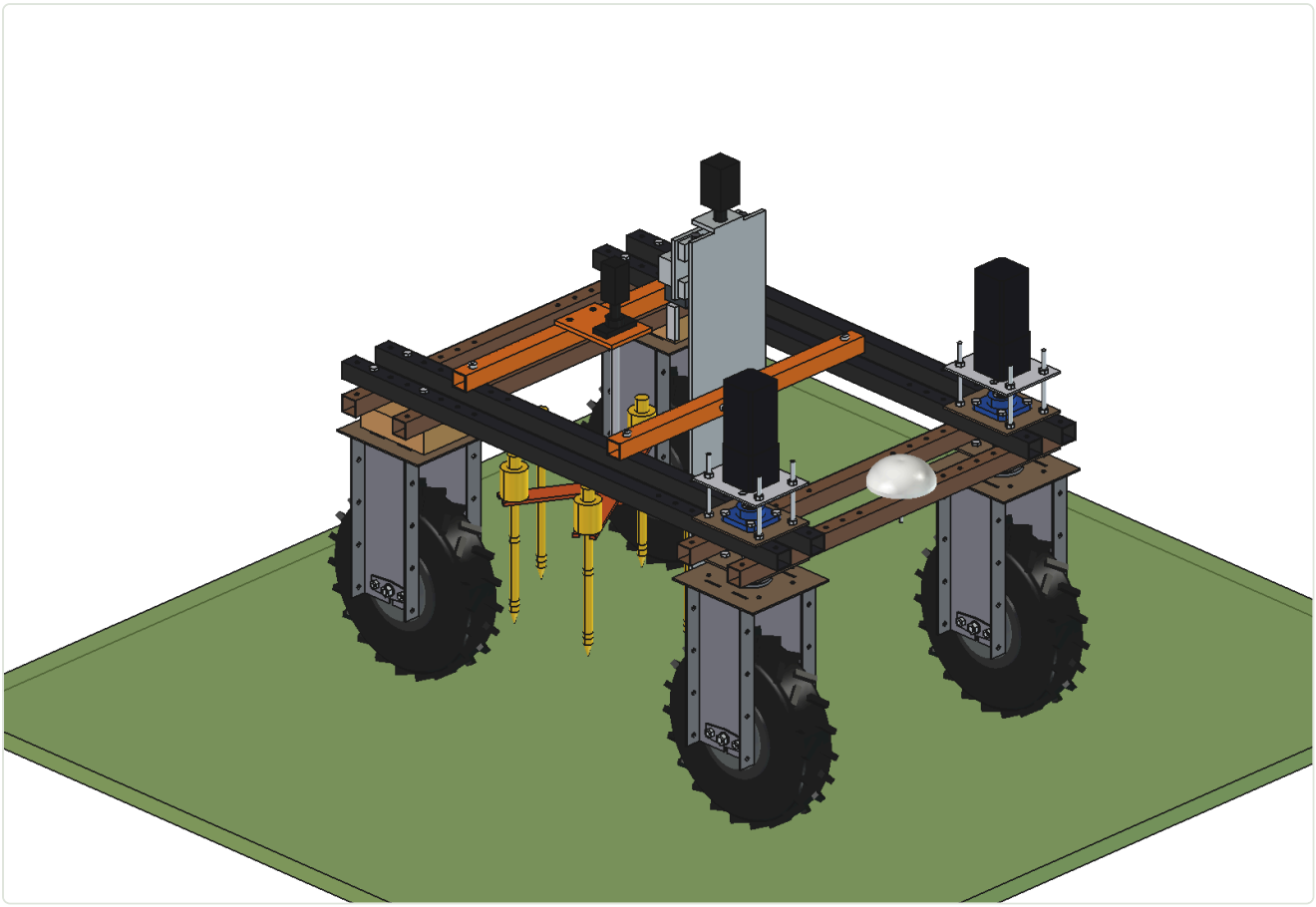


Simple

1 variant: Simple
Folder: `soil moisture placer/`

SOIL MOISTURE SENSOR PLACER · VARIANT 1 OF 1

Simple



A 24 V gripper on the soil sampler's carriage holds a sensor by its knob; a star of 6 fork arms on a shaft from the rear bridge carries the sensors. The gripper lifts a sensor out of the index fork, the star turns half a step and the ball screw pushes the O20 x 300 stem into the soil with the weight of robot and placer. Sensor: cap with LoRa radio and 2 AA lithium cells, rings at 100 and 250 mm.

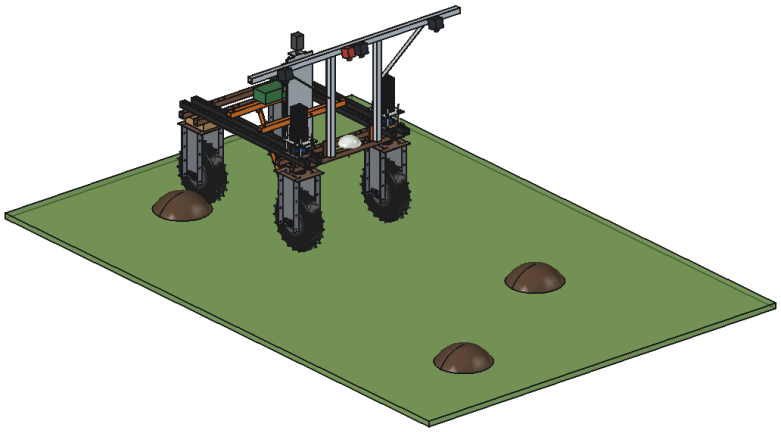
- ~850-1320 N push, 1350 N available
- ~64 s per sensor, 6 per trip (3 ha)
- 36 kg incl. 6 sensors

Folder: `soil moisture placer/simple/`

IMPLEMENT · CROP & SOIL

Mole and tunnel mapper

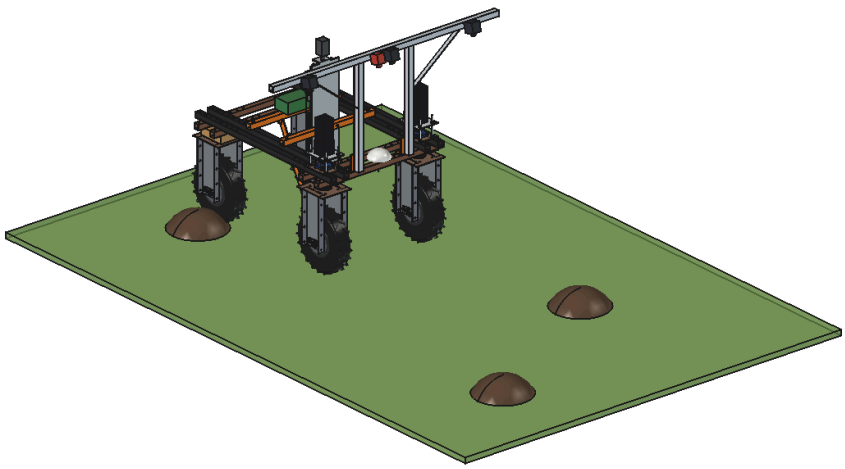
Maps fresh molehills with 3 RGB cameras and a thermal camera on a 1.4 m camera bridge, and finds the tunnels around a hill with the soil compaction meter's cone penetrometer: a resistance drop 50-250 mm deep is a tunnel.



Simple

1 variant: Simple
Folder: `mole mapper/`

Simple



Aluminium camera bridge on the front outer lower beam with 3 RGB 4K cameras and a thermal camera covering 2.5 m, plus the soil compaction meter (imported) that probes a ring of 16 points around a fresh hill.

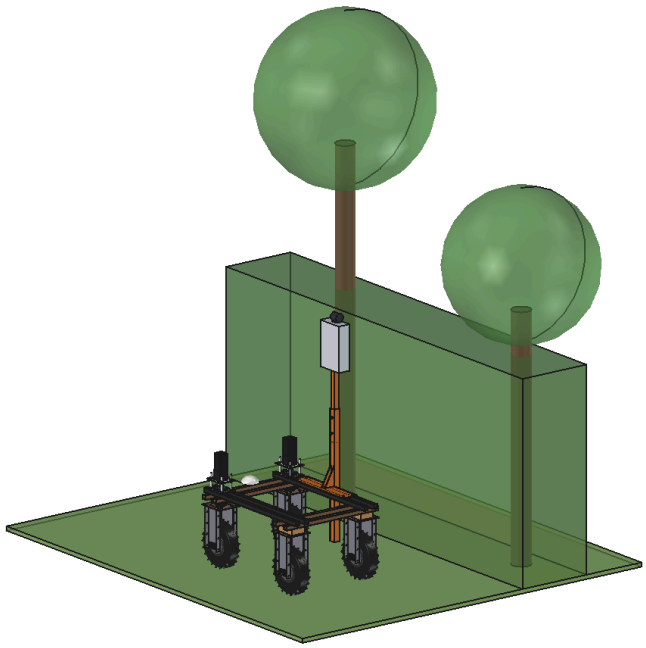
- 1.26 ha/h mapping, ~9 ha per charge
- Tunnel found ~85 % with 16 probe points
- 40 kg

Folder: `mole mapper/simple/`

IMPLEMENT · MEASURING & SENSORS

Windbreak and tree-row inspector

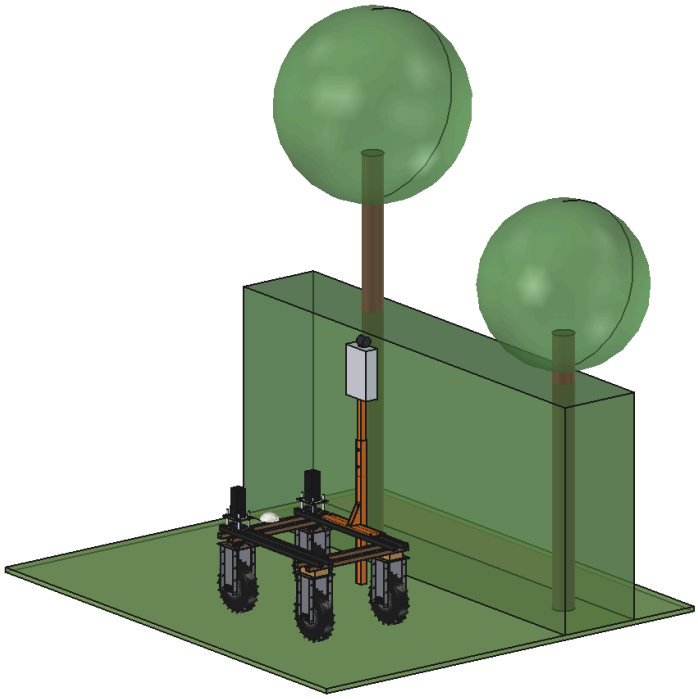
Drives along hedgerows, windbreaks and orchard rows: a 2D lidar scans the row profile (height, width, density, gaps) every 150 mm and a stereo pair plus a crown camera photograph it; each trunk gets its own GNSS point and status line.



Simple

1 variant: Simple
Folder: `windbreak inspector/`

Simple



Fence-inspector mast and base plate (imported) with an extension tube to 2.1 m and a sensor head: a 270 degree 2D lidar in a vertical plane across the row, a vertical stereo camera pair and a crown camera tilted up 35 degrees.

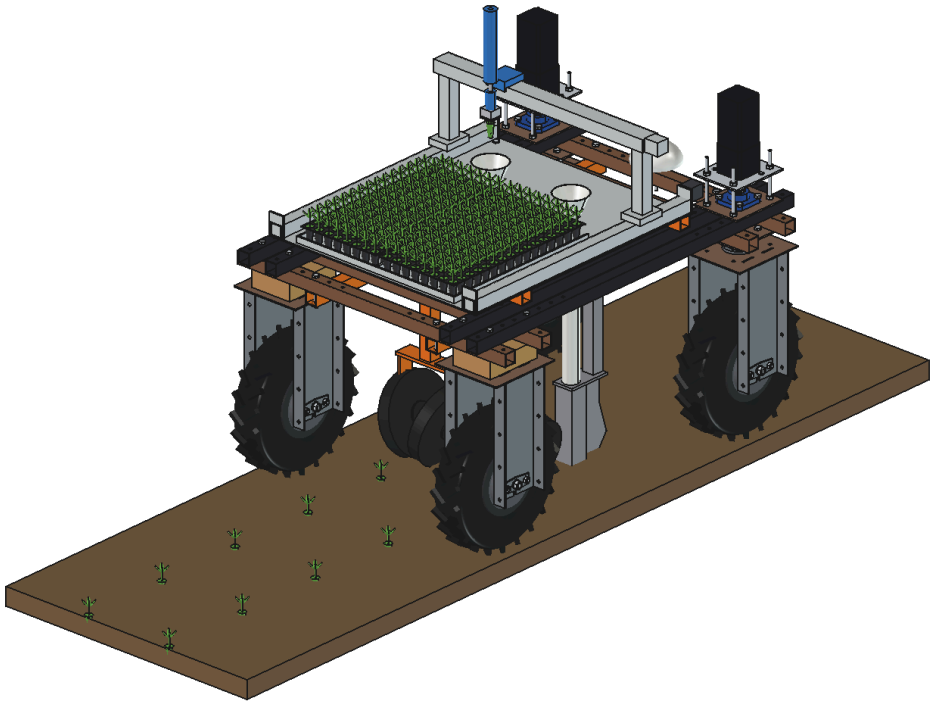
- Profile every 150 mm, up to 6 m trees
- ~40 km of row per charge
- 26 kg

Folder: `windbreak inspector/simple/`

IMPLEMENT · SOWING & PLANTING

Tray setter

Transplanter: an XYZ gantry with a needle gripper takes plug seedlings out of 2 trays on top of the robot and drops them through tubes to 2 runner shoes with press wheels under the belly (2 rows, 300 mm in the row).

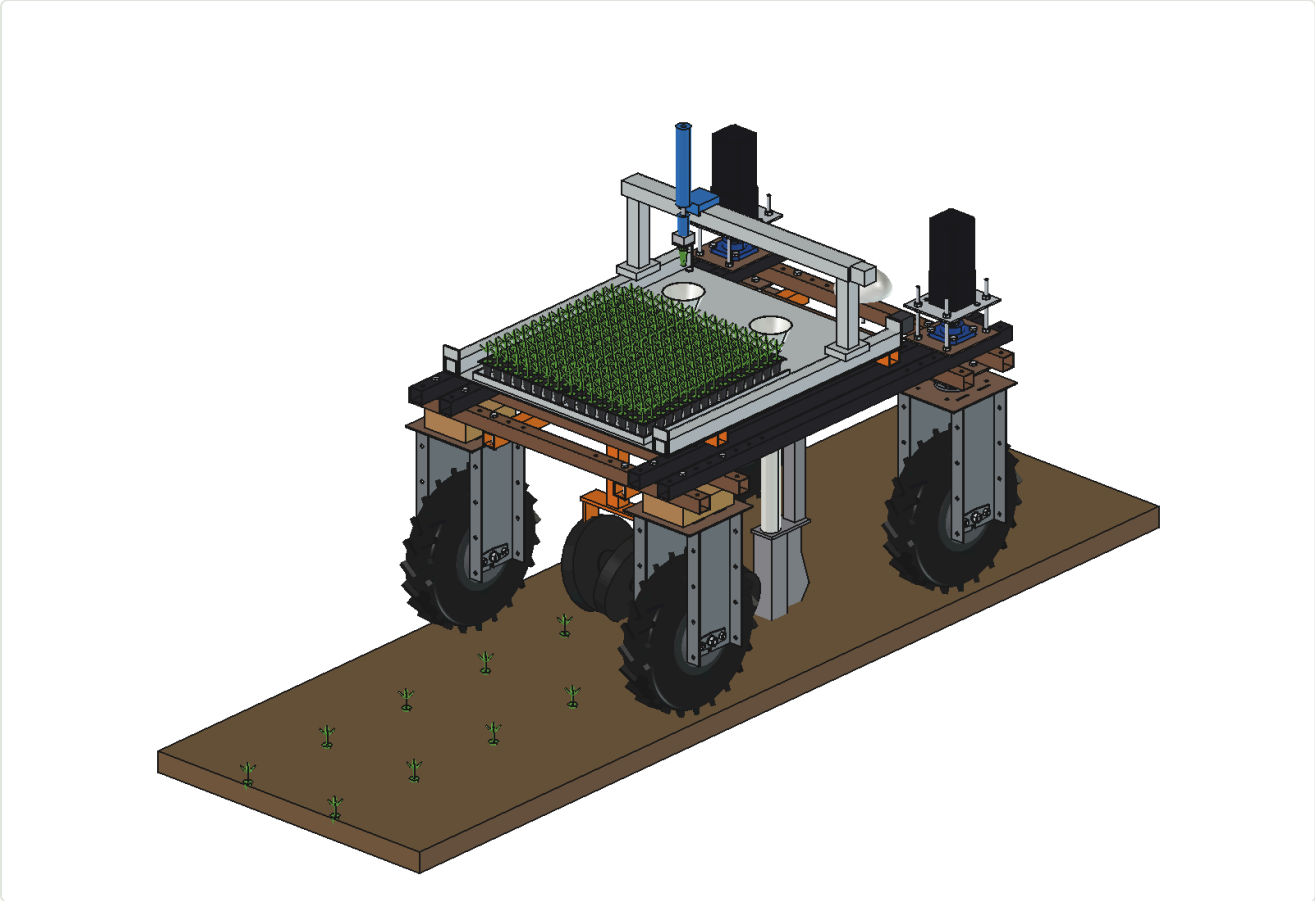


Simple

1 variant: Simple
Folder: `tray setter/`

TRAY SETTER · VARIANT 1 OF 1

Simple



Tray table on the upper beams with 2 plug trays of 128 cells; Y rails on the table edges, an X beam and a picker with a 4-needle gripper (GT2 belts, NEMA17). Each plug falls through a funnel and a O50 tube into the boot of a runner shoe that opens a 60 mm slot; 2 press wheels tilted 15 degrees close it. The shoes hang from the belly rails of the drop-on-demand boom.

1.6-3.0 s per plant, 0.048 m/s

~780 plants/h, 256 per set of trays

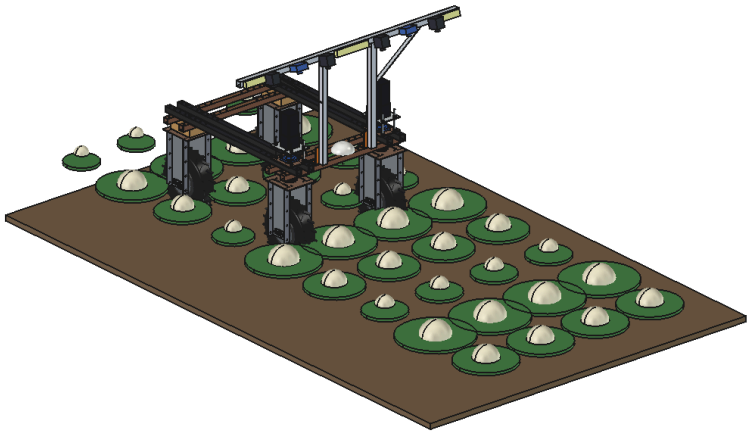
68 kg with full trays

Folder: `tray setter/simple/`

IMPLEMENT · MEASURING & SENSORS

Yield estimator

Counts and sizes heads and plants in 4 rows with RGB cameras, depth cameras and LED flashes on the mole mapper's camera bridge, and logs each plant with GNSS, size and ripeness for a harvest forecast.

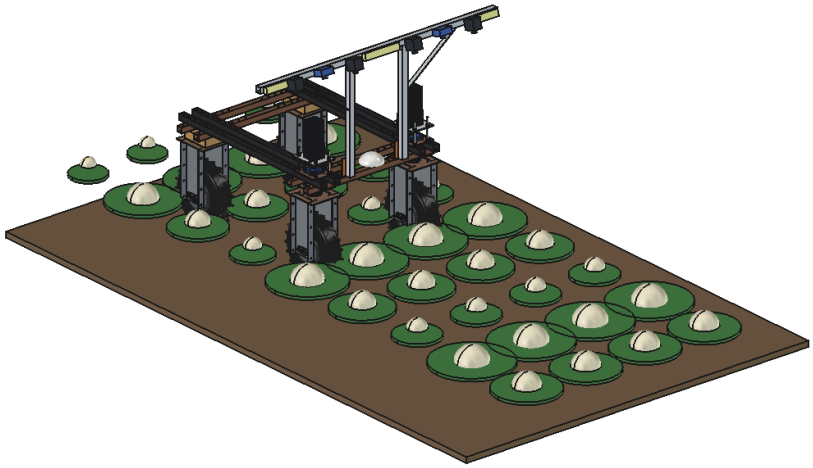


Simple

1 variant: Simple
Folder: `yield estimator/`

YIELD ESTIMATOR · VARIANT 1 OF 1

Simple



Mole-mapper camera bridge (imported) with 4 RGB 4K cameras (one per row at 450 mm), 2 ToF depth cameras and 3 LED flash bars. Head diameter to 0.25 mm/px, mass +-12 % per head, +-0.4 % on 1000 heads after calibration.

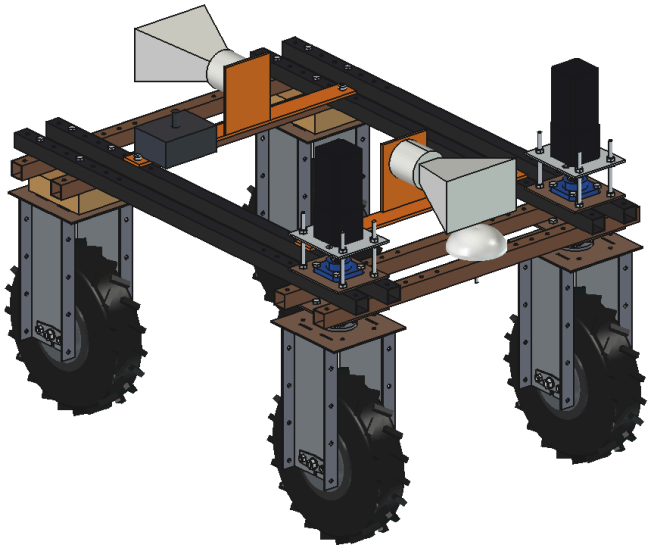
- 0.65 ha/h, ~29 000 plants/h
- ~4.9 ha per charge
- 10 kg

Folder: `yield estimator/simple/`

IMPLEMENT · TRANSPORT & GENERAL

Speech speaker

Two 30 W horn speakers (front and rear), an amplifier and a microphone: the robot warns people, reports status and lets an operator talk through it; the volume follows the distance to the nearest person.

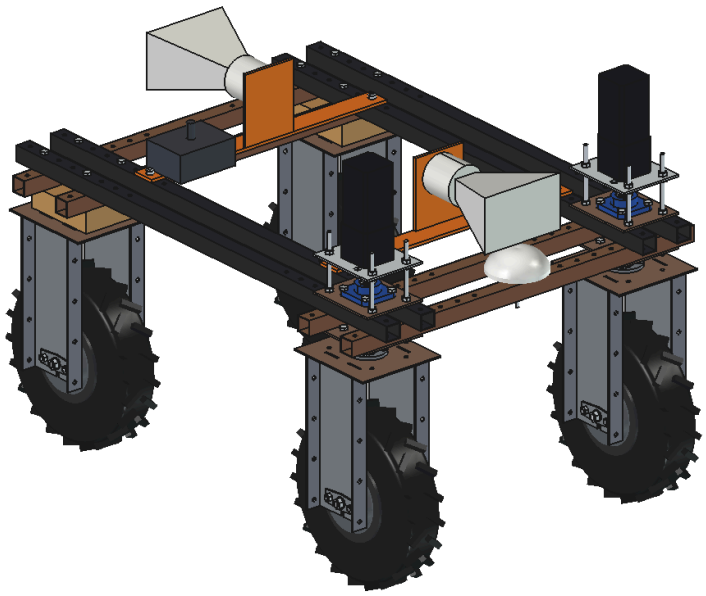


Simple

1 variant: Simple
Folder: `speaker/`

SPEECH SPEAKER · VARIANT 1 OF 1

Simple



2 flat bars across the inner upper beams, each with a horn speaker on an upright plate (the front one over the GNSS dome), an IP65 amplifier box and a microphone.

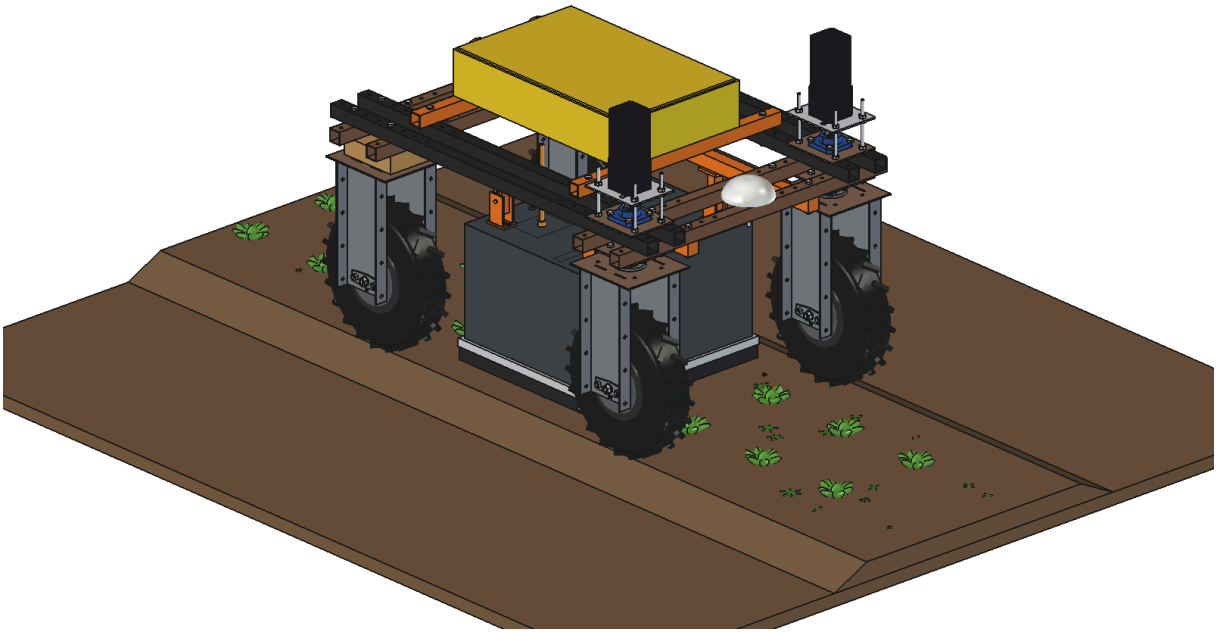
- 120 dB at 1 m
- ~300 m reach on a farmyard
- Max 0.25 W near animals

Folder: `speaker/simple/`

IMPLEMENT · WEED & CROP PROTECTION

Laser weeder

Kills weed seedlings with a laser under a light-tight hood: a camera finds each weed and a galvo scan head aims the beam of a fibre-coupled 60 W diode laser at its growing point, without chemicals or soil movement.

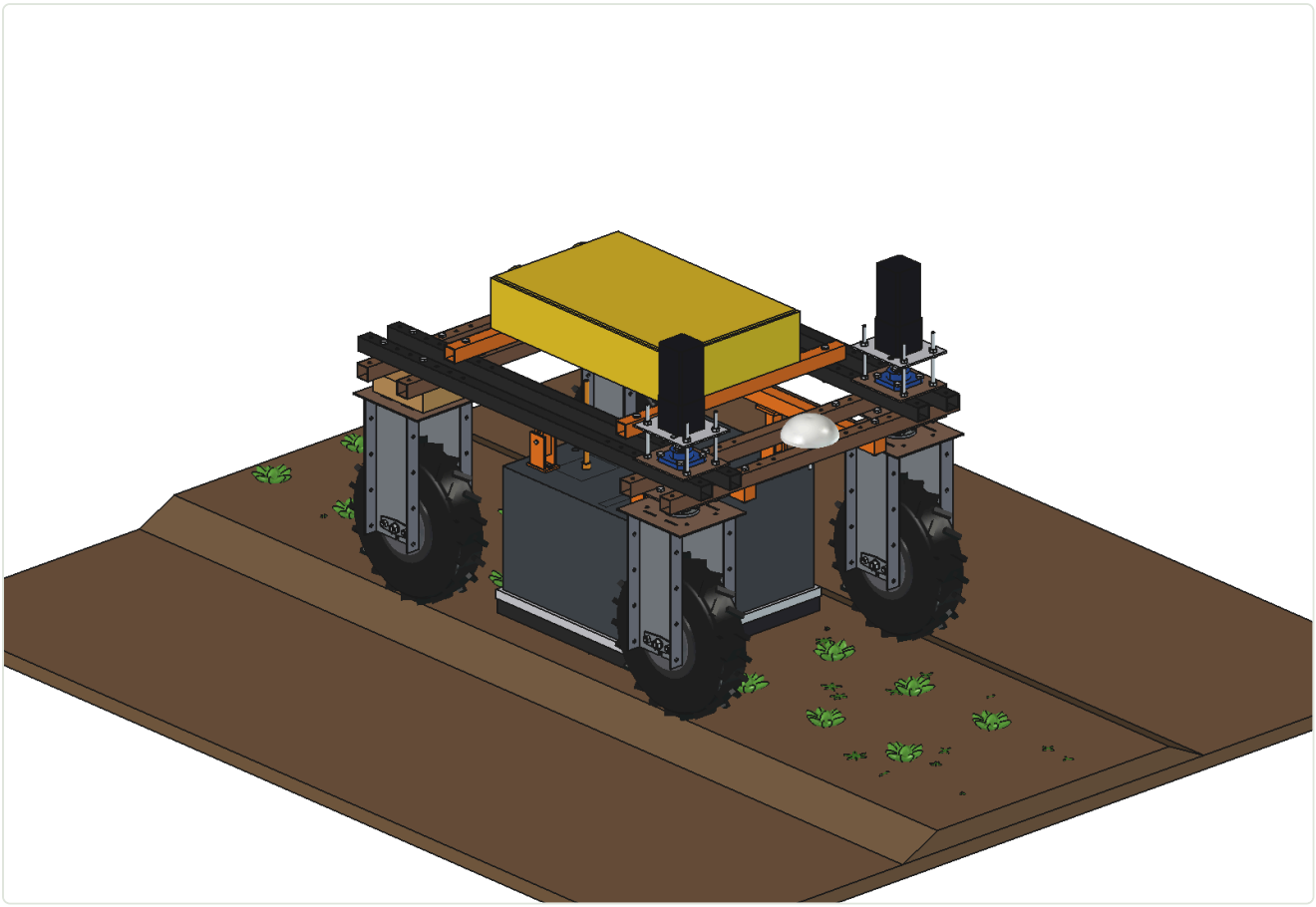


Simple

1 variant: Simple
Folder: `laser weeder/`

LASER WEEDER · VARIANT 1 OF 1

Simple



Hood, belly rails and camera of the drop-on-demand boom (imported); 2 galvo scan heads (one per 300 mm of the 0.6 m bed) and 2 x 60 W diode lasers in an air-cooled box on top, linked by armoured fibres. Class 4: fires only with the hood down and the robot level.

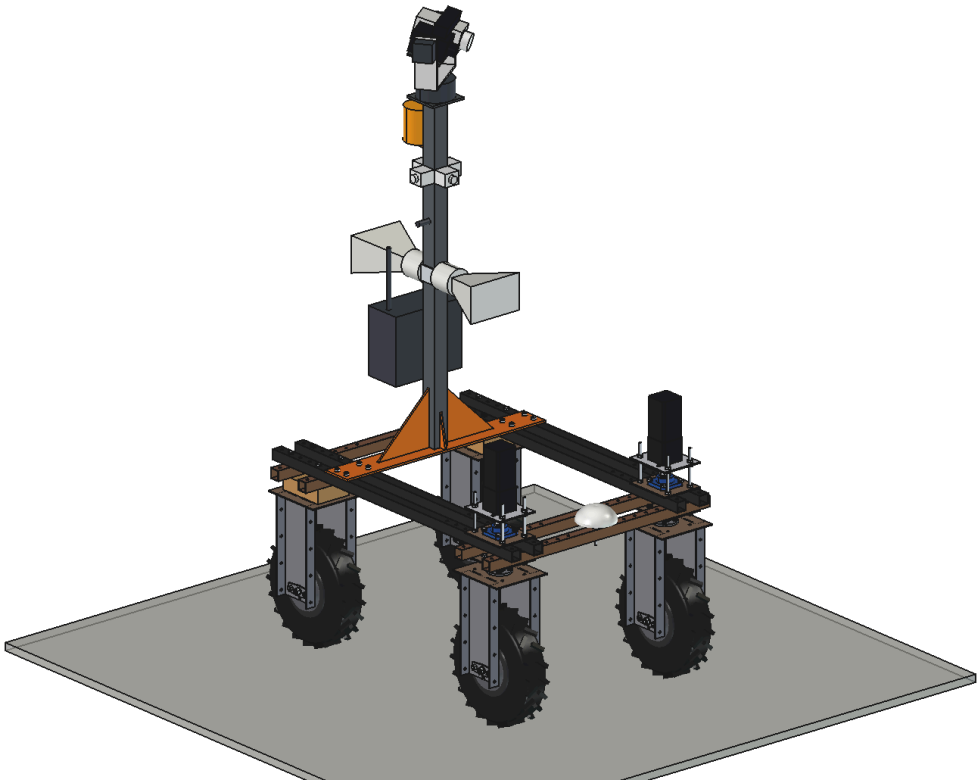
- ~19 J, 0.33 s per weed
- 0.2 m/s, ~990 m2 per charge
- ~48 kg

Folder: `laser weeder/simple/`

IMPLEMENT · MEASURING & SENSORS

Security patrol

A 2 m camera mast with a pan/tilt head (30x zoom day/night camera, thermal camera, IR light), 4 fixed all-round cameras, horns and a beacon: the robot patrols the farm at night, sweeps every viewpoint with the thermal camera and alerts the farmer.

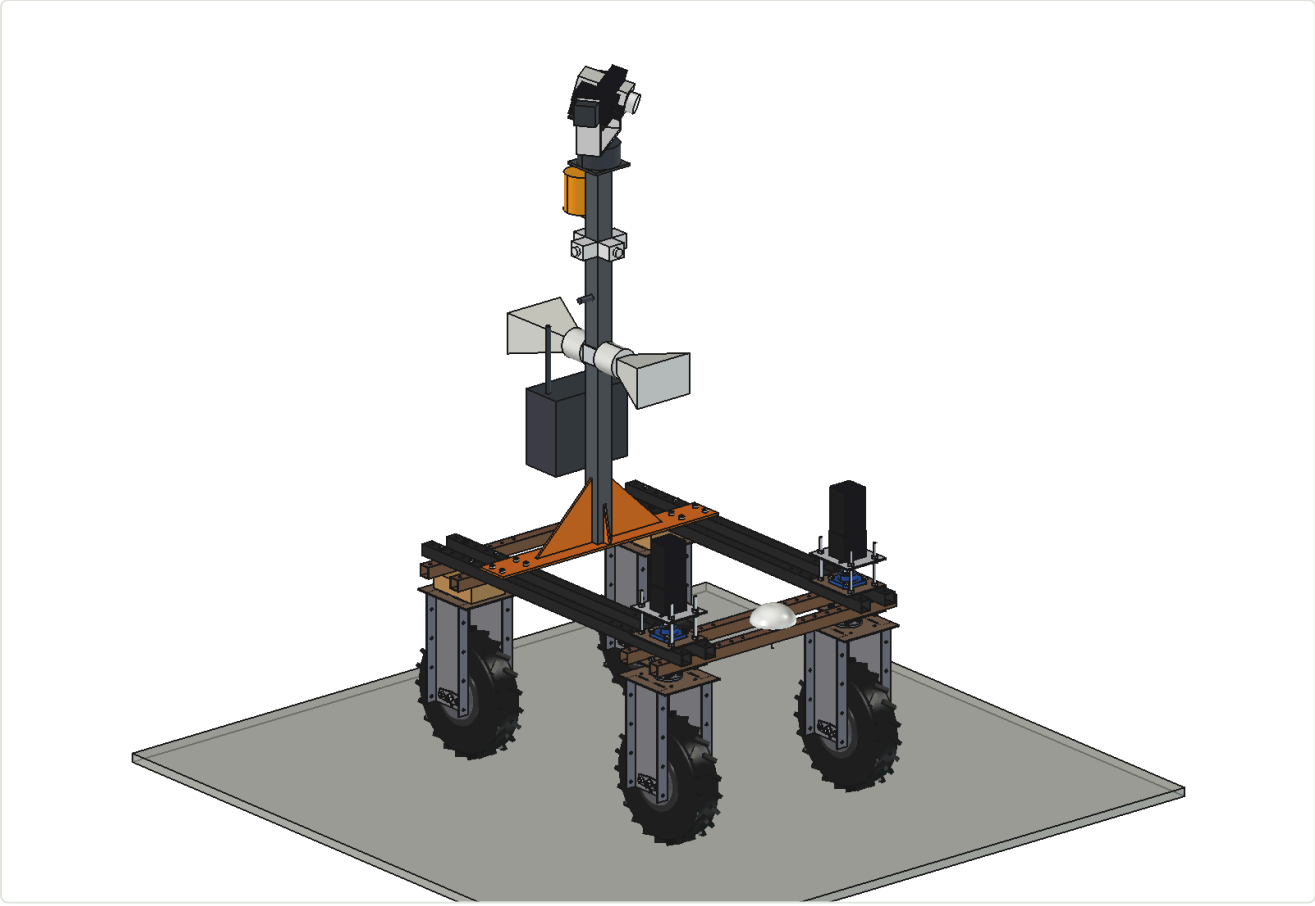


Simple

1 variant: Simple
Folder: `security patrol/`

SECURITY PATROL · VARIANT 1 OF 1

Simple



Mast 60 x 60 x 3 on an 8 mm plate across the 4 upper beams behind the centre (8 x M10); pan/tilt head at 2.23 m with a 4 MP 30x zoom camera, a 640 x 512 thermal camera and an 850 nm illuminator; 4 fixed 2 MP cameras, 2 horn speakers, beacon, microphone and an IP65 box with recorder and LTE router.

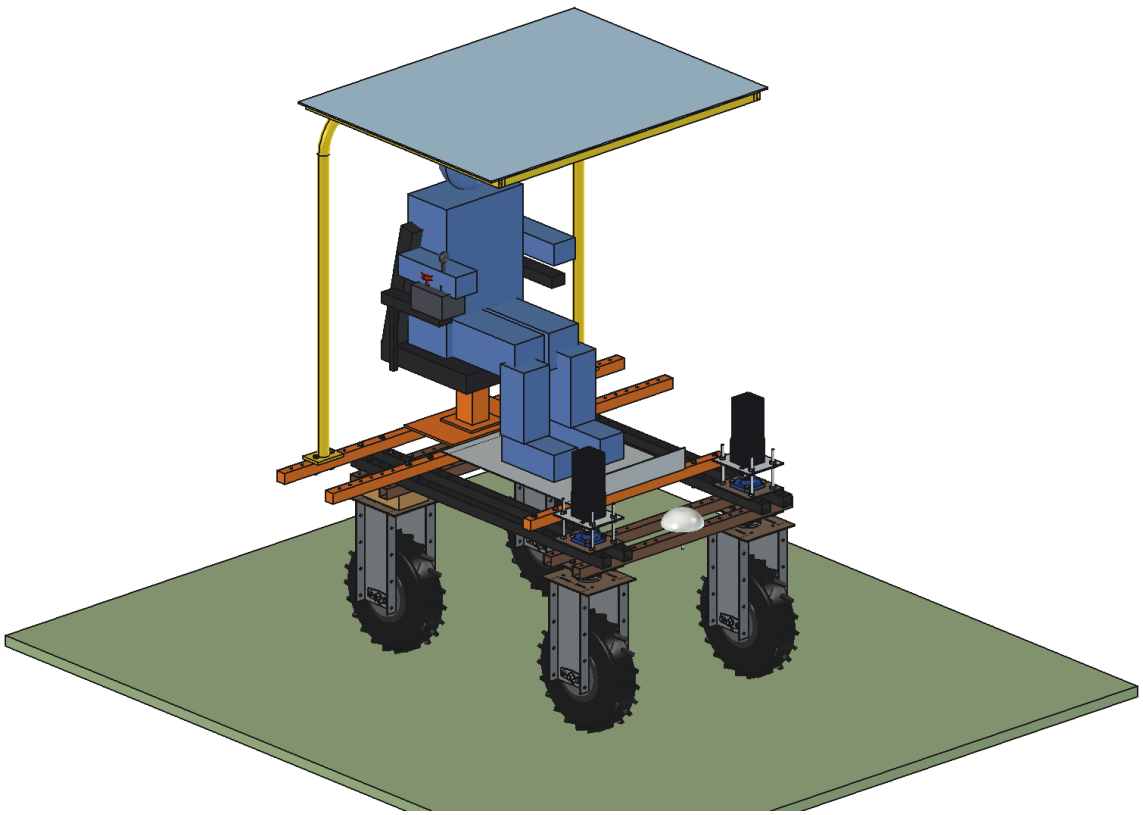
- Thermal: person detected at ~780 m
- ~12 rounds of 2 km per charge
- ~38 kg

Folder: `security patrol/simple/`

IMPLEMENT · TRANSPORT & GENERAL

Seat, canopy and frame extension

A seat so a person can ride on the robot and drive it by hand, a roll bar with a light canopy, and a pair of side extensions of the frame to x +-800 for wider implements.

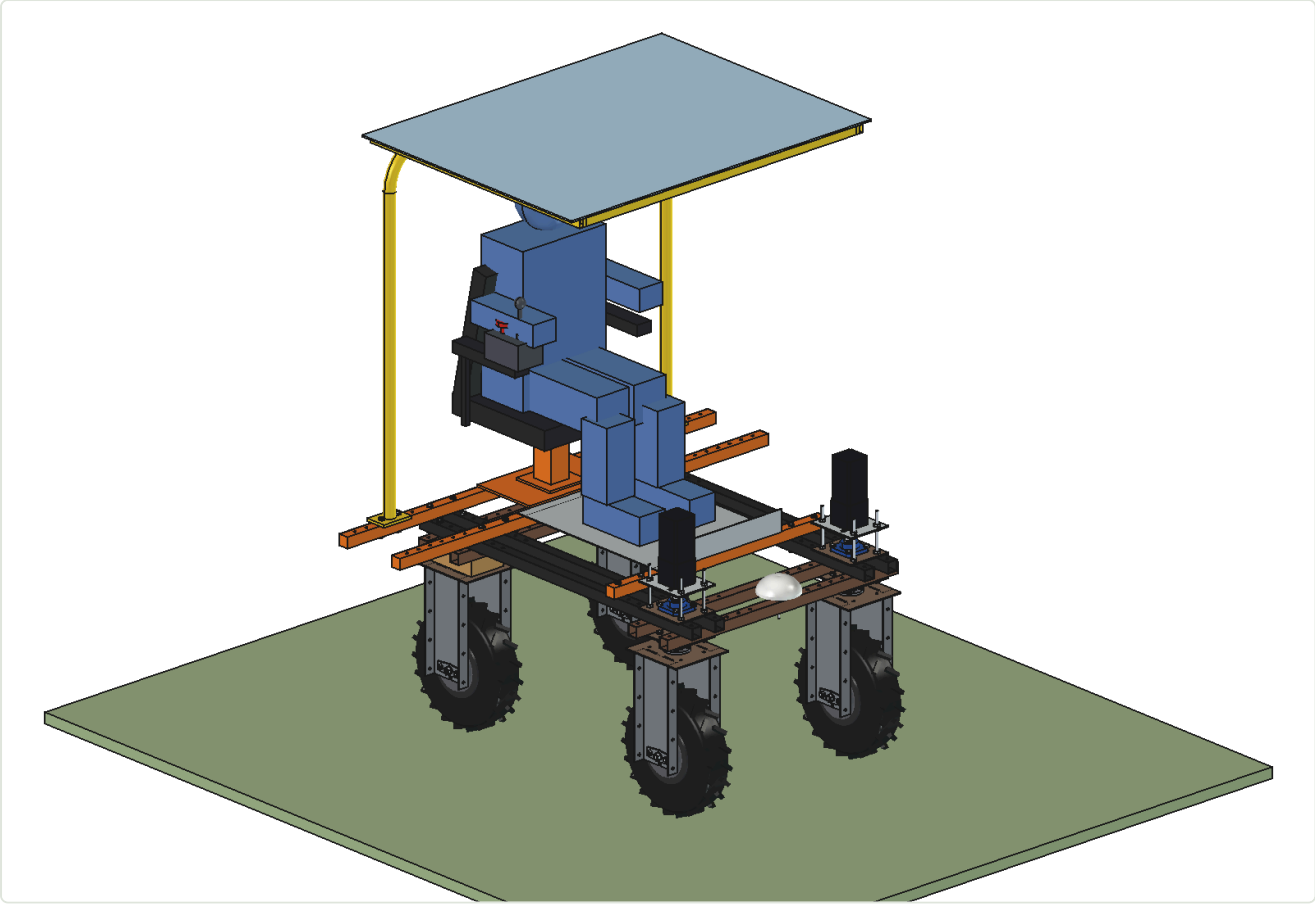


Simple

1 variant: Simple
Folder: `seat frame extension/`

SEAT, CANOPY AND FRAME EXTENSION · VARIANT 1 OF 1

Simple



Cross tubes on the upper beams; the 2 rear ones are 1600 mm long side extensions with a 50 mm hole grid. Tractor seat with a seat switch over the rear axle, a console with joystick and emergency stop on the right armrest, a checker-plate footplate, a roll bar O42 at x +-600 with a polycarbonate canopy. Manual mode only, at most 1.5 m/s.

> 94 kg; 114 / 231 kg front / rear with a 90 kg operator

> Tip 23 degrees sideways, turn margin 5.5

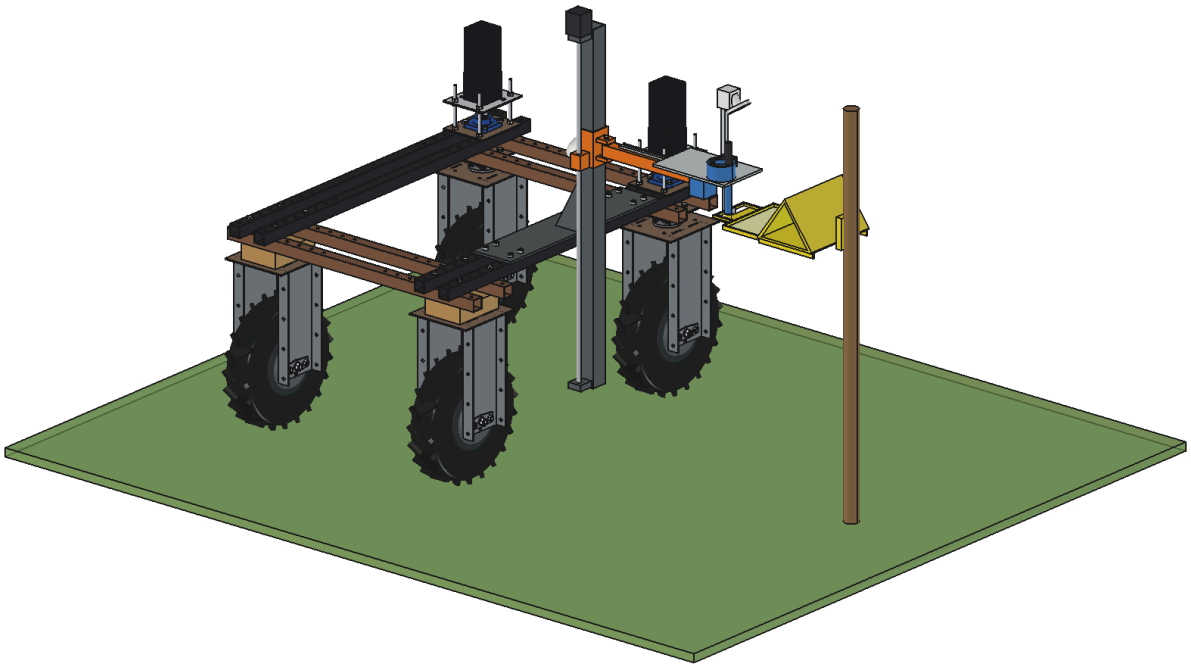
> Extensions to x +-800

Folder: `seat frame extension/simple/`

IMPLEMENT · WEED & CROP PROTECTION

Pheromone trap service

Weekly round of the pheromone and sticky traps on the field edges: the head on the fence-inspector mast pulls out the drawer of each trap, photographs the sticky card to count the catch and pushes a fresh lure capsule into the clip when it is due.

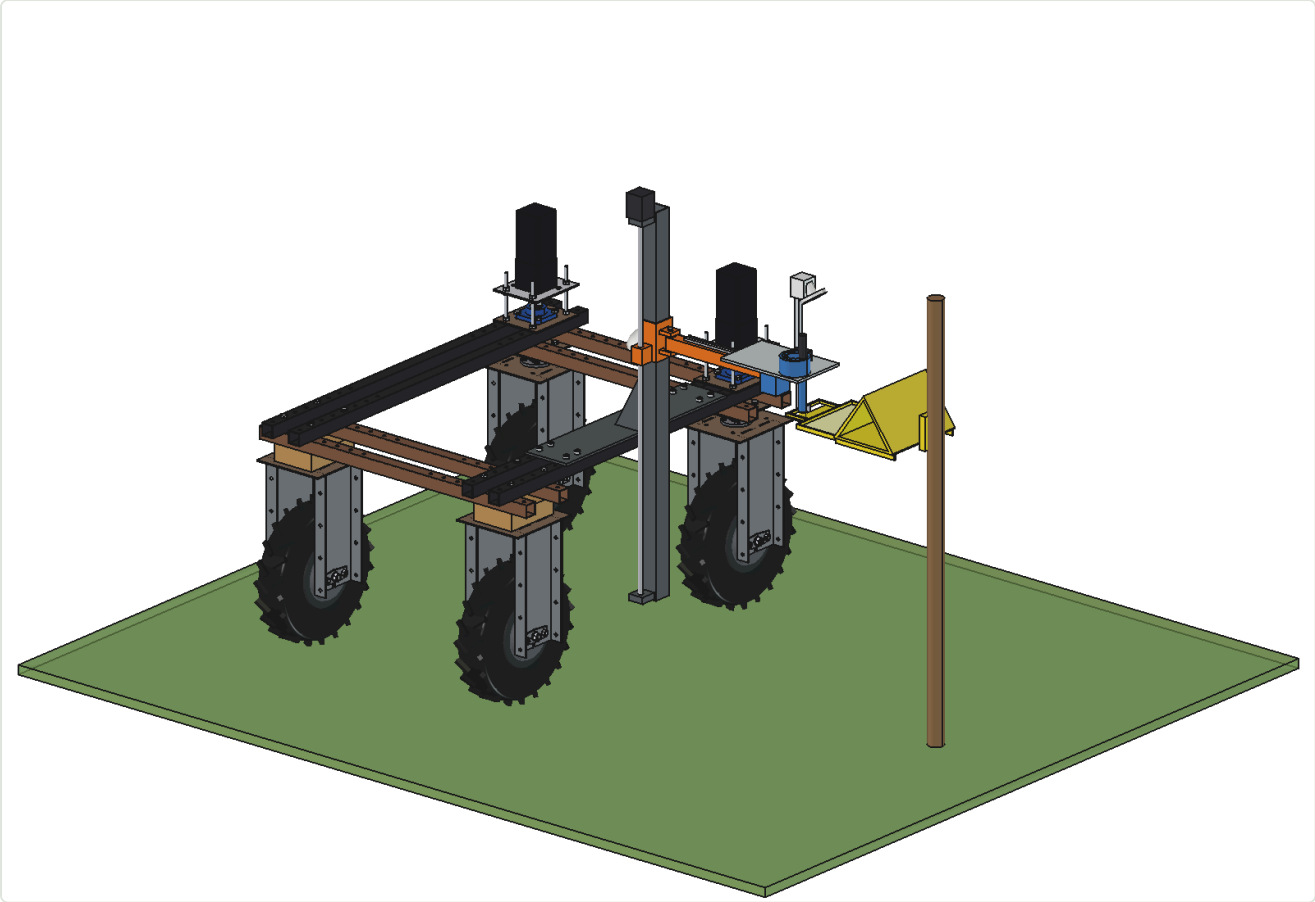


Simple

1 variant: Simple
Folder: `pheromone trap service/`

PHEROMONE TRAP SERVICE · VARIANT 1 OF 1

Simple



Fence-inspector mast and carriage (imported) with a fixed arm, MGN12 guide and 24 V actuator; the head has a hook, a 12 MP camera with LED ring and a revolver of 12 lure capsules with a plunger. Robot-friendly delta trap with a pull-out drawer.

> ~50 s per trap

> 40 traps: 2.4 h, 22 % of a charge

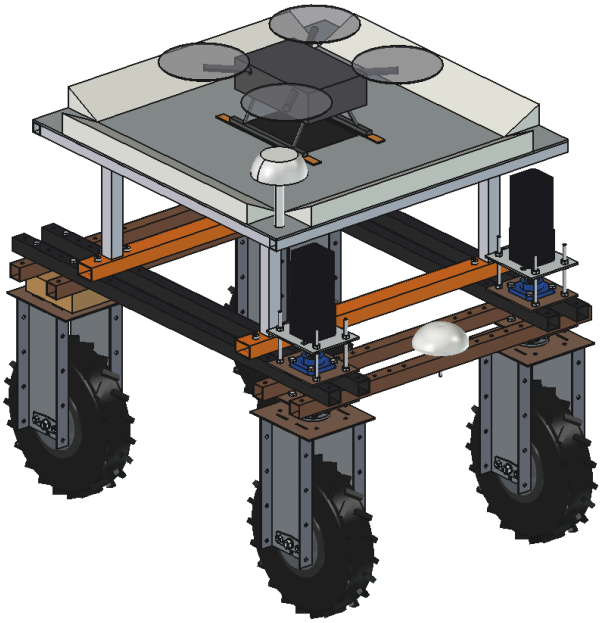
> ~27 kg

Folder: `pheromone trap service/simple/`

IMPLEMENT · COOPERATION & SWARM OPERATION

Drone combination

A 1 m landing platform above the robot so a mapping drone uses it as a mobile base station: it takes off, maps ~97 ha per flight, lands back, sends its images and charges from the robot battery.

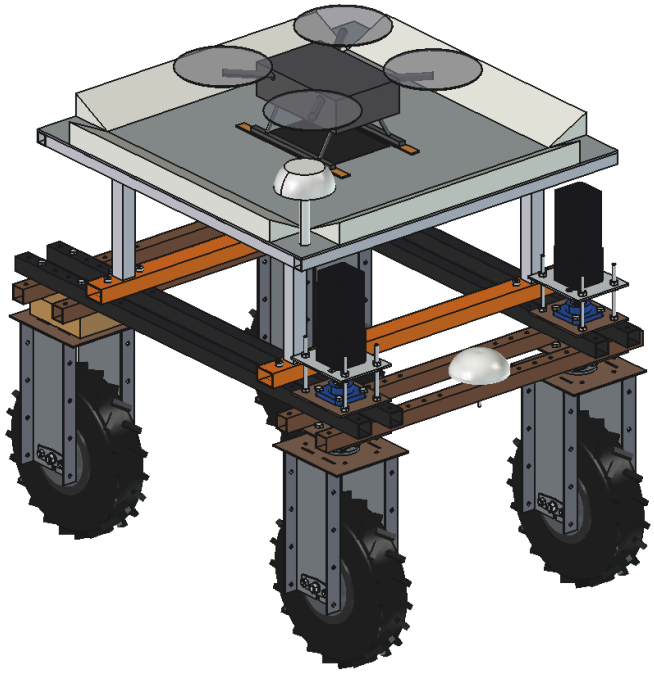


Simple

1 variant: Simple
Folder: `drone combination/`

DRONE COMBINATION · VARIANT 1 OF 1

Simple



Aluminium plate 1000 x 1000 at z 1100 on 4 posts and 2 cross bars over the upper beams, PE guide wedges for passive centring, copper charging strips with holding magnets, an AprilTag for landing, and the GNSS antenna moved to a platform corner.

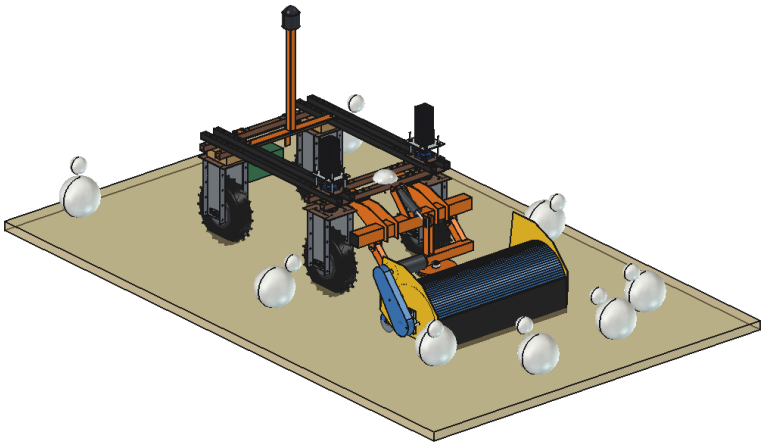
- Catches landings up to 0.25 m off
- ~60 min charge = 4.6 % of the robot battery
- 42 kg

Folder: `drone combination/simple/`

IMPLEMENT · LIVESTOCK & FARM

Chicken house robot

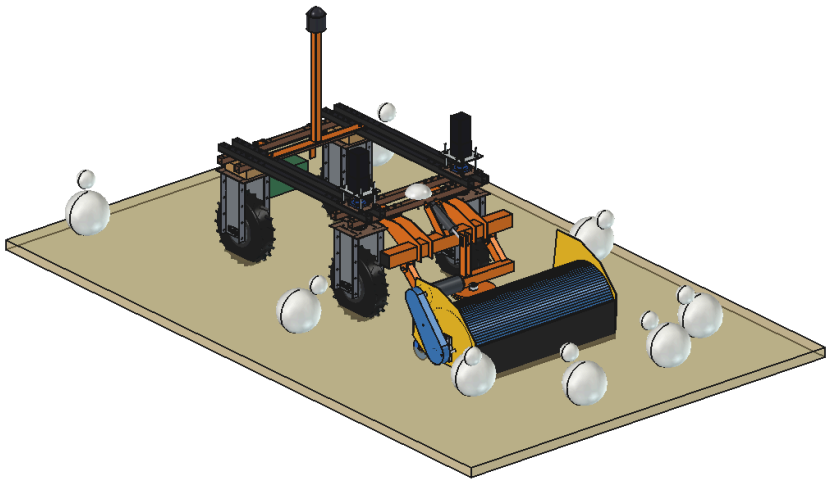
Turns the litter in a poultry house with a tine rotor in the broom sweeper's head, measures NH3, CO2, temperature and humidity at bird height and watches the flock with a 360 degree camera.



Simple

1 variant: Simple
Folder: `chicken house robot/`

Simple



Broom sweeper (imported) with 60 spring tines O6 instead of bristles (tip circle O360, 150 rpm, 10 mm above the concrete), a climate sensor box under the rear beam with an inlet at 300 mm, and a 360 degree camera on a 1.4 m post.

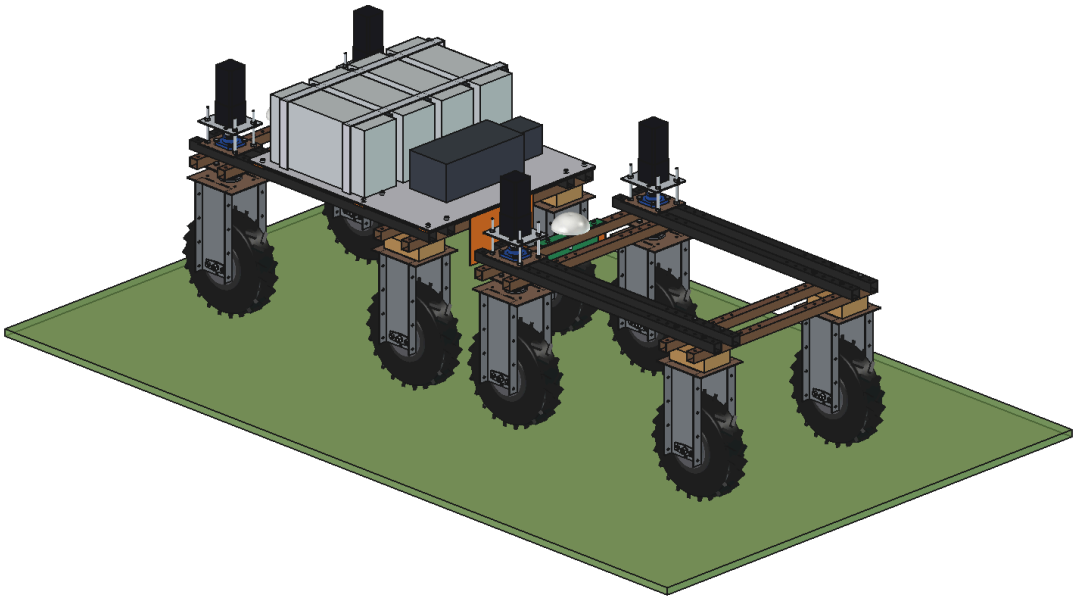
- 1500 m2 house in ~1.5 h
- ~810 W rotor, 1.4 houses per charge
- 122 kg

Folder: `chicken house robot/simple/`

IMPLEMENT · POWER & DRIVE

Charging robot

A second Bruut as a mobile charger: 4 battery-pack pods (10.2 kWh) on a deck, a 1.5 kW DC/DC charger and the charging dock's contact head on an arm at the rear, so a working robot docks and charges in the field instead of driving to the shed.

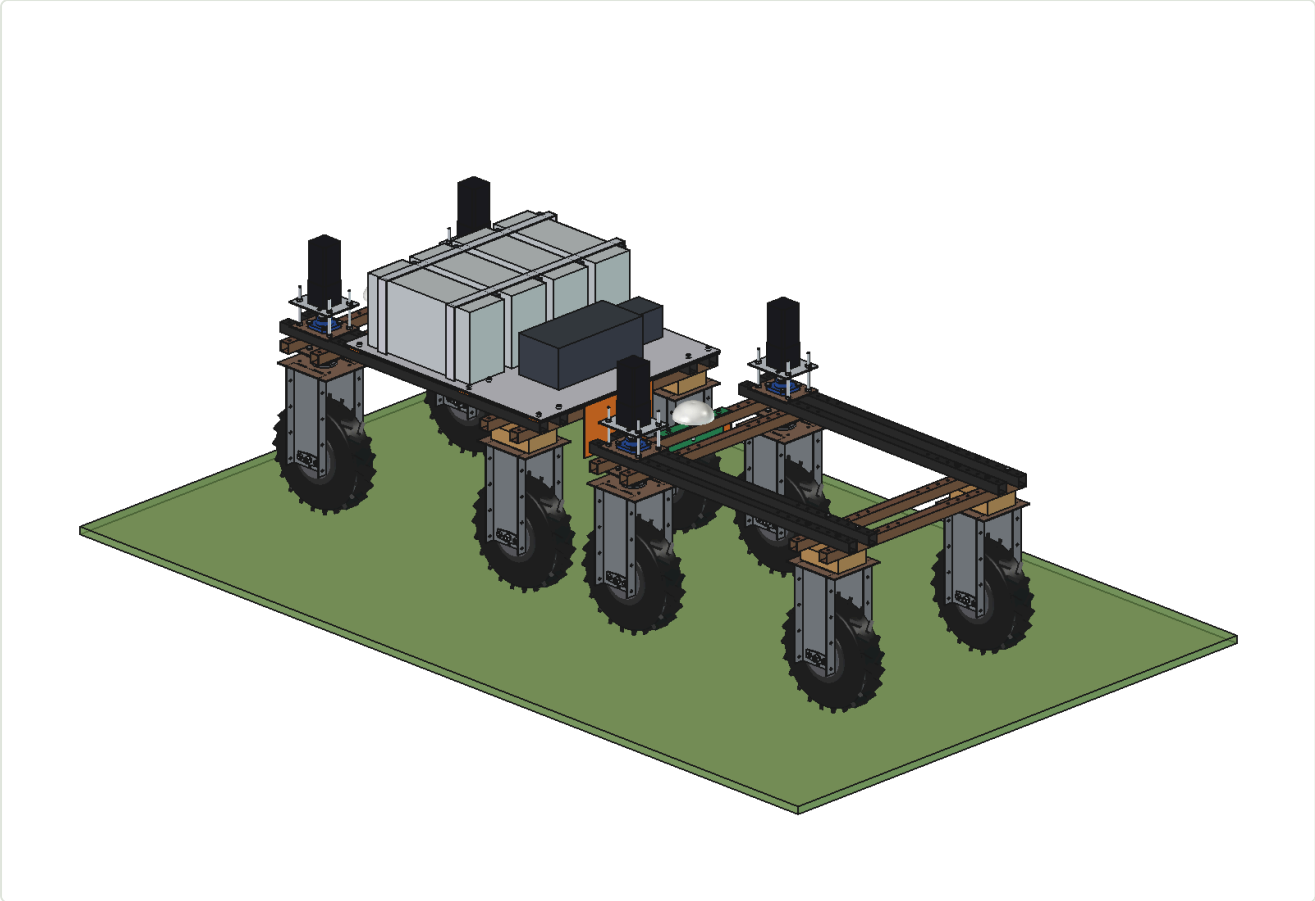


Simple

1 variant: Simple
Folder: `charging robot/`

CHARGING ROBOT · VARIANT 1 OF 1

Simple



Deck on 3 flat bars across the upper beams with 4 LiFePO4 pods (battery pack, imported), DC/DC 54 V CC/CV 1.5 kW, fuse box; rear bracket and arm with the dock's PE head, copper bars and funnel (charging dock, imported).

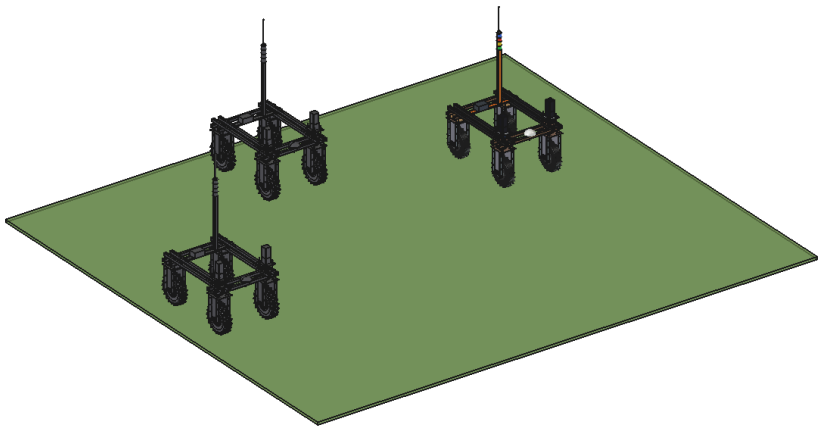
- 9.2 kWh usable, ~5.6 charges per trip
- ~73 min per charge (20 -> 100 %)
- ~135 kg on the robot

Folder: `charging robot/simple/`

IMPLEMENT · COOPERATION & SWARM OPERATION

Swarm operation

Several units work one field as a swarm: passes shared out, position and intent over a mesh radio, headland corridors reserved, a pass handed over when a unit stops. Per robot a swarm kit with radio, status light tower and computer.

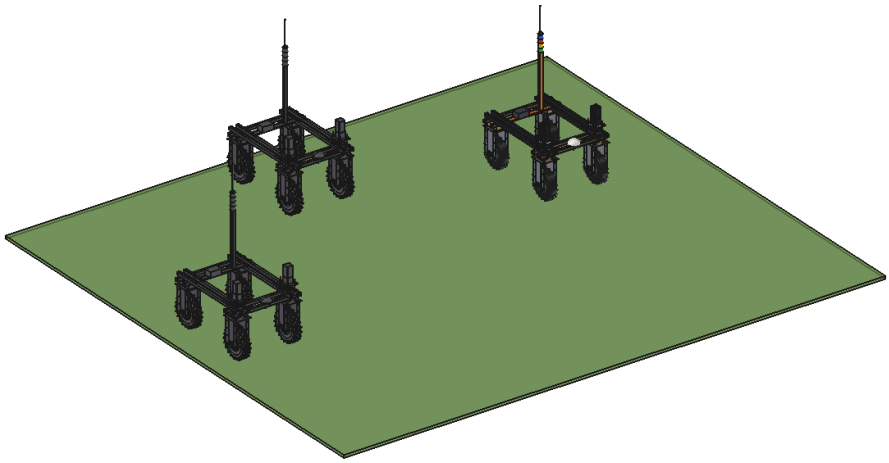


Simple

1 variant: Simple
Folder: `swarm operation/`

SWARM OPERATION · VARIANT 1 OF 1

Simple



Swarm kit: a 1.5 m post on a bar across the inner upper beams with a 868 MHz mesh radio, a 4-colour status light tower and a swarm computer box. The calculation gives the speed-up per number of units, charger sharing, radio load and soil pressure.

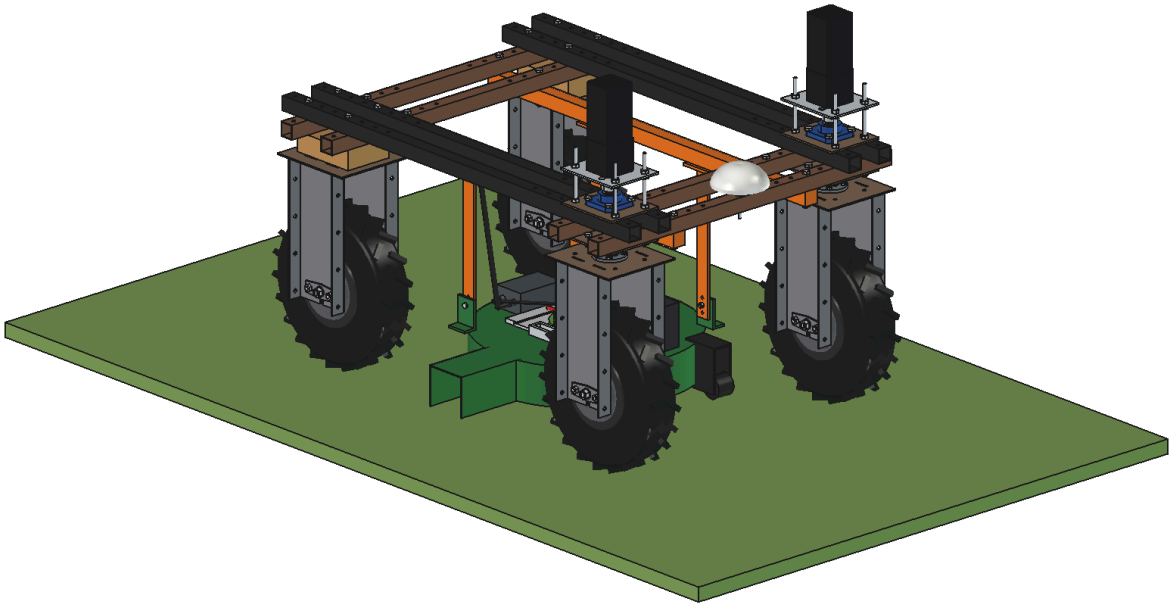
- 4 units: 3.8x faster, 95 % efficient
- One charger per ~3 units
- 16 kPa per unit vs 42 kPa tractor

Folder: `swarm operation/simple/`

IMPLEMENT · POWER & DRIVE

Heavy-duty electric motors

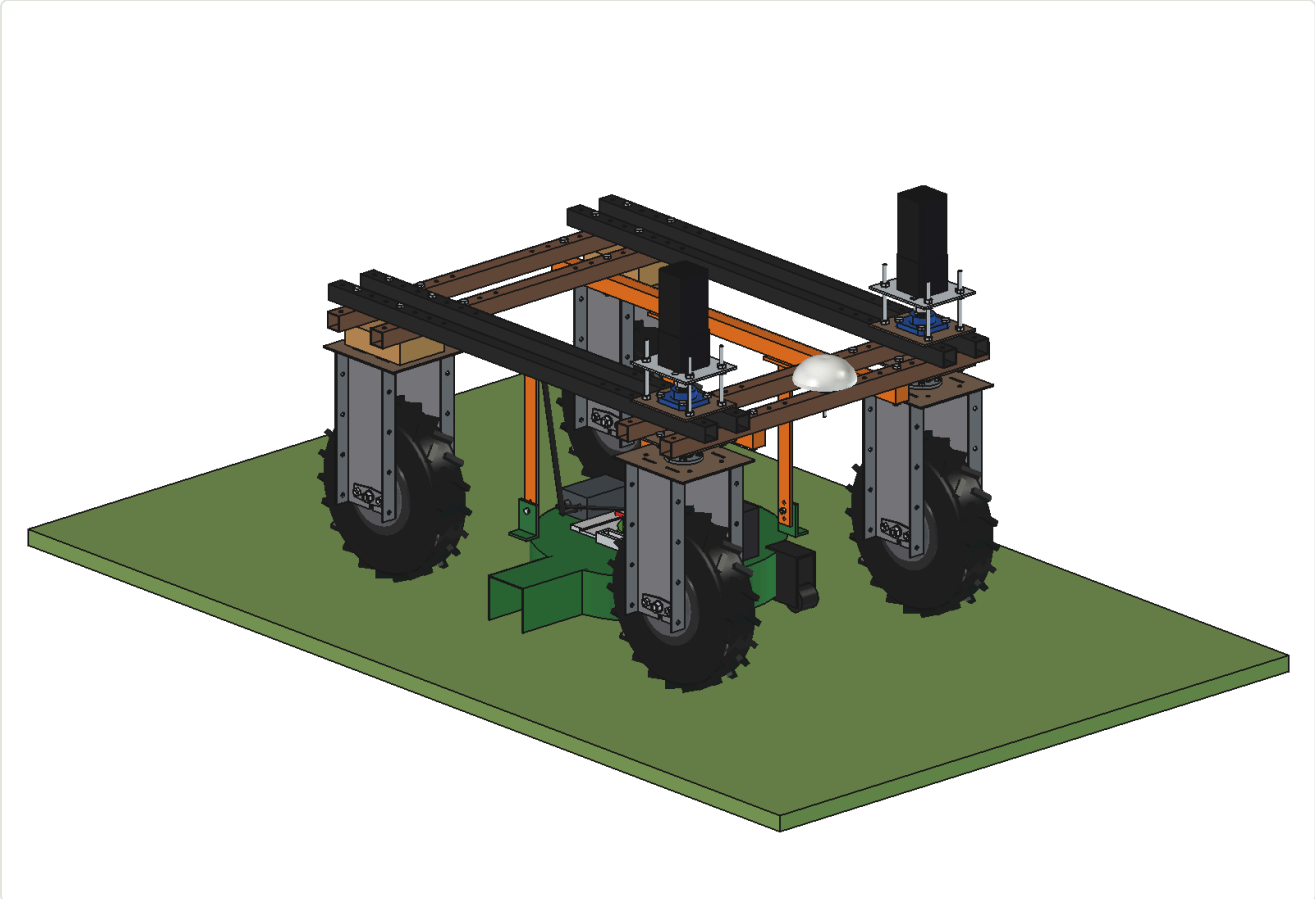
A universal quick-change module for bought battery garden-tool motors (56-60 V brushless lawnmower motors, ~2 kW) with an own BLDC controller on the robot's 48 V bus, shown on a 530 mm mid-mower deck under the belly.



Simple

1 variant: Simple
Folder: `heavy duty motors/`

Simple



Aluminium interface plate with a brand adapter ring slides into a dovetail receiver on the implement and is held by one lock pin. A 75 V-class BLDC controller in an IP67 box runs the motor from the robot's 48 V bus (the tool battery in a clip is an option). Demonstrator: mower deck O600 with side discharge, hung from the belly rails with 3 cutting heights.

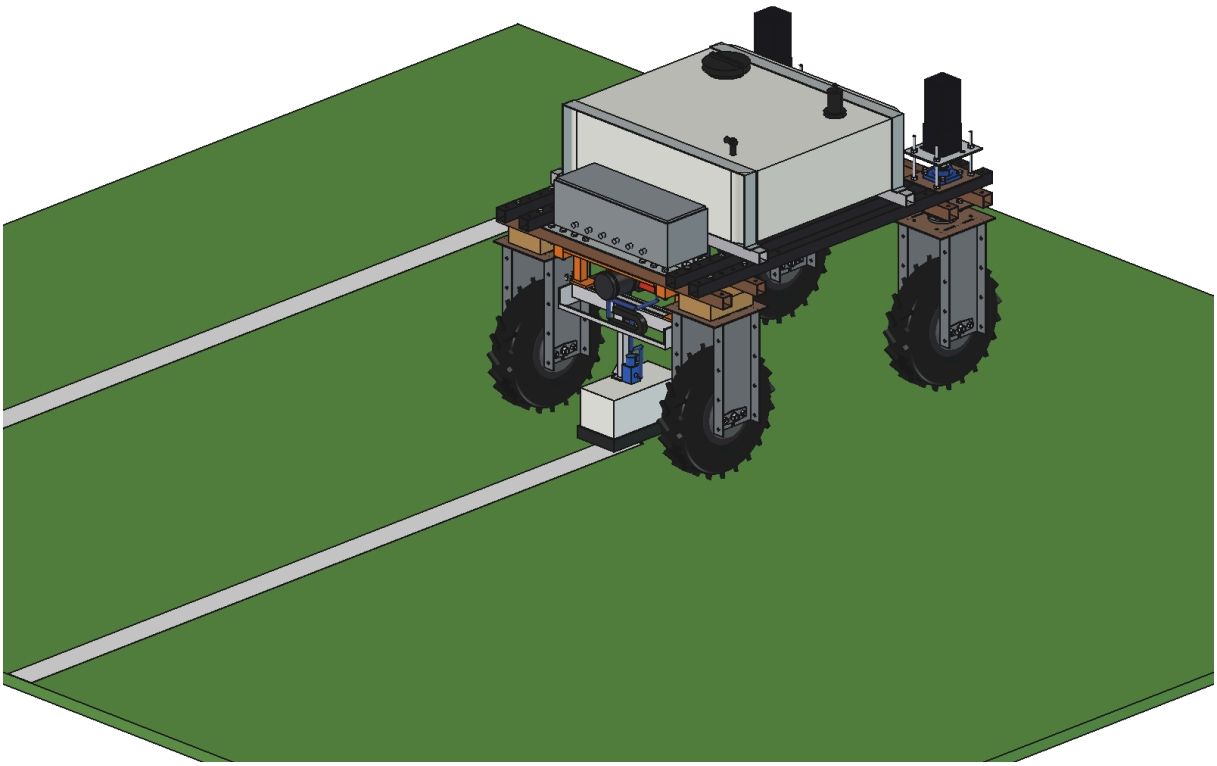
- ~1.55 kW electric mowing, 32 A at 48 V
- ~1350 m2/h, ~1750 m2 per robot charge
- Module 8 kg, deck 39 kg

Folder: `heavy duty motors/simple/`

IMPLEMENT · TRANSPORT & GENERAL

Sports field line marker

Paints the lines of a football or hockey field from a stored layout: one paint nozzle on the rear axle line follows the steered path exactly, a short slide across the robot corrects the line and draws the tight corner arcs.

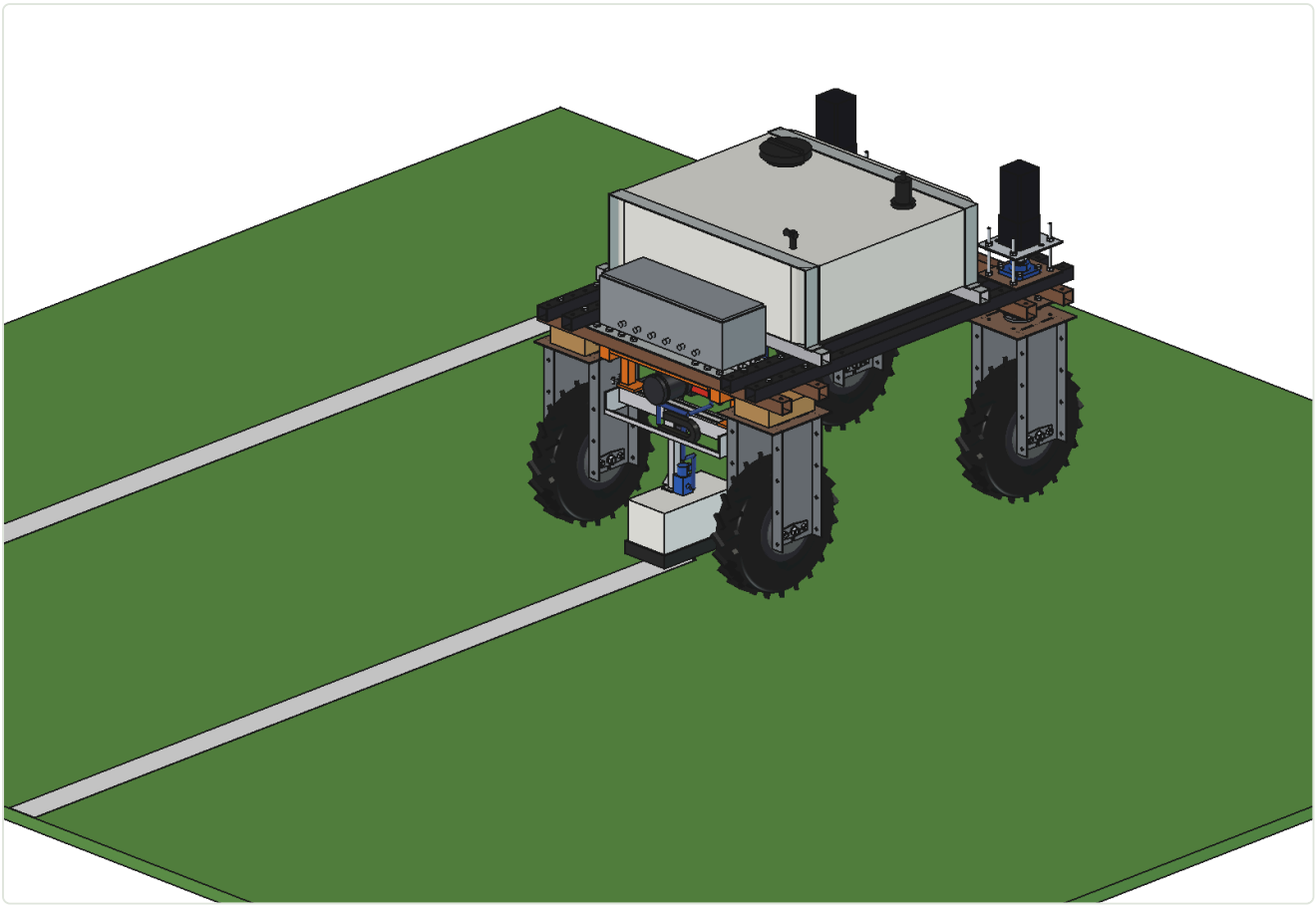


Simple

1 variant: Simple
Folder: `line marker/`

SPORTS FIELD LINE MARKER · VARIANT 1 OF 1

Simple



Belt module (stroke 340) hung from the belly rails at the rear axle line, a paint head with a 12 V valve, an even flat-fan tip and a brush-skirted shroud, a drag chain for the hose; tank, pump, regulator (agitation) and control box of the spot sprayer.

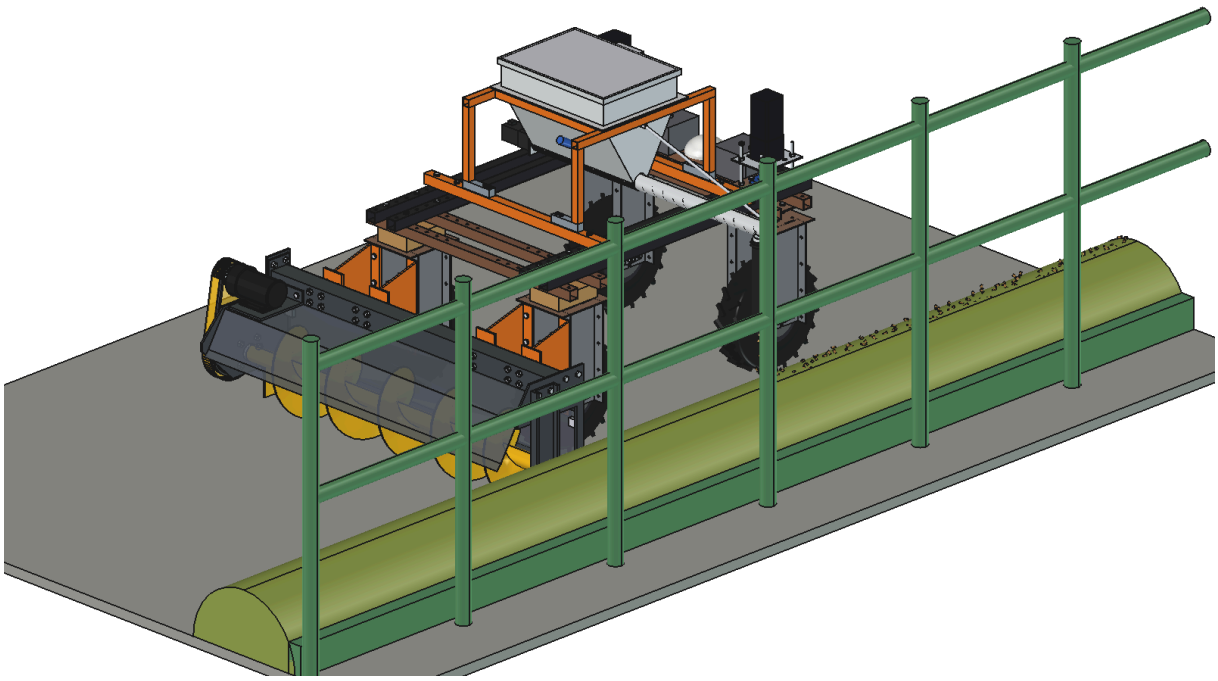
- Football field (717 m) in ~19 min, ~10 l paint
- ~18 mm (95 %) at the nozzle, estimated
- 62 kg empty

Folder: `line marker/simple/`

IMPLEMENT · LIVESTOCK & FARM

Concentrate dispenser

Doses concentrate pellets or minerals on the freshly pushed windrow while the robot pushes feed with the feed pusher auger, to draw the cows to the fence.

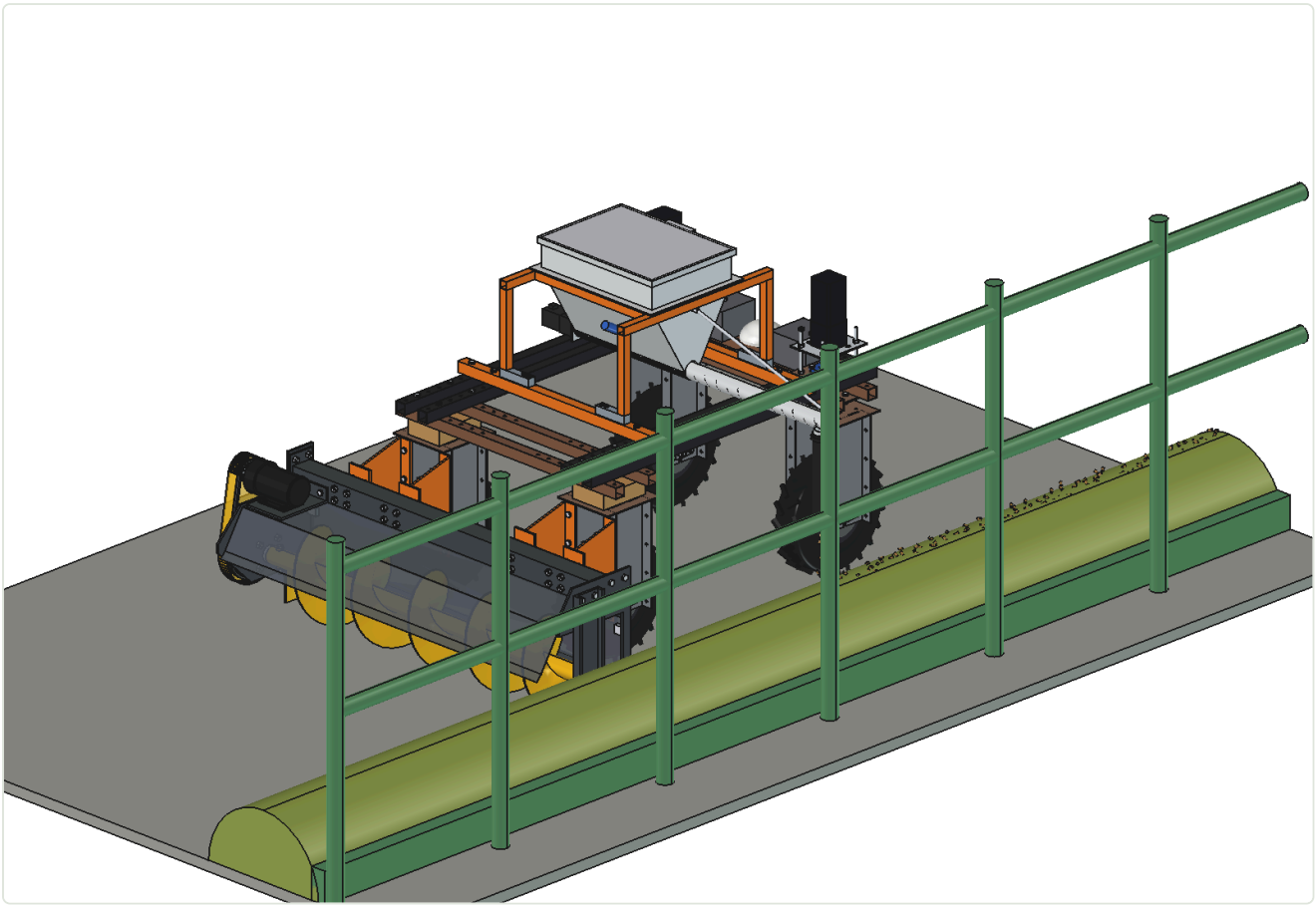


Simple

1 variant: Simple
Folder: `concentrate dispenser/`

CONCENTRATE DISPENSER · VARIANT 1 OF 1

Simple



47 l stainless hopper in the middle of the robot on 4 load cells (loss-in-weight dosing), a metering auger O54 in a tube out past the feed pusher's discharge end and a rubber spout over the windrow; NEMA 23 with a 10:1 planetary.

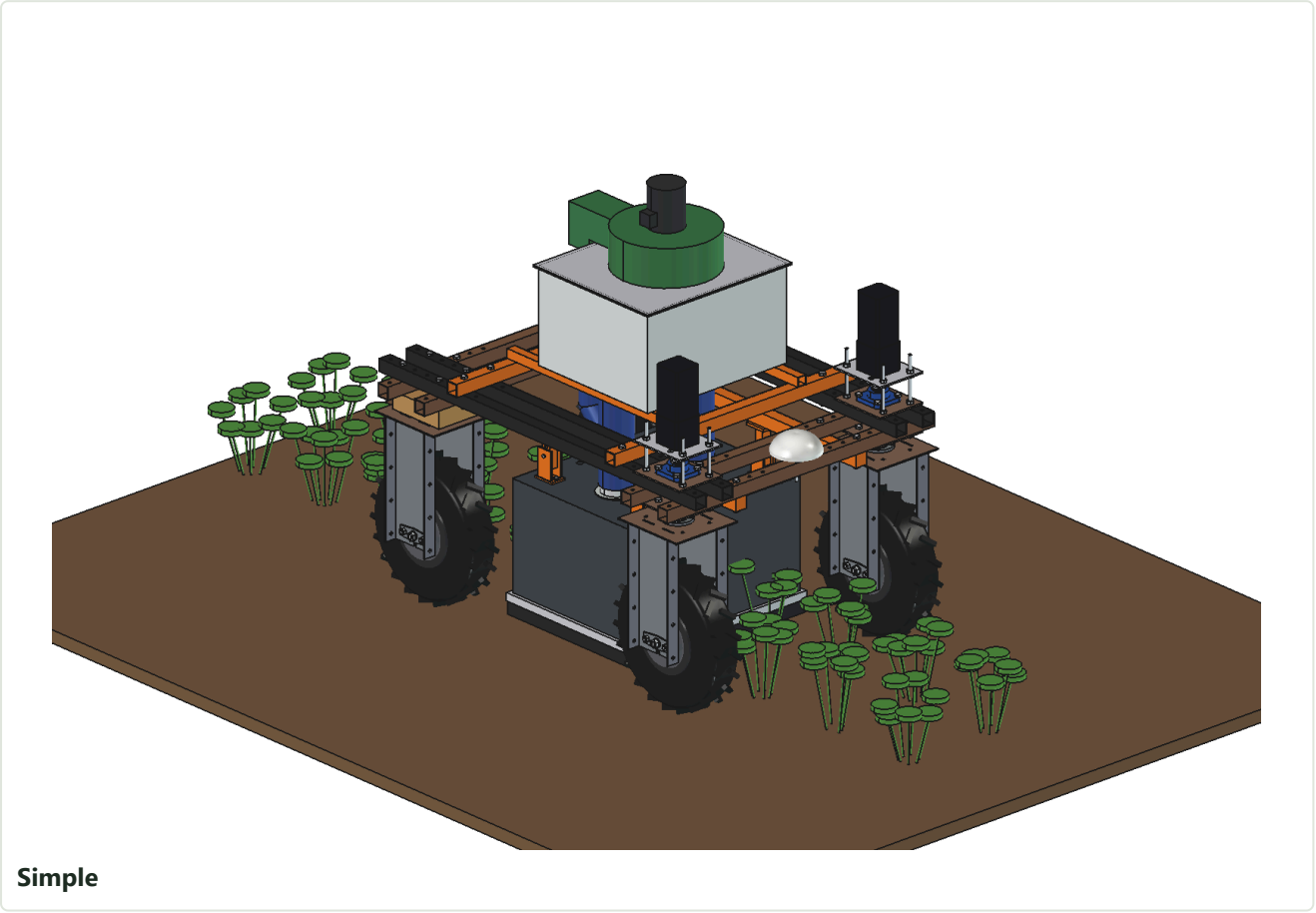
- 100 g per feeding place at 0.35 m/s (5-150 g)
- +-2 % by loss-in-weight (estimated)
- 37 kg empty, 69 kg full

Folder: `concentrate dispenser/simple/`

IMPLEMENT · WEED & CROP PROTECTION

Insect vacuum

Sucks pest insects (lygus bugs in strawberries, Colorado beetles) off the crop instead of spraying: push-pull under a closed belly hood, a mesh basket keeps the catch for counting.

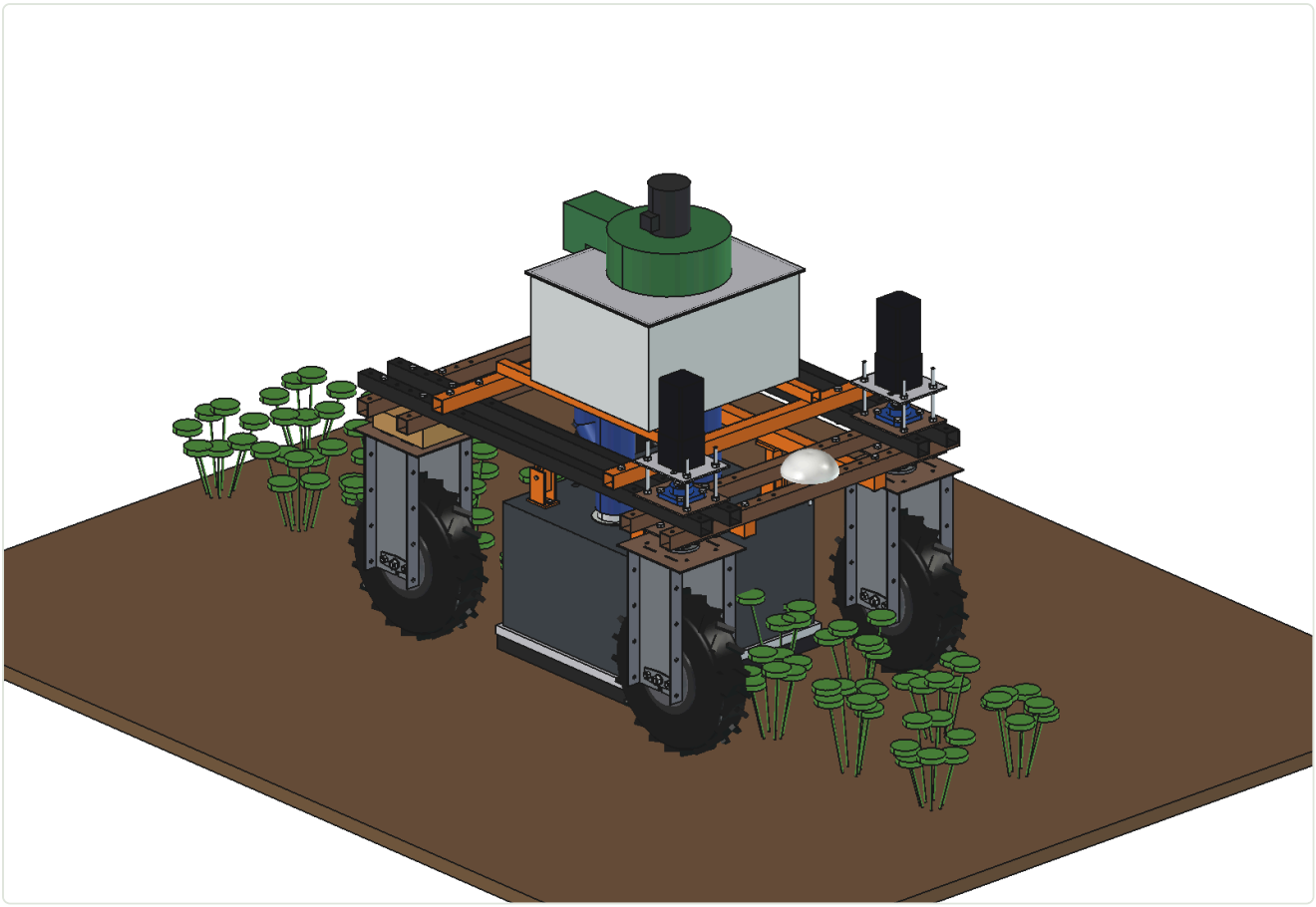


Simple

1 variant: Simple
Folder: `insect vacuum/`

INSECT VACUUM · VARIANT 1 OF 1

Simple



The drop-on-demand boom's hood with 2 air knives low along the bed (8 mm slots, blowing in and up), 2 slot nozzles over the rows and a strip curtain; hoses up into a catch box with a 0.8 mm mesh basket; the fan on the lid blows the clean air back to the knives (closed loop).

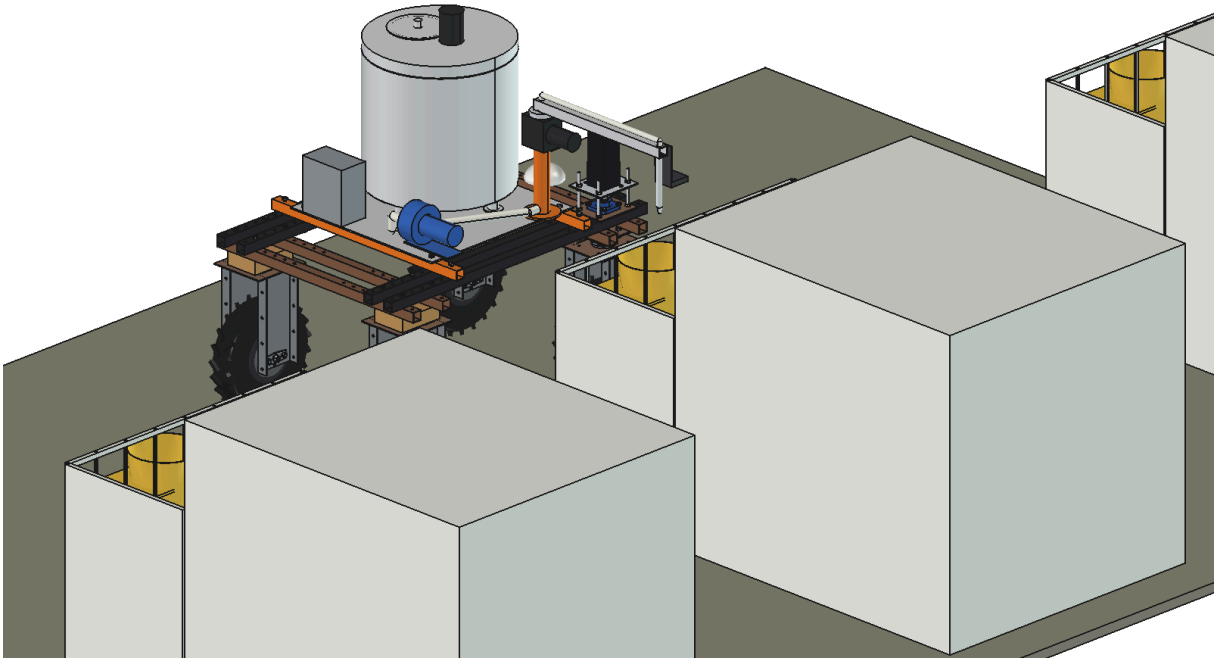
- Jet ~23 / 13 m/s at the near / far row
- ~670 W fan, ~0.43 ha per charge
- 68 kg

Folder: `insect vacuum/simple/`

IMPLEMENT · LIVESTOCK & FARM

Calf milk taxi

Brings warm milk from the calf milk kitchen to the hutches and doses each calf's portion into its teat bucket: insulated tank, peristaltic pump and a swing arm with an RFID reader.

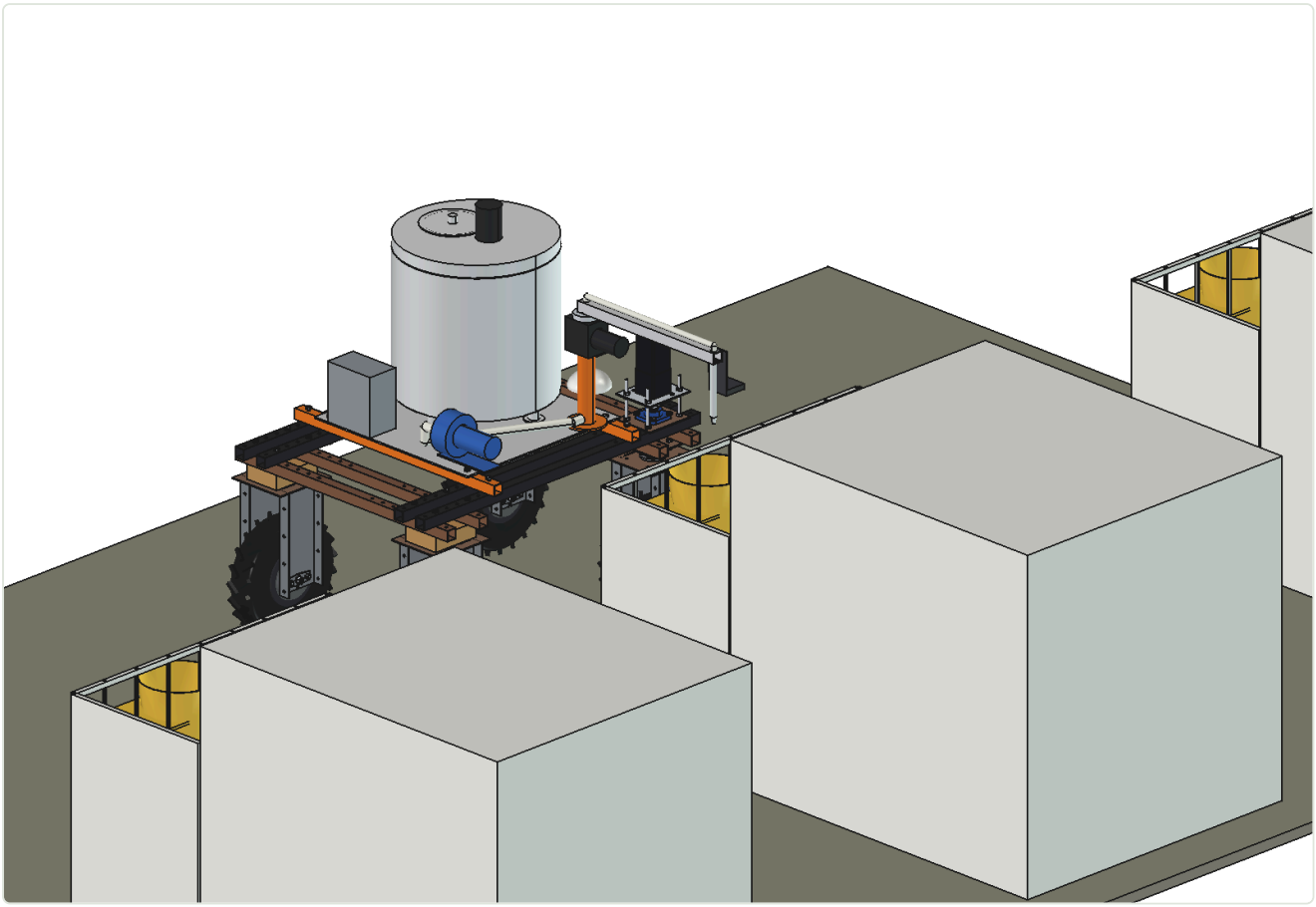


Simple

1 variant: Simple
Folder: `calf milk taxi/`

CALF MILK TAXI · VARIANT 1 OF 1

Simple



83 l insulated stainless tank with stirrer and 300 W heating mat on the upper beams, a 6 l/min peristaltic pump, and a swing arm on a hollow-shaft gearmotor that puts the nozzle over the bucket at x 920; RFID and camera at the tip.

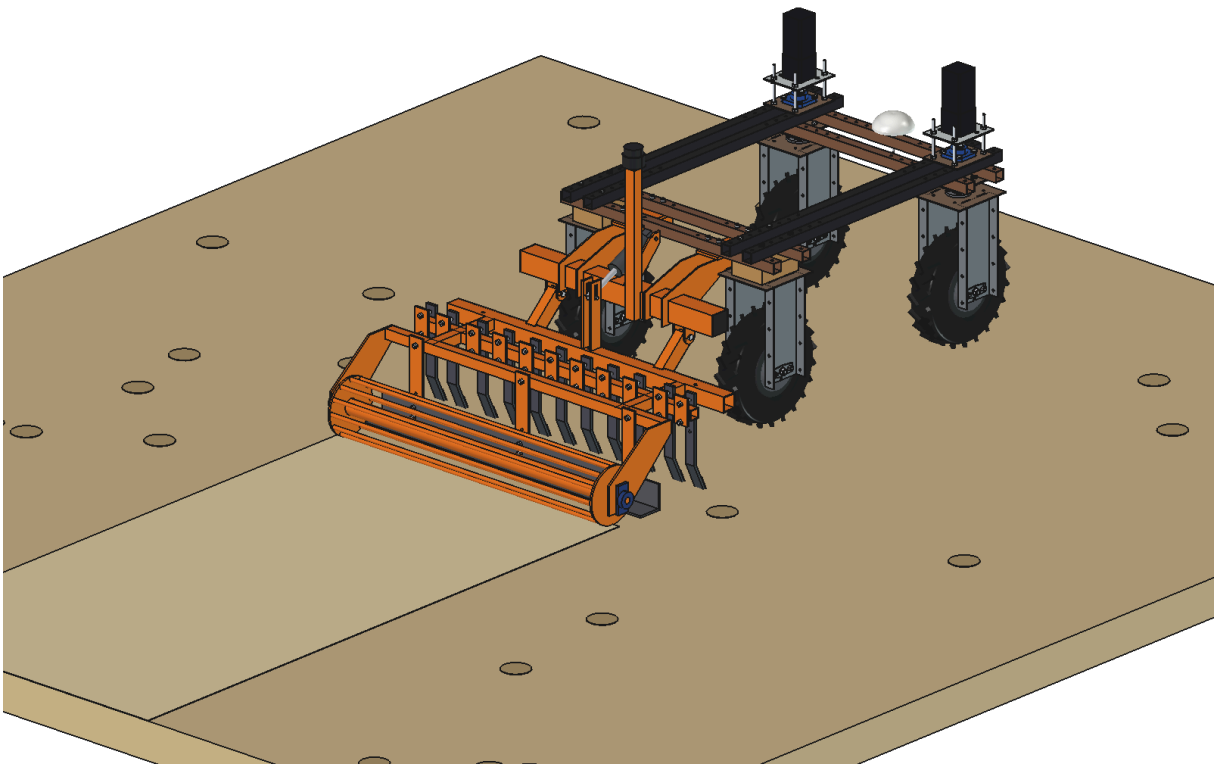
- 28 calves at 3 l per fill, ~18 min per round
- -0.4 K/h without heating
- 76 kg empty, +86 kg milk

Folder: `calf milk taxi/simple/`

IMPLEMENT · CROP & SOIL

Riding arena leveller

Loosens and levels the footing of a horse riding arena at night: rigid tines, a levelling board and a rod roller on the grassland harrow's frame, with a lidar for indoor arenas.

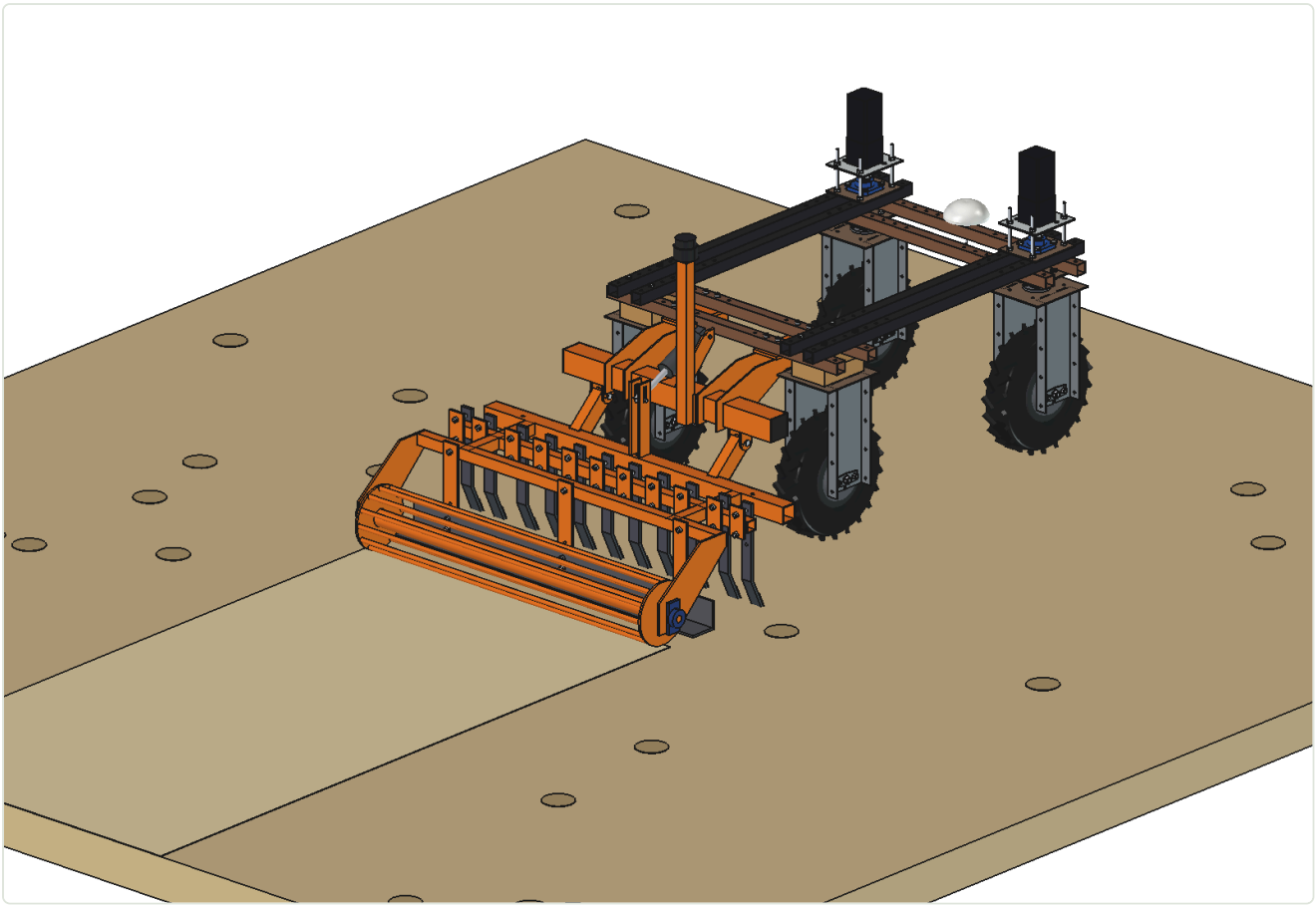


Simple

1 variant: Simple
Folder: `arena leveller/`

RIDING ARENA LEVELLER · VARIANT 1 OF 1

Simple



Grassland harrow mount, lift and frame (imported) with 11 rigid tines (35 mm deep, transport holes), a 1300 mm levelling board and a rod roller O220 that carries the frame and sets the depth; a 360 degree lidar on a post navigates on the arena walls.

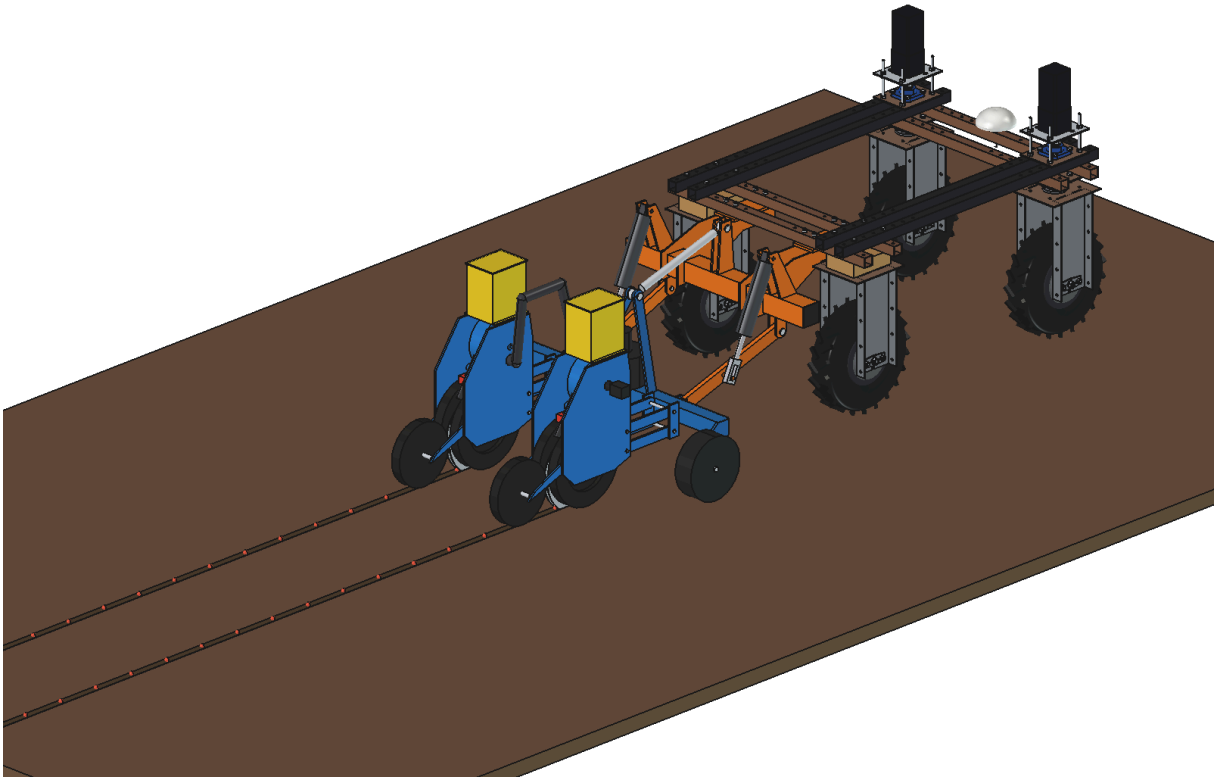
- 20 x 40 arena in ~17 min
- ~660 N draft, 44 kg front ballast for grip
- 158 kg, stays on its roller

Folder: `arena leveller/simple/`

IMPLEMENT · SOWING & PLANTING

Precision seeder

Sows row crops seed by seed and logs the RTK position of every seed, so the weeders can work up to the plant before it is visible. Two vacuum seeding units in the three-point hitch.

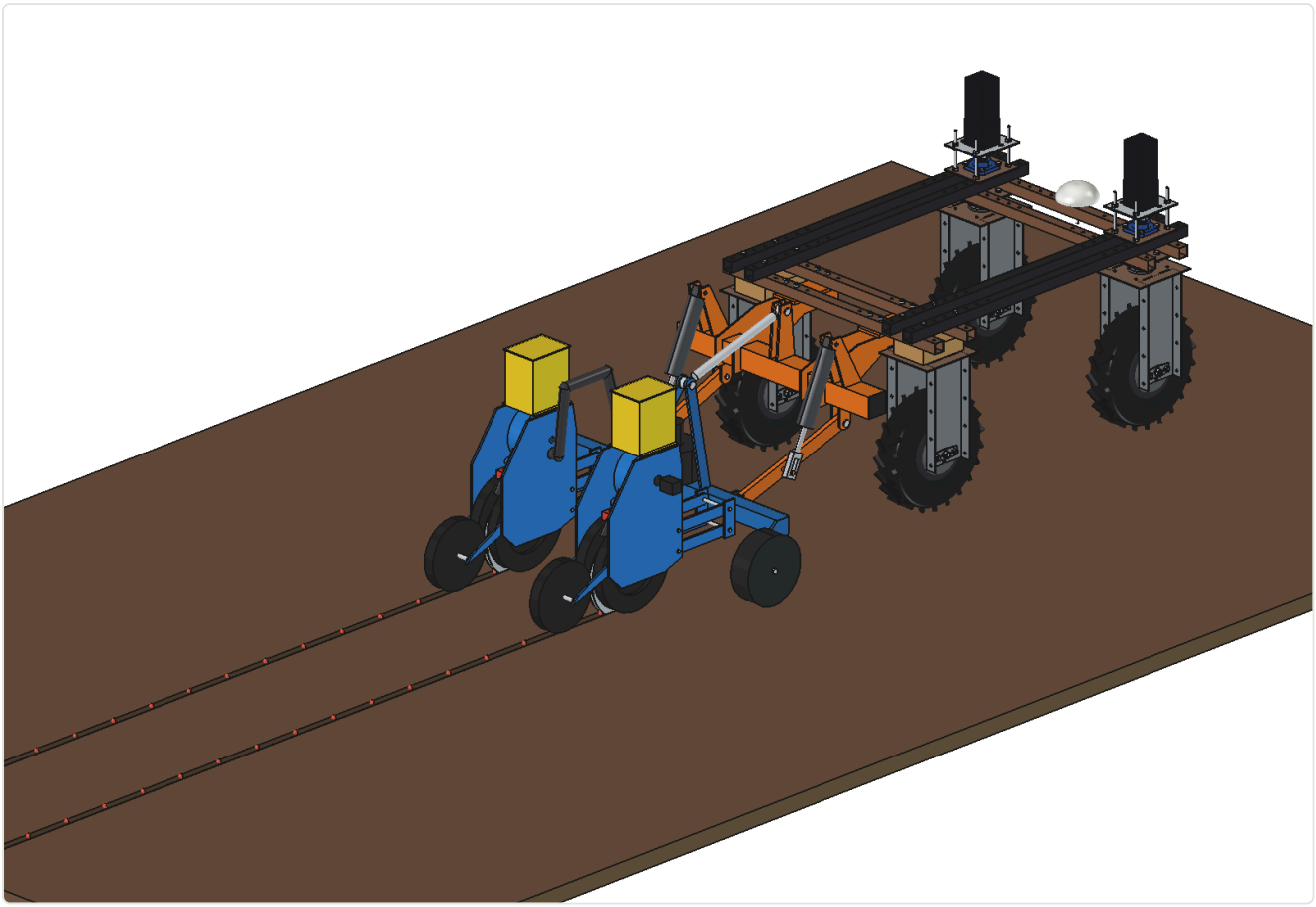


Simple

1 variant: Simple
Folder: `precision seeder/`

PRECISION SEEDER · VARIANT 1 OF 1

Simple



The three-point hitch and its toolbar frame (imported) with 2 units on parallelograms for rows at 500 mm: 4 l hopper, vacuum disc on its own stepper, seed tube with an optical seed sensor, double-disc opener between depth wheels, press wheel; a 24 V vacuum blower.

> ~13 mm (1 sigma) logged seed position, estimated

> 0.23 ha/h, ~0.9 ha per charge

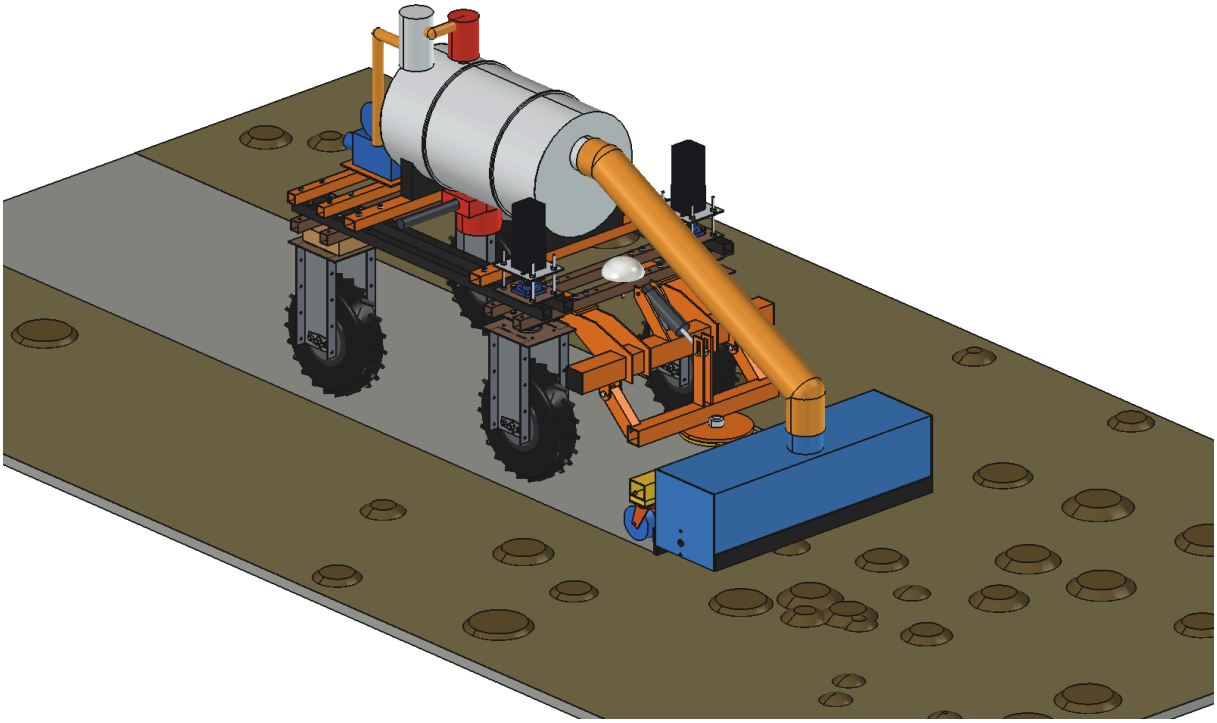
> 107 kg with the hitch

Folder: `precision seeder/simple/`

IMPLEMENT · LIVESTOCK & FARM

Manure vacuum collector

Sucks fresh manure from a solid barn floor into a 130 l vacuum tank and empties it at the pit, instead of pushing it: the debris vacuum's nozzle with a squeegee lip, a vacuum pump and a dump valve.

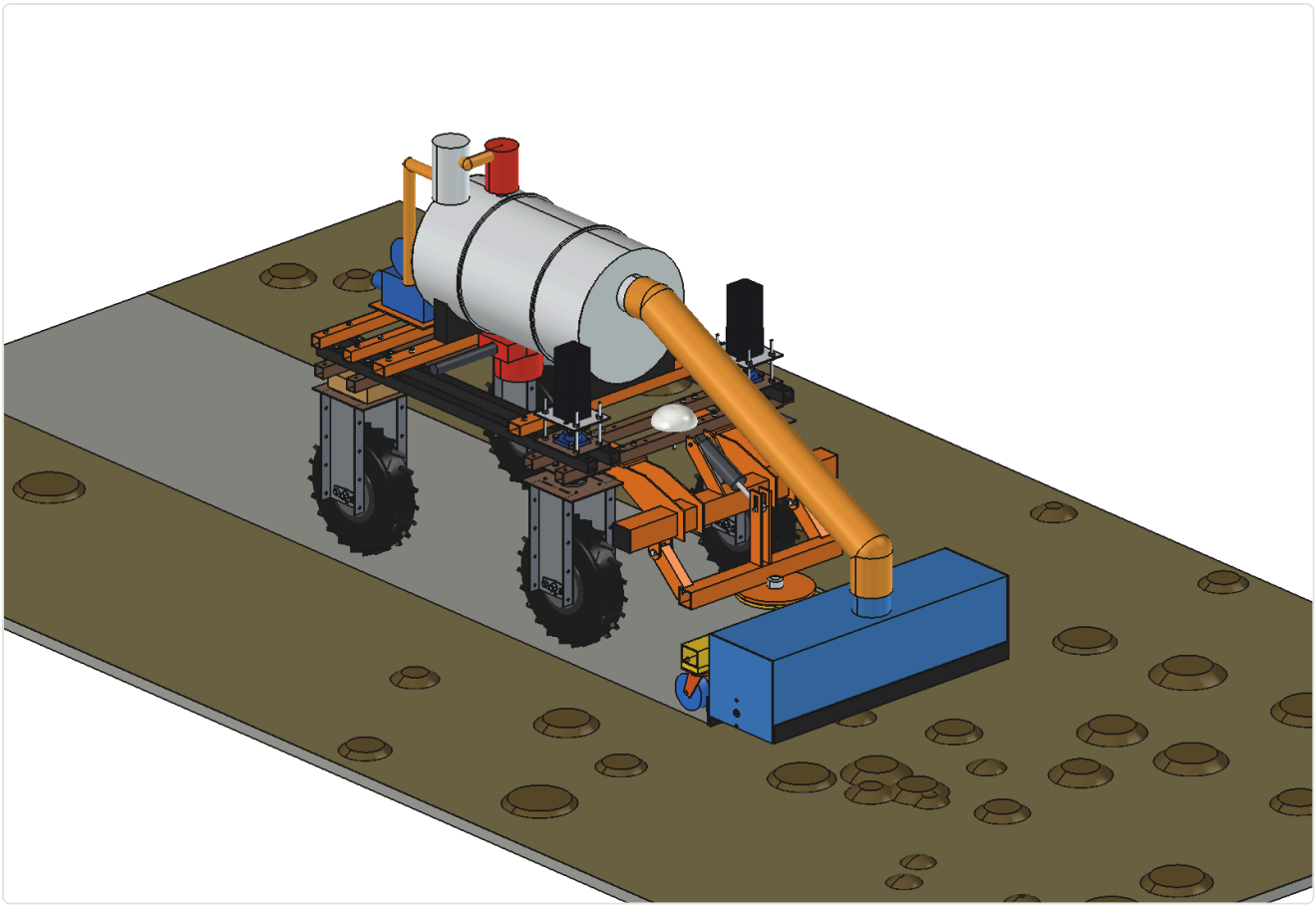


Simple

1 variant: Simple
Folder: `manure vacuum/`

MANURE VACUUM COLLECTOR · VARIANT 1 OF 1

Simple



Debris vacuum nozzle, lift and hose (imported) without the brush, a rubber squeegee lip at the rear; a stainless 130 l vacuum tank on the upper beams with float valve and moisture trap, a rotary-vane pump behind it and a knife-gate dump valve.

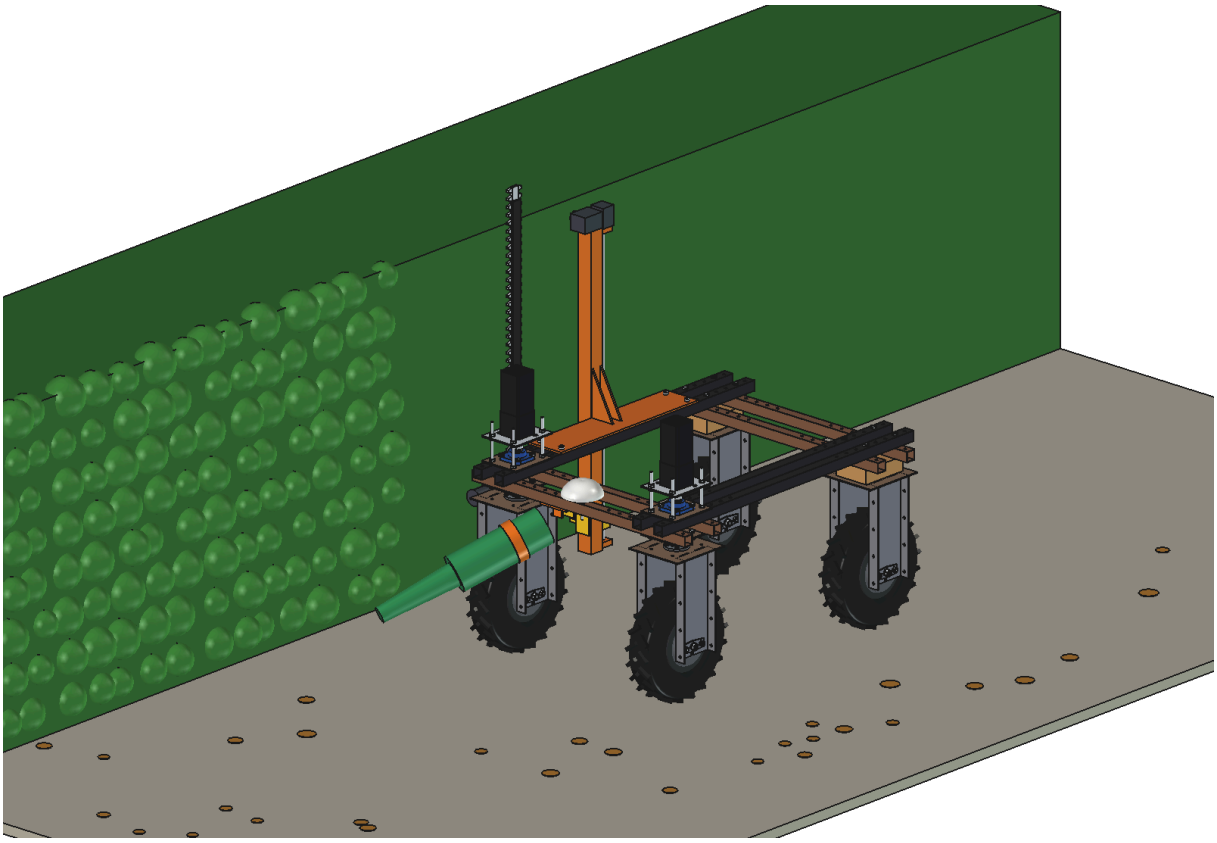
- ~87 m of alley per load
- 100 cows: ~17 loads, ~2.5 h a day
- 182 kg empty, 476 kg full

Folder: `manure vacuum/simple/`

IMPLEMENT · TRANSPORT & GENERAL

Hedge trimmer and leaf blower

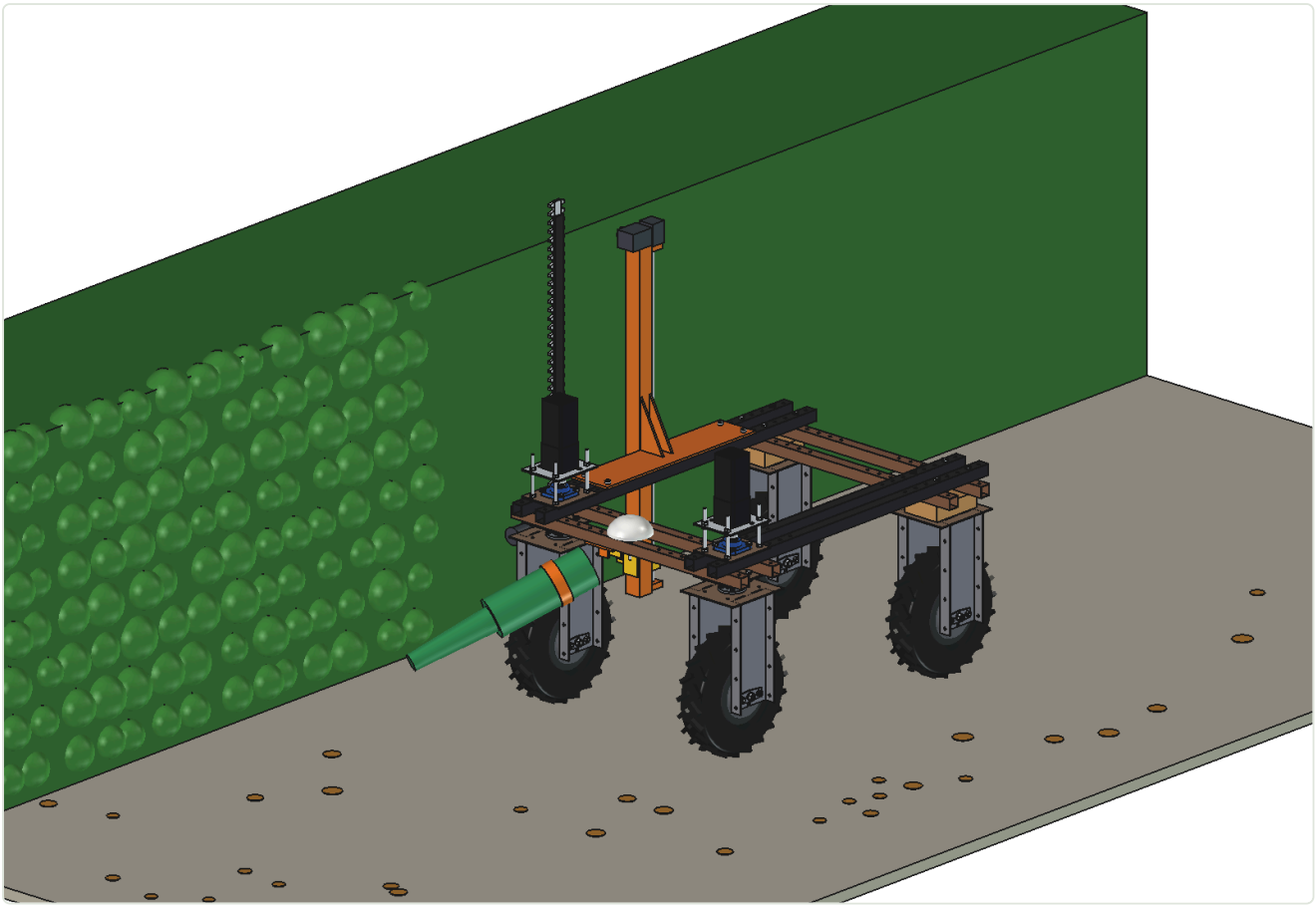
Trims hedges along a stored line and blows clippings and leaves off the path: a tilting double-sided trimmer bar on the fence inspector's mast and arm, a blower in front.



Simple

1 variant: Simple
Folder: `landscaping tools/`

Simple



Fence inspector mast, carriage and spring-hinged arm (imported) with a tilt gearmotor and a 1000 mm double-sided trimmer bar (0 = side cut, 90 = top cut, 45 = chamfer), a ToF sensor for the 60 mm gap, and a 48 V axial blower under the front beam.

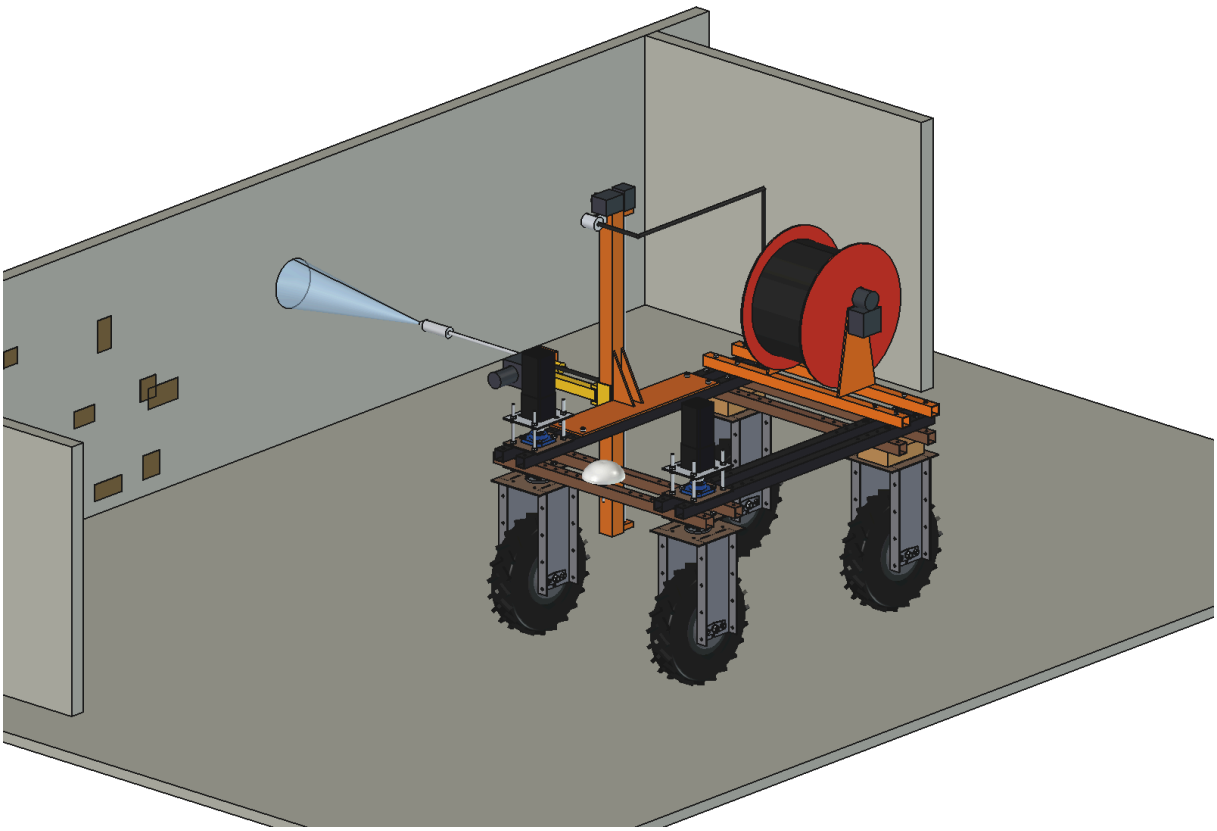
- Side cut 0.39-2.37 m, top cut up to 1.26 m
- ~180 m of hedge per hour
- 36 kg

Folder: `landscaping tools/simple/`

IMPLEMENT · LIVESTOCK & FARM

Barn pressure washer

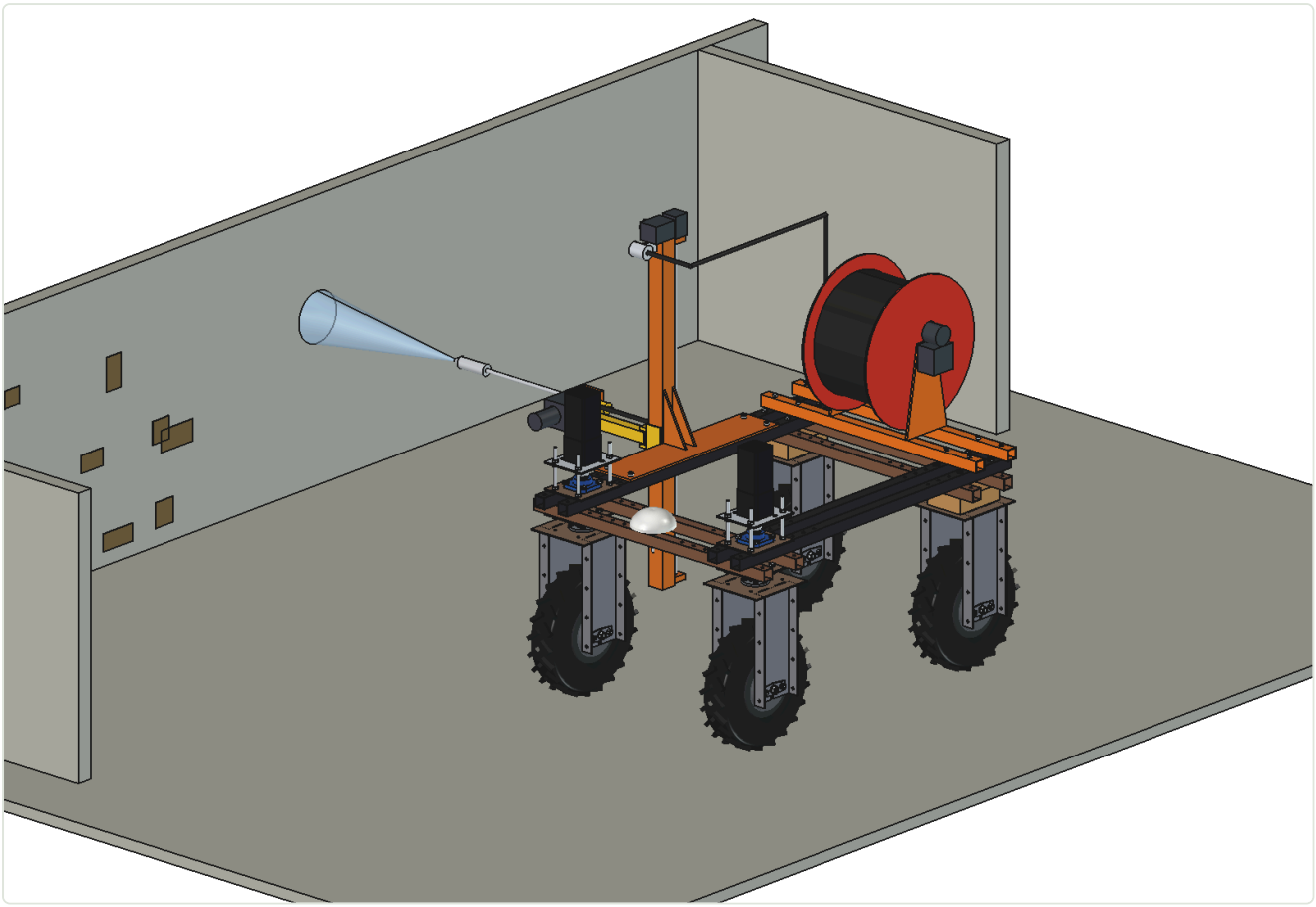
Cleans pens, walls and floors with a high-pressure turbo nozzle when the barn is empty. The HP water comes from the barn's stationary washer through 30 m of hose on a reel on the robot; the robot only aims.



Simple

1 variant: Simple
Folder: `barn pressure washer/`

Simple



Fence inspector mast and arm with the hedge trimmer's tilt head (imported) and a 400 mm lance with a rotating turbo nozzle; a powered HP hose reel (30 m, slip clutch, rotary joint) at the rear of the upper beams.

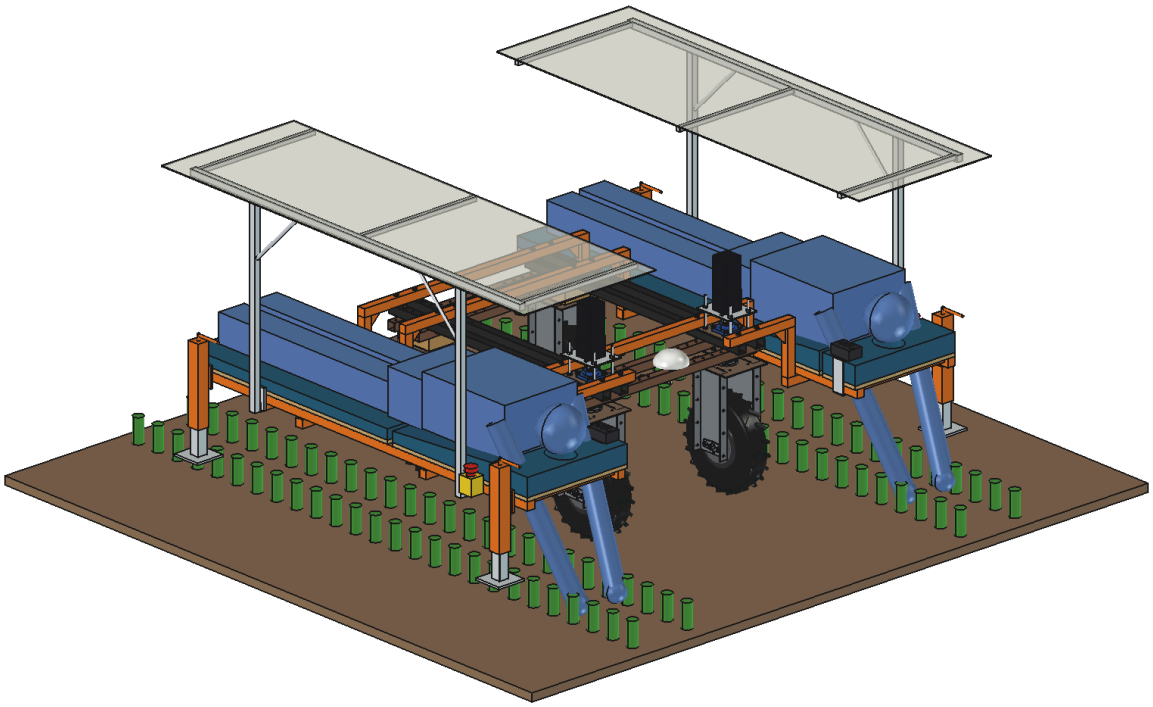
- 150 bar, 15 l/min from the barn's washer
- ~41 min per 3 x 4 m pen
- 69 kg, ~18 h per charge

Folder: `barn pressure washer/simple/`

IMPLEMENT · TRANSPORT & GENERAL

Weeding bed

A lying platform for hand weeding or harvesting: two workers lie face down on padded boards beside the robot, one on each side, and weed the row under them while the robot creeps along with hold-to-run.

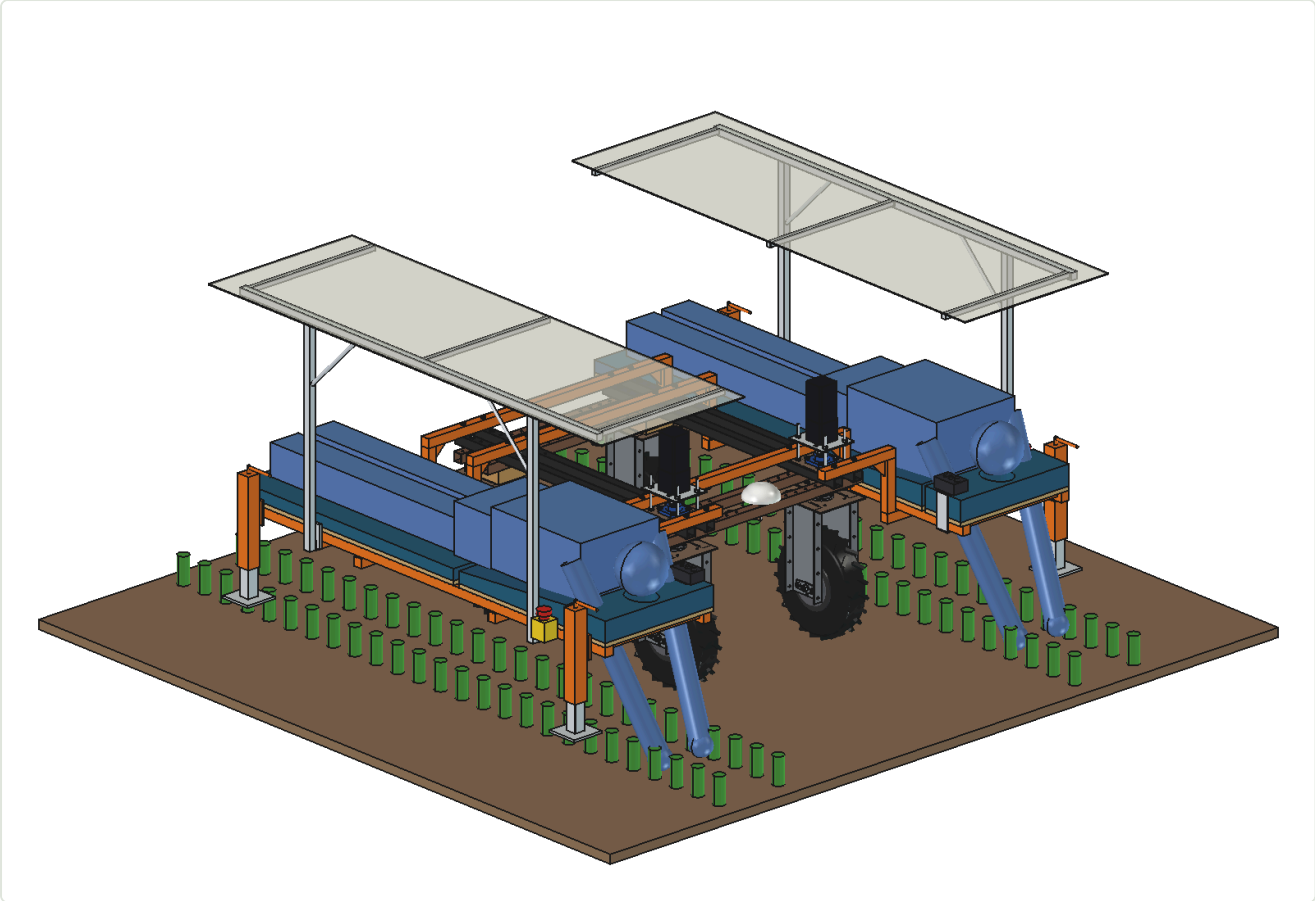


Simple

1 variant: Simple
Folder: `weeding bed/`

WEEDING BED · VARIANT 1 OF 1

Simple



4 cross tubes on the upper beams (stubs at the GNSS dome) ending in C-frames that carry a bed per side outside the steering sweep: leg board, chest wedge and face rest on plywood and foam, a hold-to-run pendant and an e-stop per worker, drop-leg parking stands for getting on, and a sun canopy per side.

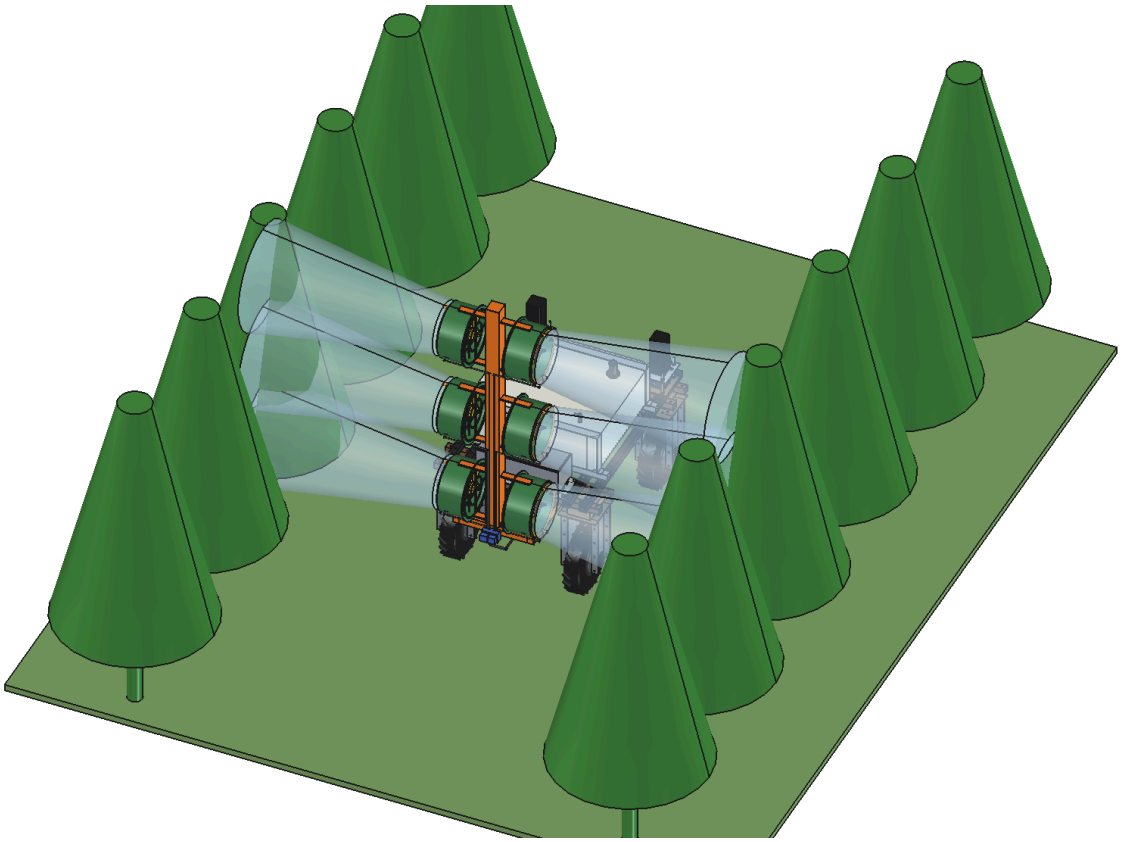
- 2 workers, 100-400 m2/h at 100-400 m/h
- ~160 W, ~10 h per charge
- 178 kg; stands down for getting on (factor 1.44)

Folder: `weeding bed/simple/`

IMPLEMENT · SPRAYING, FERTILIZING & IRRIGATION

Air-assisted orchard sprayer

Sprays low fruit trees, vines and hedges on both sides of the alley: 3 fans per side carry the droplets into the canopy, and a lidar per side switches each height section on and off.

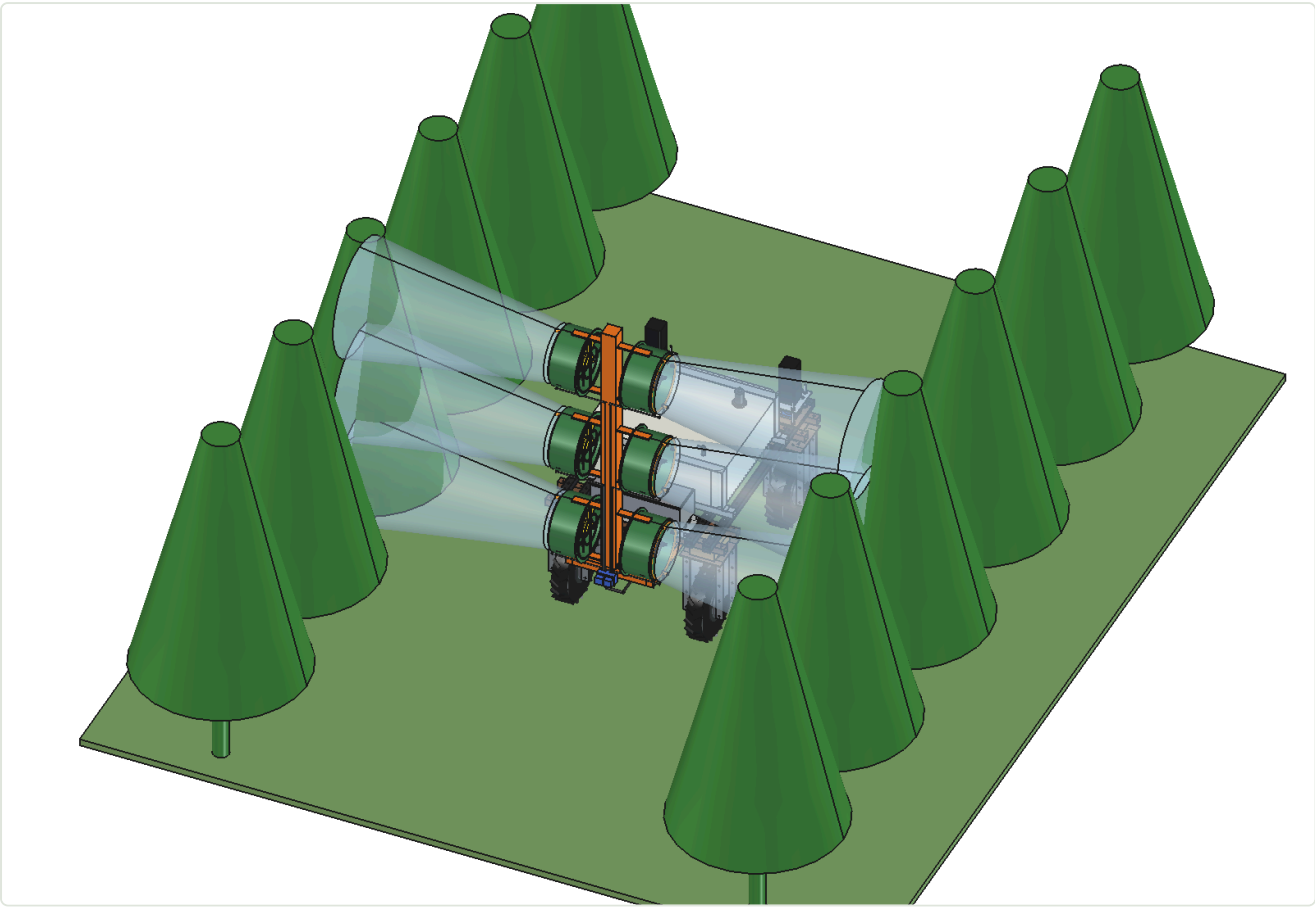


Simple

1 variant: Simple
Folder: `orchard air sprayer/`

AIR-ASSISTED ORCHARD SPRAYER · VARIANT 1 OF 1

Simple



The spot sprayer's 120 l tank, pump and control box (imported) with a tower behind the robot: arms under the rear lower beams, a mast with 6 axial fan pods O300 (48 V) at 0.8, 1.35 and 1.9 m blowing sideways, a nozzle ring with 2 hollow-cone nozzles and a section valve per pod, and a 2D lidar per side 225 mm ahead of the nozzles.

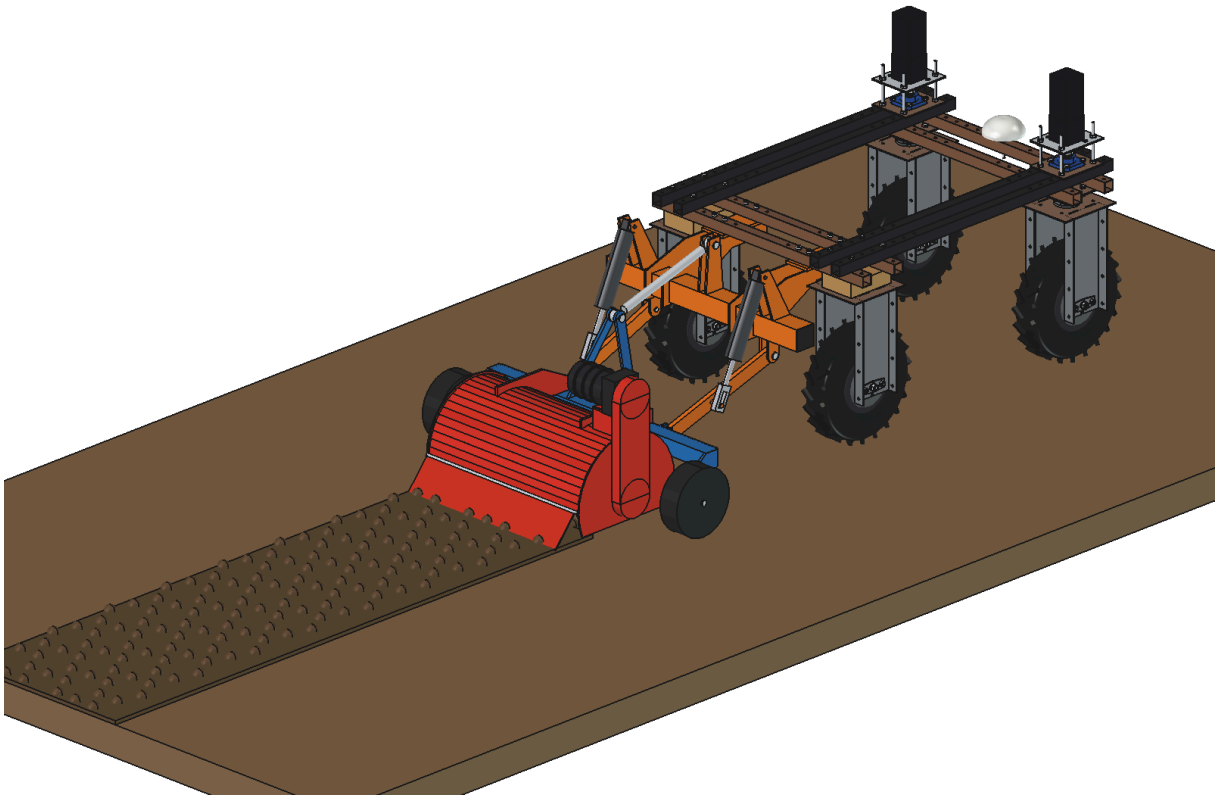
- 5100 m3/h per side, up to 0.63 m/s
- 317 l/ha, 0.68 ha/h, 48 min per tank
- ~1.7 kW: ~0.9 h per charge; 105 kg empty

Folder: `orchard air sprayer/simple/`

IMPLEMENT · CROP & SOIL

Power harrow / rotary tiller

A narrow rotary tiller for seedbeds in strips and loosening between rows, in the three-point hitch. The rotor turns forward (down-cut), so it pushes the light robot instead of pulling it back.

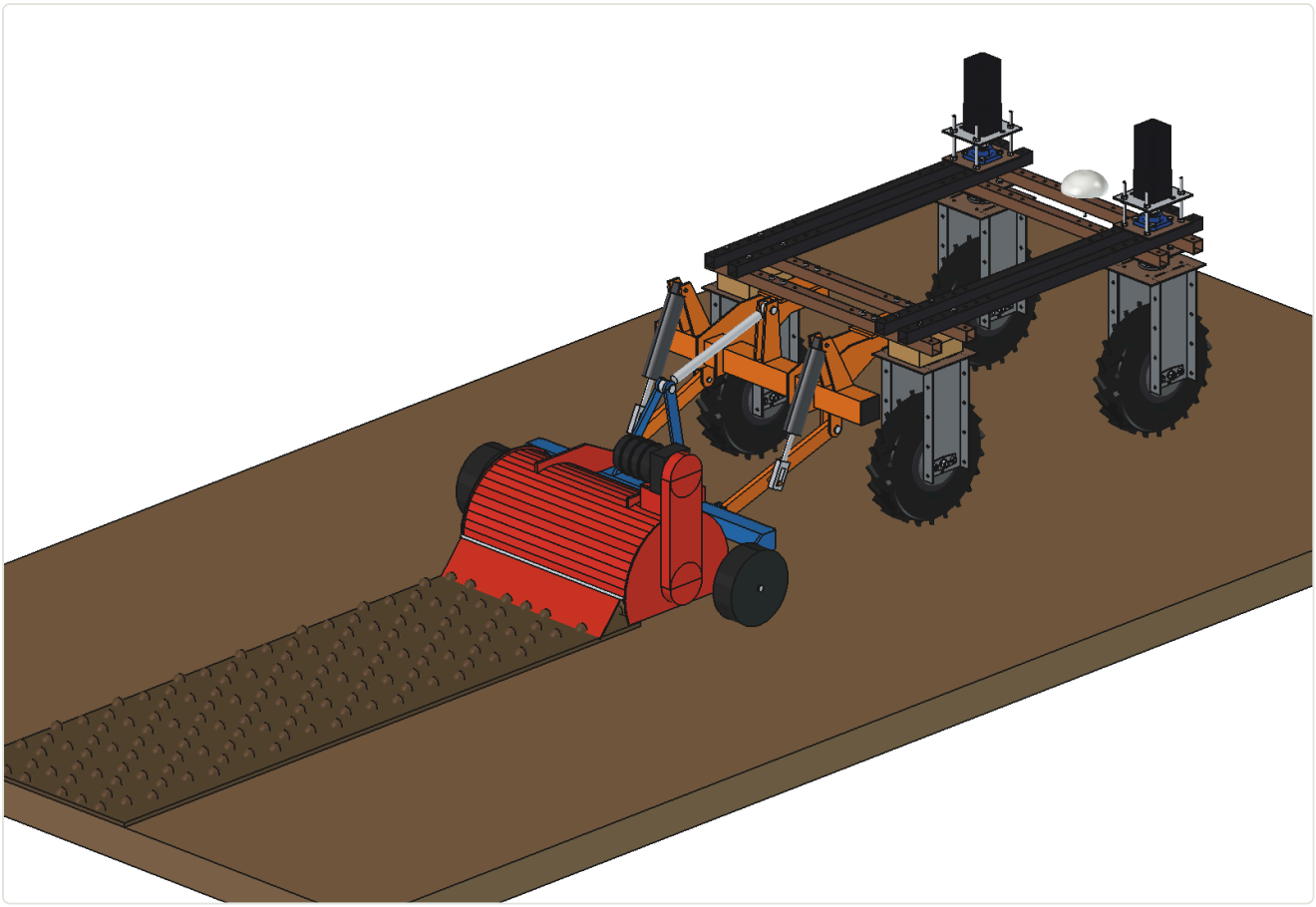


Simple

1 variant: Simple
Folder: `power harrow/`

POWER HARROW / ROTARY TILLER · VARIANT 1 OF 1

Simple



The three-point hitch's mount, links, toolbar frame and gauge wheels (imported) with a 700 mm rotor (16 L-blades, O350, 250 rpm, 60 mm deep) under a hood with side skids, a rubber front skirt and a levelling flap, driven by a 48 V 1.5 kW gearmotor through a chain case.

> ~0.6 kW at the rotor, 250-500 m2/h

> ~1.6 h, ~400 m2 per charge

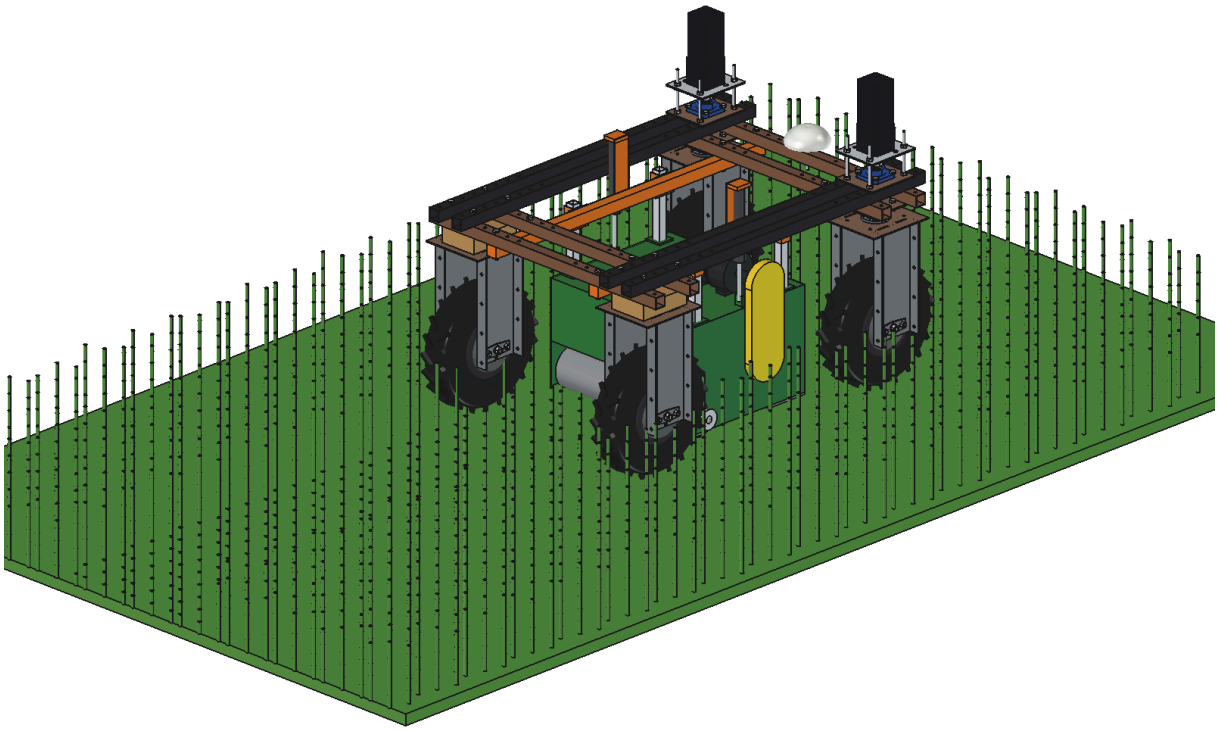
> 94 kg; lifting needs 92 kg front ballast

Folder: `power harrow/simple/`

IMPLEMENT · CROP & SOIL

Flail mulcher

Shreds tough grass, stubble, residues and thin prunings between the rows. Hung under the belly between the wheels, so its weight stays between the axles and no ballast is needed.

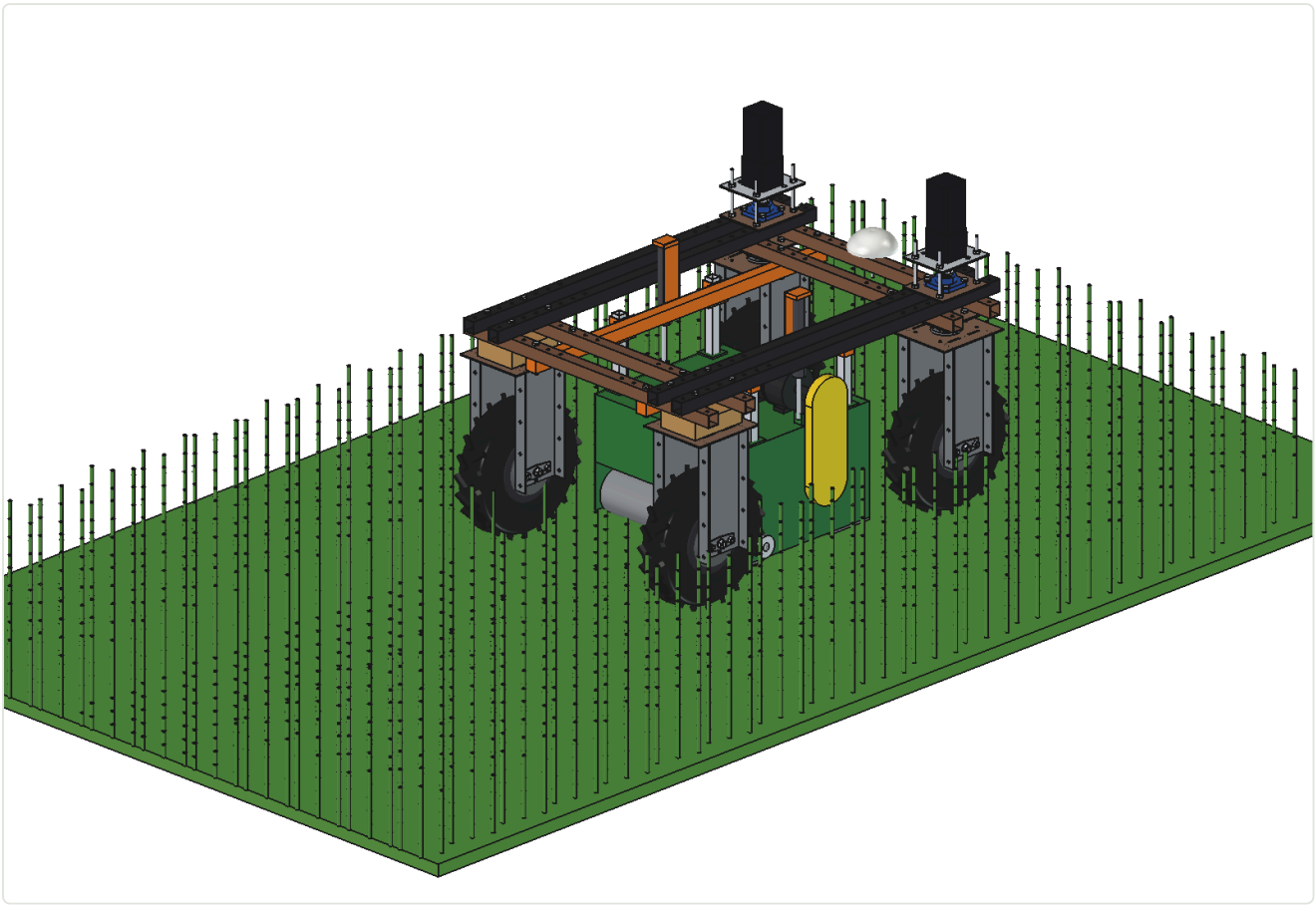


Simple

1 variant: Simple
Folder: `flail mulcher/`

FLAIL MULCHER · VARIANT 1 OF 1

Simple



Belly rails under the lower beams, a closed housing (x +-300, y +-262) with a 548 mm flail rotor (72 Y-knives, O320, 2700 rpm), a shear bar, a rear roller for the 40 mm cutting height, a 48 V 2 kW motor with a poly-V belt, and 2 actuators on guide posts that lift it 100 mm.

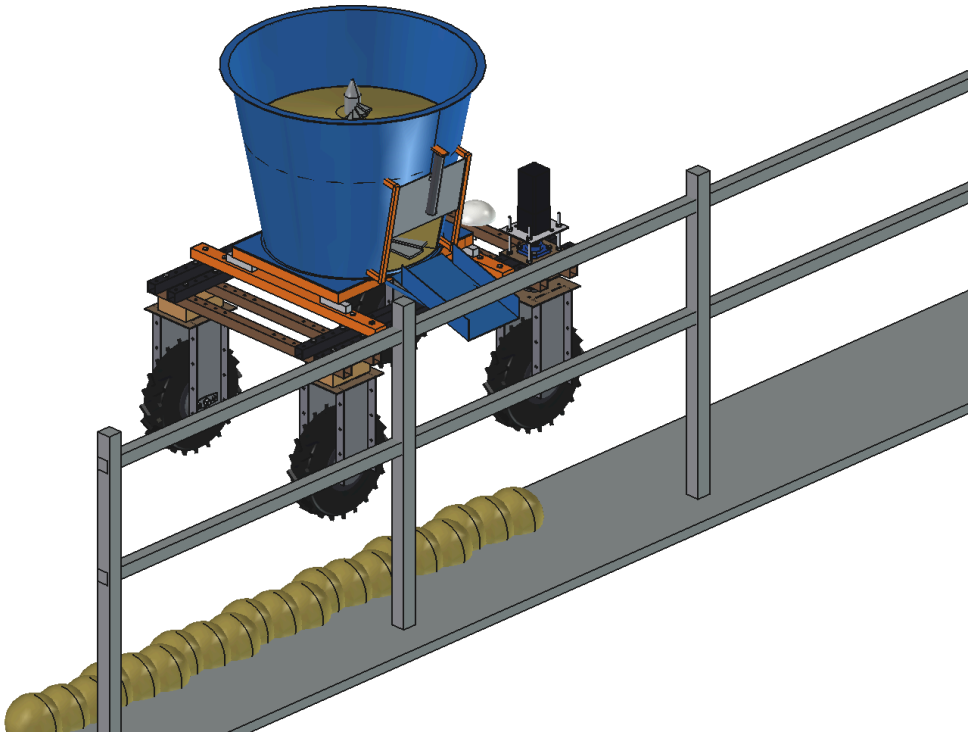
- 45 m/s tips, 360 cuts/m at 0.5 m/s
- 116 kg, front share 56 %
- Grass: ~1580 m2/h, ~1.5 h per charge

Folder: `flail mulcher/simple/`

IMPLEMENT · LIVESTOCK & FARM

Feed mixing and distribution wagon

A small vertical mixer that is loaded at a feed kitchen, weighs, mixes and lays the ration along the feed fence several times a day; sized for calves, young stock and small groups.

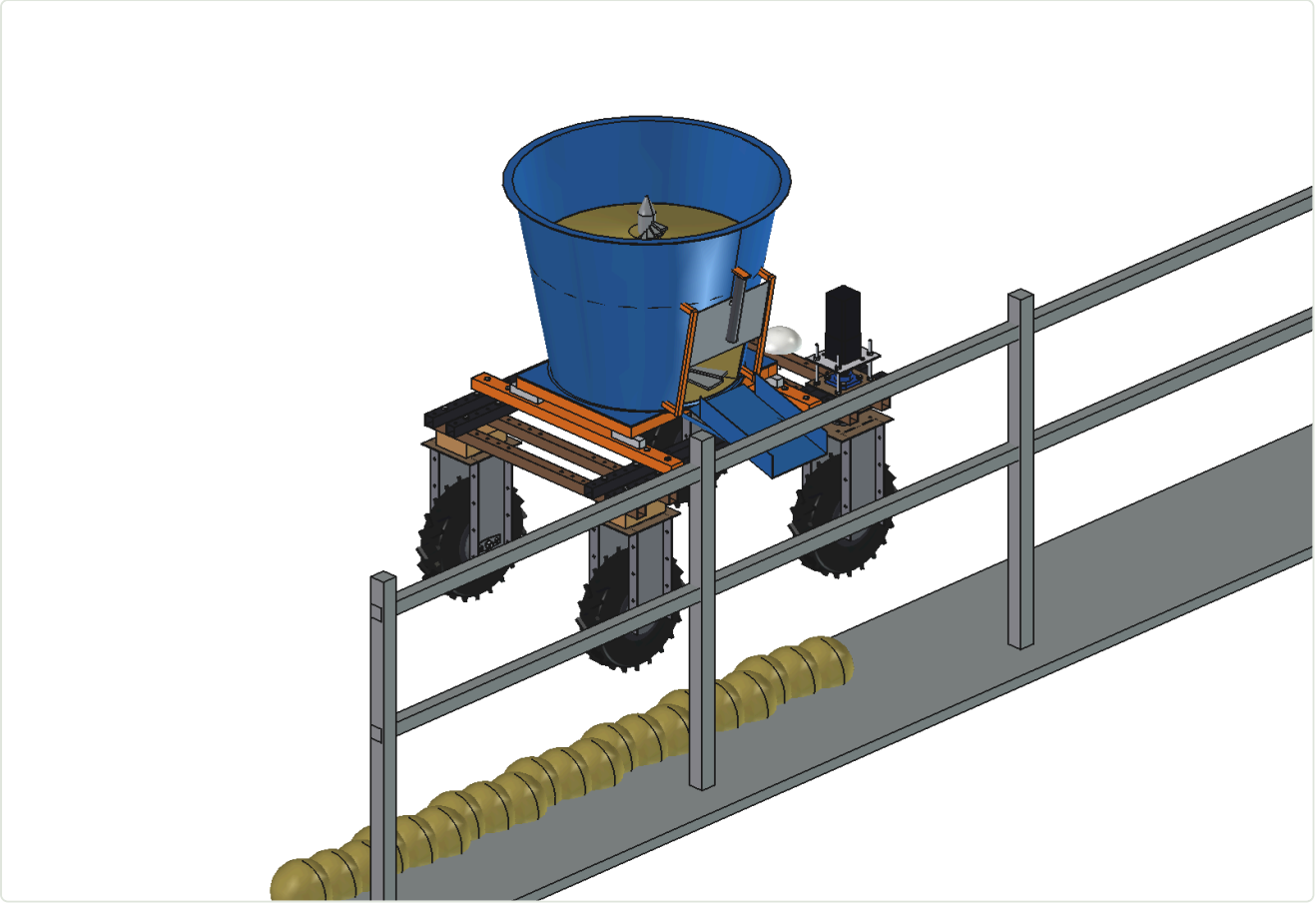


Simple

1 variant: Simple
Folder: `feed mixer wagon/`

FEED MIXING AND DISTRIBUTION WAGON · VARIANT 1 OF 1

Simple



A 0.33 m3 cone bin on 4 load cells and a sub-frame on 2 cross tubes, a vertical conical auger with 2 knives (30 rpm, planetary gearbox, 48 V 2 kW motor under the floor), a door that slides up along the cone wall and a chute to the right.

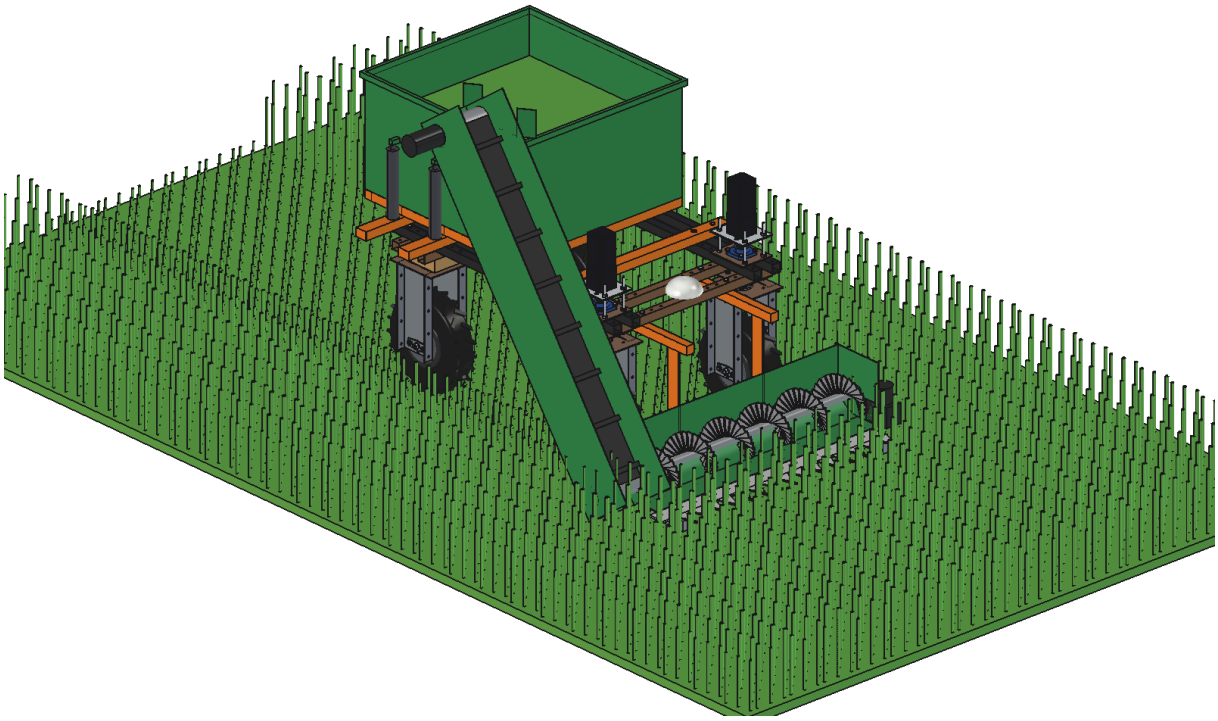
- ~92 kg per batch, ~250 Wh
- ~6 batches per charge: ~74 calves a day
- 136 kg; full 390 kg, front share 51 %

Folder: `feed mixer wagon/simple/`

IMPLEMENT · LIVESTOCK & FARM

Fresh grass harvester (zero grazing)

Mows fresh grass, lifts it into a tipping container and brings it to the feed fence, many small loads a day.

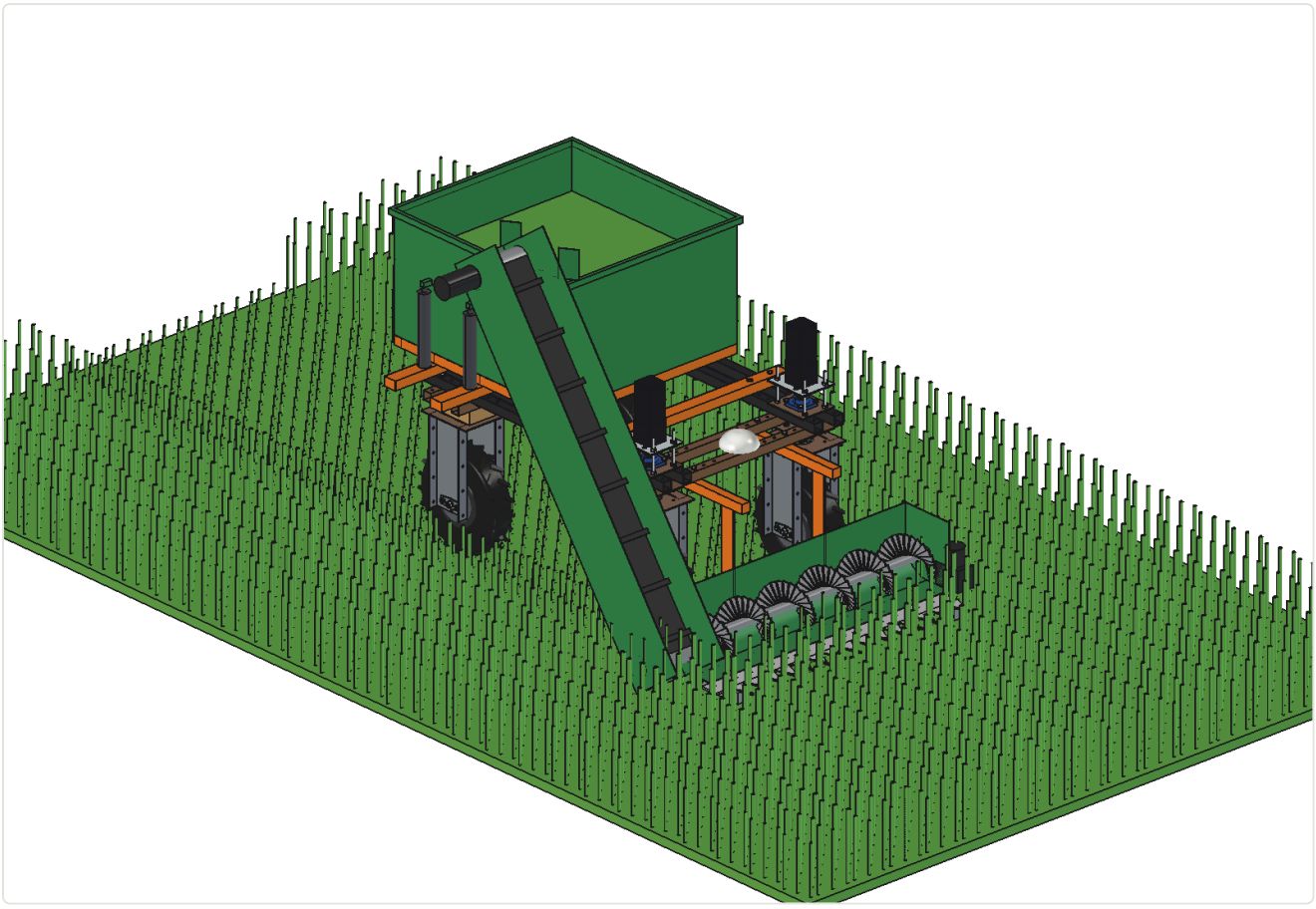


Simple

1 variant: Simple
Folder: `fresh grass harvester/`

FRESH GRASS HARVESTER (ZERO GRAZING) · VARIANT 1 OF 1

Simple



Front arms under the lower beams with a 1.2 m sickle bar and a cross auger, a slatted elevator outside the right front wheel up to 1.42 m, a 0.41 m3 container that tips 55 degrees to the left, and 60 kg rear ballast.

> ~73 kg per load, ~11 min per trip

> ~1.3 t per charge: ~21 cows

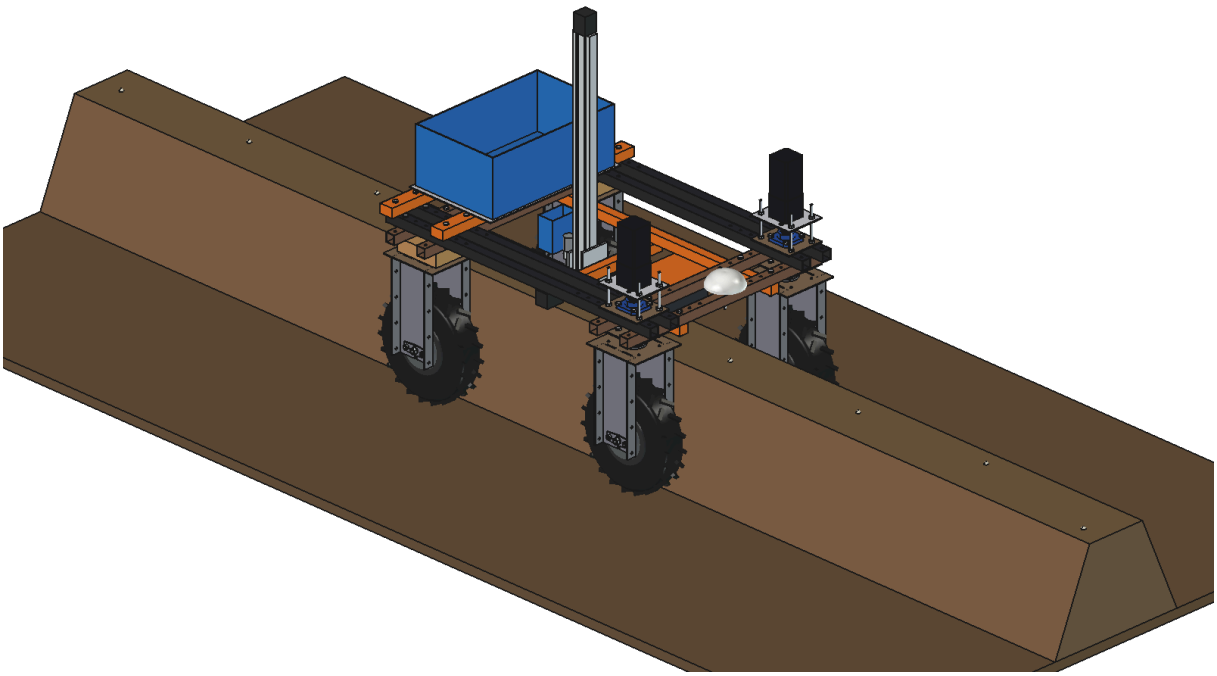
> 286 kg; front share 57 % empty, 51 % full

Folder: `fresh grass harvester/simple/`

IMPLEMENT · TRANSPORT & GENERAL

Asparagus harvester

Straddles one ridge of white asparagus, finds spear tips cracking the ridge top with a camera, cuts each spear 220 mm deep and lifts it out on an X-Z gantry under the belly.

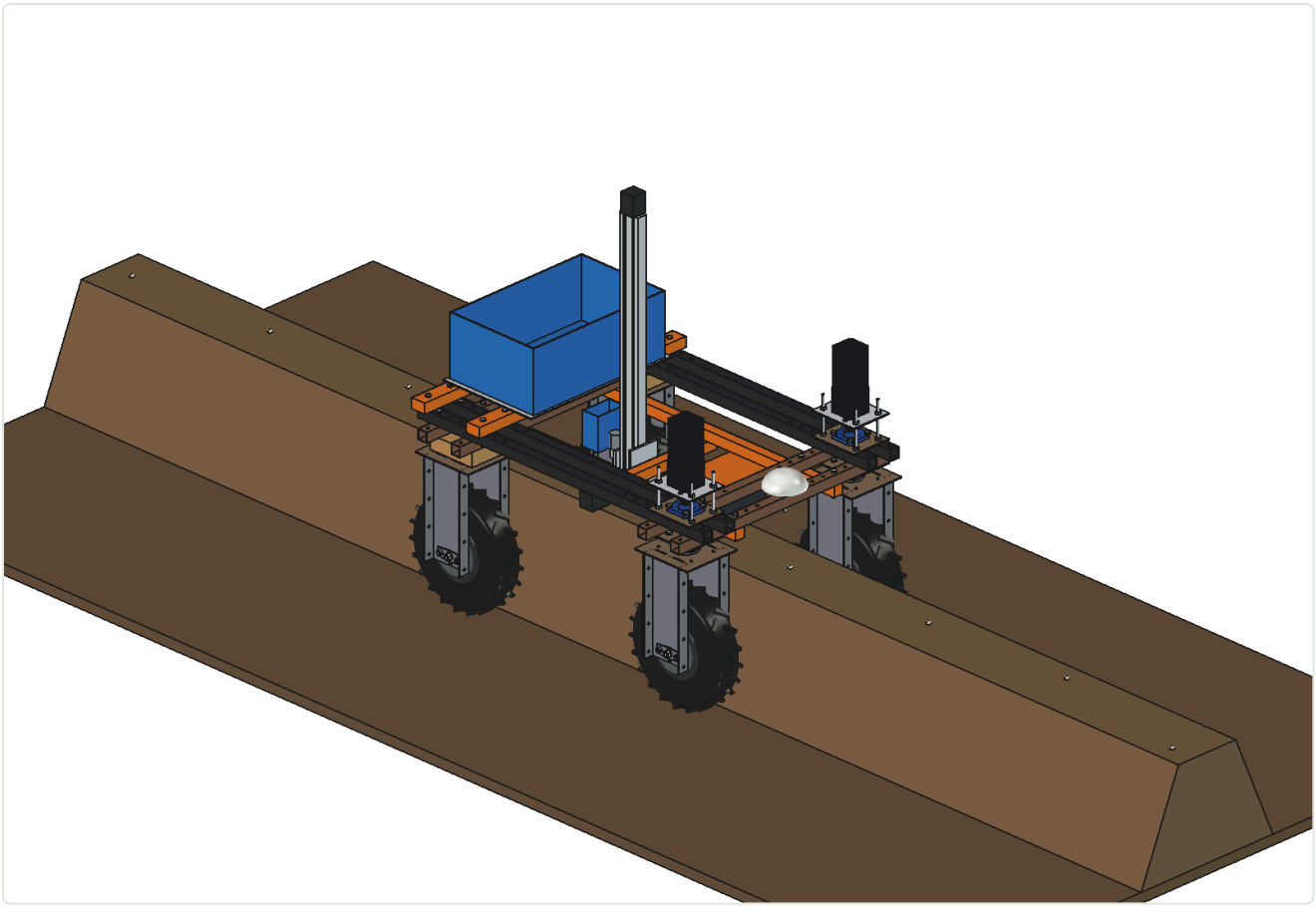


Simple

1 variant: Simple
Folder: `asparagus harvester/`

ASPARAGUS HARVESTER · VARIANT 1 OF 1

Simple



Belly rails (from the flail mulcher) with an X gantry (+-120) and a tall Z axis through the robot at y 0, a knife tube O30 with gripper fingers (tip 180-1000), a spear bin on the slide, a crate on a rear deck, and a camera hood with LEDs and a line laser at the front.

> ~3.5 s per spear

> ~270 m of ridge, ~20 kg per hour

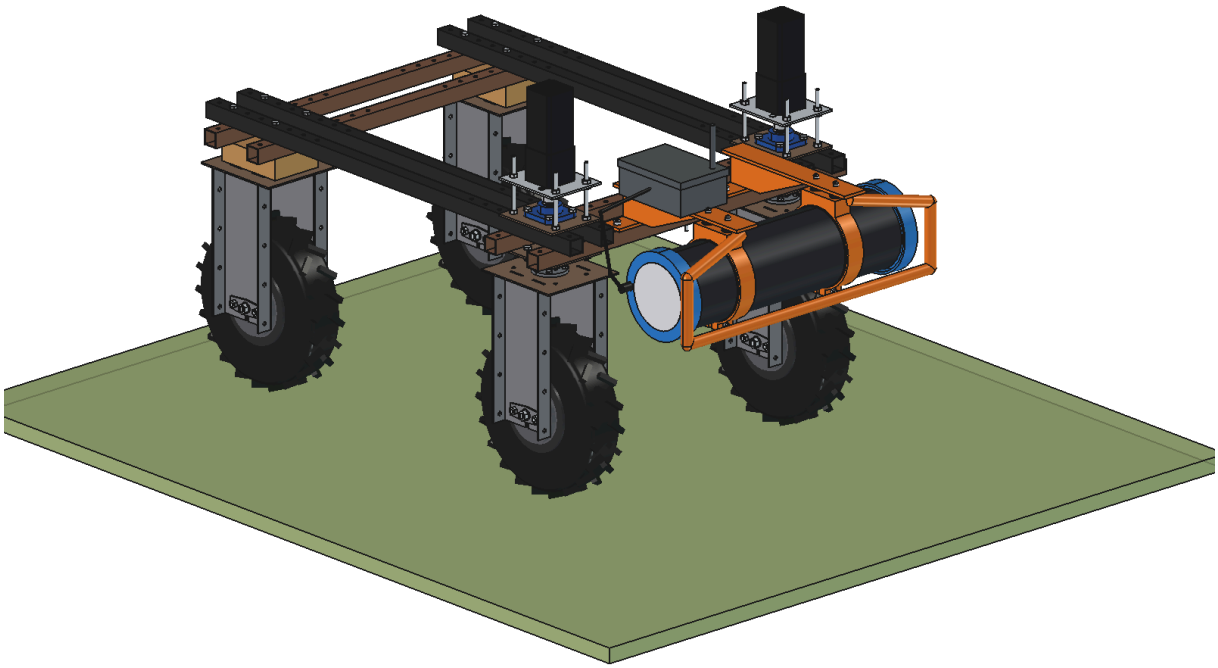
> 44 kg, ~6.5 h per charge

Folder: `asparagus harvester/simple/`

IMPLEMENT · MEASURING & SENSORS

Soil scanner

Passive gamma-ray soil scanner (as Interra Scan / SoilOptix) 0.6 m above the ground in front of the robot: counts Cs-137, U-238, K-40 and Th-232 from the topsoil in lanes of 10 m, ~830 points per ha; with calibration samples it maps texture, pH, organic matter, P, K and CEC.

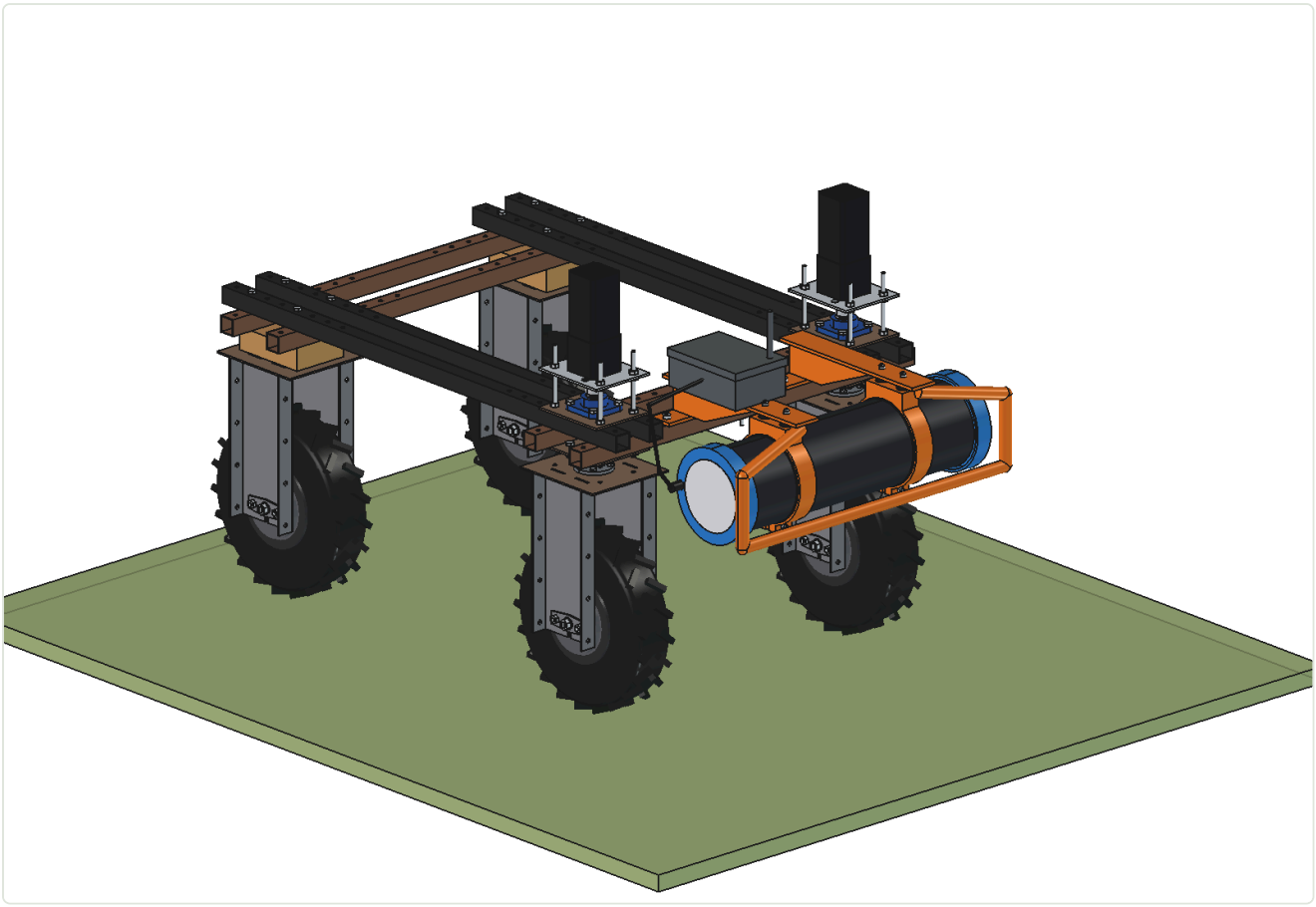


Simple

1 variant: Simple
Folder: `soil scanner/`

SOIL SCANNER · VARIANT 1 OF 1

Simple



CsI(Tl) crystal 4 l with PMT and MCA in a sealed aluminium tube O200 with blue end caps, across the robot at y 880, in 2 clamp rings on 4 rubber mounts under 2 steel brackets on the front outer lower beam, a guard tube and a processor box; optional soil sampler in the middle for the calibration samples (animated).

- 4.2 ha/h, ~36 ha per charge
- 90 % of the counts from within 4.6 m
- 45 kg; 69 % on the front axle

Folder: `soil scanner/simple/`

IMPLEMENT IDEAS

Ideas not worked out yet

0 ideas (from the website's list, own brainstorming and the robot catalogue) that have no model yet. Each has a folder in `implement ideas/` with a `README.md` describing what it does, so it can be worked out later.